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# 開放系と宇宙論の講義

マスター方程式、Schwinger-Keldysh 形式、開放系の有効場理論

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要旨：開放系は物理学の至るところに現れる。現実の多くの系は、少なくとも弱く、環境の自由度と相互作用している。これらの自由度は、あまりにも多数であったり、複雑であったり、アクセスできなかったり、未知であったりする。自由度の一部しか観測しない場合、その縮約された動力学は閉鎖系の動力学と質的に異なり、散逸、ノイズ、デコヒーレンス、記憶効果、あるいは観測しないセクターへの情報の流出を示しうる。微視的な記述も未知である場合には、関心のある自由度を系統的に扱う一方、環境と微視的構造は陽に解く代わりにパラメータ化する、開放系の有効記述へと導かれる。

この見方は重力と宇宙論においてとりわけ重要である。インフレーション、暗黒物質、暗黒エネルギーを含む宇宙論の主要な未解決問題には、時空を満たすセクターが関わっており、その微視的な正体は未知で、観測される効果は主として重力的である。同時に、重力系は時間依存性をもつため、保存エネルギーの優先的な定義が存在しないことも多く、非平衡の動力学が自然に現れる。本講義ノートでは、開放系を研究するための道具として演算子形式と Schwinger-Keldysh 経路積分を導入し、特に開放系の有効場理論とインフレーションに重点を置く。対象読者は、これらの話題に初めて取り組む修士課程の学生、博士課程の学生、および研究者である。

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## 序論

開放系の理論は、あらゆる物理学者が標準的に身につける道具の一部であるべきだ。現実の系で完全に孤立したものはない。どの系も、少なくとも弱く、自由度があまりにも多数であったり、複雑であったり、アクセスできなかったり、単に未知であったりする環境と相互作用する。これはきわめて一般的な状況であり、開放系の手法は、量子力学と量子情報から、物性物理学、統計物理学、素粒子物理学、宇宙論、一般相対論まで、事実上あらゆる物理学の分野に応用される。本ノートの目的は、ある部分系が相互作用するすべての対象の完全な微視的状态を追跡せずに、その部分系の動力学を記述するための基本的な道具をいくつか導入することである。

**重力と宇宙論への開放系からのアプローチ。** 開放系は宇宙論において特に自然に現れる。宇宙論の主要な未解決問題は、おおむねインフレーション、暗黒物質、暗黒エネルギーの問題に要約できる。数十年にわたり精力的に研究されてきたこれらの問題には、際立った共通点がある。第一に、いずれも時空を満たす成分、セクター、あるいは有効媒質に関わる。インフレーションとは、高温ビッグバン以前の宇宙を満たし、初期の加速膨張期を引き起こした物理に与えた名前であり、暗黒エネルギーは現在の加速膨張を担う成分を指す。暗黒物質も、晩期には完全に一様ではなく集積しているものの、時空を満たす成分であり、その存在はほぼ全面的に重力効果を通じて推測される。このため、これらの問題を、孤立して準備・研究できる真空上に粒子が伝播すると考える通常の素粒子物理の問題として捉えるのは自然ではない。むしろ、関心のある動力学、この場合は重力の動力学が媒質の内部で起こる、物性物理の問題に精神的に近い。

第二に、これらすべての場合で微視的な物理が未知である。これは、微視的な法則は分かっている、そこから生じる動力学を実際に解くのが困難という、QCD や複雑な物質でおなじみの状況にとどまらない。宇宙論では、微視的自由度そのものが全く分からないことが多い。インフレーション、暗黒物質、暗黒エネルギーのモデルは多数あるにもかかわらず、これらが実際に何なのかは分かっていない。このことは、有効場理論の立場をほぼ必然的に要請する。すなわち、微視的物理の詳細については可能な限り特定の仮定を置かず、関係する自由度と、それらを制約する対称性を同定するのである。

第三に、現在の観測から分かる限り、これらのセクターは可視世界と主として、あるいは重力のみを通じて相互作用する。例えば、再加熱が起こるためにはインフラトンが最終的に標準模型と相互作用するはずだと考えられるが、その相互作用を直接観測することはできない。同様に、膨大な実験的努力にもかかわらず、暗黒物質と標準模型粒子の間の非重力的相互作用を示す肯定的な証拠は、まだ得られていない。この意味で、宇宙論は自然に開放系の問題となる。重力の自由度、すなわち計量そのものがアクセス可能な系であり、宇宙を満たす未知の成分が環境として働く。より一般には、系と環境の具体的な分け方は問いに依存する。系は計量、曲率摂動、長波長モードの集合、あるいはその他のアクセス可能な自由度の集合でありうる。したがって、得られる記述は開放系の有効場理論となる。環境の微視的構造が未知であるという意味で有効理論であり、環境の自由度を無視したとき、観測する重力の自由度がユニタリーに発展するとは限らないという意味で開放系なのである。

開放系の考え方は、より広く一般相対論にも深く関わっている。重力はあらゆるものと相互作用するため、真に閉じた系の記述には、原理的には宇宙のすべての自由度を記述する必要があるが、それは実行可能でも有用でもない。さらに、物質を導入すると、重力は通常、時間依存する時空を生じさせることで応答する。そのような背景では時間的 Killing ベクトルが存在するとは限らず、したがって時間並進に対応する保存エネルギーも存在するとは限らない。これは、エネルギー保存により低エネルギーでは重い自由度が効率よく励起されないために、それらを積分消去できるという、最もなじみ深い Wilsonian な直観に対する重大な障害である。時間依存する重力背景では、この議論が成り立たないことがある。重い場が背景の発展によって励起され、その後、軽いセクターに対する環境自由度として働きうるからである。一般相対論には地平線もあり、系の一部へのアクセスに、場合によっては原理的にも、さらなる障害をもたらす。本ノートでは地平線に重点を置かないが、それらは重力物理学において開放量子系の手法を発展させる重要な追加の

動機となる。

密接に関連する論点は、開放系が一般に非平衡であることだ。背景の時間依存性が小さな摂動ではなく、まさに記述したい現象そのものである宇宙論では、これは本質的である。平衡熱力学には膨大な関心が寄せられてきており、多くの状況で適切な言葉を提供する。しかし、本ノートの動機となる宇宙論的状况は、通常、平衡状態ではない。そのため、明示的に断らない限り熱平衡への特殊化を避け、一般的な非平衡過程を記述できる柔軟性を形式論に持たせる。この点は重力物理学でも重要である。時空が時間的 Killing ベクトルをもつ場合、あるいは適切な解析接続を定義できるだけの対称性をもつ場合には、ブラックホールや適切な de Sitter パッチでのように、しばしば Euclid 幾何へ Wick 回転し、熱的な解釈を引き出せる。しかし、現実的な重力や宇宙論の多くの問題には、それほど対称性はない。それらは Lorentz 的な非平衡の動力学的問題として扱うべきである。そもそも、宇宙史を通じた構造形成が明らかに示すように、私たちの宇宙は平衡からはるかに遠い。

**ねらいと目標。** 本講義ノートの主な目的は、しばしば別々に論じられる三つの主題、すなわち演算子形式における開放量子系、Schwinger–Keldysh 経路積分、宇宙論を結びつけることである。演算子形式と量子マスター方程式は、特に量子情報や量子光学において、開放量子系に関する多くの議論の自然な言葉となる。経路積分形式は原理的には等価だが、異なる研究共同体で用いられることが多く、場の量子論、ゲージ理論、重力には欠かせない。宇宙論は、量子場、時間依存する背景、重力の拘束条件、原始揺らぎに関する観測的な問いを組み合わせるため、さらに一層の構造を加える。この三つの主題は互いに多くを教え合える。ある領域の結果や直観を別の領域へ持ち込むことで進展が得られるが、必要な内容が統一した記法と共通の教育的水準で提示されることはまれであり、それは容易ではない。

本ノートが採る分野横断的な視点は、今日とりわけ価値がある。人工知能の急速な発展は、研究者の学び方と働き方を変えつつある。物理学者が分野の境界を越え、不慣れな道具を習得し、もともとの専門領域の外で着想を試すことが、ますます現実的になっている。最近利用可能になった AI の道具により、研究者は、不慣れな文献を読み進め、計算を確認し、隣接分野の言葉を習得するうえで、ますます強力な支援を得ている。このことは、学際的研究を恐れず、野心的な問いにより大胆に取り組むよう、私たちを後押しするはずだ。互いにほとんど交流しない、ますます専門化した道筋を進む代わりに、ある問題に本当に必要な道具を見定め、それらを効率よく学ぶことができる。本ノートはその精神で書かれた。網羅性を目指すのではなく、研究共同体の橋渡しをし、学生や研究者が演算子法、経路積分、宇宙論的应用の間を行き来しやすくすることを目指す。

本ノートは、開放量子系、Schwinger–Keldysh 形式、あるいは非平衡宇宙論に初めて取り組む修士課程の学生、博士課程初期の学生、および研究者を対象とする。そのため、密度行列の基本理論や標準宇宙論の初歩など、多くの読者がすでに他で学んだかもしれない入門的な内容も含めた。その目的は、ノートを自己完結的にするだけでなく、より高度な話題に進む前に記法と規約を定めることにもある。重点は、文献を読み始めるために必要な主要な定義、方程式、例、概念的構造をそろえ、実用的な道具立てを築くことに置く。より専門的で研究段階の発展については、大部分を参考文献に委ねる。

本ノートは 2026 年に三つのスクールで行った講義のために準備された。イタリアの Padova にある Scuola Galileiana di Studi Superiori で「基本相互作用の理論物理学」を主題として開催された [Scuola Tematica](#)、「宇宙論的相関関数」を主題とする [2026 年 IHES サマースクール](#)、そしてスペインの Castro Caldelas、As Barreiras で「無秩序な宇宙」を主題として開催された [La Ricotta Summer School](#) である。したがって、教育的な利用を意図している。関心ある読者が、開放量子系、Schwinger–Keldysh 有効作用、宇宙論的应用に関する現代の文献に取り組み、やがて自らの研究でこれらの道具を使うための、確かな基礎となることを願っている。

本ノートの構成。第1節では、密度行列、複合系、部分トレース、エンタングルメント、デコヒーレンスを含む、開放系の量子力学を概観する。第2節では、縮約動力学、量子チャネル、力学写像、GKSL マスター方程式を導入し、その微視的な起源と、時間依存する写像および CP 可分性への拡張も扱う。第3節では、量子ビットと散逸調和振動子を例として、Lindblad 方程式の具体的な解を論じる。

第4節では、二重化した時間経路と Keldysh 基底から、Feynman–Vernon 影響汎関数とその一般的な整合性条件まで、Schwinger–Keldysh 形式を展開する。第5節では、局所的な Schwinger–Keldysh 作用を力学写像および完全正值性と関連づけ、ジャンプ演算子、Langevin 動力学、散逸するスカラー場を論じる。第6節では、宇宙論的背景、インフレーションの相関関数、原始非ガウス性を概観する。第7節では、インフレーションの開放有効場理論を構築し、パワースペクトルとバイスペクトルに適用する。第??節では、確率的インフレーションを導入し、Langevin 記述と Fokker–Planck 記述を用いて、de Sitter 時空における永年成長と晩期の振る舞いを研究する。付録 A では、二次の Schwinger–Keldysh 核の逆を求める際に有用な結果をまとめる。

## 参考文献と資料

ここでは、本ノートの作成に用いた参考文献をまとめ、文献に見られる数多くの資料の中から、有用なものを選んで紹介する。

密度行列の入門的な説明は、ほぼどの量子力学の教科書にも見つかる。第1節は、ケンブリッジ大学数学 Tripos の「量子力学の原理」の講義と、David Skinner による優れた講義ノート [?] に沿っている。

開放量子系の古典的かつ包括的な教科書として、Breuer と Petruccione の著書 [?] がある。第2節と第3節の記述は、一部この文献に沿っている。同書では演算子形式と確率的手法により重点が置かれているが、きわめて幅広い話題と応用が収められている。

開放系の Schwinger–Keldysh 経路積分についての優れた参考書は、Alex Kamenev の著書 [?] である。この教科書は基礎から積み上げ、量子多体系への多数の応用を提示している。量子多体物理学の言葉で書かれているため、高エネルギー物理学を背景とする読者には、慣れるまで少し時間がかかるかもしれない。第4節の一部はこの文献に基づいている。

宇宙論とインフレーションの概説は、ケンブリッジ大学数学 Tripos Part III の「宇宙論」 [?] および「宇宙論における場の理論」 [?] の講義のために私が書いた講義ノートに基づく。閉鎖系での Schwinger–Keldysh 経路積分の摂動展開については、文献 [?] が優れた参考資料である。第7節は主として原著論文 [?, ?, ?, ?, ?] に基づく。

de Sitter 時空の軽い場と確率的インフレーションについては、非常に優れたレビュー [?, ?, ?] が多数あり、本講義で提示する初歩的な議論よりもはるかに高度な側面も論じられている。確率的インフレーション、およびすべての量子効果を系統的に捉えるための最低次の記述の改良に関しては、何百もの論文がある。これらの問題はいずれも詳しくは論じず、上記のレビューとそこに挙げられた文献を読者に勧める。

開放量子系には、さまざまな観点から問題の異なる側面を論じた優れたレビューが多数ある。影響汎関数、ノイズ、デコヒーレンス、宇宙論を結びつける補完的な場の理論による取り扱い、Calzetta と Hu [?] に与えられている。現代的な Schwinger–Keldysh 有効理論と非 Markov 的量子動力学は、それぞれ文献 [?, ?] で概説されている。Hong Liu と Paolo Glorioso の先駆的な研究は、この主題についての私の理解に特に大きな影響を与えた。T. Colas によるごく最近の優れたレビュー [?] は、インフレーションと晩期宇宙の重力への、開放量子系および Schwinger–Keldysh 経路積分の応用を論じている。そこでは開放系の有効場理論の構築が強調され、この分野の最新の進展の一部も扱われている。もう一つの優れた参考資料は、L. M. Sieberer、M. Buchhold、S. Diehl によるレビュー [?] であり、駆動された開放量子系、特に冷却原子と光共振器への応用に重点が置かれている。

# 1 開放系の量子力学

量子力学の標準的な定式化は、孤立した Hilbert 空間内をユニタリーに発展する状態ベクトルで記述される閉鎖系から始まることが多い。しかし、多くの物理的状況では、この記述は実用的でないか、不完全である。微視的状态について確率的な情報しか得られない場合もあれば、より大きな複合系の部分系に関心がある場合もあり、観測する自由度が、アクセス不能な、あるいは詳細にモデル化するには複雑すぎる環境と相互作用している場合もある。こうした状況が、密度行列形式、そしてより一般には開放量子系の枠組みを動機づける。この枠組みでは、基本的な対象は必ずしも純粋状態ではなく、密度行列である。密度行列によって、純粋状態、統計混合、そして後に扱う、観測しない自由度についてトレースを取って得られる縮約状態を記述できる。

本節の目的は、本ノートの以後の部分に必要な、量子力学の基本的な道具を導入することである。まず密度行列とその解釈から始め、次に量子ビットや Bloch ベクトルなどの簡単な例を論じ、その後、複合系、テンソル積、部分トレース、エンタングルメント、デコヒーレンスに進む。これらの概念は概念的に重要であるだけでなく、開放系の動力学を定式化する自然な言葉にもなる。特に、全体系が純粋状態を保ち、ユニタリーに発展していても、部分系が純粋状態から混合状態へ発展しうることが明らかにする。この内容にすでに親しんでいる読者も多いだろう。その場合は、直接次節へ進んでもよい。

## 1.1 密度行列

多くの物理的状況では、量子系を単一の状態ベクトルで記述することは、不便であるか、不十分である。これは、系が異なる状態に確率的にしか準備されないためである場合もあれば、その正確な微視的状态を知らないためである場合もあり、あるいは後に、より大きな量子系の部分系を記述したくなるためである場合もある。こうした考察が、密度行列とも呼ばれる密度演算子の導入を動機づける。

系の量子状態について、古典的な確率情報しかもっていないとしよう。この追加の不確定性は、量子力学に標準的に備わる確率的性質に重ねて加わるものである。そこで、系が規格化された状態  $|\psi_\alpha\rangle \in \mathcal{H}$  に古典的な確率  $p_\alpha$  であると考えよう。ここで

$$0 \leq p_\alpha \leq 1, \quad \sum_\alpha p_\alpha = 1, \quad \langle \psi_\alpha | \psi_\alpha \rangle = 1. \quad (1.1)$$

状態  $|\psi_\alpha\rangle$  は互いに直交している必要はない。ここで、系が状態  $|\psi_\alpha\rangle$  の線形重ね合わせにあると言っているのではないことを強調しておく。それならば、単に Hilbert 空間内の別の一つの状態にすぎないからである。

この場合、観測量をどのように取り出せばよいだろうか。 $Q$  が物理的観測量に対応する Hilbert 空間上の線形 Hermite 演算子であるなら、その期待値は

$$\langle Q \rangle = \sum_\alpha p_\alpha \langle \psi_\alpha | Q | \psi_\alpha \rangle. \quad (1.2)$$

系の状態についてもっている情報を一つの量にまとめると便利である。次の演算子はその役割を果たすことが分かる。

$$\rho \equiv \sum_\alpha p_\alpha |\psi_\alpha\rangle \langle \psi_\alpha|, \quad (1.3)$$

$\rho$  を密度行列と呼ぶ<sup>1</sup>。

この定義の利点は、期待値が次の簡潔な形に書けることにある。

$$\langle Q \rangle = \text{Tr}_{\mathcal{H}}(\rho Q), \quad (1.4)$$

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<sup>1</sup>同義の名称として密度演算子がある。残念ながら、どちらの用語にも難点がある。「行列」という語は有限次元 Hilbert 空間を連想させるが、本ノートでは無限次元 Hilbert 空間の場合にも「線形演算子」と同じ意味で用いる。一方、「演算子」という語は物理的観測量を連想させるが、ここでは  $\rho$  は系の状態を表している。

ここで  $\text{Tr}$  はトレース操作である。有限次元 Hilbert 空間ではトレースはよく知られたものだが、無限次元空間では微妙な問題が生じる。より形式的には、トレースは、演算子の適切な線形空間  $\mathcal{A}$  上で定義された写像  $\text{Tr}$  であり、すべての  $A, B \in \mathcal{A}$  とすべての複素数  $a, b$  に対して線形性

$$\text{Tr}(aA + bB) = a \text{Tr}(A) + b \text{Tr}(B), \quad (1.5)$$

および、両方の積が定義域に属する場合の巡回性

$$\text{Tr}(AB) = \text{Tr}(BA), \quad (1.6)$$

を満たす。ここでいう定義域は  $\text{Tr}$  の定義域である。さらに正値性<sup>2</sup>

$$\text{Tr}(A^\dagger A) \geq 0. \quad (1.7)$$

も満たす。無限次元では、トレースは一般にすべての有界演算子に対して定義されるわけではなく、「トレース級」演算子のような適切な部分集合に対してのみ定義される。本ノートではこれらの問題には立ち入らず、トレースが常に適切に定義されていると仮定する。

(1.4) が  $Q$  の期待値を計算することを確認するため、 $\{|q_n\rangle\}$  を  $Q$  の固有状態からなる正規直交基底とする。したがって

$$Q = \sum_n q_n |q_n\rangle \langle q_n|. \quad (1.8)$$

すると

$$\text{Tr}_{\mathcal{H}}(\rho Q) = \sum_n \langle q_n | \rho Q | q_n \rangle \quad (1.9)$$

$$= \sum_{n,\alpha} p_\alpha \langle q_n | \psi_\alpha \rangle \langle \psi_\alpha | Q | q_n \rangle \quad (1.10)$$

$$= \sum_{n,\alpha} p_\alpha q_n |\langle q_n | \psi_\alpha \rangle|^2 \quad (1.11)$$

$$= \sum_\alpha p_\alpha \langle \psi_\alpha | Q | \psi_\alpha \rangle = \langle Q \rangle. \quad (1.12)$$

となる。このように、密度行列は、任意の観測量の期待値を計算するために必要なすべての情報を含んでいる。

特に重要な特殊例は、系が単一の規格化された状態  $|\psi\rangle$  で記述される場合である。この場合には

$$\rho = |\psi\rangle \langle \psi|, \quad (1.13)$$

となり、トレースによる公式は通常の表式

$$\text{Tr}(\rho Q) = \langle \psi | Q | \psi \rangle. \quad (1.14)$$

に帰着する。ここで、密度行列の一般的な性質をまとめよう。密度演算子  $\rho: \mathcal{H} \rightarrow \mathcal{H}$  は、次を満たさなければならない<sup>3</sup>。

$$\rho^\dagger = \rho, \quad (1.15)$$

$$\langle \phi | \rho | \phi \rangle \geq 0 \quad \text{for all } |\phi\rangle \in \mathcal{H}, \quad (1.16)$$

$$\text{Tr} \rho = 1. \quad (1.17)$$

これらの条件はそれぞれ、 $\rho$  が Hermite であること、半正定値であること、トレースが 1 となるように規格化されていることを意味する。

<sup>2</sup>忠実性、すなわち  $\text{Tr}(A^\dagger A) = 0$  ならば  $A = 0$  であることも要求する場合が多い。

<sup>3</sup>半正定値性を、略記  $\rho \geq 0$  で表すこともある。

逆に、これら三つの性質を満たす任意の演算子は、正当な密度行列である。実際、簡単のため有限次元を仮定すると、 $\rho$  は Hermite であるから、スペクトル分解

$$\rho = \sum_n p_n |n\rangle\langle n|, \quad (1.18)$$

をもつ。ここで固有値  $p_n$  は実数である。正值性から

$$p_n \geq 0, \quad (1.19)$$

が従い、一方、トレース条件から

$$\sum_n p_n = 1. \quad (1.20)$$

が得られる。したがって、 $\rho$  は常に、正規直交状態  $|n\rangle$  の確率的混合として解釈できる。 $\rho$  の行列要素、すなわち

$$\rho_{nm} := \langle n|\rho|m\rangle, \quad (1.21)$$

が任意の正規直交基底で有界であることにも注意しておくといよい。対角成分は

$$\rho_{nn} \geq 0, \quad \sum_n \rho_{nn} = 1, \quad \Rightarrow \quad \rho_{nn} \leq 1. \quad (1.22)$$

を満たす。さらに、正值性から次の上界が従う。

$$\begin{aligned} |\rho_{nm}| &:= |\langle n|\rho|m\rangle| \\ &= |\langle n|A^\dagger A|m\rangle| \quad \text{with } \rho \geq 0 \Rightarrow \rho = A^\dagger A \\ &= |\langle An|Am\rangle| \\ &\leq \sqrt{\langle An|An\rangle} \sqrt{\langle Am|Am\rangle} \quad (\text{Cauchy-Schwarz}) \\ &= \sqrt{\langle n|A^\dagger A|n\rangle} \sqrt{\langle m|A^\dagger A|m\rangle} \\ &= \sqrt{\langle n|\rho|n\rangle} \sqrt{\langle m|\rho|m\rangle} \\ &= \sqrt{\rho_{nn}\rho_{mm}}. \end{aligned} \quad (1.23)$$

したがって、 $n \neq m$  に対して  $|\rho_{nm}| \leq \frac{1}{2}$  である。

**純粋状態と混合状態** 次に、純粋状態と混合状態の区別に進む。規格化されたベクトル  $|\chi\rangle \in \mathcal{H}$  が存在して

$$\rho = |\chi\rangle\langle\chi|. \quad (1.24)$$

と書ける場合、その状態を純粋という。そうでなければ、その状態を混合という。この定義には、いくつかの等価な特徴づけがある。まず、 $\rho = |\psi\rangle\langle\psi|$  ならば

$$\rho^2 = |\psi\rangle\langle\psi|\psi\rangle\langle\psi| = |\psi\rangle\langle\psi| = \rho. \quad (1.25)$$

である。したがって、どの純粋状態も  $\rho^2 = \rho$  を満たす。逆に、

$$\rho^2 = \rho. \quad (1.26)$$

を仮定しよう。 $\rho$  の固有基底では

$$\rho|n\rangle = p_n|n\rangle, \quad (1.27)$$

となるので、条件  $\rho^2 = \rho$  は

$$p_n^2 = p_n. \quad (1.28)$$

を意味する。したがって各固有値は0または1である。 $\text{Tr } \rho = 1$ であるから、ちょうど一つの固有値が1に等しく、残りはすべてゼロである。よって、ある規格化された状態に対して

$$\rho = |\chi\rangle\langle\chi| \quad (1.29)$$

と書ける。この規格化された状態が $|\chi\rangle$ であり、状態は純粋である。以上から、次が示された。

$$\rho \text{ pure} \iff \rho^2 = \rho. \quad (1.30)$$

このことから、次の自明な系が得られる。整数 $n \geq 2$ について

$$\rho \text{ pure} \iff \rho^n = \rho. \quad (1.31)$$

である。さらに別の等価な条件は

$$\text{Tr } \rho^2 = 1. \quad (1.32)$$

である。実際、 $\rho$ が純粋ならば $\rho^2 = \rho$ であるから、

$$\text{Tr } \rho^2 = \text{Tr } \rho = 1. \quad (1.33)$$

となる。逆に、対角基底では

$$\text{Tr } \rho^2 = \sum_n p_n^2. \quad (1.34)$$

である。 $p_n \geq 0$  および  $\sum_n p_n = 1$  であるから、

$$\sum_n p_n^2 \leq 1, \quad (1.35)$$

となり、等号が成立するのは、一つの固有値が1に等しく、その他がすべてゼロの場合、かつその場合に限る。したがって

$$\text{Tr } \rho^2 = 1 \iff \rho \text{ is pure.} \quad (1.36)$$

である。より一般には、混合状態に対して

$$0 < \text{Tr } \rho^2 < 1. \quad (1.37)$$

が成り立つ。純粋性の等価な判定条件をまとめると、

$$\begin{aligned} \rho \text{ is pure} &\iff \rho = |\chi\rangle\langle\chi| \text{ for some normalized } |\chi\rangle \\ &\iff \rho^2 = \rho \\ &\iff \text{Tr } \rho^2 = 1. \end{aligned} \quad (1.38)$$

となる。これらの条件が満たされない場合、状態は混合状態である。密度行列は、開放量子系を論じる際の自然な出発点となる。

**von Neumann エントロピー** 密度行列 $\rho$ が与えられたとき、その von Neumann エントロピーを次で定義する。

$$S(\rho) := -\text{Tr}(\rho \ln \rho). \quad (1.39)$$

一見すると恣意的に見えるこの操作は、古典的な情報理論に根ざしており、Shannon エントロピーと密接に関連する。ある密度行列の von Neumann エントロピーは、対応する量子状態の本質的な性質を捉える。これは決して唯一のエントロピーの概念ではないが、実際、次の小節でもさらにいくつかに出会うことになるが、量子情報理論と開放系の研究で、とりわけ重要な役割を果たす。

$\rho$  は Hermite 演算子なので、対角化できる。 $\rho$  のスペクトル分解が

$$\rho = \sum_n p_n |n\rangle\langle n|, \quad p_n \geq 0, \quad \sum_n p_n = 1, \quad (1.40)$$

であるなら、 $\rho$  の台の上で  $\ln \rho = \sum_n (\ln p_n) |n\rangle\langle n|$  であることから、エントロピーは

$$S(\rho) = - \sum_n p_n \ln p_n, \quad (1.41)$$

となる。ここでは、 $x \rightarrow 0^+$  のとき  $x \ln x \rightarrow 0$  とする標準的な規約を用いる。したがって、von Neumann エントロピーは  $\rho$  の固有値のみに依存し、計算に用いる基底には依存しない。表式 (1.41) は、von Neumann エントロピーを Shannon エントロピーの量子版とみなせることを明示している。密度行列の固有値は  $0 \leq p_n \leq 1$  を満たすため、 $\ln p_n \leq 0$  であり、したがって

$$S(\rho) \geq 0. \quad (1.42)$$

となる。さらに、

$$S(\rho) = 0 \quad \Longleftrightarrow \quad \rho \text{ is pure.} \quad (1.43)$$

が成り立つ。実際、 $\rho$  が純粋ならば、その固有値は  $1, 0, 0, \dots$  であるから、

$$S(\rho) = -1 \cdot \ln 1 = 0. \quad (1.44)$$

となる。逆に、 $S(\rho) = 0$  ならば

$$0 = - \sum_n p_n \ln p_n. \quad (1.45)$$

である。 $0 \leq x \leq 1$  に対して  $-x \ln x \geq 0$  なので、和の各項は非負である。よって、各項はすべてゼロでなければならない。しかし、 $-p_n \ln p_n = 0$  となるのは  $p_n = 0$  または  $p_n = 1$  の場合に限られる。固有値の和は 1 なので、ちょうど一つの固有値が 1 に等しく、その他はすべてゼロでなければならない。したがって、 $\rho$  は純粋である。

Hilbert 空間が有限次元であり、

$$d := \dim \mathcal{H} < \infty, \quad (1.46)$$

ならば、エントロピーは上から

$$S(\rho) \leq \ln d. \quad (1.47)$$

で抑えられる。この上界は、規格化の拘束条件  $\text{Tr } \rho = 1$  のもとでエントロピーの極値を求めることで得られる。Lagrange 未定乗数  $\lambda$  を導入して、次の極値を求める。

$$\Phi[\rho, \lambda] := S(\rho) - \lambda(1 - \text{Tr } \rho). \quad (1.48)$$

$\rho$  について変分すると

$$0 = \delta \Phi = - \text{Tr} [\delta \rho (\ln \rho + 1 - \lambda)]. \quad (1.49)$$

を得る。これは任意の  $\delta \rho$  に対して成り立たなければならないため、

$$\ln \rho + 1 - \lambda = 0, \quad (1.50)$$

が得られ、したがって

$$\rho = e^{\lambda-1} \mathbb{1}_{\mathcal{H}}. \quad (1.51)$$

となる。 $\text{Tr } \rho = 1$  を課すと

$$1 = \text{Tr } \rho = e^{\lambda-1} \text{Tr } \mathbb{1}_{\mathcal{H}} = e^{\lambda-1} d, \quad (1.52)$$

を得るので、

$$\rho_{\max} = \frac{1}{d} \mathbb{1}_{\mathcal{H}}. \quad (1.53)$$

である。この状態でエントロピーを評価すると、

$$S_{\max} = -\text{Tr}\left(\frac{1}{d}\mathbb{1}_{\mathcal{H}} \ln \frac{1}{d}\mathbb{1}_{\mathcal{H}}\right) = -\text{Tr}\left(\frac{1}{d}\mathbb{1}_{\mathcal{H}}\right) \ln \frac{1}{d} = \ln d. \quad (1.54)$$

を得る。このように、von Neumann エントロピーは最大混合状態

$$\rho = \frac{1}{d}\mathbb{1}_{\mathcal{H}}, \quad (1.55)$$

で最大となる。この状態では、すべての状態が等しい確率をもつ。この結果は、エントロピーが大きいほど、系の正確な量子状態についての不確定性が大きい、あるいは同じことだが情報が少ない、という直観を与える。

**熱的密度行列と Gibbs 分布** 物理学で際立った役割を果たす密度行列として、熱的密度行列がある。この特定の選択は非常に重要な役割を果たすため、ときには、ある密度行列が非熱的であるとわざわざ断る必要を感じることもさえる。実際には、測度ゼロの集合を除くすべての密度行列が非熱的であるにもかかわらず、である。密度行列形式は量子統計力学に不可欠である。ここでは、ごく手短かに述べるにとどめる。規格化条件

$$\text{Tr} \rho = 1, \quad (1.56)$$

に加えて、平均エネルギーも

$$\text{Tr}(\rho H) = U. \quad (1.57)$$

に固定するとしよう。ここで、両方の拘束条件のもとでエントロピーの極値を求める。Lagrange 未定乗数  $\lambda$  と  $\beta$  を導入して、次を考える。

$$\Phi[\rho, \lambda, \beta] := S(\rho) - \lambda(1 - \text{Tr} \rho) - \beta(\text{Tr}(\rho H) - U). \quad (1.58)$$

その変分は

$$\begin{aligned} 0 &= \delta\Phi = \delta S(\rho) + \lambda\delta(\text{Tr} \rho) - \beta\delta(\text{Tr}(\rho H)) \\ &= -\text{Tr}[\delta\rho(\ln \rho + 1)] + \lambda \text{Tr}(\delta\rho) - \beta \text{Tr}(\delta\rho H) \\ &= -\text{Tr}[\delta\rho(\ln \rho + 1 - \lambda + \beta H)]. \end{aligned} \quad (1.59)$$

である。これは任意の  $\delta\rho$  について成り立たなければならないため、

$$\ln \rho + 1 - \lambda + \beta H = 0, \quad (1.60)$$

を得る。したがって

$$\rho = e^{\lambda-1} e^{-\beta H}. \quad (1.61)$$

である。定数  $e^{\lambda-1}$  は、規格化条件によって次のように決まる。

$$1 = \text{Tr} \rho = e^{\lambda-1} \text{Tr}(e^{-\beta H}). \quad (1.62)$$

よって

$$\rho = \frac{e^{-\beta H}}{\text{Tr}(e^{-\beta H})} =: \frac{e^{-\beta H}}{Z(\beta)}, \quad (1.63)$$

である。ここで

$$Z(\beta) := \text{Tr}(e^{-\beta H}) \quad (1.64)$$

は分配関数であり、ここでは単にトレースの規格化因子として現れる。 $H|n\rangle = E_n|n\rangle$  を満たすエネルギー固有状態の基底では、

$$\rho = \frac{1}{Z(\beta)} \sum_n e^{-\beta E_n} |n\rangle\langle n|. \quad (1.65)$$

となる。これが熱的密度行列、または Gibbs 状態である。パラメータ  $\beta$  は、条件

$$\text{Tr}(\rho H) = U, \quad (1.66)$$

によって陰に決まり、熱力学では逆温度

$$\beta = \frac{1}{T} \quad (1.67)$$

と同定される。ここでは Boltzmann 定数を 1 とする単位系を用いた。最後に、熱的密度行列が平衡状態を表すのは、Hamiltonian が時間に依存しない場合、または断熱近似のもとで非常にゆっくり変化する場合には限ることに注意しておこう。

## 1.2 例：量子ビットと Bloch ベクトル

おもちゃのモデルを用いて、上述の概念への理解を固めることは重要である。ここでは、量子力学的な系としておそらく最も簡単なもの、量子ビットを論じる。量子ビットは二次元 Hilbert 空間  $\mathcal{H} = \mathbb{C}^2$  をもち、これは量子力学における最も単純な非自明な可能性である。この Hilbert 空間の二つの基底要素を、上向きと下向きの矢印で表す。

$$\mathbb{C}^2 = \text{Span}(|\uparrow\rangle, |\downarrow\rangle), \quad (1.68)$$

したがって、任意の純粋状態は次の複素線形結合で与えられる。

$$|\psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle. \quad (1.69)$$

規格化条件は

$$|\alpha|^2 + |\beta|^2 = 1. \quad (1.70)$$

である。したがって、この純粋状態に対応する密度行列は

$$\begin{aligned} \rho &:= |\psi\rangle\langle\psi| \\ &= (\alpha|\uparrow\rangle + \beta|\downarrow\rangle)(\alpha^*\langle\uparrow| + \beta^*\langle\downarrow|) \\ &= |\alpha|^2|\uparrow\rangle\langle\uparrow| + \alpha\beta^*|\uparrow\rangle\langle\downarrow| + \alpha^*\beta|\downarrow\rangle\langle\uparrow| + |\beta|^2|\downarrow\rangle\langle\downarrow|. \end{aligned} \quad (1.71)$$

となる。基底  $\{|\uparrow\rangle, |\downarrow\rangle\}$  では、これは

$$\rho = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}. \quad (1.72)$$

となる。この行列が Hermite であり、トレースが 1 であり、さらに

$$\rho^2 = \rho, \quad (1.73)$$

を満たすことは直ちに確かめられる。これは純粋状態に期待される性質である。

次に混合状態を考えよう。例えば、系が確率  $1/2$  で状態  $|\uparrow\rangle$  にあり、確率  $1/2$  で状態  $|\downarrow\rangle$  にあるとする。すると

$$\rho := \frac{1}{2}|\uparrow\rangle\langle\uparrow| + \frac{1}{2}|\downarrow\rangle\langle\downarrow| = \frac{1}{2}\mathbf{1}_{\mathcal{H}}. \quad (1.74)$$

となる。行列表現では

$$\rho = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (1.75)$$

である。この状態は純粋ではない。なぜなら

$$\rho^2 = \frac{1}{4}\mathbf{1}_{\mathcal{H}} \neq \rho, \quad (1.76)$$

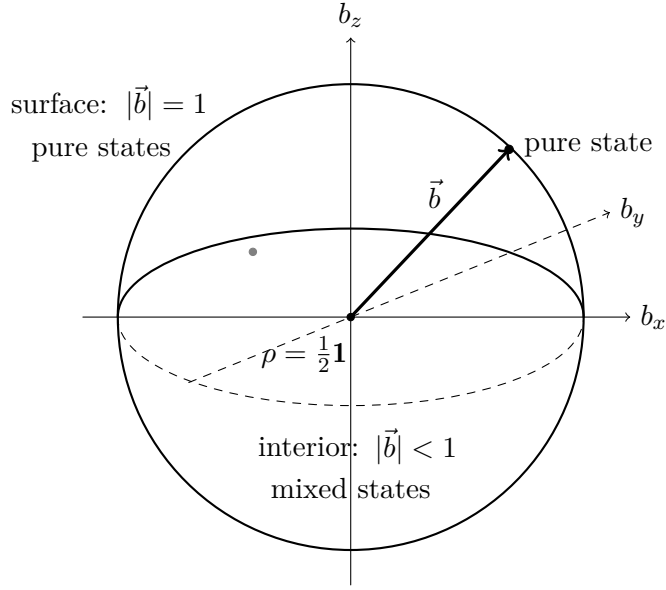


図 1: 二準位量子系の Bloch 球体。任意の量子ビットの密度行列は、 $|\vec{b}| \leq 1$  を満たす  $\rho := \frac{1}{2}(\mathbf{1} + \vec{b} \cdot \vec{\sigma})$  として書ける。表面上の点、すなわち  $|\vec{b}| = 1$  となる点は純粋状態に対応し、内部の点、すなわち  $|\vec{b}| < 1$  となる点は混合状態に対応する。中心  $\vec{b} = 0$  は最大混合状態である。

あるいは、等価に、

$$\text{Tr } \rho^2 = \frac{1}{2} < 1. \quad (1.77)$$

だからである。これは実際、二次元 Hilbert 空間のどの方向にも等しい重みを与えるという意味で、最大混合状態である。

より一般に、任意の Hermite な  $2 \times 2$  行列は、単位行列と Pauli 行列からなる基底で展開できる。

$$\sigma_x := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y := \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (1.78)$$

密度行列は Hermite で、トレースが 1 なので、常に次の形に書ける。

$$\rho := \frac{1}{2}(\mathbf{1}_{\mathcal{H}} + \vec{b} \cdot \vec{\sigma}), \quad (1.79)$$

ここで

$$\vec{b} := (b_x, b_y, b_z) \in \mathbb{R}^3, \quad \vec{\sigma} := (\sigma_x, \sigma_y, \sigma_z). \quad (1.80)$$

行列表現では、これは

$$\rho = \frac{1}{2} \begin{pmatrix} 1 + b_z & b_x - ib_y \\ b_x + ib_y & 1 - b_z \end{pmatrix}. \quad (1.81)$$

となる。ベクトル  $\vec{b}$  を Bloch ベクトルと呼ぶ。  $\rho$  の正値性は、そのノルムを制約する。実際、

$$\begin{aligned} \det \rho &= \frac{1}{4} \left[ (1 + b_z)(1 - b_z) - (b_x - ib_y)(b_x + ib_y) \right] \\ &= \frac{1}{4} (1 - b_x^2 - b_y^2 - b_z^2) \\ &= \frac{1}{4} (1 - |\vec{b}|^2) \geq 0. \end{aligned} \quad (1.82)$$

である。よって

$$|\vec{b}| \leq 1. \quad (1.83)$$

となる。したがって、量子ビットの密度行列の空間は  $\mathbb{R}^3$  の単位球体であり、Bloch 球体と呼ばれる。

この言葉を用いると、純粋状態と混合状態の区別に簡単な幾何学的解釈が与えられる。恒等式

$$(\vec{b} \cdot \vec{\sigma})^2 = |\vec{b}|^2 \mathbf{1}_{\mathcal{H}}, \quad (1.84)$$

を用いると、

$$\begin{aligned} \rho^2 &= \frac{1}{4} (\mathbf{1}_{\mathcal{H}} + 2\vec{b} \cdot \vec{\sigma} + (\vec{b} \cdot \vec{\sigma})^2) \\ &= \frac{1}{4} ((1 + |\vec{b}|^2) \mathbf{1}_{\mathcal{H}} + 2\vec{b} \cdot \vec{\sigma}). \end{aligned} \quad (1.85)$$

を得る。したがって、 $\rho^2 = \rho$  が成り立つのは

$$|\vec{b}| = 1. \quad (1.86)$$

の場合、かつその場合に限る。よって、純粋状態は Bloch 球体の表面、すなわち Bloch 球面上にあり、混合状態はその内部にある。最大混合状態は中心に対応する。

$$\vec{b} = 0, \quad \rho = \frac{1}{2} \mathbf{1}_{\mathcal{H}}. \quad (1.87)$$

スピン演算子の期待値は Bloch ベクトルと直接関係している。実際、

$$\langle \sigma_i \rangle := \text{Tr}(\rho \sigma_i) = b_i, \quad i = x, y, z. \quad (1.88)$$

となる。このように、Bloch ベクトルは、量子ビットの状態を幾何学的にパラメータ化する便利な方法を与える。

量子ビットの von Neumann エントロピーを、Bloch ベクトルを用いて直接表すことも示唆的である。

$$\rho := \frac{1}{2} (\mathbf{1}_{\mathcal{H}} + \vec{b} \cdot \vec{\sigma}), \quad (1.89)$$

および  $(\vec{b} \cdot \vec{\sigma})^2 = |\vec{b}|^2 \mathbf{1}_{\mathcal{H}}$  より、 $\rho$  の固有値は

$$p_{\pm} = \frac{1 \pm |\vec{b}|}{2}. \quad (1.90)$$

である。実際、適切なユニタリーな基底変換を行えば、 $\vec{b}$  を常に  $z$  軸にそろえることができ、

$$\rho \sim \frac{1}{2} \begin{pmatrix} 1 + |\vec{b}| & 0 \\ 0 & 1 - |\vec{b}| \end{pmatrix}, \quad (1.91)$$

となるので、固有値は直ちに読み取れる。von Neumann エントロピーは密度行列の固有値にしか依存しないため、

$$S(\rho) = -\frac{1 + |\vec{b}|}{2} \ln \frac{1 + |\vec{b}|}{2} - \frac{1 - |\vec{b}|}{2} \ln \frac{1 - |\vec{b}|}{2}. \quad (1.92)$$

を得る。このように、量子ビットのエントロピーは  $\vec{b}$  の方向には依存せず、その大きさだけに依存する。この公式によって、Bloch 球体の幾何学がさらに明瞭になる。Bloch 球体の表面では  $|\vec{b}| = 1$  であり、固有値は  $(1, 0)$  なので、

$$S(\rho) = 0, \quad (1.93)$$

となる。これは純粋状態に期待されるとおりである。Bloch 球体の中心では  $\vec{b} = 0$  で、固有値はいずれも  $1/2$  に等しく、エントロピーは最大値

$$S(\rho) = -2 \cdot \frac{1}{2} \ln \frac{1}{2} = \ln 2. \quad (1.94)$$

に達する。より一般には、 $|\vec{b}|$  が 0 から 1 へ増加するにつれて、エントロピーは単調に減少する。したがって、Bloch 球体の中心に近い状態ほど混合の度合いが大きく、エントロピーも大きい。一方、表面に近い状態ほど混合の度合いが小さく、エントロピーも小さい。量子ビットでは、Bloch 球体の中心からの距離が、純粋性、そして等価にエントロピーの小ささを直接測る幾何学的な指標となる。

### 1.3 複合系、テンソル積、部分トレース

これまでは単一の量子系の状態を論じてきた。しかし、物理的に興味ある多くの状況では、考えている系はいくつかの部分から構成される。これは、典型的には環境  $B$  と相互作用する部分系、例えば  $A$  と記す系に関心をもつ開放量子系で、特に重要となる。このような複合系に適した数学的枠組みは、Hilbert 空間のテンソル積である。

$\mathcal{H}_A$  と  $\mathcal{H}_B$  を、それぞれ二つの量子系  $A$  と  $B$  に対応する Hilbert 空間とする。結合した系の Hilbert 空間は

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B. \quad (1.95)$$

である。 $\{|e_a\rangle\}$  が  $\mathcal{H}_A$  の正規直交基底であり、 $\{|f_\alpha\rangle\}$  が  $\mathcal{H}_B$  の正規直交基底であるなら、 $\mathcal{H}_A \otimes \mathcal{H}_B$  の基底はテンソル積ベクトル

$$|e_a\rangle \otimes |f_\alpha\rangle, \quad (1.96)$$

で与えられ、任意の状態  $|\Psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$  は次のように展開できる。

$$|\Psi\rangle = \sum_{a,\alpha} c_{a\alpha} |e_a\rangle \otimes |f_\alpha\rangle. \quad (1.97)$$

複合系の一般の状態は、単一のテンソル積

$$|\Psi\rangle \neq |\phi\rangle \otimes |\chi\rangle \quad \text{in general.} \quad (1.98)$$

としては書けないことを強調しておく。実際、テンソル積 Hilbert 空間の次元は

$$\dim(\mathcal{H}_A \otimes \mathcal{H}_B) = \dim(\mathcal{H}_A) \times \dim(\mathcal{H}_B). \quad (1.99)$$

である。(1.98) の因子化された形に書ける状態を積状態、または分離可能状態と呼ぶ。この形に書けない状態をエンタングルした状態と呼ぶ。エンタングルした状態も積状態も、混合状態ではなく純粋状態であることに注意されたい。

テンソル積空間上の内積は、基底要素について

$$(\langle e_a| \otimes \langle f_\alpha|)(|e_b\rangle \otimes |f_\beta\rangle) = \langle e_a|e_b\rangle \langle f_\alpha|f_\beta\rangle, \quad (1.100)$$

と定義し、線形性によって任意の状態へ拡張する。Dirac 記法は、同じ部分系に属するブラとケットを結びつけるよう視覚的に導いてくれる。

複合系上の演算子も同様に構成される。

$$\mathcal{O}_A : \mathcal{H}_A \rightarrow \mathcal{H}_A, \quad \mathcal{O}_B : \mathcal{H}_B \rightarrow \mathcal{H}_B, \quad (1.101)$$

ならば、テンソル積演算子

$$\mathcal{O}_A \otimes \mathcal{O}_B : \mathcal{H}_A \otimes \mathcal{H}_B \rightarrow \mathcal{H}_A \otimes \mathcal{H}_B \quad (1.102)$$

を次によって定義する。

$$(\mathcal{O}_A \otimes \mathcal{O}_B)(|\phi\rangle \otimes |\chi\rangle) = (\mathcal{O}_A|\phi\rangle) \otimes (\mathcal{O}_B|\chi\rangle). \quad (1.103)$$

複合系上の一般の演算子は、このような積演算子の線形結合であり、単純な積  $\mathcal{O}_A \otimes \mathcal{O}_B$  の形にはならない。実際、 $\{|e_a\rangle\}$  が  $\mathcal{H}_A$  の基底で、 $\{|f_\alpha\rangle\}$  が  $\mathcal{H}_B$  の基底ならば、階数 1 の演算子

$$|e_a\rangle\langle e_b| \otimes |f_\alpha\rangle\langle f_\beta| \quad (1.104)$$

は、 $\mathcal{H}_A \otimes \mathcal{H}_B$  上の線形演算子の空間の基底をなす。したがって、複合 Hilbert 空間上の任意の演算子  $\mathcal{O}$  は

$$\mathcal{O} = \sum_{a,b,\alpha,\beta} \mathcal{O}_{a\alpha,b\beta} \left( |e_a\rangle\langle e_b| \otimes |f_\alpha\rangle\langle f_\beta| \right). \quad (1.105)$$

と書ける。この意味で、積演算子は、テンソル積 Hilbert 空間上の一般の演算子を構成する基本要素である。

後に重要となる特殊な場合として、部分系  $A$  のみに作用する演算子がある。これは

$$\mathcal{O}_A \otimes \mathbf{1}_B, \quad (1.106)$$

と表される。一方、部分系  $B$  のみに作用する演算子は

$$\mathbf{1}_A \otimes \mathcal{O}_B. \quad (1.107)$$

と表される。このような演算子は可換である。

$$[\mathcal{O}_A \otimes \mathbf{1}_B, \mathbf{1}_A \otimes \mathcal{O}_B] = 0. \quad (1.108)$$

これは単に、異なる部分系に作用する操作が独立であるという事実を表している。

**部分トレースと縮約密度行列** ここで、開放系の中心的な問いの一つを述べる準備が整った。部分系  $A$  に作用する演算子だけに関心があるとしよう。これにはさまざまな理由がありうる。例えば、系  $B$  が大きすぎて、実際には測定できないのかもしれない。あるいは、複雑すぎたり記述が困難だったりするために、技術的に正確な予言ができないのかもしれない。では、一般には系  $B$  とやり取りし、相互作用しうる系  $A$  の量子力学について、どのように数学的記述を定式化すればよいだろうか。その答えは、部分系  $B$  の自由度について部分トレースを取って得られる縮約密度行列である。

$\{|n\rangle_B\}$  を  $\mathcal{H}_B$  の正規直交基底とする。部分系  $A$  の縮約密度行列を次で定義する。

$$\rho_A \equiv \text{Tr}_B(\rho_{AB}) = \sum_n {}_B\langle n | \rho_{AB} | n \rangle_B. \quad (1.109)$$

同様に、部分系  $B$  の縮約密度行列は

$$\rho_B \equiv \text{Tr}_A(\rho_{AB}). \quad (1.110)$$

である。部分トレースの定義は、選んだ正規直交基底を使って書かれることが多いが、部分トレースそのものは基底に依存しない。

縮約密度行列が適切な対象であるのは、部分系  $A$  のみに作用するすべての観測量の期待値を再現するからである。観測量が

$$Q = Q_A \otimes \mathbf{1}_B, \quad (1.111)$$

の形をもつならば、

$$\langle Q \rangle = \text{Tr}_{A \otimes B}(\rho_{AB}(Q_A \otimes \mathbf{1}_B)) = \text{Tr}_A(\rho_A Q_A). \quad (1.112)$$

となる。したがって、部分系  $A$  に対する測定に関する限り、全状態  $\rho_{AB}$  に含まれるすべての情報は、縮約密度行列  $\rho_A$  に符号化されている。

この公式を具体的に確かめよう。 $\mathcal{H}_B$  の基底  $\{|n\rangle_B\}$  を用いて部分トレースを書くと、

$$\begin{aligned} \text{Tr}_{A \otimes B}(\rho_{AB}(Q_A \otimes \mathbf{1}_B)) &= \sum_n \text{Tr}_A \left( {}_B\langle n | \rho_{AB}(Q_A \otimes \mathbf{1}_B) | n \rangle_B \right) \\ &= \sum_n \text{Tr}_A \left( {}_B\langle n | \rho_{AB} | n \rangle_B Q_A \right) \\ &= \text{Tr}_A \left( \left( \sum_n {}_B\langle n | \rho_{AB} | n \rangle_B \right) Q_A \right) \\ &= \text{Tr}_A(\rho_A Q_A). \end{aligned} \quad (1.113)$$

となる。縮約密度行列は、開放量子系研究の中心的な対象である。本ノートでは、その多くの性質のいくつかと、その動力学を記述する異なる形式論を論じる。

次節では、部分トレースが開放量子系にとって非常に重要であることを見る。その文脈では、通常、全 Hilbert 空間を

$$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E, \quad (1.114)$$

と分解する。ここで  $S$  は関心のある系を、 $E$  はその環境を表す。全体系は、標準的な閉鎖系の量子力学と同様、ユニタリーに発展していてもよいが、観測者が測定するのは部分系  $S$  だけである。このとき重要な対象は全密度行列  $\rho_{SE}$  ではなく、縮約密度行列

$$\rho_S = \text{Tr}_E(\rho_{SE}). \quad (1.115)$$

である。 $\rho_{SE}$  が純粋状態を記述していても、系が環境とエンタングルすると、 $\rho_S$  は一般に混合状態となる。まもなく見るように、初めには存在しなくても、エンタングルメントは時間発展によって生成されうる。

**例：量子ビットについてトレースを取る。** 量子ビットを用いた例を論じよう。二つの量子ビット  $A$  と  $B$  のテンソル積からなる全体系と、分離可能な二量子ビット状態

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)_A \otimes |\downarrow\rangle_B. \quad (1.116)$$

を考える。この純粋な積状態を記述する全密度行列は

$$\rho_{AB} = |\Psi\rangle\langle\Psi|. \quad (1.117)$$

である。部分系  $B$  についてトレースを取ると

$$\begin{aligned} \rho_A &= \text{Tr}_B(\rho_{AB}) \\ &= {}_B\langle\uparrow|\rho_{AB}|\uparrow\rangle_B + {}_B\langle\downarrow|\rho_{AB}|\downarrow\rangle_B. \end{aligned} \quad (1.118)$$

となる。状態は  $|\downarrow\rangle_B$  上にしか台をもたないので、第二項だけが寄与し、

$$\rho_A = \frac{1}{2}(|\uparrow\rangle + |\downarrow\rangle)_A (\langle\uparrow| + \langle\downarrow|)_A. \quad (1.119)$$

を得る。基底  $\{|\uparrow\rangle, |\downarrow\rangle\}$  での行列表現では、

$$\rho_A = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}. \quad (1.120)$$

となる。これは依然として純粋状態である。まもなく完全に一般的な形で証明するように、全状態が積状態であれば、それぞれの部分系も純粋状態にある。直観的には、二つの部分系の間にエンタングルメントがなかったため、一方を知らないことが、もう一方を正確に知る妨げにはならなかったのである。

今度は、代わりにエンタングルした純粋状態

$$|\text{EPR}\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle_A |\downarrow\rangle_B - |\downarrow\rangle_A |\uparrow\rangle_B), \quad (1.121)$$

を考える。これは Bell 状態として知られている<sup>4</sup>。全状態は純粋であり、

$$\rho_{AB} = |\text{EPR}\rangle\langle\text{EPR}|. \quad (1.122)$$

である。しかし、部分系  $B$  をトレースで消去すると、

$$\rho_A = \text{Tr}_B(\rho_{AB}) = \frac{1}{2}(|\uparrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow|)_A, \quad (1.123)$$

<sup>4</sup> 「EPR」は Einstein、Podolsky、Rosen の頭文字である。

あるいは行列表現で、

$$\rho_A = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}_A. \quad (1.124)$$

を得る。この縮約密度行列は混合状態である。

$$\rho_A^2 \neq \rho_A. \quad (1.125)$$

したがって、全状態が純粋であっても、より大きな系の部分系は混合状態にありうる。これはエンタングルメントの基本的な特徴の一つである。

#### 1.4 エンタングルメント、Schmidt 分解、エントロピー

エンタングルメントは、量子力学を最もよく特徴づける性質の一つである。開放系の文脈では、全状態が純粋でも、部分系があたかも混合状態にあるかのように振る舞う原因が、まさにエンタングルメントである。

ほとんどの場合、全体系は純粋状態  $|\Psi\rangle$  にあり、したがって  $\rho_{AB} := |\Psi\rangle\langle\Psi|$  であると仮定する。二つの部分系の一方についてトレースを取って得られる縮約密度行列は、

$$\rho_A := \text{Tr}_B \rho_{AB}, \quad \rho_B := \text{Tr}_A \rho_{AB}. \quad (1.126)$$

である。 $\rho_A$  または  $\rho_B$  に対応する von Neumann エントロピーをエンタングルメント・エントロピーと呼び、純粋状態  $|\Psi\rangle$  のエンタングルメントを測る。まもなく見るように、

$$S_A = -\text{Tr} \rho_A \log \rho_A = -\text{Tr} \rho_B \log \rho_B = S_B. \quad (1.127)$$

が成り立つ。これを詳しく論じる前に、非常に有用な技術的結果を二つ説明しておく必要がある。

**Schmidt 分解と Schmidt 数** 二部分系の純粋状態に関する基本的な定理が Schmidt 分解である。

$$|\Psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B. \quad (1.128)$$

とする。このとき、 $\mathcal{H}_A$  内の正規直交集合  $\{|i_A\rangle\}$  と  $\mathcal{H}_B$  内の正規直交集合  $\{|i_B\rangle\}$ 、および非負実数  $\lambda_i \geq 0$  が存在して、

$$|\Psi\rangle = \sum_{i=1}^r \lambda_i |i_A\rangle \otimes |i_B\rangle, \quad \sum_{i=1}^r \lambda_i^2 = 1. \quad (1.129)$$

と書ける。この結果の証明の概略をまもなく示す。整数

$$r := \#\{i : \lambda_i \neq 0\} \quad (1.130)$$

を、状態の *Schmidt* 数、または Schmidt 階数と呼ぶ。Schmidt 数  $r$  は、添字  $i_A$  と  $i_B$  の両方に関する和の範囲を定める。このため、二つの部分系の一方、例えば  $B$  が  $A$  よりもはるかに大きく、 $\dim \mathcal{H}_A \ll \dim \mathcal{H}_B$  である場合でも、 $B$  から  $\dim \mathcal{H}_A$  を超える本数の正規直交ベクトルを使う必要は決していない。

Schmidt 分解は次のようにして得られる。 $\mathcal{H}_A$  の正規直交基底  $\{|e_a\rangle\}$  と  $\mathcal{H}_B$  の正規直交基底  $\{|f_\alpha\rangle\}$  を選び、

$$|\Psi\rangle = \sum_{a,\alpha} c_{a\alpha} |e_a\rangle \otimes |f_\alpha\rangle. \quad (1.131)$$

と展開する。係数  $c_{a\alpha}$  を行列  $C$  とみなすことができる。線形代数の特異値分解を思い出そう。これによると、大きさ  $\dim \mathcal{H}_A \times \dim \mathcal{H}_A$  のユニタリー正方行列  $U$  と、大きさ  $\dim \mathcal{H}_B \times \dim \mathcal{H}_B$  のユニタリー正方行列  $V$  が存在して、

$$C = UDV^\dagger, \quad (1.132)$$

と書ける。ここで  $D$  は大きさ  $\dim \mathcal{H}_A \times \dim \mathcal{H}_B$  の長方形対角行列で、対角成分  $D_{ii} = \lambda_i$  は非負である。次を定義すると、

$$|i_A\rangle := \sum_a U_{ai} |e_a\rangle, \quad |i_B\rangle := \sum_\alpha V_{\alpha i}^* |f_\alpha\rangle, \quad (1.133)$$

(1.129) を得る。 $|\Psi\rangle$  の規格化から

$$1 = \langle \Psi | \Psi \rangle = \sum_i \lambda_i^2. \quad (1.134)$$

が従う。Schmidt 分解は、二部分系の純粋状態の構造を完全に明らかにする。状態が積状態であることと、その Schmidt 数が  $r = 1$  であることは同値である。 $r > 1$  ならば、その状態はエンタングルしている。

**Schmidt 形式での縮約密度行列** Schmidt 分解を用いると、全体系の密度行列は

$$\rho_{AB} = |\Psi\rangle\langle\Psi| = \sum_{i,j} \lambda_i \lambda_j (|i_A\rangle\langle j_A|) \otimes (|i_B\rangle\langle j_B|). \quad (1.135)$$

となる。部分系  $B$  についてトレースを取ると、

$$\begin{aligned} \rho_A &:= \text{Tr}_B \rho_{AB} = \sum_{i,j} \lambda_i \lambda_j |i_A\rangle\langle j_A| \text{Tr}(|i_B\rangle\langle j_B|) \\ &= \sum_{i,j} \lambda_i \lambda_j |i_A\rangle\langle j_A| \langle j_B | i_B \rangle = \sum_i \lambda_i^2 |i_A\rangle\langle i_A|. \end{aligned} \quad (1.136)$$

を得る。同様に、

$$\rho_B = \sum_i \lambda_i^2 |i_B\rangle\langle i_B|. \quad (1.137)$$

となる。ここから、部分トレースを論じる際に Schmidt 分解が重要である理由が分かる。この基底では、縮約密度行列は自動的に対角になるからである。したがって、 $\rho_A$  と  $\rho_B$  のゼロでない固有値は共通であり、

$$p_i := \lambda_i^2, \quad p_i \geq 0, \quad \sum_i p_i = 1. \quad (1.138)$$

である。このことから、直ちにいくつかの帰結が得られる。

第一に、 $\rho_A$  と  $\rho_B$  の階数は Schmidt 数に等しい。

$$\text{rank}(\rho_A) = \text{rank}(\rho_B) = r. \quad (1.139)$$

第二に、 $\rho_A$  と  $\rho_B$  のすべてのスペクトル量は一致する。特に、von Neumann エントロピーは等しい。

$$S(\rho_A) = S(\rho_B). \quad (1.140)$$

実際、

$$S(\rho_A) := -\text{Tr}(\rho_A \ln \rho_A) = -\sum_i p_i \ln p_i, \quad (1.141)$$

であり、同様に

$$S(\rho_B) = -\sum_i p_i \ln p_i. \quad (1.142)$$

となる。二部分系の純粋状態では、この共通のエントロピーをエンタングルメント・エントロピーと呼ぶ。

$$S_E(|\Psi\rangle) := S(\rho_A) = S(\rho_B). \quad (1.143)$$

縮約状態が純粋であることと、積状態であることの同値性 ここで、エンタングルメントと縮約状態の混合性の間の基本的な関係を証明する。  
まず、 $|\Psi\rangle$  が積状態であるとしよう。

$$|\Psi\rangle = |\phi\rangle_A \otimes |\chi\rangle_B. \quad (1.144)$$

すると

$$\rho_{AB} = (|\phi\rangle\langle\phi|)_A \otimes (|\chi\rangle\langle\chi|)_B, \quad (1.145)$$

であり、 $B$  についてトレースを取ると

$$\rho_A = |\phi\rangle\langle\phi| \operatorname{Tr}(|\chi\rangle\langle\chi|) = |\phi\rangle\langle\phi|. \quad (1.146)$$

を得る。したがって、 $\rho_A$  は純粋である。同様に、 $\rho_B$  も純粋である。  
逆に、 $\rho_A$  が純粋であるとしよう。 $\rho_A$  のゼロでない固有値は  $\lambda_i^2$  なので、純粋性より、ちょうど一つの固有値が 1 に等しく、その他はすべてゼロとなる。したがって、ゼロでない Schmidt 係数は一つだけであり、Schmidt 数は  $r = 1$  となって、

$$|\Psi\rangle = |1_A\rangle \otimes |1_B\rangle. \quad (1.147)$$

である。このように、元の純粋状態は積状態である。  
以上により、次が示された。

$$\rho_A \text{ pure} \iff |\Psi\rangle \text{ is a product state.} \quad (1.148)$$

等価に、

$$|\Psi\rangle \text{ entangled} \iff \rho_A \text{ is mixed,} \quad (1.149)$$

であり、 $\rho_B$  についても同様である。Schmidt 数で述べれば、これは単に

$$r = 1 \iff \text{product state,} \quad (1.150)$$

であるという主張であり、一方、

$$r > 1 \iff \text{entangled state.} \quad (1.151)$$

が成り立つ。例えば、Bell 状態

$$|\text{EPR}\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle_A |\downarrow\rangle_B - |\downarrow\rangle_A |\uparrow\rangle_B), \quad (1.152)$$

に対して、Schmidt 係数は

$$\lambda_1 = \lambda_2 = \frac{1}{\sqrt{2}}, \quad (1.153)$$

であるから、

$$S_E = -2 \cdot \frac{1}{2} \ln \frac{1}{2} = \ln 2. \quad (1.154)$$

となる。

純度、線形エントロピー、Rényi エントロピー von Neumann エントロピーに加えて、状態の混合性を調べる他の量を考えることも、ときに有用である。  
密度行列の純度を、次で定義する。

$$\mathcal{P}(\rho) := \text{Tr}(\rho^2). \quad (1.155)$$

これは

$$0 < \mathcal{P}(\rho) \leq 1, \quad (1.156)$$

を満たし、さらに

$$\mathcal{P}(\rho) = 1 \iff \rho \text{ is pure.} \quad (1.157)$$

が成り立つ。固有値が  $p_i$  である状態では、

$$\mathcal{P}(\rho) = \sum_i p_i^2. \quad (1.158)$$

となる。密接に関連する量として、次で定義される線形エントロピーがある。

$$S_L(\rho) := 1 - \text{Tr}(\rho^2) = 1 - \mathcal{P}(\rho). \quad (1.159)$$

これは純粋状態ではゼロとなり、混合状態では正となる。技術的には、von Neumann エントロピーよりも簡単な場合が多い<sup>5</sup>。

より一般に、Rényi エントロピーは、 $n > 0$ 、 $n \neq 1$  に対して次で定義される。

$$S_n(\rho) := \frac{1}{1-n} \ln \text{Tr}(\rho^n). \quad (1.161)$$

固有値  $p_i$  を用いれば、

$$S_n(\rho) = \frac{1}{1-n} \ln \left( \sum_i p_i^n \right). \quad (1.162)$$

となる。全体の規格化は、極限  $n \rightarrow 1$  が von Neumann エントロピーを再現するように選ばれている。

$$\lim_{n \rightarrow 1} S_n(\rho) = -\text{Tr}(\rho \ln \rho) = S(\rho). \quad (1.163)$$

$n = 2$  の場合は純度と直接関係している。

$$S_2(\rho) = -\ln \text{Tr}(\rho^2) = -\ln \mathcal{P}(\rho). \quad (1.164)$$

このように、純度、線形エントロピー、Rényi エントロピー、von Neumann エントロピーは、いずれも混合性を定量化する異なる方法を与える。

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<sup>5</sup>厳密には、von Neumann エントロピーは純粋状態の周りで Taylor 展開できない。 $\rho$  がゼロ固有値をもつ場合、 $S(\rho)$  は非解析的だからであり、「線形」という語はいくらか誤解を招く。むしろ、 $\sum_i p_i = 1$  かつ  $\epsilon \ll 1$  で、固有値が  $\{1 - \epsilon, \epsilon p_1, \dots, \epsilon p_{d-1}\}$  となるほぼ純粋な状態では、

$$S(\rho) = -\epsilon \ln \epsilon + \mathcal{O}(\epsilon), \quad S_L(\rho) = 1 - \text{Tr}(\rho^2) = 2\epsilon + \mathcal{O}(\epsilon^2).$$

を得る。したがって、状態が純粋状態に近づくとき、どちらのエントロピーもゼロになるが、 $S$  は特徴的な非解析的振る舞い  $-\epsilon \ln \epsilon$  を示す。実際には、 $S_L$  は最大混合状態の近傍で  $S$  を近似する。 $\text{Tr}(\delta\rho) = 0$  として  $\rho = \frac{1}{d} + \delta\rho$  と書き、対数を展開すると、

$$S(\rho) = \ln d - \frac{d}{2} \text{Tr}(\delta\rho^2) + \dots, \quad 1 - \text{Tr}(\rho^2) = 1 - \frac{1}{d} - \text{Tr}(\delta\rho^2). \quad (1.160)$$

が得られる。したがって、加法定数と乗法定数を除けば、 $S_L$  は  $S$  の最大値からのずれの主要項を測る。

**劣加法性と相互情報量** 量子情報理論という美しい分野には、二度だけ短く立ち入ることにする。ここでは、劣加法性と相互情報量を扱う。密度行列  $\rho_{AB}$  と縮約状態  $\rho_A$ 、 $\rho_B$  をもつ二部分系では、von Neumann エントロピーは劣加法性の不等式

$$S(\rho_{AB}) \leq S(\rho_A) + S(\rho_B). \quad (1.165)$$

を満たす。等価に、

$$S(\rho_A) + S(\rho_B) - S(\rho_{AB}) \geq 0. \quad (1.166)$$

である。この非負量を相互情報量と呼ぶ。

$$I(A : B) := S(\rho_A) + S(\rho_B) - S(\rho_{AB}). \quad (1.167)$$

これは、二つの部分系の間、古典的相関と量子的相関の両方を含む相関の総量を測る。積状態  $\rho_{AB} = \rho_A \otimes \rho_B$  に対しては、

$$I(A : B) = 0. \quad (1.168)$$

である。二部分系の純粋状態では、 $S(\rho_{AB}) = 0$  なので、相互情報量は

$$I(A : B) = 2S_E. \quad (1.169)$$

となる。ここで、相対エントロピーの正値性と Jensen の不等式に基づいて、劣加法性の証明の概略を示そう。相対エントロピーを次で定義する。

$$S(\rho \parallel \sigma) := \text{Tr}(\rho \ln \rho) - \text{Tr}(\rho \ln \sigma). \quad (1.170)$$

Klein の不等式とも呼ばれる標準的な結果によると、

$$S(\rho \parallel \sigma) \geq 0. \quad (1.171)$$

が成り立つ。その証明の一つは、関数

$$f(x) := x \ln x, \quad (1.172)$$

の凸性を、Jensen の不等式<sup>6</sup>とともに用い、具体的には  $\sigma$  の固有基底で適用するものである。劣加法性を証明するには、

$$\sigma := \rho_A \otimes \rho_B. \quad (1.174)$$

と選ぶ。このとき、相対エントロピーの正値性から

$$S(\rho_{AB} \parallel \rho_A \otimes \rho_B) \geq 0. \quad (1.175)$$

を得る。これを展開すると、

$$\begin{aligned} 0 &\leq \text{Tr}(\rho_{AB} \ln \rho_{AB}) - \text{Tr}(\rho_{AB} \ln(\rho_A \otimes \rho_B)) \\ &= \text{Tr}(\rho_{AB} \ln \rho_{AB}) - \text{Tr}(\rho_{AB}(\ln \rho_A \otimes \mathbb{1}_B)) - \text{Tr}(\rho_{AB}(\mathbb{1}_A \otimes \ln \rho_B)). \end{aligned} \quad (1.176)$$

である。部分トレースの定義上の性質を用いて、

$$\text{Tr}(\rho_{AB}(\ln \rho_A \otimes \mathbb{1}_B)) = \text{Tr}_A(\rho_A \ln \rho_A), \quad (1.177)$$

および  $B$  についての同様の関係を使うと、

$$0 \leq \text{Tr}(\rho_{AB} \ln \rho_{AB}) - \text{Tr}_A(\rho_A \ln \rho_A) - \text{Tr}_B(\rho_B \ln \rho_B). \quad (1.178)$$

を得る。 $-1$  を掛ければ、

$$S(\rho_{AB}) \leq S(\rho_A) + S(\rho_B), \quad (1.179)$$

となり、これが求める結果である。

<sup>6</sup>Jensen の不等式は、凸関数  $f$  と、 $\sum_i p_i = 1$  を満たす非負の重み  $p_i$  に対して、

$$f\left(\sum_i p_i x_i\right) \leq \sum_i p_i f(x_i). \quad (1.173)$$

が成り立つと述べる。

**Jensen の不等式から得られるエントロピーの凹性** もう一つの重要な性質は、von Neumann エントロピーが凹であることである。

$$S\left(\sum_{\alpha} p_{\alpha} \rho_{\alpha}\right) \geq \sum_{\alpha} p_{\alpha} S(\rho_{\alpha}). \quad (1.180)$$

有用な特殊例は、密度行列を対角化して

$$\rho = \sum_i p_i |i\rangle\langle i|, \quad (1.181)$$

とした場合であり、このとき

$$S(\rho) = -\sum_i p_i \ln p_i. \quad (1.182)$$

となる。関数

$$g(x) := -\ln x \quad (1.183)$$

は  $x > 0$  上で凸であるため、Jensen の不等式から

$$-\ln\left(\sum_i p_i x_i\right) \leq \sum_i p_i (-\ln x_i). \quad (1.184)$$

が従う。この初等的な不等式は、相対エントロピーの正值性や古典的な劣加法性を含む、多くのエントロピー不等式の基礎となっている。

**純粋化** これまでは、全体系の状態から出発し、部分トレースを取ることで縮約密度行列を得る方法を見てきた。逆方向に進む方法があるか、もちろん一意ではないとしても、と問うのは自然である。これは実際に可能であり、Hilbert 空間  $\mathcal{H}_A$  上の任意の混合状態は、常に、より大きな Hilbert 空間内の純粋状態の縮約状態とみなせることが分かる。この構成を純粋化と呼ぶ。

次を、

$$\rho_A = \sum_i p_i |i_A\rangle\langle i_A| \quad (1.185)$$

$\mathcal{H}_A$  上の密度行列のスペクトル分解とし、 $p_i \geq 0$  および  $\sum_i p_i = 1$  とする。 $\rho_A$  の階数以上の次元をもち、正規直交基底  $\{|i_R\rangle\}$  を備えた補助 Hilbert 空間  $\mathcal{H}_R$  を導入する。このとき、状態

$$|\Psi\rangle := \sum_i \sqrt{p_i} |i_A\rangle \otimes |i_R\rangle \in \mathcal{H}_A \otimes \mathcal{H}_R \quad (1.186)$$

は純粋であり、

$$\mathrm{Tr}_R(|\Psi\rangle\langle\Psi|) = \rho_A. \quad (1.187)$$

を満たす。実際、

$$|\Psi\rangle\langle\Psi| = \sum_{i,j} \sqrt{p_i p_j} |i_A\rangle\langle j_A| \otimes |i_R\rangle\langle j_R|, \quad (1.188)$$

なので、 $R$  についてトレースを取ると

$$\begin{aligned} \mathrm{Tr}_R(|\Psi\rangle\langle\Psi|) &= \sum_{i,j} \sqrt{p_i p_j} |i_A\rangle\langle j_A| \langle j_R|i_R\rangle \\ &= \sum_i p_i |i_A\rangle\langle i_A| = \rho_A. \end{aligned} \quad (1.189)$$

を得る。したがって、どの混合状態も、拡大された系とのエンタングルメントから生じると考えられる。

純粋化 (1.186) は一意ではない。異なる純粋化は、補助 Hilbert 空間  $\mathcal{H}_R$  上に作用するユニタリー変換によって関係づけられる。この自由度は、異なる多数の、より大きな純粋状態から、同じ縮約密度行列が生じうることを反映している。

以上をまとめると、Schmidt 分解は、すべての二部分系の純粋状態が、非負の Schmidt 係数の集合によって特徴づけられることを示す。これらの係数は、縮約密度行列、エンタングルメント・エントロピー、Schmidt 数、純度、そして Rényi エントロピーの族全体を決定する。特に、縮約状態が純粋であることと、元の状態が積状態であることは同値であり、縮約状態が混合であることと、元の状態がエンタングルしていることも同値である。最後に、純粋化は、どの混合状態も、より大きな純粋な系に埋め込めることを示す。この見方は、開放量子系の研究で中心的なものとなる。

## 1.5 デコヒーレンス

ここで、複合系におけるエンタングルメントの中心的な物理的帰結の一つ、デコヒーレンスに進む。ここでは、非常に単純ながら広範な含意をもつ基本的な考え方だけを論じる。結合した系がユニタリーに発展し、純粋状態を保っている、部分系は、観測されない自由度とエンタングルするため、純粋状態から混合状態へ発展しうる。この意味で、デコヒーレンスは量子力学の破綻ではなく、より大きな系に量子力学を適用しつつ、その一部だけを観測することの直接の帰結である。

そこで、Hilbert 空間

$$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B, \quad (1.190)$$

をもつ二部分系を考える。 $A$  は関心のある系、 $B$  はその環境、測定装置、あるいは浴である。時刻  $t = 0$  で、全状態は純粋かつ分離可能であり、

$$|\Psi_0\rangle = |\phi\rangle_A \otimes |\chi\rangle_B, \quad (1.191)$$

したがって

$$\rho_{AB}(0) = |\Psi_0\rangle\langle\Psi_0|, \quad \rho_A(0) = |\phi\rangle\langle\phi|. \quad (1.192)$$

とする。特に、部分系  $A$  の初期縮約状態は純粋である。

$$\rho_A(0)^2 = \rho_A(0). \quad (1.193)$$

次節で見るように、また標準的な量子力学でもおなじみであろうように、全体系はある演算子  $U_{AB}(t)$  のもとでユニタリーに発展する。

$$\rho_{AB}(t) = U_{AB}(t) \rho_{AB}(0) U_{AB}^\dagger(t). \quad (1.194)$$

$A$  と  $B$  の間に相互作用がなく、発展が

$$U_{AB}(t) = U_A(t) \otimes U_B(t), \quad (1.195)$$

と因子化するならば、 $A$  の縮約密度行列は、すべての時刻で純粋なままである。なぜなら

$$\rho_A(t) = U_A(t) \rho_A(0) U_A^\dagger(t). \quad (1.196)$$

だからである。したがって、部分系が環境とエンタングルしなければ、純粋状態は純粋なまま保たれる。

$A$  が  $B$  と相互作用すると、状況は質的に変わる。この場合、全体のユニタリー発展は因子化するとは限らず、縮約密度行列は

$$\rho_A(t) := \text{Tr}_B \rho_{AB}(t) = \text{Tr}_B \left( U_{AB}(t) \rho_{AB}(0) U_{AB}^\dagger(t) \right). \quad (1.197)$$

となる。上述の初期積状態を仮定すると、これを有用な形に書き直せる。 $\mathcal{H}_B$  の正規直交基底  $\{|\beta\rangle\}$  を選び、次を定義する。

$$M_\beta(t) := \langle\beta|U_{AB}(t)|\chi\rangle, \quad (1.198)$$

ここで  $M_\beta(t)$  は、Kraus 演算子と呼ばれる、 $\mathcal{H}_A$  上の演算子である。すると

$$\rho_A(t) = \sum_{\beta} M_\beta(t) \rho_A(0) M_\beta^\dagger(t). \quad (1.199)$$

となる。これは、縮約時間発展の演算子和表現、または Kraus 表現である。演算子  $M_\beta(t)$  は、

$$\sum_{\beta} M_\beta^\dagger(t) M_\beta(t) = \mathbb{1}_A, \quad (1.200)$$

を満たす。これは  $U_{AB}(t)$  のユニタリー性の帰結である。実際、

$$\begin{aligned} \sum_{\beta} M_\beta^\dagger M_\beta &= \sum_{\beta} \langle \chi | U_{AB}^\dagger | \beta \rangle \langle \beta | U_{AB} | \chi \rangle \\ &= \langle \chi | U_{AB}^\dagger \left( \sum_{\beta} | \beta \rangle \langle \beta | \right) U_{AB} | \chi \rangle \\ &= \langle \chi | U_{AB}^\dagger U_{AB} | \chi \rangle = \mathbb{1}_A. \end{aligned} \quad (1.201)$$

である。したがって、縮約時間発展は  $\rho_A$  のトレースを保存する。

物理的に重要なのは、相互作用する系では、縮約状態  $\rho_A(t)$  が一般にはもはや純粋ではないことである。

$$\rho_A(t)^2 \neq \rho_A(t). \quad (1.202)$$

この純粋性の喪失は、 $A$  と  $B$  の間に生成されたエンタングルメントを反映している。このとき、部分系  $A$  にデコヒーレンスが生じたという。実際には、デコヒーレンスは、ある選ばれた基底における密度行列の非対角成分の抑制として現れることが多い。

**簡単な測定モデル**  $A$  を基底  $\{|\uparrow\rangle, |\downarrow\rangle\}$  をもつ量子ビットとし、 $B$  を正規直交する指針状態  $\{|0\rangle, |1\rangle, |2\rangle\}$  をもつ測定装置とすると、特に示唆的な例が得られる。装置の初期状態を  $|0\rangle$  とし、次を満たす理想化された「1 ステップ」のユニタリー発展を考える。

$$U(|\uparrow\rangle \otimes |0\rangle) = |\uparrow\rangle \otimes (\sqrt{1-p}|0\rangle + \sqrt{p}|1\rangle), \quad (1.203)$$

$$U(|\downarrow\rangle \otimes |0\rangle) = |\downarrow\rangle \otimes (\sqrt{1-p}|0\rangle + \sqrt{p}|2\rangle), \quad (1.204)$$

ここで  $0 \leq p \leq 1$  である。言葉で言えば、装置は準備状態  $|0\rangle$  にとどまることもあれば、 $|1\rangle$  へ移ることで結果  $|\uparrow\rangle$  を、あるいは  $|2\rangle$  へ移ることで結果  $|\downarrow\rangle$  を記録することもある。

環境は  $|0\rangle$  から始まるため、Kraus 演算子は

$$M_\beta = \langle \beta | U | 0 \rangle, \quad \beta = 0, 1, 2. \quad (1.205)$$

となる。上述の  $U$  の定義から、直ちに

$$M_0 = \sqrt{1-p} \mathbb{1}_A, \quad M_1 = \sqrt{p} |\uparrow\rangle \langle \uparrow|, \quad M_2 = \sqrt{p} |\downarrow\rangle \langle \downarrow|. \quad (1.206)$$

を得る。次の関係は容易に確かめられる。

$$M_0^\dagger M_0 + M_1^\dagger M_1 + M_2^\dagger M_2 = (1-p) \mathbb{1}_A + p |\uparrow\rangle \langle \uparrow| + p |\downarrow\rangle \langle \downarrow| = \mathbb{1}_A. \quad (1.207)$$

量子ビットの初期密度行列を

$$\rho_A = \begin{pmatrix} \rho_{\uparrow\uparrow} & \rho_{\uparrow\downarrow} \\ \rho_{\downarrow\uparrow} & \rho_{\downarrow\downarrow} \end{pmatrix}. \quad (1.208)$$

とする。相互作用の後、縮約密度行列は

$$\rho_A \mapsto \sum_{\beta=0}^2 M_{\beta} \rho_A M_{\beta}^{\dagger}. \quad (1.209)$$

となる。Kraus 演算子の具体的な形を用いると、

$$\rho_A \mapsto \begin{pmatrix} \rho_{\uparrow\uparrow} & (1-p)\rho_{\uparrow\downarrow} \\ (1-p)\rho_{\downarrow\uparrow} & \rho_{\downarrow\downarrow} \end{pmatrix}. \quad (1.210)$$

を得る。このように、対角成分は変化せず、非対角成分は因子  $(1-p)$  によって抑制される。これこそがデコヒーレンスの特徴である。二つの結果の確率は保たれる一方で、 $|\uparrow\rangle$  と  $|\downarrow\rangle$  の間のコヒーレンスが徐々に失われる。

**位相減衰** 上述の 1 ステップ過程が短い時間間隔  $\delta t$  ごとに繰り返し作用するとし、次の率を定義する。

$$\Gamma := \frac{p}{\delta t}. \quad (1.211)$$

$N$  ステップ後、経過時間の合計が  $t = N\delta t$  のとき、非対角項には

$$(1-p)^N = \left(1 - \frac{\Gamma t}{N}\right)^N \approx e^{-\Gamma t} \quad \text{for } N \gg 1. \quad (1.212)$$

が掛かる。したがって、縮約密度行列は近似的に

$$\rho_A(t) = \begin{pmatrix} \rho_{\uparrow\uparrow}(0) & e^{-\Gamma t} \rho_{\uparrow\downarrow}(0) \\ e^{-\Gamma t} \rho_{\downarrow\uparrow}(0) & \rho_{\downarrow\downarrow}(0) \end{pmatrix}. \quad (1.213)$$

と発展する。この過程は位相減衰、または位相緩和と呼ばれることが多い。量子マスター方程式の解として、再びこの過程に出会うことになる。

例えば、量子ビットの初期状態が純粋であり、

$$|\psi\rangle = a|\uparrow\rangle + b|\downarrow\rangle, \quad (1.214)$$

であるなら、初めは

$$\rho_A(0) = \begin{pmatrix} |a|^2 & ab^* \\ a^*b & |b|^2 \end{pmatrix}, \quad (1.215)$$

であるのに対し、十分時間が経つと

$$\lim_{t \rightarrow \infty} \rho_A(t) = \begin{pmatrix} |a|^2 & 0 \\ 0 & |b|^2 \end{pmatrix}. \quad (1.216)$$

となる。最終状態は、装置によって選ばれた基底、すなわち  $\{|\uparrow\rangle, |\downarrow\rangle\}$  において対角である。この基底は、選択基底、あるいは指針基底と呼ばれることが多い。系を上向きまたは下向きに見いだす確率は変わっていない。変わったのは、量子干渉に典型的な、上向きと下向きの両方で同時にあるかのような振る舞いを、系がもはや示さないことである。代わりに、上向きである古典的確率  $|a|^2$  と、下向きである古典的確率  $|b|^2$  だけが存在する。これら二つの状態の間のコヒーレンスの喪失により、量子力学は古典的な統計集団へと帰着する。

この振る舞いの原因は明らかである。部分系  $A$  は装置  $B$  とエンタングルし、 $B$  の自由度を無視すると、 $A$  の縮約状態は位相のコヒーレンスを失う。したがって、デコヒーレンスは、巨視的な状況でコヒーレントな重ね合わせを観測しにくい理由を説明する。環境との相互作用は、その環境への支配的な結合によって選ばれた基底で、非対角項を急速に抑制するのである。

## 2 量子マスター方程式

密度行列と複合量子系の基本的な運動学を導入したので、ここから動力学に進む。全体系がユニタリーに発展していても、部分系の縮約密度行列は一般にそうではない。環境自由度についてトレースを取ると、部分系は、散逸、デコヒーレンス、ノイズを含みうる有効的な発展法則によって記述される。本節の中心的な目的は、そのような縮約動力学を、数学的に整合的で、物理的な意味が明瞭な形で、どのように特徴づけられるかを理解することである。

まず、全体系のユニタリー発展と開放部分系の有効的な発展を対比し、次に縮約力学写像、その Kraus 表現、および正值性、完全正值性、トレース保存性の概念を導入する。続いて、連続的な時間発展が Liouvillian によって生成され、Markov 極限で普遍的な Gorini–Kossakowski–Sudarshan–Lindblad 形式を取る条件を調べる。最後に、この記述を導く微視的な仮定を論じ、宇宙論でしばしば重要となるように生成子が陽に時間に依存するとき、通常の半群の描像をどう一般化すべきかを説明する。

### 2.1 全体系と開放系の時間発展

これまでの、密度行列を量子系の状態を符号化する運動学的対象として論じてきた。ここから、その時間発展を論じる。まず、全体系を制御でき、量子力学の標準的なユニタリー発展を用いられると仮定しよう。系がユニタリーに発展する場合、Schrödinger 描像で純粋状態は

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle, \quad (2.1)$$

と発展する。ここで  $U(t)$  は、次を満たすユニタリー演算子である。

$$U^\dagger(t)U(t) = U(t)U^\dagger(t) = \mathbf{1}_{\mathcal{H}}. \quad (2.2)$$

このことから直ちに、密度行列は次に従って発展する。

$$\rho(t) = U(t)\rho(0)U^\dagger(t). \quad (2.3)$$

実際、純粋状態  $\rho(0) = |\psi(0)\rangle\langle\psi(0)|$  に対して、

$$\begin{aligned} \rho(t) &:= |\psi(t)\rangle\langle\psi(t)| \\ &= U(t)|\psi(0)\rangle\langle\psi(0)|U^\dagger(t) \\ &= U(t)\rho(0)U^\dagger(t). \end{aligned} \quad (2.4)$$

である。線形性により、同じ公式は混合状態にも成り立つ。

式 (2.3) は、密度行列を定義するすべての性質を保存する。Hermite 性が保存されるのは、

$$\rho^\dagger(t) = (U(t)\rho(0)U^\dagger(t))^\dagger = U(t)\rho^\dagger(0)U^\dagger(t) = \rho(t). \quad (2.5)$$

だからである。正值性も保存される。任意の  $|\phi\rangle \in \mathcal{H}$  について、

$$\langle\phi|\rho(t)|\phi\rangle = \langle\phi|U(t)\rho(0)U^\dagger(t)|\phi\rangle = \langle\phi'(t)|\rho(0)|\phi'(t)\rangle \geq 0, \quad (2.6)$$

だからである。ここでは  $|\phi'(t)\rangle := U^\dagger(t)|\phi\rangle$  と定義した。最後に、巡回性によりトレースも保存される。

$$\mathrm{Tr}\rho(t) = \mathrm{Tr}(U(t)\rho(0)U^\dagger(t)) = \mathrm{Tr}(\rho(0)U^\dagger(t)U(t)) = \mathrm{Tr}\rho(0) = 1. \quad (2.7)$$

したがって、ユニタリー発展は密度行列を密度行列へ写す。

Hamiltonian が時間に依存しない場合、ユニタリー時間発展演算子は

$$U(t) = e^{-iHt}, \quad (2.8)$$

である。ここでは  $\hbar = 1$  とした。(2.3) を時間で微分すると、von Neumann 方程式

$$i \frac{d\rho}{dt} = [H, \rho]. \quad (2.9)$$

を得る。実際、

$$\begin{aligned} \frac{d\rho}{dt} &= \frac{dU}{dt} \rho(0) U^\dagger + U \rho(0) \frac{dU^\dagger}{dt} \\ &= (-iHU) \rho(0) U^\dagger + U \rho(0) (iU^\dagger H) \\ &= -iH\rho(t) + i\rho(t)H, \end{aligned} \quad (2.10)$$

であり、これは (2.9) と等価である。この方程式は Schrödinger 方程式の密度行列版であり、古典力学の Liouville 方程式の量子版とみなせる<sup>7</sup>。

ユニタリー発展の有用な帰結として、観測量  $Q$  の期待値を、異なる描像で等価に計算できる。Schrödinger 描像では状態が発展し、演算子は固定されるので、

$$\langle Q \rangle_t = \text{Tr}(\rho_S(t) Q_S). \quad (2.12)$$

となる。Heisenberg 描像では状態が固定され、演算子は

$$Q_H(t) := U^\dagger(t) Q_S U(t), \quad \rho_H := \rho(0). \quad (2.13)$$

と発展する。トレースの巡回性を用いると、

$$\text{Tr}(\rho_S(t) Q_S) = \text{Tr}(U(t) \rho(0) U^\dagger(t) Q_S) = \text{Tr}(\rho(0) U^\dagger(t) Q_S U(t)) = \text{Tr}(\rho_H Q_H(t)). \quad (2.14)$$

を得る。このように、閉鎖系では密度行列形式は量子力学の通常の定式化と完全に等価であり、しかも純粋状態と混合状態を統一的に扱えるという利点がある。

最後に、(2.3) は相似変換であるため、ユニタリー発展は  $\rho$  のスペクトルを保存する。したがって、 $\text{Tr}(\rho^n)$  や von Neumann エントロピー  $S(\rho)$  のように、 $\rho$  の固有値だけに依存する量は、閉鎖系の時間発展のもとで一定に保たれる。特に、ユニタリー発展のもとで純粋状態は純粋なまま、混合状態は混合のままである。後に部分系だけに注目し、観測しない自由度をトレースで消去して初めて、純度やエントロピーが非自明に変化できるようになる。

**開放系と縮約動力学** これまでは、閉じた量子系のユニタリー時間発展を論じてきた。しかし、多くの物理的状況では、関心のある系は孤立していない。むしろ、未知であったり、制御されなかったり、観測されなかったり、単に詳細に記述するには複雑すぎたりする追加の自由度と相互作用している。このとき、開放量子系という。基本的な方針は、「系」と呼ぶ部分系  $S$  を同定し、残りの自由度を、一般に「環境」と呼ぶ別の部分系  $E$  へ分離することである。このとき、全 Hilbert 空間を

$$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E. \quad (2.15)$$

と書く。結合した系  $S + E$  がユニタリーに発展していても、部分系  $S$  だけでは一般にそうならない。

環境という語は最も一般的なものであり、以下でも主にこれを用いる。文献では浴やリザーバーという語にもよく出会う。大まかに言えば、リザーバーは通常、 $S$  との相互作用によって巨視的

<sup>7</sup>古典統計力学では、Liouville 方程式は、Hamilton 流のもとでの位相空間上の確率密度  $f(q, p, t)$  の時間発展を支配する。

$$\frac{\partial f}{\partial t} = \{H, f\}, \quad (2.11)$$

ここで  $\{\cdot, \cdot\}$  は Poisson 括弧を表す。これは、Hamilton 方程式の解  $(q(t), p(t))$  に沿って  $f(q(t), p(t), t)$  が一定であるという主張と等価である。von Neumann 方程式との類似点は、両者とも Hamilton 動力学のもとでの統計的状態の決定論的な発展を記述することである。古典の場合は位相空間上の確率分布、量子の場合は Hilbert 空間上の密度行列である。

な性質が実質的に影響を受けないほど大きな環境を示唆する。一方、浴はしばしば、熱浴の場合の熱平衡のように、追加の物理的性質をもつ環境を指す。こうした区別は物理的に有用な場合が多いが、以下の形式的な展開で重要なのは、関心のある部分系の外部に自由度があり、それらを陽には追跡しないという点だけである。

全体系の密度行列を  $\rho_{SE}(t)$  とする。そのユニタリー時間発展は

$$\rho_{SE}(t) = U(t) \rho_{SE}(0) U^\dagger(t), \quad (2.16)$$

である。ここで  $U(t)$  は  $\mathcal{H}_S \otimes \mathcal{H}_E$  上に作用する。部分系  $S$  にとって物理的に重要な対象は、縮約密度行列

$$\rho_S(t) := \text{Tr}_E \rho_{SE}(t). \quad (2.17)$$

である。この表式だけでも、開放系の動力学が一般に閉鎖系の動力学より複雑になる理由が明らかになる。 $\rho_{SE}$  の発展はユニタリーで、したがって可逆であっても、環境についてトレースを取って得られる  $\rho_S$  の発展は、ユニタリーでも可逆でもあるとは限らない。

初期密度行列が因子化しているならば<sup>8</sup>、

$$\rho_{SE}(0) = \rho_S(0) \otimes \rho_E(0), \quad (2.18)$$

縮約時間発展は、系の初期状態から終状態への写像として書ける。

$$\rho_S(t) = \Phi_t[\rho_S(0)], \quad (2.19)$$

ここで

$$\Phi_t[\rho] := \text{Tr}_E \left( U(t) (\rho \otimes \rho_E(0)) U^\dagger(t) \right). \quad (2.20)$$

である。写像の族  $\{\Phi_t\}$  を、部分系の量子力学写像、または縮約力学写像と呼ぶ。これは開放系理論の基本的な対象である。

仮定 (2.18) は、第一印象よりも強い。代わりに初期状態  $\rho_{SE}(0)$  が  $S$  と  $E$  の間の相関をすでに含んでいる場合、後の時刻の縮約状態は  $\rho_S(0)$  だけでは決まらないことがある。その場合、部分系の発展は一般に、(2.20) の形の、状態に依存しない写像では記述できない。したがって、開放系の動力学を明確に定義された写像  $\Phi_t$  で記述できるということ自体が、すでに系と環境の初期相関についての仮定に基づいている。

いくつか注意しておこう。第一に、ユニタリー発展も部分トレースも線形操作なので、写像  $\Phi_t$  は  $\rho_S(0)$  に関して線形である。第二に、全体系のユニタリー発展と異なり、部分系の縮約時間発展は  $\rho_S$  の純度とエントロピーを変えうる。これはまさに、系と環境の間を情報が流れうること、そして構成上、環境自由度を追跡しないことによる。これが、縮約された記述におけるデコヒーレンス、散逸、エントロピー生成の形式的な起源である。

最後に、写像 (2.20) は密度行列を密度行列へ写す。すなわち、Hermite 性、正值性、トレースが 1 であることを保存する。この段階でそれらの性質を抽象的に証明するより、まず写像をより具体的な形に書き直すと有用である。そうすると自然に Kraus 表現に至り、これらの構造的性質が明示される。

## 2.2 縮約力学写像の Kraus 表現

ここで、縮約力学写像 (2.20) を、より具体的な演算子の形に書き直す。これによって構造がはるかに明瞭になるとともに、許容される量子写像についての、より抽象的な議論への準備にもなる。本小節の論理的な内容は、全体系のユニタリー発展と環境についてのトレース操作から、力学写像が Kraus 表現に書けることが従う、というものである。

$$\text{Partial trace over environment} \Rightarrow \text{Kraus representation.} \quad (2.21)$$

<sup>8</sup>この仮定は、初めに因子化した純粋状態  $\rho_{SE}(0) = |\psi_S\rangle\langle\psi_S| \otimes |\chi_E\rangle\langle\chi_E|$  を取ることよりも一般的である。ここでは、 $\rho_S(0)$  と  $\rho_E(0)$  自身は混合状態でもよい。以下の導出で重要なのは、初期の系と環境の相関がないこと、すなわち密度行列が因子化していることだけである。

出発点は写像

$$\Phi_t[\rho] := \text{Tr}_E \left( U(t) (\rho \otimes \rho_E) U^\dagger(t) \right), \quad (2.22)$$

である。ここでは記法を簡単にするため、 $\rho_E(0)$  の代わりに  $\rho_E$  と書いた。この段階では、初期密度行列が (2.18) と因子化していることだけを仮定する。 $\rho_E$  は  $\mathcal{H}_E$  上の任意の密度行列であり、混合状態でもよい。

環境の密度行列を対角化しよう。

$$\rho_E = \sum_{\beta} p_{\beta} |\beta\rangle\langle\beta|, \quad p_{\beta} \geq 0, \quad \sum_{\beta} p_{\beta} = 1, \quad (2.23)$$

ここで  $\{|\beta\rangle\}$  は、 $\rho_E$  の固有ベクトルからなる  $\mathcal{H}_E$  の正規直交基底である。また、部分トレースを実行するため、 $\mathcal{H}_E$  の任意の正規直交基底  $\{|\alpha\rangle\}$  を選ぶ。すると

$$\begin{aligned} \Phi_t[\rho] &= \text{Tr}_E \left( U(t) (\rho \otimes \rho_E) U^\dagger(t) \right) \\ &= \sum_{\alpha} \langle \alpha | U(t) (\rho \otimes \rho_E) U^\dagger(t) | \alpha \rangle \\ &= \sum_{\alpha, \beta} p_{\beta} \langle \alpha | U(t) (\rho \otimes |\beta\rangle\langle\beta|) U^\dagger(t) | \alpha \rangle. \end{aligned} \quad (2.24)$$

となる。 $\rho$  は系の Hilbert 空間にしか作用しないため、これは

$$\begin{aligned} \Phi_t[\rho] &= \sum_{\alpha, \beta} \left( \sqrt{p_{\beta}} \langle \alpha | U(t) | \beta \rangle \right) \rho \left( \sqrt{p_{\beta}} \langle \beta | U^\dagger(t) | \alpha \rangle \right) \\ &= \sum_{\alpha, \beta} M_{\alpha\beta}(t) \rho M_{\alpha\beta}^\dagger(t), \end{aligned} \quad (2.25)$$

と書き直せる。ここで、*Kraus* 演算子を次で定義した。

$$M_{\alpha\beta}(t) := \sqrt{p_{\beta}} \langle \alpha | U(t) | \beta \rangle. \quad (2.26)$$

ここで  $M_{\alpha\beta}(t)$  は、系の Hilbert 空間  $\mathcal{H}_S$  にのみ作用する。力学写像  $\Phi_t$  の表式 (2.25) が、この写像の *Kraus* 表現である。

*Kraus* 演算子が満たす規格化条件は、 $U(t)$  のユニタリー性と  $\text{Tr} \rho_E = 1$  という事実から直接従う。実際、

$$\begin{aligned} \sum_{\alpha, \beta} M_{\alpha\beta}^\dagger(t) M_{\alpha\beta}(t) &= \sum_{\alpha, \beta} p_{\beta} \langle \beta | U^\dagger(t) | \alpha \rangle \langle \alpha | U(t) | \beta \rangle \\ &= \sum_{\beta} p_{\beta} \langle \beta | U^\dagger(t) \left( \sum_{\alpha} |\alpha\rangle\langle\alpha| \right) U(t) | \beta \rangle \\ &= \sum_{\beta} p_{\beta} \langle \beta | U^\dagger(t) U(t) | \beta \rangle = \sum_{\beta} p_{\beta} \mathbf{1}_S = \mathbf{1}_S. \end{aligned} \quad (2.27)$$

である。組  $(\alpha, \beta)$  は単に *Kraus* 演算子のラベルであり、常に一つの複合添字  $\mu := (\alpha, \beta)$  にまとめられる。すると力学写像は、より一般的に用いられる形

$$\Phi_t[\rho] = \sum_{\mu} M_{\mu}(t) \rho M_{\mu}^\dagger(t), \quad (2.28)$$

に書き直せ、規格化条件は

$$\sum_{\mu} M_{\mu}^\dagger(t) M_{\mu}(t) = \mathbf{1}_S. \quad (2.29)$$

となる。この規格化条件は、トレースではなく演算子としての等式であることに注意しよう。したがって、二添字版と一添字版は構造的に異なるものではなく、同じ演算子表現にラベルをつける二つの方法にすぎない。二添字の形は、初期環境状態のスペクトル分解と、終状態の環境基底についての部分トレースから、Kraus 演算子がどのように生じるかを追跡できるため有用である。Kraus 表現は一意ではないことを強調しておく。異なる Kraus 演算子の集合が、同じ力学写像を記述しうる。特に、微視的な導出を素直に行ってある Kraus 演算子が得られたとしても、より少数の演算子で同じ写像を書き直せる場合がある。したがって、添字の数そのものは写像の基本的な性質ではなく、特定の表現に属するものである。  
特殊な場合として、環境が純粋状態

$$\rho_E = |\chi\rangle\langle\chi|. \quad (2.30)$$

から始まるとしよう。このときスペクトル分解には、1 に等しいゼロでない固有値が一つしかなく、 $\beta$  についての和は一項になる。Kraus 演算子は

$$M_\alpha(t) := \langle\alpha|U(t)|\chi\rangle, \quad (2.31)$$

に帰着し、縮約写像はより簡単な形

$$\Phi_t[\rho] = \sum_\alpha M_\alpha(t) \rho M_\alpha^\dagger(t). \quad (2.32)$$

を取る。これは初等的な議論でよく書かれる形である。初期密度行列が (2.22) と因子化しているという特殊な場合に有効である。

本小節の結論は、初めに因子化した密度行列と全体系のユニタリー発展から得られる任意の縮約力学写像は、常に Kraus 表現をもつ、ということである。

### 2.3 正写像、完全正写像、トレース保存写像

縮約力学写像の Kraus 表現を導出したので、ここで一步引いて、より抽象的な問いを考える。単に、系の Hilbert 空間上の演算子に作用する線形写像

$$\Phi: \mathcal{L}(\mathcal{H}_S) \rightarrow \mathcal{L}(\mathcal{H}_S) \quad (2.33)$$

が与えられたとしよう。密度行列の物理的に許容される発展を記述するために、このような写像はどのような性質を満たすべきだろうか。その答えから、正值性、完全正值性、トレース保存性の概念へと導かれる。

**正值性** 最初の明らかな要請は、 $\Phi$  が正の演算子を正の演算子へ写すことである。密度行列  $\rho$  が正であるとは、

$$\langle\psi|\rho|\psi\rangle \geq 0 \quad \text{for all } |\psi\rangle \in \mathcal{H}_S. \quad (2.34)$$

を満たすことであった。したがって、線形写像  $\Phi$  は、

$$\rho \geq 0 \quad \implies \quad \Phi[\rho] \geq 0. \quad (2.35)$$

を満たすとき正と呼ばれる。 $\Phi$  が密度行列を密度行列へ写すためには、明らかにこれは必要である。しかし、正值性だけでは不十分であることが分かる。系  $S$  自身がより大きな系の一部でありうるため、 $S$  に物理的に作用する写像は、複合 Hilbert 空間の一つの因子にだけ適用した場合でも、物理的に意味のあるものでなければならないからである。

**完全正值性**  $\mathcal{H}_A$  を、しばしばアンシラと呼ばれる補助 Hilbert 空間とし、 $\mathbf{1}_A$  を  $\mathcal{H}_A$  上の演算子に作用する恒等写像とする。 $\mathcal{L}(\mathcal{H}_S)$  上の写像  $\Phi$  が与えられると、拡張された写像

$$\mathbf{1}_A \otimes \Phi \quad (2.36)$$

を定義できる。これは、アンシラを変化させず、系にだけ  $\Phi$  を適用することで、 $\mathcal{H}_A \otimes \mathcal{H}_S$  上の演算子に作用する。

写像  $\Phi$  が完全正であるとは、あらゆる補助 Hilbert 空間  $\mathcal{H}_A$  に対し、

$$X \geq 0 \implies (\mathbf{1}_A \otimes \Phi)[X] \geq 0 \quad (2.37)$$

が、 $\mathcal{H}_A \otimes \mathcal{H}_S$  上のすべての正の演算子  $X$  について成り立つことをいう。したがって、完全正值性は、系単独での正值性だけでなく、任意に大きい量子系の一部として系を見る場合にも正值性を要求する。

この要請は物理的に自然である。 $\Phi$  が部分系  $S$  の許容される発展を記述するならば、 $S$  が、それ自身は発展させない傍観的な自由度とたまたまエンタングルしていても、非物理的な演算子を生じてはならない。完全正写像はすべて正写像であるが、その逆は成り立たない。

有名な反例が転置写像である。固定した基底で、これは

$$T[\rho] := \rho^T. \quad (2.38)$$

と定義される。正行列を転置しても固有値は同じで、依然として正であるため、この写像は正である。しかし、完全正ではない。これを見るため、最大エンタングルした Bell 状態

$$|\Phi^+\rangle := \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad \rho_{\Phi^+} := |\Phi^+\rangle\langle\Phi^+|. \quad (2.39)$$

を考える。第二の部分系だけに転置を適用すると、得られる演算子

$$(\mathbf{1} \otimes T)[\rho_{\Phi^+}] \quad (2.40)$$

は  $\rho_{\Phi^+}$  の部分転置であり、負の固有値をもつ。したがって、これは正の演算子ではない。これにより、 $T$  は単独の系では正であるが、より大きな系へ自明に拡張すると正でなくなること、したがって完全正ではないことが分かる。

**トレース保存性** 第二の基本的な要請は、全確率が規格化されたままであることである。写像  $\Phi$  はトレースを保存すべきである。

$$\text{Tr } \Phi[\rho] = \text{Tr } \rho. \quad (2.41)$$

この性質をもつ写像をトレース保存写像と呼ぶ。 $\Phi$  がトレースを保存し、密度行列を正の演算子へ写すなら、それは密度行列を密度行列へ写す。

完全正で、かつトレースを保存する線形写像を *CPTP* 写像と呼ぶ。有限次元の量子情報理論では、こうした写像は量子チャネルともよく呼ばれる。これが、量子系の物理的に許容される時間発展写像を抽象的に捉える適切な概念である。

力学写像が密度行列の Hermite 性を保存すべきことは要求していない点に注意しよう。実際、これは、より強い要請である正值性からすでに従い、したがって完全正值性からも従う。実際、任意の Hermite 演算子  $H$  は、二つの正の演算子の差

$$H = H_+ - H_-, \quad (2.42)$$

として書ける。例えば、スペクトル分解の正の部分と負の部分を分ければよい。 $\Phi$  が線形な正写像なら、 $\Phi[H_+]$  も  $\Phi[H_-]$  も正であり、したがって Hermite である。線形性より、

$$\Phi[H] = \Phi[H_+] - \Phi[H_-], \quad (2.43)$$

となり、これもまた Hermite である。このように、正值性だけで、Hermite 演算子が Hermite 演算子へ写されることが保証される。

**Kraus 表現から CPTP 性が従う** 前小節で導いた Kraus 表現の重要性は、これらの性質が直ちに分かることである。実際、写像が Kraus 表現 (2.28) で書かれているとしよう。まず正值性を示す。 $\rho \geq 0$  ならば、任意の  $|\psi\rangle \in \mathcal{H}_S$  に対して、

$$\begin{aligned}\langle\psi|\Phi[\rho]|\psi\rangle &= \sum_{\mu} \langle\psi|M_{\mu}\rho M_{\mu}^{\dagger}|\psi\rangle \\ &= \sum_{\mu} \langle\phi_{\mu}|\rho|\phi_{\mu}\rangle \geq 0,\end{aligned}\tag{2.44}$$

となる。ここでは

$$|\phi_{\mu}\rangle := M_{\mu}^{\dagger}|\psi\rangle.\tag{2.45}$$

と定義した。したがって

$$\rho \geq 0 \quad \implies \quad \Phi[\rho] \geq 0.\tag{2.46}$$

である。実際、同じ議論から直ちに完全正值性も示される。任意の補助 Hilbert 空間  $\mathcal{H}_A$  について、拡張写像は

$$(\mathbf{1}_A \otimes \Phi)[X] = \sum_{\mu} (\mathbf{1}_A \otimes M_{\mu}) X (\mathbf{1}_A \otimes M_{\mu}^{\dagger}).\tag{2.47}$$

の形を取る。 $X \geq 0$  ならば、任意のベクトル  $|\Psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_S$  に対して、

$$\langle\Psi|(\mathbf{1}_A \otimes \Phi)[X]|\Psi\rangle = \sum_{\mu} \langle\Psi_{\mu}|X|\Psi_{\mu}\rangle \geq 0,\tag{2.48}$$

となる。ここで

$$|\Psi_{\mu}\rangle := (\mathbf{1}_A \otimes M_{\mu}^{\dagger})|\Psi\rangle.\tag{2.49}$$

である。よって

$$X \geq 0 \quad \implies \quad (\mathbf{1}_A \otimes \Phi)[X] \geq 0,\tag{2.50}$$

であり、完全正值性が証明された。

トレース保存性は、トレースの巡回性から直接従う。

$$\mathrm{Tr} \Phi[\rho] = \mathrm{Tr} \left( \sum_{\mu} M_{\mu} \rho M_{\mu}^{\dagger} \right) = \mathrm{Tr} \left( \rho \sum_{\mu} M_{\mu}^{\dagger} M_{\mu} \right) = \mathrm{Tr} \rho,\tag{2.51}$$

最後の等号では (2.29) を用いた。このように、Kraus 形式に書かれ、規格化条件 (2.29) を満たす写像は、自動的に CPTP である。

系として、全体系のユニタリー発展と環境についてのトレース操作から得られる力学写像は、完全正かつトレース保存であると結論できる。

以上の結果は、Kraus 形式から完全正值性とトレース保存性が従うことを示す。逆も真である。有限次元 Hilbert 空間上のすべての CPTP 写像は、規格化された Kraus 表現をもつ。これは量子論の基本的な構造定理である。ここでは完全な詳細にわたる証明は行わないが、以下では繰り返し用いる。したがって、有限次元では、CPTP 写像と Kraus 写像の概念は等価である。

論理的なつながりをまとめると、

$$\text{Partial trace over env.} \quad \Rightarrow \quad \text{Kraus representation} \quad \Leftrightarrow \quad \text{CPTP}.\tag{2.52}$$

## 2.4 1パラメータ力学写像、半群、Liouvillian

これまでの、 $\Phi_t$  を時刻  $t = 0$  からある時刻  $t$  への単一の写像と考えてきた。しかし実際には、任意の時刻から任意の時刻へ発展させたい。この単純な考察から、深い物理的含意をもつ重要な数学的性質が見えてくる。まず数学から始めよう。考えたいのは、写像の1パラメータ族

$$\{\Phi_t\}_{t \geq 0}, \quad (2.53)$$

である。各  $\Phi_t$  は系の密度行列に作用し、初期時刻 0 から後の時刻  $t$  への発展を記述する。

$$\rho(t) = \Phi_t[\rho(0)]. \quad (2.54)$$

前の小節では、系を環境に結合し、全体系をユニタリーに発展させ、環境についてトレースを取ることで、そのような写像を微視的に動機づけた。ここではその構成から離れて抽象化し、族  $\{\Phi_t\}$  の性質を直接調べる。

基本的な整合性条件は、何もしなければ状態は変わらない、ということである。したがって、

$$\Phi_0 = \mathbf{1}, \quad (2.55)$$

を要求する。ここで  $\mathbf{1}$  は、演算子上の恒等写像を表す。さらに、次の半群性を考える。

$$\Phi_{t+s} = \Phi_t \circ \Phi_s, \quad t, s \geq 0. \quad (2.56)$$

これは、まず時間  $s$  だけ発展させ、さらに時間  $t$  だけ発展させることが、合計時間  $t+s$  だけ一度に発展させることと等価であることを意味する。この性質は、環境の性質に関する一連の近似と仮定から導くことになる。今のところは、この性質が何を述べているかを考えるにとどめよう。時間発展の法則は時間間隔の長さだけに依存し、絶対的な初期時刻には依存しないのである。

式 (2.56) は、閉鎖系のユニタリー発展でおなじみの群の性質に対応する、開放系での自然な類似物である。閉鎖系では

$$U(t+s) = U(t)U(s), \quad (2.57)$$

であり、逆  $U(-t) = U^\dagger(t)$  も存在する。対照的に、開放系では、縮約された記述において環境へ情報が失われうるため、写像  $\Phi_t$  は可逆であるとは限らない。したがって一般には、完全な群ではなく、半群だけが得られる。

半群性には重要な物理的解釈がある。将来の発展が、系の詳細な過去の履歴ではなく、現在の状態だけに依存するという考えを表しているのである。これは、次のステップの確率が、それまでのステップの履歴ではなく、現在の配置だけに依存する古典的確率過程の Markov 性に対応する量子的な性質である。すべての開放系の時間発展がこの性質を厳密に満たすわけではないが、これは中心的な理想化であり、Lindblad 定理の出発点である。

ここで、族  $\{\Phi_t\}$  が時間に連続的に依存すると仮定すると、その無限小生成子を定義できる。これは演算子に作用する線形写像  $\mathcal{L}$  であり、次で定義される。

$$\mathcal{L}[\rho] := \lim_{\delta t \rightarrow 0^+} \frac{\Phi_{\delta t}[\rho] - \rho}{\delta t}. \quad (2.58)$$

写像  $\mathcal{L}$  を *Liouvillian*、Liouville 超演算子<sup>9</sup>、あるいは単に力学半群の生成子と呼ぶ。これは、閉鎖系で Hamiltonian が果たす役割を、開放系で果たす。

<sup>9</sup>状態を Hilbert 空間上のベクトル、観測量を演算子とするなじみ深い数学的構造を踏まえると、この名称は、 $\Phi_t$  がベクトルからベクトルへの写像ではなく、演算子から演算子への写像であることを思い出させる。式で書けば、 $\mathcal{B}(\mathcal{H})$  が Hilbert 空間  $\mathcal{H}$  上の演算子の空間ならば、超演算子は  $\mathcal{L} : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$  と写す。 $\mathcal{L}$  は密度行列だけでなく、一般の演算子に作用することに注意しよう。実際、密度行列の空間はベクトル空間ではなく、 $\mathcal{B}(\mathcal{H})$  のコンパクトな凸部分集合である。

半群性を用いると、時間発展の局所的な微分方程式を導ける。実際、

$$\begin{aligned}
\frac{d}{dt}\rho(t) &= \lim_{\delta t \rightarrow 0^+} \frac{\rho(t + \delta t) - \rho(t)}{\delta t} \\
&= \lim_{\delta t \rightarrow 0^+} \frac{\Phi_{t+\delta t}[\rho(0)] - \Phi_t[\rho(0)]}{\delta t} \\
&= \lim_{\delta t \rightarrow 0^+} \frac{\Phi_{\delta t}[\Phi_t[\rho(0)]] - \Phi_t[\rho(0)]}{\delta t} \\
&= \mathcal{L}[\rho(t)].
\end{aligned} \tag{2.59}$$

となる。したがって、縮約動力学はマスター方程式

$$\frac{d\rho}{dt} = \mathcal{L}[\rho]. \tag{2.60}$$

を満たす。これが時間一様な量子マスター方程式の一般形である。  
形式的には、(2.60) の解は

$$\rho(t) = e^{t\mathcal{L}}[\rho(0)], \tag{2.61}$$

と書ける。したがって

$$\Phi_t = e^{t\mathcal{L}}. \tag{2.62}$$

となる。ここでの指数関数は、演算子の空間に作用する線形超演算子の指数関数を意味する。常微分方程式の場合と全く同様に、生成子  $\mathcal{L}$  が半群全体を決定する。  
閉鎖系では、Liouvillian は Hamiltonian との交換子に帰着する。

$$\mathcal{L}[\rho] = -i[H, \rho], \tag{2.63}$$

そのため、マスター方程式は von Neumann 方程式 (2.9) となる。しかし、開放系では、Liouvillian はデコヒーレンス、散逸、ノイズを記述する追加の項を含みうる。そこで本質的な問いは、すべての  $t \geq 0$  について写像  $\Phi_t = e^{t\mathcal{L}}$  が完全正かつトレース保存となるような、 $\mathcal{L}$  の最も一般的な形は何か、ということである。その答えを与えるのが、これから述べる Gorini–Kossakowski–Sudarshan–Lindblad 定理である。

## 2.5 GKSL 定理と Lindblad 方程式

ここで、Markov 的な開放系動力学の中心的な構造定理を述べる準備が整った。これは Gorini、Kossakowski、Sudarshan、Lindblad によるもので、GKSL 定理と呼ばれる<sup>10</sup>。この定理は、1 パラメータ量子力学半群の最も一般的な生成子の特徴づける。

**GKSL 定理**  $\mathcal{H}_S$  を有限次元 Hilbert 空間とし、 $\{\Phi_t\}_{t \geq 0}$  を  $\mathcal{L}(\mathcal{H}_S)$  上の線形写像の 1 パラメータ族で、次を満たすものとする。

1.  $\Phi_0 = \mathbb{1}$ 。
2. すべての  $t, s \geq 0$  に対して  $\Phi_{t+s} = \Phi_t \circ \Phi_s$ 。
3. 各  $\Phi_t$  は完全正かつトレース保存である。
4. この族は  $t$  について連続である。

<sup>10</sup> この定理は、1976 年の二つの独立した論文にさかのぼる。Gorini、Kossakowski、Sudarshan[?] は有限次元の  $N$  単位系の結果を与え、Lindblad[?] は、それと密接に関連する、完全正な量子力学半群の生成子の演算子代数的な特徴づけを与えた。当時、Sudarshan は Texas で指導的立場にあり、Gorini と Kossakowski は Texas を訪問中の若手ないし中堅の数理論理学者で、Lindblad は Stockholm で博士号を取得したばかりだった。

このとき、線形超演算子  $\mathcal{L}$  が存在して、

$$\Phi_t = e^{t\mathcal{L}}, \quad \frac{d\rho}{dt} = \mathcal{L}[\rho], \quad (2.64)$$

と書ける。さらに、生成子  $\mathcal{L}$  は必ず次の形を取る。

$$\mathcal{L}[\rho] = -i[H, \rho] + \sum_{a,b} c_{ab} \left( F_a \rho F_b^\dagger - \frac{1}{2} \{F_b^\dagger F_a, \rho\} \right). \quad (2.65)$$

ここで  $H = H^\dagger$  は  $\mathcal{H}_S$  上の Hermite 演算子、 $F_a$  は  $\mathcal{H}_S$  上のトレースがゼロの演算子の基底要素であり、 $\{.,.\}$  は反交換子を表す。また、*Kossakowski* 行列と呼ばれる  $c_{ab}$  は半正定値である。逆に、(2.65) の形の任意の生成子は、完全正かつトレース保存の半群を定義する。

定理の数学的仮定と、微視的な導出でよく用いられる物理的近似を区別することが重要である。定理自体は、初めから半群による発展と CPTP 性を仮定する。一方、環境に結合した系からマスター方程式を導く場合には、通常、弱結合、大きな環境、短い環境相関時間、そしてしばしば Born 近似、Markov 近似、永年近似など、追加の物理的仮定を用いる。これらは、定理が適用される数学的枠組みを正当化するためのものである。本節では数学に専念し、これらの仮定の物理的根拠には後で戻る。

**対角 Lindblad 形式** 行列  $c_{ab}$  は半正定値なので、ユニタリー変換で対角化できる。したがって、(2.65) を、等価でよりなじみ深い次の形に書き直せる。

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_{\mu} \left( L_{\mu} \rho L_{\mu}^\dagger - \frac{1}{2} \{L_{\mu}^\dagger L_{\mu}, \rho\} \right). \quad (2.66)$$

これが *Lindblad* 方程式、あるいは *Lindblad* マスター方程式である。演算子  $L_{\mu}$  をジャンプ演算子、*Lindblad* 演算子、あるいはノイズ演算子と呼ぶ。

各項の解釈は明快である。交換子の項

$$-i[H, \rho] \quad (2.67)$$

は、発展の Hamilton 的な部分である。これだけならば、通常のユニタリー時間発展を生成する。残りの項は、開放系に固有の効果を表す。項

$$L_{\mu} \rho L_{\mu}^\dagger \quad (2.68)$$

は、環境に誘起される遷移、ジャンプ、あるいはノイズを表す。反交換子の項

$$-\frac{1}{2} \{L_{\mu}^\dagger L_{\mu}, \rho\} \quad (2.69)$$

は、トレースを保存し、生成子全体が完全正值性と両立するために必要である。実際、(2.66) がトレースを保存することは容易に確かめられる。

$$\frac{d}{dt} \text{Tr} \rho = -i \text{Tr}[H, \rho] + \sum_{\mu} \left( \text{Tr}(L_{\mu} \rho L_{\mu}^\dagger) - \frac{1}{2} \text{Tr}(L_{\mu}^\dagger L_{\mu} \rho) - \frac{1}{2} \text{Tr}(\rho L_{\mu}^\dagger L_{\mu}) \right) = 0, \quad (2.70)$$

ここではトレースの巡回性を用いた。Hermite 性が保存されることも明らかである。定理のきわめて非自明な点は、半群全体の完全正值性を保証するために必要な形が、まさに (2.66) であるということである。

**有限次元での導出の概略** 生成子がなぜ (2.65) の形でなければならないかについて、物理学的な導出の概略を簡潔に示す。 $\Phi_t$  は半群なので、その生成子は写像の短時間での振る舞いから決まる。非常に小さい時間ステップ  $\delta t$  について、

$$\Phi_{\delta t} = \mathbf{1} + \delta t \mathcal{L} + O(\delta t^2). \quad (2.71)$$

と書ける。各  $\Phi_{\delta t}$  は CPTP なので、Kraus 表現

$$\Phi_{\delta t}[\rho] = \sum_{\mu} M_{\mu}(\delta t) \rho M_{\mu}^{\dagger}(\delta t), \quad \sum_{\mu} M_{\mu}^{\dagger}(\delta t) M_{\mu}(\delta t) = \mathbf{1}. \quad (2.72)$$

をもつ。ここでの考え方は、小さい  $\delta t$  について Kraus 演算子を展開することである。一つの Kraus 演算子、例えば  $M_0$  に主要な恒等演算子の寄与を含め、

$$M_0 = \mathbf{1} + \delta t K + O(\delta t^2). \quad (2.73)$$

と書く。残りの Kraus 演算子はすべて、 $\sqrt{\delta t}$  の次数から始まらなければならない。

$$M_{\mu} = \sqrt{\delta t} L_{\mu} + O(\delta t) \quad (\mu \geq 1). \quad (2.74)$$

このスケーリングは規格化条件から要請される。非自明な Kraus 演算子の寄与は、1 の次数ではなく、 $\delta t$  の次数で現れなければならない。

これらの展開を演算子和の公式に代入すると、

$$\begin{aligned} \Phi_{\delta t}[\rho] &= M_0 \rho M_0^{\dagger} + \sum_{\mu \geq 1} M_{\mu} \rho M_{\mu}^{\dagger} \\ &= \rho + \delta t (K \rho + \rho K^{\dagger}) + \delta t \sum_{\mu \geq 1} L_{\mu} \rho L_{\mu}^{\dagger} + O(\delta t^2). \end{aligned} \quad (2.75)$$

を得る。したがって、生成子は暫定的に次の形をもつ。

$$\mathcal{L}[\rho] = K \rho + \rho K^{\dagger} + \sum_{\mu \geq 1} L_{\mu} \rho L_{\mu}^{\dagger}. \quad (2.76)$$

ここでトレース保存性を課す必要がある。

$$\sum_{\mu} M_{\mu}^{\dagger} M_{\mu} = \mathbf{1} \quad (2.77)$$

を  $\delta t$  の一次まで展開すると、

$$K + K^{\dagger} + \sum_{\mu \geq 1} L_{\mu}^{\dagger} L_{\mu} = 0. \quad (2.78)$$

を得る。 $K$  を Hermite 部分と反 Hermite 部分に分けると、一般解は

$$K = -\frac{1}{2} \sum_{\mu \geq 1} L_{\mu}^{\dagger} L_{\mu} - iH, \quad (2.79)$$

と書けることが分かる。ここで  $H = H^{\dagger}$  は Hermite である。これを生成子に代入し直すと、

$$\begin{aligned} \mathcal{L}[\rho] &= -i[H, \rho] + \sum_{\mu \geq 1} \left( L_{\mu} \rho L_{\mu}^{\dagger} - \frac{1}{2} L_{\mu}^{\dagger} L_{\mu} \rho - \frac{1}{2} \rho L_{\mu}^{\dagger} L_{\mu} \right) \\ &= -i[H, \rho] + \sum_{\mu \geq 1} \left( L_{\mu} \rho L_{\mu}^{\dagger} - \frac{1}{2} \{L_{\mu}^{\dagger} L_{\mu}, \rho\} \right). \end{aligned} \quad (2.80)$$

を得る。これはまさに Lindblad 形式である。この導出は、定理の基本的な論理を説明している。Hamiltonian の交換子は、主要な Kraus 演算子への短時間補正の反 Hermite 部分から生じ、散逸項は  $\sqrt{\delta t}$  の次数の Kraus 演算子から生じる。完全な定理はこの議論を精密化し、固定した演算子基底での係数行列が半正定値でなければならないことを示す。また、Lindblad 方程式で生成される写像は CPTP 半群であるという予想が正しいことも証明する。時間に依存しない Lindblad 生成子では、半群性は  $\Phi_t = e^{t\mathcal{L}}$  から直ちに従い、トレース保存性は  $\text{Tr } \mathcal{L}[\rho] = 0$  から直接従う。定理の真に非自明な点は、この形が発展全体の完全正値性も保証することにある。通常の正値性は、その帰結として従う。

いくつか補足する。第一に、Lindblad 表現は一意ではなく、異なる Lindblad 演算子の選択が同じ力学半群を生成しうる。第二に、この定理は時間一様な Markov 的發展に適用される。より一般的な時間依存マスター方程式も物理学で頻繁に現れ、それらが半群を定義するとは限らない。必要に応じて、後でこの区別に戻る。今のところ、(2.66) が連続時間の Markov 的 CPTP 發展を記述する最も一般的なマスター方程式を与える。

必要ならば、ジャンプ演算子を再スケールし、 $\gamma_\mu \geq 0$  および規格化条件に従って  $L_\mu = \sqrt{\gamma_\mu} A_\mu$  と書ける。

$$\text{Tr}(A_\mu^\dagger A_\mu) = 1. \quad (2.81)$$

すると Lindblad 方程式は次の形になる。

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_\mu \gamma_\mu \left( A_\mu \rho A_\mu^\dagger - \frac{1}{2} \{A_\mu^\dagger A_\mu, \rho\} \right). \quad (2.82)$$

このようにすると、演算子  $A_\mu$  は散逸過程の種類、すなわち系に作用する遷移、デコヒーレンスのチャンネル、あるいはノイズ機構を特徴づける。一方、係数  $\gamma_\mu$  は対応する率、つまりその過程が作用する時間尺度や強さを測る。この分離は、ジャンプの構造と、それが時間発展にどれだけ速く影響するかを制御する正の係数とを切り分けるため、物理的に便利な場合が多い。

有用な特殊例に触れておこう。ジャンプ演算子が Hermite であり、 $L^\dagger = L$  ならば、その散逸の寄与は二重交換子

$$\frac{d\rho}{dt} = -i[H, \rho] - \frac{1}{2} [L, [L, \rho]]. \quad (2.83)$$

として書き直せる。この形は、位相緩和と拡散の構造をより明瞭にするため、しばしば有用である。また、二重交換子は単位演算子をゼロに写すため、ユニタリ性も明らかになる。<sup>11</sup>

**Lindblad 表現の冗長性** 生成子の Lindblad 形式は一意ではない。第一の冗長性は、ジャンプ演算子の間のユニタリー回転である。

$$L_\mu \rightarrow L'_\mu = \sum_\nu U_{\mu\nu} L_\nu, \quad U^\dagger U = 1, \quad (2.85)$$

とすると、Lindblad 方程式は変わらない。実際、

$$\sum_\mu L'_\mu \rho L'^\dagger_\mu = \sum_\mu L_\mu \rho L^\dagger_\mu, \quad \sum_\mu L'^\dagger_\mu L'_\mu = \sum_\mu L^\dagger_\mu L_\mu. \quad (2.86)$$

である。第二の冗長性は、ジャンプ演算子に単位演算子の定数倍を加えることである。

$$L_\mu \rightarrow L'_\mu = L_\mu + \alpha_\mu, \quad \alpha_\mu \in \mathbb{C}, \quad (2.87)$$

<sup>11</sup> より一般に、Hermite 演算子の族  $L^\dagger_\mu = L_\mu$  と、実対称な半正定値係数行列  $c_{\mu\nu}$  があるならば、

$$\sum_{\mu,\nu} c_{\mu\nu} \left( L_\mu \rho L_\nu - \frac{1}{2} \{L_\nu L_\mu, \rho\} \right) = -\frac{1}{2} \sum_{\mu,\nu} c_{\mu\nu} [L_\mu, [L_\nu, \rho]]. \quad (2.84)$$

としたとき、同時に Hamiltonian を

$$H \rightarrow H' = H - \frac{i}{2} \sum_{\mu} \left( \alpha_{\mu}^* L_{\mu} - \alpha_{\mu} L_{\mu}^{\dagger} \right). \quad (2.88)$$

とずらせば、同じ時間発展が得られる。簡単な計算から、

$$-i[H', \rho] + \sum_{\mu} \left( L'_{\mu} \rho L_{\mu}^{\dagger} - \frac{1}{2} \{L_{\mu}^{\dagger} L'_{\mu}, \rho\} \right) = -i[H, \rho] + \sum_{\mu} \left( L_{\mu} \rho L_{\mu}^{\dagger} - \frac{1}{2} \{L_{\mu}^{\dagger} L_{\mu}, \rho\} \right). \quad (2.89)$$

が示される。これは、コヒーレントな Hamilton 的发展の一部をジャンプ演算子に移すことも、その逆もできる、という主張を明確にするものである。第三の自明な冗長性として、Hamiltonian には単位演算子の定数倍を加えてよい。

これらの任意性は、しばしば組み合わせた変換

$$L_{\mu} \rightarrow L'_{\mu} = \sum_{\nu} U_{\mu\nu} L_{\nu} + \alpha_{\mu} \mathbf{1}, \quad (2.90)$$

$$H \rightarrow H' = H - \frac{i}{2} \sum_{\mu} \left( \alpha_{\mu}^* L_{\mu} - \alpha_{\mu} L_{\mu}^{\dagger} \right) + c \mathbf{1}, \quad (2.91)$$

としてまとめられる。この変換のもとで Liouvillian は不変である。この冗長性を取り除くため、有限次元では、それらのトレースをゼロとするか、より一般には単位演算子に直交するように選べる。

**非対角形式と Kossakowski 行列** 対角 Lindblad 形式 (2.66) だけが、生成子の散逸部分を書く有用な方法ではない。 $F_a$  を、系の Hilbert 空間上のトレースがゼロの演算子の固定した基底、例えば Hilbert–Schmidt 内積に関する正規直交基底とする<sup>12</sup>。このとき、最も一般的な散逸項は

$$\mathcal{D}[\rho] = \sum_{a,b} c_{ab} \left( F_a \rho F_b^{\dagger} - \frac{1}{2} \{F_b^{\dagger} F_a, \rho\} \right), \quad (2.92)$$

と書ける。行列  $C = (c_{ab})$  を Kossakowski 行列と呼ぶ。発展が完全正であるという条件は、この行列の正値性

$$C \geq 0. \quad (2.93)$$

に符号化されている。この形では、ラベル  $a, b$  はまだ独立した物理的ジャンプ過程に対応していない。むしろ、生成子の散逸部分を展開するための基底演算子を指している。 $C$  の非対角成分は、異なる演算子の方向の間の相関、あるいは干渉を記述する。

$C$  は半正定値なので、対角化できる。

$$c_{ab} = \sum_{\mu} U_{\mu a} \gamma_{\mu} U_{\mu b}^*, \quad \gamma_{\mu} \geq 0. \quad (2.94)$$

と書けば、次を定義できる。

$$L_{\mu} = \sqrt{\gamma_{\mu}} \sum_a U_{\mu a} F_a. \quad (2.95)$$

この表式を (2.92) に代入すると、対角 Lindblad 形式 (2.66) が再現される。したがって、通常のジャンプ演算子は、正行列  $C$  の固有ベクトルに、対応する固有値の平方根を掛けたものである。固有値  $\gamma_{\mu}$  が、対応する正の率となる。

<sup>12</sup>以下の議論では、(2.65) と同様、単位演算子成分が散逸演算子の基底から分離されていると仮定する。ジャンプ演算子に単位演算子の成分を許すと、以下で論じる冗長性により、散逸のように見える寄与の一部を Hamiltonian に移せる。そのため、Kossakowski 行列を定義する際には、トレースがゼロの基底演算子  $F_a$  を用いるのが最も明快な場合が多い。

これにより、ジャンプ演算子の個数の意味も明らかになる。 $C$  のいくつかの固有値がゼロなら、対応する  $L_\mu$  もゼロとなり、マスター方程式に寄与しない。したがって、対角 Lindblad 形式におけるゼロでないジャンプ演算子の最小個数は

$$\#_{\text{jumps}} = \text{rank}(C). \quad (2.96)$$

である。等価に、Kossakowski 行列の階数は、演算子空間における独立した散逸方向の数を数える。選んだ基底  $F_a$  では、マスター方程式に多数の結合した散逸項があるように見えても、 $C$  を対角化すると、独立したジャンプ演算子は  $\text{rank}(C)$  個だけ残る。

**Heisenberg 描像でのマスター方程式** これまでは、観測量を固定し、密度行列が時間発展する Schrödinger 描像で Lindblad 方程式を論じてきた。同じ動力学を、状態を固定して観測量を発展させる Heisenberg 描像で記述することも、同様に有用な場合が多い。観測量  $\mathcal{O}$  の期待値は

$$\langle \mathcal{O} \rangle_t := \text{Tr}(\rho(t) \mathcal{O}). \quad (2.97)$$

である。この時間依存性を、状態ではなく観測量に帰すことができるかを、ここで問う。そのために、随伴超演算子  $\mathcal{L}^\dagger$  を、次の関係で定義する。

$$\text{Tr}(\mathcal{L}[\rho] \mathcal{O}) = \text{Tr}(\rho \mathcal{L}^\dagger[\mathcal{O}]) \quad (2.98)$$

これはすべての密度行列  $\rho$  と観測量  $\mathcal{O}$  について成り立つものとする。これは単に、Hilbert–Schmidt 内積に関する通常の演算子の随伴に対応する、超演算子版である。(2.80) を用いると、

$$\begin{aligned} \text{Tr}(\mathcal{L}[\rho] \mathcal{O}) &= -i \text{Tr}([H, \rho] \mathcal{O}) + \sum_\mu \text{Tr}(L_\mu \rho L_\mu^\dagger \mathcal{O}) - \frac{1}{2} \sum_\mu \text{Tr}(\{L_\mu^\dagger L_\mu, \rho\} \mathcal{O}) \\ &= i \text{Tr}(\rho [H, \mathcal{O}]) + \sum_\mu \text{Tr}(\rho L_\mu^\dagger \mathcal{O} L_\mu) - \frac{1}{2} \sum_\mu \text{Tr}(\rho \{L_\mu^\dagger L_\mu, \mathcal{O}\}), \end{aligned} \quad (2.99)$$

を得る。ここではトレースの巡回性を用いた。したがって、随伴 Liouvillian は観測量に対して次のように作用する。

$$\mathcal{L}^\dagger[\mathcal{O}] = i[H, \mathcal{O}] + \sum_\mu \left( L_\mu^\dagger \mathcal{O} L_\mu - \frac{1}{2} \{L_\mu^\dagger L_\mu, \mathcal{O}\} \right). \quad (2.100)$$

Heisenberg 描像のマスター方程式は、

$$\frac{d\mathcal{O}}{dt} = \mathcal{L}^\dagger[\mathcal{O}]. \quad (2.101)$$

となる。この方程式は、閉鎖系の通常の Heisenberg 方程式

$$\frac{d\mathcal{O}}{dt} = i[H, \mathcal{O}] \quad (2.102)$$

に直接対応する開放系版である。違いは、Lindblad 演算子を含む追加の項の存在であり、それらは散逸、ノイズ、デコヒーレンスが観測量に与える効果を符号化する。Schrödinger 描像と Heisenberg 描像の等価性は、(2.98) から直ちに従う。

直ちに得られる帰結の一つ述べておくと有用である。Schrödinger 描像の発展は密度行列のトレースを保存するため、単位演算子は Heisenberg 描像の発展の固定点でなければならない、 $\mathcal{L}^\dagger[1] = 0$  となる。これは (2.100) から直接、容易に確認できる。これが、Schrödinger 描像におけるトレース保存性の、Heisenberg 描像での表れである。

後のために、Heisenberg 方程式を等価な次の形にも書いておくとも便利である。

$$\frac{d\mathcal{O}}{dt} = i[H, \mathcal{O}] + \frac{1}{2} \sum_{\mu} \left( L_{\mu}^{\dagger} [\mathcal{O}, L_{\mu}] + [L_{\mu}^{\dagger}, \mathcal{O}] L_{\mu} \right). \quad (2.103)$$

この形は、散逸項を単に並べ替えることで得られ、解釈をより明瞭にする場合がある。応用では、占有数、スピン成分、相関関数など、観測量の期待値の発展方程式を導く最も効率的な方法が、Heisenberg 描像のマスター方程式である場合がある。全密度行列を解いてからトレースを計算する代わりに、関心のある観測量を直接発展させればよい。

## 2.6 Lindblad 方程式を導く微視的な仮定

これまでは Lindblad 方程式を構造的な観点から論じてきた。それは、連続的な Markov 的 CPTP 半群の最も一般的な生成子である。ここでは、より微視的な水準で、環境に結合した系の基礎にあるユニタリー発展から、どのような物理的仮定のもとでこの方程式が生じるかを説明する。

微視的な設定 全 Hilbert 空間

$$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E \quad (2.104)$$

と、次の形の全 Hamiltonian を考える。

$$H_{\text{tot}} = H_S + H_E + H_{\text{int}}, \quad (2.105)$$

ここで  $H_S$  は系に、 $H_E$  は環境に作用し、相互作用 Hamiltonian は

$$H_{\text{int}} = \sum_{\alpha} S_{\alpha} \otimes X_{\alpha}. \quad (2.106)$$

とする。演算子  $S_{\alpha}$  は  $\mathcal{H}_S$  上に、演算子  $X_{\alpha}$  は  $\mathcal{H}_E$  上に作用する。全体系の密度行列は、次に従ってユニタリーに発展する。

$$\frac{d\rho_{SE}}{dt} = -i[H_{\text{tot}}, \rho_{SE}]. \quad (2.107)$$

目標は、ここから縮約密度行列についての閉じた方程式を導くことである。

$$\rho_S(t) := \text{Tr}_E \rho_{SE}(t). \quad (2.108)$$

相互作用描像と厳密な縮約動力学 自由 Hamiltonian に関する相互作用描像を用いると便利である。

$$H_0 := H_S + H_E. \quad (2.109)$$

次を定義すると、

$$\tilde{\rho}_{SE}(t) := e^{iH_0 t} \rho_{SE}(t) e^{-iH_0 t}, \quad \tilde{H}_{\text{int}}(t) := e^{iH_0 t} H_{\text{int}} e^{-iH_0 t}, \quad (2.110)$$

時間発展方程式は

$$\frac{d\tilde{\rho}_{SE}}{dt} = -i[\tilde{H}_{\text{int}}(t), \tilde{\rho}_{SE}(t)]. \quad (2.111)$$

となる。一度積分して代入し直すと、厳密な積分微分方程式

$$\frac{d\tilde{\rho}_{SE}}{dt} = -i[\tilde{H}_{\text{int}}(t), \tilde{\rho}_{SE}(0)] - \int_0^t ds [\tilde{H}_{\text{int}}(t), [\tilde{H}_{\text{int}}(s), \tilde{\rho}_{SE}(s)]]. \quad (2.112)$$

を得る。さらに環境についてトレースを取ると、 $\tilde{\rho}_S(t)$  に対する、厳密だが閉じていない方程式が得られる。この厳密な形では、系の動力学は一般に非 Markov 的であり、積分核と積分中の全状態  $\tilde{\rho}_{SE}(s)$  を通じて、過去の記憶を保持している。

**因子化された初期状態** 第一の標準的な仮定は、初期密度行列が因子化していることである。

$$\rho_{SE}(0) = \rho_S(0) \otimes \rho_E. \quad (2.113)$$

先に論じたように、これは初期の系と環境の相関を排除する。これだけで、縮約時間発展が  $\rho_S(0)$  上の状態に依存しない写像で記述できることが保証される。これは因子化された純粋状態を仮定するよりも一般的である。環境状態  $\rho_E$  は混合状態でもよく、多くの物理的応用では熱的である。さらに、環境状態は自由な環境 Hamiltonian に関して定常であると仮定する。

$$[H_E, \rho_E] = 0. \quad (2.114)$$

This ensures that environment correlation functions depend only on time differences and not on absolute time. It is also convenient to shift the interaction so that

$$\text{Tr}_E(\rho_E X_\alpha) = 0, \quad (2.115)$$

which removes the first-order term in the reduced evolution.

**Born approximation** The next assumption is weak coupling between system and environment. Physically, one assumes that the environment is so large that its state is only weakly perturbed by the interaction, and that system–environment correlations remain small. One then replaces the exact total state inside the memory kernel by the factorized approximation

$$\tilde{\rho}_{SE}(s) \approx \tilde{\rho}_S(s) \otimes \rho_E. \quad (2.116)$$

This is the *Born approximation*. Substituting (2.116) into the exact reduced equation produces an equation that is already closed in  $\tilde{\rho}_S$ , but still nonlocal in time:

$$\frac{d\tilde{\rho}_S}{dt} = - \int_0^t ds \text{Tr}_E \left( [\tilde{H}_{\text{int}}(t), [\tilde{H}_{\text{int}}(s), \tilde{\rho}_S(s) \otimes \rho_E]] \right). \quad (2.117)$$

At this stage one has a weak-coupling master equation with memory.

**Markov approximation** The next step is the Markov approximation. The physical idea is that the environment has a very short correlation time  $\tau_E$ , while the system evolves on a much longer timescale  $\tau_S$ . Equivalently, the bath quickly forgets the information it acquires from the system. Then, inside the memory kernel, one may approximate

$$\tilde{\rho}_S(s) \approx \tilde{\rho}_S(t) \quad (2.118)$$

whenever the bath correlation functions are appreciable only for  $t - s \lesssim \tau_E$ . As a second step, we also extend the upper limit of the integral to infinity:

$$\int_0^t ds \longrightarrow \int_0^\infty ds. \quad (2.119)$$

With these approximations, the evolution becomes local in time:

$$\frac{d\tilde{\rho}_S}{dt} = - \int_0^\infty ds \text{Tr}_E \left( [\tilde{H}_{\text{int}}(t), [\tilde{H}_{\text{int}}(t-s), \tilde{\rho}_S(t) \otimes \rho_E]] \right). \quad (2.120)$$

This is the *Born–Markov master equation*. It is already first order in time and time local, but it is not yet guaranteed to be in Lindblad form.

**Decomposition into energy differences** To go further, one resolves the system operators into components of definite energy difference. Writing the spectral decomposition of the system Hamiltonian as

$$H_S = \sum_n E_n |n\rangle\langle n|, \quad (2.121)$$

one decomposes each system operator  $S_\alpha$  as

$$S_\alpha = \sum_\omega A_\alpha(\omega), \quad (2.122)$$

where

$$A_\alpha(\omega) := \sum_{E_n - E_m = \omega} |m\rangle\langle m| S_\alpha |n\rangle\langle n|. \quad (2.123)$$

By construction, these operators satisfy

$$[H_S, A_\alpha(\omega)] = -\omega A_\alpha(\omega). \quad (2.124)$$

Equivalently, in the interaction picture,

$$e^{iH_S t} A_\alpha(\omega) e^{-iH_S t} = e^{-i\omega t} A_\alpha(\omega). \quad (2.125)$$

Thus the interaction-picture system operators split into harmonics labeled by the energy differences of the system.

**Secular approximation** After inserting the decomposition (2.122) into the Born–Markov equation, one encounters terms oscillating as

$$e^{i(\omega - \omega')t}. \quad (2.126)$$

If distinct energy differences are well separated compared with the dissipative rates, these rapidly oscillating terms average to nearly zero over the coarse-grained timescale relevant to the reduced dynamics. One therefore discards all contributions with  $\omega \neq \omega'$ , keeping only the resonant terms with equal energy differences. This is the *secular approximation*.

Physically, the secular approximation means that only processes with the same energy transfer interfere coherently over long times. Mathematically, it is this step that typically ensures that the generator takes genuine GKSL form and that populations and coherences decouple in the energy basis.

**Emergence of the Lindblad equation** After the Born, Markov, and secular approximations, one arrives at a master equation of the form

$$\frac{d\tilde{\rho}_S}{dt} = \sum_\omega \sum_{\alpha, \beta} \gamma_{\alpha\beta}(\omega) \left( A_\beta(\omega) \tilde{\rho}_S A_\alpha^\dagger(\omega) - \frac{1}{2} \left\{ A_\alpha^\dagger(\omega) A_\beta(\omega), \tilde{\rho}_S \right\} \right), \quad (2.127)$$

The coefficients  $\gamma_{\alpha\beta}(\omega)$  are determined by the environment two-point correlation matrix

$$C_{\alpha\beta}(t) := \text{Tr}_E \left( \rho_E X_\alpha^\dagger(t) X_\beta(0) \right), \quad X_\alpha(t) := e^{iH_E t} X_\alpha e^{-iH_E t}. \quad (2.128)$$

Its Fourier transform at the transition frequency  $\omega$  gives

$$\gamma_{\alpha\beta}(\omega) := \int_{-\infty}^{+\infty} dt e^{i\omega t} C_{\alpha\beta}(t). \quad (2.129)$$

The matrix  $\gamma_{\alpha\beta}(\omega)$  is positive semidefinite for every fixed  $\omega$ . Indeed, for any complex numbers  $v_\alpha$ ,

$$\sum_{\alpha,\beta} v_\alpha^* \gamma_{\alpha\beta}(\omega) v_\beta = \int_{-\infty}^{+\infty} dt e^{i\omega t} \text{Tr}_E \left( \rho_E X^\dagger(t) X(0) \right), \quad X := \sum_{\beta} v_\beta X_\beta, \quad (2.130)$$

and by Bochner's theorem this Fourier transform is non-negative because the kernel

$$f(t) := \text{Tr}_E \left( \rho_E X^\dagger(t) X(0) \right) \quad (2.131)$$

is of positive type. This positivity guarantees that the dissipative part can be written in Lindblad form. These dissipative terms come together with an additional Hamiltonian correction, the so-called *Lamb-shift term*,

$$-i[H_{\text{LS}}, \tilde{\rho}_S]. \quad (2.132)$$

The Lamb-shift Hamiltonian may be written explicitly in terms of the transition-frequency operators as

$$H_{\text{LS}} = \sum_{\omega} \sum_{\alpha,\beta} S_{\alpha\beta}(\omega) A_\alpha^\dagger(\omega) A_\beta(\omega), \quad (2.133)$$

where the coefficients  $S_{\alpha\beta}(\omega)$  are determined by the imaginary, or principal-value, part of the bath correlation functions. Thus the same operators  $A_\alpha(\omega)$  that appear in the dissipator also determine the Hamiltonian renormalization induced by the environment.

Transforming back to the Schrödinger picture, one obtains

$$\frac{d\rho_S}{dt} = -i[H_S + H_{\text{LS}}, \rho_S] + \sum_{\omega} \sum_{\alpha,\beta} \gamma_{\alpha\beta}(\omega) \left( A_\beta(\omega) \rho_S A_\alpha^\dagger(\omega) - \frac{1}{2} \left\{ A_\alpha^\dagger(\omega) A_\beta(\omega), \rho_S \right\} \right). \quad (2.134)$$

This is of GKSL form, provided that the matrices  $\gamma_{\alpha\beta}(\omega)$  are positive semidefinite. In microscopic derivations, these coefficients are determined by Fourier transforms of bath correlation functions, and their positivity follows from positivity properties of those correlators.

**Summary of assumptions** Let us summarize the microphysical assumptions that typically lead from unitary system–environment dynamics to the Lindblad equation:

1. *Factorized initial state:*

$$\rho_{SE}(0) = \rho_S(0) \otimes \rho_E. \quad (2.135)$$

2. *Stationary environment state:*

$$[H_E, \rho_E] = 0. \quad (2.136)$$

3. *Weak coupling / Born approximation:* the environment is only weakly perturbed and one neglects the build-up of system–environment correlations,

$$\tilde{\rho}_{SE}(t) \approx \tilde{\rho}_S(t) \otimes \rho_E. \quad (2.137)$$

4. *Short bath memory / Markov approximation:* the bath correlation time is much shorter than the system evolution time, so the memory kernel becomes local in time.

5. *Secular approximation*: rapidly oscillating terms with different energy differences are dropped, keeping only resonant contributions.

The first four assumptions already lead to a time-local Born–Markov master equation. The last one is what typically turns that equation into manifest Lindblad form. Thus the Lindblad equation should be understood not as the exact reduced dynamics of a generic open system, but as the effective description obtained when the environment is weakly coupled, rapidly forgetting, and the system dynamics is viewed on coarse-grained timescales for which non-resonant oscillatory terms average out.

## 2.7 Time-dependent dynamical maps, composition law and CP divisibility

In cosmology and gravity, explicit time dependence is the rule rather than the exception. The background geometry evolves in time, the Hamiltonian of the subsystem is generally time dependent, and the same is true, in general, for the effective couplings induced by the environment. For this reason, the notion of a one-parameter dynamical map is too restrictive:  $\Phi$  does not only depend on the length of the time interval  $t_1 - t_0$  by which we evolve, but also on the starting time  $t_0$ . Therefore, we need to discuss the more general framework of two-parameter dynamical maps.

The basic point is already visible in the simplest unitary setting. Consider a closed system with time-dependent Hamiltonian  $H(t)$ . Its density matrix obeys

$$\rho(t) = U(t, t_0) \rho(t_0) U^\dagger(t, t_0), \quad (2.138)$$

with

$$U(t, t_0) = T \exp \left( -i \int_{t_0}^t ds H(s) \right). \quad (2.139)$$

This defines a *two*-parameter family of maps

$$\Phi(t, t_0)[\rho] := U(t, t_0) \rho U^\dagger(t, t_0). \quad (2.140)$$

These maps always satisfy the *composition law*

$$\Phi(t_2, t_0) = \Phi(t_2, t_1) \circ \Phi(t_1, t_0), \quad t_2 \geq t_1 \geq t_0. \quad (2.141)$$

However, in general they do not define a semigroup. Indeed, the semigroup property would require that the map depend only on the elapsed time,

$$\Phi(t, t_0) = \Phi(t - t_0), \quad (2.142)$$

and hence satisfy

$$\Phi(t + s) = \Phi(t) \circ \Phi(s). \quad (2.143)$$

This is lost as soon as the generator depends explicitly on time.

This motivates the following general framework. Let  $\Phi(t, t_0)$  be a differentiable family of linear maps acting on density matrices, satisfying

$$\rho(t) = \Phi(t, t_0)[\rho(t_0)]. \quad (2.144)$$

Assuming differentiability with respect to the upper time argument, one defines the time-local generator  $\mathcal{L}_t$  by

$$\partial_t \Phi(t, t_0) = \mathcal{L}_t \circ \Phi(t, t_0). \quad (2.145)$$

Formally, the solution is the time-ordered exponential

$$\Phi(t, t_0) = T \exp \left( \int_{t_0}^t ds \mathcal{L}_s \right). \quad (2.146)$$

Equation (2.146) is the natural generalization of  $e^{(t-t_0)\mathcal{L}}$  when the generator is time dependent. In the special case in which  $\mathcal{L}_t$  is actually independent of time, one recovers the semigroup case studied previously.

**CP divisibility** The natural time-dependent replacement of the semigroup notion is *CP divisibility*. One says that a family of maps  $\Phi(t, t_0)$  is CP divisible if, for every triple of times  $t_2 \geq t_1 \geq t_0$ , there exists an intermediate map  $\Phi(t_2, t_1)$  such that

$$\Phi(t_2, t_0) = \Phi(t_2, t_1) \circ \Phi(t_1, t_0), \quad (2.147)$$

and such that the intermediate map  $\Phi(t_2, t_1)$  is completely positive and trace preserving. In other words, not only the full evolution from  $t_0$  to  $t_2$ , but every time step in between, in particular infinitesimal time steps, must itself define a physical quantum channel.

This notion is especially useful because it admits a precise characterization in terms of the time-local generator. Under the assumptions of differentiability and invertibility of the map, one has the following theorem: the family  $\Phi(t, t_0)$  is CP divisible if and only if the generator  $\mathcal{L}_t$  is of GKSL form at every time  $t$ . Explicitly, this means that for each  $t$ ,

$$\mathcal{L}_t[\rho] = -i[H(t), \rho] + \sum_{\alpha, \beta} c_{\alpha\beta}(t) \left( L_\alpha(t) \rho L_\beta^\dagger(t) - \frac{1}{2} \{L_\beta^\dagger(t) L_\alpha(t), \rho\} \right), \quad (2.148)$$

with the Kossakowski matrix  $c_{\alpha\beta}(t)$  a positive semidefinite matrix for every  $t$ . Equivalently, after diagonalizing  $c_{\alpha\beta}(t)$ , one may write

$$\mathcal{L}_t[\rho] = -i[H(t), \rho] + \sum_{\mu} \gamma_{\mu}(t) \left( A_{\mu}(t) \rho A_{\mu}^\dagger(t) - \frac{1}{2} \{A_{\mu}^\dagger(t) A_{\mu}(t), \rho\} \right), \quad (2.149)$$

with  $\gamma_{\mu}(t) \geq 0$ . Thus the time-dependent generalization of Lindblad evolution consists simply in allowing both the Hamiltonian and the jump operators, or equivalently the canonical rates, to depend explicitly on time.

From the point of view of cosmology, this is the appropriate framework. One should generically expect a two-parameter propagator  $\Phi(t, t_0)$ , not a one-parameter semigroup. If the reduced dynamics is sufficiently local in time and the instantaneous generator is of GKSL form with non-negative rates, then the evolution is CP divisible. These considerations will also be important later from the Schwinger–Keldysh point of view. A local-in-time effective action naturally leads to a time-local generator, but in a cosmological background that generator will in general be explicitly time dependent.

### 3 Solutions of the Lindblad equation

Having derived the general form of Markovian open-system evolution, we now turn to solving the Lindblad equation in concrete situations. This is the stage at which the abstract structure of the previous section becomes physically transparent: one can identify stationary states, track the evolution of populations and coherences, and understand explicitly how dissipation, decoherence, and noise manifest themselves in time. Even simple examples already display a rich interplay between the Hamiltonian part of the evolution and the genuinely open-system effects encoded in the jump operators.

The aim of this section is twofold. First, we develop some general intuition for the qualitative behaviour of Lindblad evolution, including the role of stationary states and unitality. Second, we illustrate these ideas in explicit examples, focusing on the qubit and the dissipative harmonic oscillator. These examples will serve as a bridge between the general formalism of the master equation and the field-theoretic language introduced later through the Schwinger–Keldysh formalism.

### 3.1 Perturbative generators and secular resummation

In this subsection, we critically distinguish between knowing the Liouvillian generator to a given order in a perturbative expansion and knowing the corresponding solution to the same order. The two notions need not coincide at late times, because perturbative solutions may contain secular terms that grow with time and eventually invalidate the fixed-order expansion. We will explain how solving a truncated generator non-perturbatively can resum a distinguished class of such secular contributions.

In many applications, the Liouvillian governing the reduced dynamics is known only as an expansion in some small parameter. We therefore write

$$\mathcal{L} = \epsilon \mathcal{L}_1 + \epsilon^2 \mathcal{L}_2 + \epsilon^3 \mathcal{L}_3 + \dots, \quad (3.1)$$

where  $\epsilon \ll 1$  may denote a weak system-environment coupling, a ratio of energy/time scales, or some other effective expansion parameter. For simplicity, we begin by assuming that the generators  $\mathcal{L}_n$  are time independent. The master equation is

$$\frac{d\rho}{dt} = \mathcal{L}\rho. \quad (3.2)$$

One possible approach is to solve this equation perturbatively by expanding

$$\rho(t) = \rho^{(0)}(t) + \epsilon \rho^{(1)}(t) + \epsilon^2 \rho^{(2)}(t) + \dots. \quad (3.3)$$

For a time-independent generator and initial condition  $\rho(0) = \rho_0$ , the exact formal solution is

$$\rho(t) = e^{t\mathcal{L}} \rho_0. \quad (3.4)$$

Expanding both the generator and the exponential gives

$$\rho(t) = \rho_0 + \epsilon t \mathcal{L}_1 \rho_0 + \epsilon^2 \left( \frac{t^2}{2} \mathcal{L}_1^2 + t \mathcal{L}_2 \right) \rho_0 + \dots. \quad (3.5)$$

The terms proportional to increasing powers of time are called *secular terms*. At any fixed time they are suppressed by powers of  $\epsilon$ , but the perturbative expansion breaks down when

$$\epsilon t \sim 1. \quad (3.6)$$

In this regime, all terms of the form

$$\frac{(\epsilon t)^n}{n!} \mathcal{L}_1^n \rho_0 \quad (3.7)$$

are of comparable size and should be resummed.

A natural and often useful approximation is therefore to truncate the generator at leading order,

$$\mathcal{L} \simeq \epsilon \mathcal{L}_1, \quad (3.8)$$

but solve the resulting evolution equation exactly:

$$\boxed{\rho_{\text{LO}}(t) = e^{\epsilon t \mathcal{L}_1} \rho_0.} \quad (3.9)$$

This expression resums all repeated insertions of the leading generator  $\mathcal{L}_1$ . It does not, however, determine the contributions involving  $\mathcal{L}_2, \mathcal{L}_3, \dots$ . In particular, at second order it contains

$$\frac{\epsilon^2 t^2}{2} \mathcal{L}_1^2 \rho_0, \quad (3.10)$$

but not

$$\epsilon^2 t \mathcal{L}_2 \rho_0. \quad (3.11)$$

The leading resummation is therefore useful in a regime in which

$$\epsilon t = O(1), \quad \epsilon^2 t \ll 1. \quad (3.12)$$

In this window, the repeated action of  $\mathcal{L}_1$  has an order-one effect, while the corrections generated by the omitted higher-order Liouvillians remain perturbatively small.

**Elementary decay.** The simplest and canonical example of this phenomenon is the (classical, deterministic) equation for decay

$$\dot{p}(t) = -\epsilon \gamma p(t), \quad \gamma > 0. \quad (3.13)$$

Comparing this very simple example to (3.1) we see that only the leading order generator  $-\gamma$  is non-vanishing. Keeping only the first correction in a strict perturbative expansion gives

$$p(t) \simeq p_0 (1 - \epsilon \gamma t), \quad (3.14)$$

which breaks down for  $t \sim (\epsilon \gamma)^{-1}$  and eventually becomes negative. Solving exactly the equation with the leading (and only) generator instead gives

$$p(t) = p_0 e^{-\epsilon \gamma t}. \quad (3.15)$$

In this example, the leading equation is the full equation, so the resummation happens to be exact.

**A nonlinear higher-order correction.** To see more clearly what the leading resummation does and does not capture, consider

$$\dot{p}(t) = -\epsilon \gamma p(t) - \epsilon^2 \delta p(t)^2, \quad p(0) = p_0, \quad (3.16)$$

with  $\gamma$  and  $\delta$  some parameters and  $\epsilon$  an expansion parameter. If the generator is truncated at leading order in  $\epsilon$ , one finds

$$p_{\text{LO}}(t) = p_0 e^{-\epsilon \gamma t}. \quad (3.17)$$

This resums all repeated effects of the linear term proportional to  $\gamma$ , but it does not include the nonlinear evolution generated by the term proportional to  $\delta$ .

The full equation is still simple enough to solve exactly<sup>13</sup>

$$p(t) = \frac{p_0 e^{-\epsilon\gamma t}}{1 + \frac{\epsilon\delta p_0}{\gamma} (1 - e^{-\epsilon\gamma t})}. \quad (3.20)$$

Expanding around the leading resummed result gives

$$p(t) = p_{\text{LO}}(t) \left[ 1 - \frac{\epsilon\delta p_0}{\gamma} (1 - e^{-\epsilon\gamma t}) + O(\epsilon^2) \right]. \quad (3.21)$$

Thus the exponential resummation captures the secular terms generated by the leading linear decay, but it does not reconstruct the genuinely new nonlinear dynamics entering at higher order. In this example the higher-order correction remains uniformly perturbative and does not generate a new secular combination proportional to  $\epsilon^2 t$ .

**An exactly soluble nonlinear quantum example.** The same idea applies directly to the master equation of a density matrix. Consider a harmonic oscillator with number operator  $N := a^\dagger a$ , and take both the Hamiltonian and a Hermitian jump operator to be arbitrary functions of  $N$ :

$$H = h(N), \quad L = f(N). \quad (3.22)$$

The Lindblad equation is

$$\dot{\rho} = -i[h(N), \rho] + \gamma \left[ f(N)\rho f(N) - \frac{1}{2} \{f(N)^2, \rho\} \right]. \quad (3.23)$$

Although  $h$  and  $f$  may be nonlinear functions, the equation is exactly soluble because  $H$  and  $L$  are diagonal in the number basis. Defining

$$h_n := h(n), \quad f_n := f(n), \quad (3.24)$$

one finds that each matrix element evolves independently

$$\dot{\rho}_{nm} = \left[ -i(h_n - h_m) - \frac{\gamma}{2} (f_n - f_m)^2 \right] \rho_{nm}. \quad (3.25)$$

Therefore

$$\rho_{nm}(t) = \exp \left[ -i(h_n - h_m)t - \frac{\gamma t}{2} (f_n - f_m)^2 \right] \rho_{nm}(0). \quad (3.26)$$

The matrix units  $|n\rangle\langle m|$  are eigenoperators of the Liouvillian, and the exponential resums the repeated action of the full nonlinear generator on each of them.

For example, one may choose

$$h(N) = \omega N + \frac{\chi}{2} N(N-1), \quad f(N) = N^2. \quad (3.27)$$

The Hamiltonian then contains a so-called Kerr interaction, while the jump operator produces nonlinear dephasing. The exact solution is nevertheless obtained immediately from the formula

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<sup>13</sup>Defining  $q(t) := 1/p(t)$ , one obtains

$$\dot{q}(t) = \epsilon\gamma q(t) + \epsilon^2 \delta. \quad (3.18)$$

It follows that

$$q(t) = e^{\epsilon\gamma t} \left[ \frac{1}{p_0} + \frac{\epsilon\delta}{\gamma} (1 - e^{-\epsilon\gamma t}) \right], \quad (3.19)$$

above. This example illustrates that an interacting or nonlinear Liouvillian can be explicitly resummed whenever it admits a sufficiently simple algebraic structure, such as a basis of Liouvillian eigenoperators.

**Time-dependent generators.** The discussion above assumed a time-independent Liouvillian. The generalization to an explicitly time-dependent generator,

$$\frac{d\rho}{dt} = \mathcal{L}(t)\rho, \quad (3.28)$$

is conceptually straightforward. If the generator commutes with itself at different times,

$$[\mathcal{L}(t), \mathcal{L}(t')] = 0, \quad (3.29)$$

the evolution map is

$$\rho(t) = \exp \left[ \int_{t_0}^t dt' \mathcal{L}(t') \right] \rho(t_0). \quad (3.30)$$

This is the case in the decay equation above. In this case, the relevant secular quantity is then the integrated rate

$$\epsilon \int_{t_0}^t dt' \gamma(t'), \quad (3.31)$$

rather than simply  $\epsilon(t - t_0)$ .

For a general time-dependent Liouvillian, the generators at different times need not commute. The solution is then written as the time-ordered exponential

$$\rho(t) = \mathcal{T} \exp \left[ \int_{t_0}^t dt' \mathcal{L}(t') \right] \rho(t_0). \quad (3.32)$$

If

$$\mathcal{L}(t) = \epsilon \mathcal{L}_1(t) + \epsilon^2 \mathcal{L}_2(t) + \dots, \quad (3.33)$$

solving the equation with  $\mathcal{L}_1(t)$  retained exactly resums all time-ordered repeated insertions of the leading generator, while it does not include contributions involving  $\mathcal{L}_2(t), \mathcal{L}_3(t), \dots$ . When the resummed generator, for example  $\mathcal{L}_1(t)$ , has instantaneous GKSL form with non-negative rates at every time, the resulting evolution map is completely positive and divisible, as discussed in Section 2.7. This will be an important consideration in justifying the positivity-improved form of Schwinger-Keldysh effective actions.

### 3.2 Stationary states and unitality

A central notion in the study of open-system dynamics is that of a stationary state, or fixed point, of the evolution. Given a Lindblad generator  $\mathcal{L}$ , a density matrix  $\rho_*$  is called stationary if it satisfies

$$\mathcal{L}[\rho_*] = 0. \quad (3.34)$$

As a consequence  $\dot{\rho}_* = 0$ . In terms of the dynamical semigroup  $\Phi_t = e^{t\mathcal{L}}$ , this is the same as

$$\Phi_t[\rho_*] = \rho_* \quad \text{for all } t \geq 0. \quad (3.35)$$

Thus stationary states are the fixed points of the open-system evolution. For a closed system, stationarity reduces to the familiar condition

$$-i[H, \rho_*] = 0, \quad (3.36)$$

so that  $\rho_*$  must commute with the Hamiltonian. In particular, any energy eigenstate, and more generally any density matrix diagonal in the energy basis, is stationary under purely unitary evolution. For an open system, by contrast, the dissipative part of the Liouvillian plays an equally important role, and the stationary state is determined by the full equation

$$-i[H, \rho_*] + \sum_{\mu} \left( L_{\mu} \rho_* L_{\mu}^{\dagger} - \frac{1}{2} \{ L_{\mu}^{\dagger} L_{\mu}, \rho_* \} \right) = 0. \quad (3.37)$$

The resulting stationary state need not commute with  $H$ , and in general reflects a balance between coherent dynamics driven by  $H$  and environmental effects driven by  $L_{\mu}$ .

The existence and uniqueness of stationary states is an important qualitative property of the Lindblad evolution. If there exists a unique stationary state  $\rho_*$  and if the dynamics is sufficiently relaxing, then one expects to find the system in that state after a long time for a large class of initial states. In that case the open system loses memory of its initial condition and approaches a universal late-time state. If instead there are several stationary states, or conserved quantities that obstruct relaxation, then the late-time behavior may retain partial memory of the initial state.

**Unital maps and the non-decrease of entropy** The special case in which the stationary state is the identity operator deserves particular attention. A linear map  $\Phi$  acting on operators on a finite-dimensional Hilbert space is called *unital* if it preserves the identity operator,

$$\Phi[\mathbb{1}] = \mathbb{1}. \quad (3.38)$$

Recall that, if the Hilbert space has dimension  $d$ , then the maximally mixed state is  $\mathbb{1}/d$ . Thus unitality means that a state carrying no information, namely the maximally mixed state, remains unchanged under the action of the map.

This property is not required for a physical map to be admissible. Many physically important channels are not unital. A basic example, to be discussed at the end of this section, is relaxation of a qubit toward its ground state: such a process is trace preserving and completely positive, but it does not preserve the maximally mixed state and is therefore not unital. By contrast, pure dephasing is unital, since it suppresses coherences without selecting a preferred pure state as late-time attractor. In this sense, unitality is closely related to the distinction between decoherence and dissipation. Unital channels typically describe mixing or dephasing dynamics, while non-unital channels are needed to describe genuine relaxation toward a non-maximally mixed stationary state.

It is useful to compare unitality with trace preservation. Trace preservation is the statement

$$\text{Tr } \Phi[\rho] = \text{Tr } \rho, \quad (3.39)$$

and expresses conservation of total probability in the Schrödinger picture. Unitality is, in a precise sense, the dual notion. Let  $\Phi^{\dagger}$  denote the adjoint map with respect to the Hilbert–Schmidt pairing,

$$\text{Tr}(\Phi[\rho]\mathcal{O}) = \text{Tr}(\rho\Phi^{\dagger}[\mathcal{O}]). \quad (3.40)$$

Then, choosing  $\mathcal{O} = \mathbb{1}$  we find that

$$\Phi \text{ trace preserving} \quad \Longleftrightarrow \quad \Phi^{\dagger}[\mathbb{1}] = \mathbb{1}, \quad (3.41)$$

while

$$\Phi \text{ unital} \iff \Phi^\dagger \text{ trace preserving.} \quad (3.42)$$

Thus, in the Heisenberg picture, unitality of the Schrödinger-picture map corresponds to conservation of the trace for the adjoint map acting on observables.

In the special case of a one-parameter semigroup generated by a Liouvillian  $\mathcal{L}$ , unitality is equivalent to  $\mathcal{L}[\mathbb{1}] = 0$ . For a Lindblad generator

$$\mathcal{L}[\rho] = -i[H, \rho] + \sum_{\mu} \left( L_{\mu} \rho L_{\mu}^{\dagger} - \frac{1}{2} \{L_{\mu}^{\dagger} L_{\mu}, \rho\} \right), \quad (3.43)$$

this condition becomes

$$\sum_{\mu} [L_{\mu}, L_{\mu}^{\dagger}] = 0. \quad (3.44)$$

Therefore a Lindblad evolution is unital precisely when the dissipative terms preserve the identity operator.

A particularly important consequence of unitality in finite dimensions is that it prevents the von Neumann entropy from decreasing. The detailed proof of the statement is involved, so we will present an abridged version. Let the Hilbert space dimension be  $d$ , and define the maximally mixed state

$$\tau := \frac{\mathbb{1}}{d}. \quad (3.45)$$

For any density matrix  $\rho$ , the relative entropy with respect to  $\tau$  is

$$\begin{aligned} D(\rho \parallel \tau) &:= \text{Tr}(\rho \ln \rho - \rho \ln \tau) = \text{Tr}(\rho \ln \rho) - \text{Tr}\left(\rho \ln \frac{\mathbb{1}}{d}\right) \\ &= \text{Tr}(\rho \ln \rho) + (\ln d) \text{Tr} \rho = -S(\rho) + \ln d. \end{aligned} \quad (3.46)$$

Now let  $\Phi$  be a unital CPTP map. Since  $\Phi$  is CPTP, quantum relative entropy is monotone under  $\Phi$ :

$$D(\rho \parallel \sigma) \geq D(\Phi[\rho] \parallel \Phi[\sigma]). \quad (3.47)$$

This is a powerful result, which we invoke without proof. Applying this with  $\sigma = \tau$ , and using unitality, we obtain

$$D(\rho \parallel \tau) \geq D(\Phi[\rho] \parallel \tau). \quad (3.48)$$

Using (3.46) on both sides gives

$$\ln d - S(\rho) \geq \ln d - S(\Phi[\rho]), \quad (3.49)$$

and therefore

$$S(\Phi[\rho]) \geq S(\rho). \quad (3.50)$$

Thus a unital CPTP map cannot decrease the von Neumann entropy.

This result gives a sharp information-theoretic characterization of unital evolution. In finite dimensions, unital maps are precisely those CPTP maps that leave the maximally mixed state unchanged. They cannot decrease the von Neumann entropy and, when relaxing, may drive states toward more mixed configurations rather than toward a preferred pure state. This is why unitality naturally appears in the discussion of decoherence and entropy growth, whereas non-unitality is the hallmark of dissipative relaxation toward distinguished stationary states.

### 3.3 Populations, the Pauli master equation, and dissipation

We now return to the Lindblad equation and ask how it acts on the diagonal and off-diagonal entries of the density matrix in a physically distinguished basis. A natural choice is the energy eigenbasis of the system Hamiltonian,

$$H|n\rangle = E_n|n\rangle. \quad (3.51)$$

Other choices of basis may be justified by physical considerations, but we will stick to energy because of its universal appeal. In this basis, the diagonal matrix elements

$$p_n(t) := \rho_{nn}(t) = \langle n|\rho(t)|n\rangle \quad (3.52)$$

are interpreted as the *populations* of the corresponding energy levels, while the off-diagonal elements

$$\rho_{mn}(t) := \langle m|\rho(t)|n\rangle, \quad m \neq n, \quad (3.53)$$

represent the *coherences*.

Starting from the Lindblad equation, we first derive an equation for the populations. Taking the matrix element  $\langle n|\cdots|n\rangle$ , one finds

$$\dot{p}_n = -i\langle n|[H, \rho]|n\rangle + \sum_{\mu} \left( \langle n|L_{\mu}\rho L_{\mu}^{\dagger}|n\rangle - \frac{1}{2}\langle n|\{L_{\mu}^{\dagger}L_{\mu}, \rho\}|n\rangle \right). \quad (3.54)$$

Since the Hamiltonian is diagonal in this basis,

$$\langle n|[H, \rho]|n\rangle = E_n\rho_{nn} - \rho_{nn}E_n = 0, \quad (3.55)$$

so the Hamiltonian part does not contribute directly to the evolution of the populations, as expected. In general, the term in parentheses on the right-hand side of (3.54) still depends on the off-diagonal components of  $\rho$ .

A major simplification occurs when the diagonal sector is closed under the evolution, so that the populations evolve independently of the coherences. This is precisely what happens when the Lindblad operators may be chosen to induce transitions between definite energy levels. Let us assume that the jump operators are proportional to transitions between individual energy eigenstates <sup>14</sup>,

$$L_{nm} := \sqrt{W_{nm}}|n\rangle\langle m|, \quad n \neq m, \quad (3.57)$$

where  $W_{nm} \geq 0$  is the rate for the transition  $|m\rangle \rightarrow |n\rangle$ . In microphysical derivations of the Lindblad equation, this assumption follows from the so-called secular approximation, as discussed around (2.126). Taking the diagonal matrix element  $\langle n|\cdots|n\rangle$  of the dissipator, the first term gives

$$\sum_{k \neq \ell} \langle n|L_{k\ell}\rho L_{k\ell}^{\dagger}|n\rangle = \sum_{k \neq \ell} W_{k\ell} \langle n|k\rangle\langle \ell|\rho|\ell\rangle\langle k|n\rangle = \sum_{\ell \neq n} W_{n\ell} \rho_{\ell\ell} = \sum_{m \neq n} W_{nm} p_m. \quad (3.58)$$

Similarly, since

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<sup>14</sup>More generally, this closure holds whenever the jump operators have a definite charge (i.e. energy) with respect to the Hamiltonian, in the absence of degeneracies, namely when they satisfy

$$[H, L_{\mu}] = -\Omega_{\mu}L_{\mu}, \quad (3.56)$$

so that they connect only energy eigenspaces whose energies differ by the fixed amount  $\Omega_{\mu}$ .

$$L_{k\ell}^\dagger L_{k\ell} = W_{k\ell} |\ell\rangle\langle\ell|, \quad (3.59)$$

the anticommutator term contributes

$$\begin{aligned} -\frac{1}{2} \sum_{k \neq \ell} \langle n | \{L_{k\ell}^\dagger L_{k\ell}, \rho\} | n \rangle &= -\frac{1}{2} \sum_{k \neq \ell} W_{k\ell} \langle n | \{|\ell\rangle\langle\ell|, \rho\} | n \rangle \\ &= -\frac{1}{2} \sum_{k \neq \ell} W_{k\ell} (\delta_{n\ell} \rho_{\ell n} + \rho_{n\ell} \delta_{\ell n}) \\ &= -\sum_{k \neq n} W_{kn} \rho_{nn} = -\sum_{m \neq n} W_{mn} p_n. \end{aligned} \quad (3.60)$$

Thus the diagonal sector is closed: the right-hand side depends only on the populations and not on the coherences. Combining gain and loss terms gives

$$\dot{p}_n = \sum_m (W_{nm} p_m - W_{mn} p_n). \quad (3.61)$$

This is the *Pauli master equation*. It has the same structure as an ordinary classical rate equation: the population of level  $n$  increases by inflow from other levels  $m$  and decreases by outflow from level  $n$  to those same levels.

The normalization of the density matrix is preserved automatically. Indeed,

$$\sum_n \dot{p}_n = \sum_{n,m} (W_{nm} p_m - W_{mn} p_n) = 0. \quad (3.62)$$

Equation (3.61) makes the notion of *dissipation* very concrete. Dissipation refers to the irreversible redistribution of populations among the energy levels, typically accompanied by exchange of energy with the environment. In the presence of dissipation, an initially excited state need not remain excited. Instead, the population flows among levels according to the transition rates  $W_{nm}$ . In many physically relevant situations, especially for a cold environment, the net flow is from more excited states toward less excited ones. In that sense dissipation is directly associated with the decay of excited-state populations and the relaxation of the system toward a stationary state.

**Decoherence** The energy-basis discussion also clarifies the role of coherences. When the jump operators are as in (3.57), the coherences decouple from the populations and satisfy separate linear equations. In many common cases one finds

$$\dot{\rho}_{mn} = [-i(E_m - E_n) - \Gamma_{mn}] \rho_{mn}, \quad m \neq n, \quad (3.63)$$

so that

$$\rho_{mn}(t) = e^{-i(E_m - E_n)t} e^{-\Gamma_{mn}t} \rho_{mn}(0). \quad (3.64)$$

Thus the off-diagonal elements oscillate with the energy difference and decay with decoherence rates  $\Gamma_{mn}$ , while the populations obey the classical-looking rate equation (3.61). This is the sense in which, under suitable conditions, the Lindblad equation separates into a dissipative evolution for populations and a decohering evolution for coherences. Note also that when decoherence has suppressed all off-diagonal elements, the dynamics is effectively classical, only affecting the populations.

**Thermal bath, detailed balance, and KMS-type relations** We now specialize further and ask what happens when the environment is thermal. Suppose the environment is a thermal bath at inverse temperature  $\beta$ . One then expects the open-system dynamics to admit the Gibbs state

$$\rho_\beta := \frac{e^{-\beta H}}{Z}, \quad Z := \text{Tr}(e^{-\beta H}), \quad (3.65)$$

as a stationary state, at least in the absence of external driving and under suitable ergodicity assumptions. In the energy eigenbasis, the corresponding stationary populations are

$$p_n^{(\beta)} = \frac{e^{-\beta E_n}}{Z}. \quad (3.66)$$

Demanding that these populations solve the Pauli master equation imposes constraints on the transition rates. A sufficient local condition is detailed balance,

$$W_{nm} p_m^{(\beta)} = W_{mn} p_n^{(\beta)}. \quad (3.67)$$

Substituting the Gibbs weights gives<sup>15</sup>

$$\frac{W_{nm}}{W_{mn}} = e^{-\beta(E_n - E_m)}. \quad (3.68)$$

Thus upward transitions are Boltzmann suppressed relative to downward ones. This is the precise sense in which a thermal bath favors relaxation toward lower energies while still allowing thermally activated upward jumps. The physical content is transparent. A thermal bath does not merely cause arbitrary dissipation: it constrains the relative strength of excitation and relaxation processes in such a way that the system can settle into thermal equilibrium at the temperature of the bath.

### 3.4 Example: the qubit as an open system

We now return to the qubit and study how the general Lindblad equation acts in this simplest non-trivial example. Without loss of generality<sup>16</sup>, we take the Hamiltonian to be

$$H = \frac{\omega_0}{2} \sigma_z, \quad (3.69)$$

so that

$$H|\uparrow\rangle = \frac{\omega_0}{2}|\uparrow\rangle, \quad H|\downarrow\rangle = -\frac{\omega_0}{2}|\downarrow\rangle. \quad (3.70)$$

Thus  $|\uparrow\rangle$  is the excited state and  $|\downarrow\rangle$  the ground state. In this basis the density matrix is

$$\rho = \begin{pmatrix} \rho_{\uparrow\uparrow} & \rho_{\uparrow\downarrow} \\ \rho_{\downarrow\uparrow} & \rho_{\downarrow\downarrow} \end{pmatrix}, \quad \rho_{\uparrow\uparrow} + \rho_{\downarrow\downarrow} = 1. \quad (3.71)$$

The diagonal entries are the populations, while the off-diagonal entries are the coherences. For a qubit, a generic jump operator is simply an arbitrary complex  $2 \times 2$  matrix. Equivalently, it may be expanded in the basis

$$\mathbb{1}, \quad \sigma_+ := |\uparrow\rangle\langle\downarrow| = \frac{\sigma_x + i\sigma_y}{2}, \quad \sigma_- := |\downarrow\rangle\langle\uparrow| = \frac{\sigma_x - i\sigma_y}{2}, \quad \sigma_z, \quad (3.72)$$

<sup>15</sup>Relation (3.68) is the population-level manifestation of the more general Kubo–Martin–Schwinger, or KMS, condition satisfied by equilibrium correlation functions of a thermal bath, to be discussed later.

<sup>16</sup>Any Hermitian  $2 \times 2$  Hamiltonian can be written as  $H = a\mathbb{1} + \vec{h} \cdot \vec{\sigma}$ . The term proportional to the identity only shifts all energies by the same constant and has no effect on the dynamics, while a suitable unitary change of basis can always rotate the vector  $\vec{h}$  to the  $z$ -axis.

as

$$L = a \mathbb{1} + b \sigma_+ + c \sigma_- + d \sigma_z. \quad (3.73)$$

The identity piece is dynamically inessential, since it may be absorbed into a redefinition of the Hamiltonian and of the traceless part of the jump operator. It is therefore natural to single out the physically most relevant processes by choosing jump operators adapted to the energy basis. The three elementary cases are excitation, relaxation, and dephasing:

$$L_\uparrow := \sqrt{\gamma_\uparrow} \sigma_+, \quad L_\downarrow := \sqrt{\gamma_\downarrow} \sigma_-, \quad L_\phi := \sqrt{\gamma_\phi} \sigma_z. \quad (3.74)$$

Here  $\gamma_\uparrow$ ,  $\gamma_\downarrow$ , and  $\gamma_\phi$  are non-negative coefficients. The first raises the energy of the qubit, the second lowers it, and the third produces dephasing in the energy basis without changing the populations.

It is convenient to include all three at once. The Lindblad equation then takes the form

$$\begin{aligned} \frac{d\rho}{dt} = & -i[H, \rho] + \gamma_\downarrow \left( \sigma_- \rho \sigma_+ - \frac{1}{2} \{ \sigma_+ \sigma_-, \rho \} \right) + \gamma_\uparrow \left( \sigma_+ \rho \sigma_- - \frac{1}{2} \{ \sigma_- \sigma_+, \rho \} \right) \\ & + \gamma_\phi (\sigma_z \rho \sigma_z - \rho). \end{aligned} \quad (3.75)$$

We now discuss the physical role of each term, first individually and then in combination.

**Relaxation** Suppose first that only

$$L = \sqrt{\gamma_\downarrow} \sigma_- \quad (3.76)$$

is present. Then the master equation becomes

$$\frac{d\rho}{dt} = -i[H, \rho] + \gamma_\downarrow \left( \sigma_- \rho \sigma_+ - \frac{1}{2} \{ \sigma_+ \sigma_-, \rho \} \right). \quad (3.77)$$

A straightforward matrix computation gives

$$\dot{\rho}_{\uparrow\uparrow} = -\gamma_\downarrow \rho_{\uparrow\uparrow}, \quad (3.78)$$

$$\dot{\rho}_{\downarrow\downarrow} = \gamma_\downarrow \rho_{\uparrow\uparrow}, \quad (3.79)$$

$$\dot{\rho}_{\uparrow\downarrow} = -\left(i\omega_0 + \frac{\gamma_\downarrow}{2}\right) \rho_{\uparrow\downarrow}, \quad (3.80)$$

$$\dot{\rho}_{\downarrow\uparrow} = \left(i\omega_0 - \frac{\gamma_\downarrow}{2}\right) \rho_{\downarrow\uparrow}. \quad (3.81)$$

Thus

$$\rho_{\uparrow\uparrow}(t) = e^{-\gamma_\downarrow t} \rho_{\uparrow\uparrow}(0), \quad (3.82)$$

while trace preservation implies

$$\rho_{\downarrow\downarrow}(t) = 1 - \rho_{\uparrow\uparrow}(t) = 1 - e^{-\gamma_\downarrow t} \rho_{\uparrow\uparrow}(0). \quad (3.83)$$

The coherences oscillate and decay as

$$\rho_{\uparrow\downarrow}(t) = e^{-i\omega_0 t} e^{-\gamma_\downarrow t/2} \rho_{\uparrow\downarrow}(0). \quad (3.84)$$

This is the simplest example of dissipation: population is irreversibly transferred from the excited state to the ground state, and the stationary state is

$$\rho_* = |\downarrow\rangle\langle\downarrow|. \quad (3.85)$$

If the initial state is pure and excited, the entropy initially increases as the state becomes mixed, then decreases again and asymptotes zero at late times, when the system settles into the pure ground state. Thus the entropy does not change monotonically under relaxation. The case of just  $L_\uparrow$  is similar with the excited and ground states switched.

**Pure dephasing** Now suppose that only  $L = \sqrt{\gamma_\phi} \sigma_z$  is present. Then the dissipative contribution is

$$\gamma_\phi(\sigma_z \rho \sigma_z - \rho), \quad (3.86)$$

and one finds

$$\dot{\rho}_{\uparrow\uparrow} = 0, \quad (3.87)$$

$$\dot{\rho}_{\downarrow\downarrow} = 0, \quad (3.88)$$

$$\dot{\rho}_{\uparrow\downarrow} = -(i\omega_0 + 2\gamma_\phi) \rho_{\uparrow\downarrow}, \quad (3.89)$$

$$\dot{\rho}_{\downarrow\uparrow} = (i\omega_0 - 2\gamma_\phi) \rho_{\downarrow\uparrow}. \quad (3.90)$$

Note that this induces a unital map since  $\rho_{\downarrow\downarrow} = \rho_{\uparrow\uparrow} = 1/2$  and  $\rho_{\uparrow\downarrow} = \rho_{\downarrow\uparrow} = 0$  is a solution. We see that the populations remain constant, while the coherences oscillate and decay as

$$\rho_{\uparrow\downarrow}(t) = e^{-i\omega_0 t} e^{-2\gamma_\phi t} \rho_{\uparrow\downarrow}(0). \quad (3.91)$$

This is the cleanest example of decoherence without dissipation. There is no energy exchange with the environment and hence no redistribution of populations, but phase information is lost. The preferred basis, or pointer basis, is precisely the energy basis  $\{|\uparrow\rangle, |\downarrow\rangle\}$  singled out by  $\sigma_z$ . In this case the late-time state is

$$\rho(\infty) = \begin{pmatrix} \rho_{\uparrow\uparrow}(0) & 0 \\ 0 & \rho_{\downarrow\downarrow}(0) \end{pmatrix}, \quad (3.92)$$

which is generally mixed unless the initial state already happened to be an energy eigenstate. Therefore pure dephasing typically increases the entropy from zero to a positive asymptotic value and does so monotonically for a generic initial pure superposition.

**All three processes together** Let us now combine the three elementary processes. From (3.75), one finds

$$\dot{\rho}_{\uparrow\uparrow} = -\gamma_\downarrow \rho_{\uparrow\uparrow} + \gamma_\uparrow \rho_{\downarrow\downarrow}, \quad (3.93)$$

$$\dot{\rho}_{\downarrow\downarrow} = \gamma_\downarrow \rho_{\uparrow\uparrow} - \gamma_\uparrow \rho_{\downarrow\downarrow}, \quad (3.94)$$

$$\dot{\rho}_{\uparrow\downarrow} = -\left[i\omega_0 + \frac{\gamma_\downarrow + \gamma_\uparrow}{2} + 2\gamma_\phi\right] \rho_{\uparrow\downarrow}, \quad (3.95)$$

$$\dot{\rho}_{\downarrow\uparrow} = \left[i\omega_0 - \frac{\gamma_\downarrow + \gamma_\uparrow}{2} - 2\gamma_\phi\right] \rho_{\downarrow\uparrow}. \quad (3.96)$$

The populations therefore obey the Pauli master equation

$$\dot{p}_\uparrow = -\gamma_\downarrow p_\uparrow + \gamma_\uparrow p_\downarrow, \quad \dot{p}_\downarrow = \gamma_\downarrow p_\uparrow - \gamma_\uparrow p_\downarrow, \quad (3.97)$$

with

$$p_{\uparrow} := \rho_{\uparrow\uparrow}, \quad p_{\downarrow} := \rho_{\downarrow\downarrow}. \quad (3.98)$$

This is precisely the classical rate equation for a two-state Markov process. The diagonal sector has become autonomous. At the same time, the coherences decay according to

$$\rho_{\uparrow\downarrow}(t) = e^{-i\omega_0 t} \exp\left[-\left(\frac{\gamma_{\downarrow} + \gamma_{\uparrow}}{2} + 2\gamma_{\phi}\right)t\right] \rho_{\uparrow\downarrow}(0). \quad (3.99)$$

Thus the coherence time is controlled by both the dephasing channel and the population-changing channels.

The stationary populations follow from setting the left-hand side of (3.97) to zero:

$$\gamma_{\downarrow} p_{\uparrow}^{(*)} = \gamma_{\uparrow} p_{\downarrow}^{(*)}, \quad p_{\uparrow}^{(*)} + p_{\downarrow}^{(*)} = 1. \quad (3.100)$$

Hence

$$p_{\uparrow}^{(*)} = \frac{\gamma_{\uparrow}}{\gamma_{\uparrow} + \gamma_{\downarrow}}, \quad p_{\downarrow}^{(*)} = \frac{\gamma_{\downarrow}}{\gamma_{\uparrow} + \gamma_{\downarrow}}. \quad (3.101)$$

Since the coherences decay to zero, the stationary density matrix is diagonal,

$$\rho_* = \frac{\gamma_{\uparrow}}{\gamma_{\uparrow} + \gamma_{\downarrow}} |\uparrow\rangle\langle\uparrow| + \frac{\gamma_{\downarrow}}{\gamma_{\uparrow} + \gamma_{\downarrow}} |\downarrow\rangle\langle\downarrow|. \quad (3.102)$$

This is the generic mixed stationary state of the qubit in the presence of both upward and downward jumps.

**Thermal stationary state and detailed balance** A particularly important case occurs when the environment is thermal. Then the stationary state should be the Gibbs state

$$\rho_{\beta} = \frac{e^{-\beta H}}{\text{Tr}(e^{-\beta H})} = \frac{e^{-\beta\omega_0/2} |\uparrow\rangle\langle\uparrow| + e^{+\beta\omega_0/2} |\downarrow\rangle\langle\downarrow|}{e^{-\beta\omega_0/2} + e^{+\beta\omega_0/2}}. \quad (3.103)$$

This requires

$$\frac{p_{\uparrow}^{(*)}}{p_{\downarrow}^{(*)}} = e^{-\beta\omega_0}. \quad (3.104)$$

Comparing with (3.101), we obtain the detailed-balance condition

$$\frac{\gamma_{\uparrow}}{\gamma_{\downarrow}} = e^{-\beta\omega_0}. \quad (3.105)$$

Thus a thermal bath does not permit arbitrary excitation and relaxation rates: they must be tuned relative to one another so that upward jumps are Boltzmann suppressed with respect to downward jumps. This is the qubit version of the more general KMS relation discussed earlier. At zero temperature,  $\beta \rightarrow \infty$ , one has

$$\gamma_{\uparrow} \rightarrow 0, \quad (3.106)$$

so the excited state decays irreversibly to the ground state. At infinite temperature,  $\beta \rightarrow 0$ , one finds

$$\gamma_{\uparrow} = \gamma_{\downarrow}, \quad (3.107)$$

and the stationary state becomes maximally mixed,

$$\rho_* = \frac{1}{2} \mathbb{1}. \quad (3.108)$$

**Bloch-vector form** It is useful to translate the above discussion into Bloch-vector language. Writing

$$\rho = \frac{1}{2}(\mathbb{1} + b_x\sigma_x + b_y\sigma_y + b_z\sigma_z), \quad (3.109)$$

one finds

$$\dot{b}_x = -\left(\frac{\gamma_\downarrow + \gamma_\uparrow}{2} + 2\gamma_\phi\right)b_x - \omega_0 b_y, \quad (3.110)$$

$$\dot{b}_y = -\left(\frac{\gamma_\downarrow + \gamma_\uparrow}{2} + 2\gamma_\phi\right)b_y + \omega_0 b_x, \quad (3.111)$$

$$\dot{b}_z = -(\gamma_\downarrow + \gamma_\uparrow)b_z + (\gamma_\uparrow - \gamma_\downarrow). \quad (3.112)$$

Thus the transverse components  $b_x$  and  $b_y$  decay exponentially while precessing around the  $z$ -axis, whereas the longitudinal component  $b_z$  relaxes toward the stationary value

$$b_z^{(*)} = \frac{\gamma_\uparrow - \gamma_\downarrow}{\gamma_\uparrow + \gamma_\downarrow}. \quad (3.113)$$

This makes the geometry very transparent: dissipation pushes the Bloch vector toward a preferred point on the  $z$ -axis, while decoherence shrinks its transverse projection. The stationary point lies on the south pole for zero-temperature relaxation, at the center for an infinite-temperature bath, and at an intermediate point for finite temperature.

### 3.5 Example: the dissipative harmonic oscillator

As a second example, let us consider a single quantum harmonic oscillator with Hamiltonian

$$H = \omega \left( a^\dagger a + \frac{1}{2} \right), \quad (3.114)$$

where the annihilation and creation operators satisfy

$$[a, a^\dagger] = 1. \quad (3.115)$$

It is often convenient to introduce the number operator  $N := a^\dagger a$ , whose eigenstates  $|n\rangle$  obey

$$N|n\rangle = n|n\rangle, \quad a|n\rangle = \sqrt{n}|n-1\rangle, \quad a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle. \quad (3.116)$$

The energy eigenstates of the oscillator therefore coincide with the number eigenstates. This makes the oscillator particularly well suited for illustrating, in a simple but non-trivial setting, the notions of dissipation, excitation, dephasing, and thermalization.

The most important jump operators are the lowering operator  $a$ , the raising operator  $a^\dagger$ , and the number operator  $N$ . Respectively, these describe loss of quanta, gain of quanta, and pure dephasing in the number basis. We will discuss the corresponding processes in turn.

**Pure damping** Let us begin with the simplest dissipative choice,

$$L = \sqrt{\kappa} a, \quad (3.117)$$

with  $\kappa \geq 0$ . The Lindblad equation is

$$\frac{d\rho}{dt} = -i[H, \rho] + \kappa \left( a\rho a^\dagger - \frac{1}{2}\{a^\dagger a, \rho\} \right). \quad (3.118)$$

This is the oscillator analogue of qubit relaxation. It describes irreversible loss of excitations to the environment.

A particularly efficient way to understand the dynamics is through the Heisenberg-picture master equation. Using the adjoint Liouvillian, one finds

$$\frac{da}{dt} = -i[a, H] + \kappa \left( a^\dagger a a - \frac{1}{2} \{a^\dagger a, a\} \right) = - \left( i\omega + \frac{\kappa}{2} \right) a, \quad (3.119)$$

$$\frac{da^\dagger}{dt} = \left( i\omega - \frac{\kappa}{2} \right) a^\dagger. \quad (3.120)$$

Therefore

$$\langle a(t) \rangle = e^{-i\omega t} e^{-\kappa t/2} \langle a(0) \rangle. \quad (3.121)$$

Thus the oscillator amplitude rotates with frequency  $\omega$  and decays with rate  $\kappa/2$ . The occupation number obeys

$$\begin{aligned} \frac{dN}{dt} &= \kappa \left( a^\dagger N a - \frac{1}{2} \{a^\dagger a, N\} \right) \\ &= -\kappa N, \end{aligned} \quad (3.122)$$

so that

$$\langle N(t) \rangle = e^{-\kappa t} \langle N(0) \rangle. \quad (3.123)$$

Thus the mean number of quanta decays to zero exponentially in time. The unique stationary state is therefore the vacuum,

$$\rho_* = |0\rangle\langle 0|. \quad (3.124)$$

This is a genuinely dissipative process: it changes the populations in the energy basis and drives the system toward the lowest-energy state.

The master equation also induces a closed rate equation for the diagonal matrix elements

$$p_n(t) := \langle n | \rho(t) | n \rangle. \quad (3.125)$$

A short computation gives

$$\dot{p}_n = \kappa [(n+1)p_{n+1} - np_n]. \quad (3.126)$$

This is the Pauli master equation for damping. Population flows from higher occupation numbers to lower ones, and eventually accumulates in the vacuum state.

The effect on the off-diagonal elements is also simple. Writing

$$\rho_{mn}(t) := \langle m | \rho(t) | n \rangle, \quad (3.127)$$

the Hamiltonian part gives the phase factor  $e^{-i\omega(m-n)t}$ , while the dissipator damps the matrix elements. In particular, coherences between widely separated number states decay faster than low-lying ones. Thus damping produces both dissipation and decoherence.

**Thermal bath** The natural extension of the previous example is to include both lowering and raising jumps,

$$L_- := \sqrt{\kappa(N_{\text{th}} + 1)} a, \quad L_+ := \sqrt{\kappa N_{\text{th}}} a^\dagger, \quad (3.128)$$

The real parameter  $N_{\text{th}} \geq 0$ , which characterizes the tuning between these two operators, will later be identified with the thermal occupation number of the bath at frequency  $\omega$ . The Lindblad equation becomes

$$\begin{aligned} \frac{d\rho}{dt} = & -i[H, \rho] + \kappa(N_{\text{th}} + 1) \left( a\rho a^\dagger - \frac{1}{2}\{a^\dagger a, \rho\} \right) \\ & + \kappa N_{\text{th}} \left( a^\dagger \rho a - \frac{1}{2}\{aa^\dagger, \rho\} \right). \end{aligned} \quad (3.129)$$

This is the oscillator analogue of the qubit with both relaxation and excitation. Using the Heisenberg-picture equation, one finds again

$$\frac{da}{dt} = -\left(i\omega + \frac{\kappa}{2}\right) a, \quad (3.130)$$

so the mean oscillator amplitude still decays with the same rate  $\kappa/2$ . The number operator, however, now obeys

$$\begin{aligned} \frac{dN}{dt} &= \kappa(N_{\text{th}} + 1) \left( a^\dagger N a - \frac{1}{2}\{a^\dagger a, N\} \right) + \kappa N_{\text{th}} \left( a N a^\dagger - \frac{1}{2}\{a a^\dagger, N\} \right) \\ &= \kappa(N_{\text{th}} + 1) \left( a^\dagger N a - N^2 \right) + \kappa N_{\text{th}} \left( a N a^\dagger - N(N + 1) \right) \\ &= \kappa(N_{\text{th}} + 1) \left( a^\dagger a (N - 1) - N^2 \right) + \kappa N_{\text{th}} \left( a a^\dagger N + a a^\dagger - N(N + 1) \right) \\ &= \kappa(N_{\text{th}} + 1) \left( N(N - 1) - N^2 \right) + \kappa N_{\text{th}} \left( (N + 1)N + (N + 1) - N(N + 1) \right) \\ &= -\kappa(N_{\text{th}} + 1)N + \kappa N_{\text{th}}(N + 1) = -\kappa N + \kappa N_{\text{th}} \\ &= -\kappa(N - N_{\text{th}}). \end{aligned} \quad (3.131)$$

and therefore

$$\langle N(t) \rangle = N_{\text{th}} + e^{-\kappa t} (\langle N(0) \rangle - N_{\text{th}}). \quad (3.132)$$

Thus the system no longer relaxes to the vacuum, but instead to a stationary state with mean occupation number  $N_{\text{th}}$ .

The diagonal matrix elements satisfy the rate equation

$$\begin{aligned} \dot{p}_n &= \kappa(N_{\text{th}} + 1) [(n + 1)p_{n+1} - np_n] \\ &+ \kappa N_{\text{th}} [np_{n-1} - (n + 1)p_n]. \end{aligned} \quad (3.133)$$

The first line describes downward transitions, while the second describes upward transitions. This is a direct infinite-dimensional analogue of the two-level Pauli equation studied for the qubit.

The stationary solution is geometric,

$$p_n^{(*)} = \frac{1}{N_{\text{th}} + 1} \left( \frac{N_{\text{th}}}{N_{\text{th}} + 1} \right)^n, \quad (3.134)$$

which is precisely the Gibbs distribution

$$\rho_\beta = \frac{e^{-\beta H}}{\text{Tr}(e^{-\beta H})} \quad (3.135)$$

provided one identifies

$$N_{\text{th}} = \frac{1}{e^{\beta\omega} - 1}. \quad (3.136)$$

Equivalently,

$$\frac{\kappa N_{\text{th}}}{\kappa(N_{\text{th}} + 1)} = e^{-\beta\omega}. \quad (3.137)$$

This is the oscillator version of detailed balance. Upward and downward jumps are not independent if the bath is thermal.

In the zero-temperature limit one has

$$N_{\text{th}} \rightarrow 0, \quad (3.138)$$

and the thermal master equation reduces to pure damping, which we studied previously. In the high-temperature limit,  $N_{\text{th}} \gg 1$ , upward and downward rates become nearly equal and the stationary state is highly mixed.

**Pure dephasing** A third instructive process is obtained by taking

$$L_\phi := \sqrt{\gamma_\phi} N, \quad (3.139)$$

where  $\gamma_\phi \geq 0$ . The corresponding Lindblad equation is

$$\frac{d\rho}{dt} = -i[H, \rho] + \gamma_\phi \left( N\rho N - \frac{1}{2}\{N^2, \rho\} \right). \quad (3.140)$$

Since  $N$  is diagonal in the energy basis, this process does not change the populations. Indeed,

$$\dot{p}_n = 0. \quad (3.141)$$

However, the off-diagonal elements decay:

$$\frac{d}{dt}\rho_{mn} = -i\omega(m-n)\rho_{mn} - \frac{\gamma_\phi}{2}(m-n)^2\rho_{mn}. \quad (3.142)$$

Hence

$$\rho_{mn}(t) = e^{-i\omega(m-n)t} e^{-\frac{\gamma_\phi}{2}(m-n)^2 t} \rho_{mn}(0). \quad (3.143)$$

This is the clean oscillator analogue of qubit dephasing. There is no dissipation, since the energy populations remain fixed, but there is decoherence, because the off-diagonal elements in the number basis are exponentially suppressed. The pointer basis is therefore the number basis.

**Gaussian Lindblad equation** It is useful to briefly mention the most general *Gaussian* Lindblad equation for a single oscillator. Here Gaussian means that Gaussian states are mapped into Gaussian states, or equivalently that the evolution of the state is completely determined by the first and second moments of the canonical variables  $x$  and  $p$ . This happens when the Hamiltonian is at most quadratic in  $x$  and  $p$ , and the jump operators are at most linear in them. In that case, after suitable redefinitions of the quadratic Hamiltonian, the most general one-mode Gaussian Lindblad equation may be written as<sup>17</sup>

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<sup>17</sup>Here we are using the fact that the jump operators are Hermitian to express the dissipator in the double commutator form (2.83).

$$\begin{aligned} \frac{d\rho}{dt} = & -i[H, \rho] - \frac{i\kappa}{2}[x, \{p, \rho\}] - D_{pp}[x, [x, \rho]] - D_{xx}[p, [p, \rho]] \\ & + D_{xp}([x, [p, \rho]] + [p, [x, \rho]]), \end{aligned} \quad (3.144)$$

where  $H$  is quadratic in  $x$  and  $p$ ,  $\kappa$  is a friction coefficient, and the three real coefficients  $D_{xx}$ ,  $D_{xp} = D_{px}$ , and  $D_{pp}$  form the symmetric diffusion matrix. The terminology Gaussian reflects the fact that this equation closes on the mean values and covariance matrix of  $x$  and  $p$ , without generating higher connected moments if they are absent initially. Complete positivity imposes a constraint on the diffusion matrix, namely

$$D_{pp}D_{xx} - D_{xp}^2 \geq \frac{\kappa^2}{16}, \quad (3.145)$$

in units with  $\hbar = 1$ . The simple damping and thermal examples discussed above are special cases of this general Gaussian structure. For example, pure damping corresponds to

$$D_{xx} = \frac{\kappa}{8\omega}, \quad D_{pp} = \frac{\kappa\omega}{8}, \quad D_{xp} = 0, \quad (3.146)$$

which saturates the complete-positivity bound. For the thermal bath one finds

$$D_{xx} = \frac{\kappa}{8\omega}(2N_{\text{th}} + 1), \quad D_{pp} = \frac{\kappa\omega}{8}(2N_{\text{th}} + 1), \quad D_{xp} = 0, \quad (3.147)$$

satisfying but not saturating the bound for  $N_{\text{th}} \neq 0$ . Dephasing instead is not Gaussian.

**Entropy and stationary states** These three cases illustrate very clearly the distinction between dissipation and decoherence. Damping by  $a$  changes the populations and drives the system toward the pure vacuum, so the entropy need not be monotonic: it may first increase as the state becomes mixed and later decrease as the vacuum is approached. The thermal bath with both  $a$  and  $a^\dagger$  drives the system toward a mixed Gibbs state, and the late-time entropy approaches the thermal entropy. Pure dephasing leaves all populations unchanged but reduces coherence, so it typically increases the entropy unless the initial state was already diagonal in the number basis.

It is also worth emphasizing the relation to the discussion of populations and coherences developed above. In the oscillator example, the number basis is the natural energy basis, and the jump operators  $a$  and  $a^\dagger$  induce downward and upward transitions between adjacent levels. The resulting Pauli equation has exactly the expected birth–death form. By contrast, the jump operator  $N$  carries zero transition frequency and therefore generates pure dephasing without any redistribution of populations.

## 4 The Schwinger-Keldysh formalism

So far we have described open-system dynamics in the *operator language* of density matrices and master equations. We now introduce a complementary and extremely powerful formulation based on *path integrals*: the Schwinger–Keldysh, or closed-time-path, formalism [?, ?]. This framework is designed to compute real-time expectation values and reduced density matrices, and is therefore ideally suited both to closed systems evolving unitarily and to open systems obtained after integrating out an environment. In particular, it will provide the natural bridge between the quantum-mechanical discussion of the previous sections and the effective field theory perspective that we will later apply to cosmology.

The main idea is to double the fields and evolve them along a forward and a backward time contour, so that probabilities, expectation values, and reduced dynamics can all be encoded in

a single functional integral. In this section we first review the construction for closed systems and explain the meaning of the closed-time path and the Keldysh basis. We then show how integrating out an environment leads to the Feynman–Vernon influence functional and discuss the general constraints that unitarity, Hermiticity, and (complete) positivity impose on the resulting effective action. Finally, we illustrate the formalism in simple examples, including the damped harmonic oscillator, Gaussian environments, the semiclassical limit, and a dissipative scalar field.

#### 4.1 Closed systems, unitary evolution, and the doubled contour

We now begin our discussion of the Schwinger–Keldysh formalism. The first important point is that this formalism is not intrinsically tied to open systems. It already arises naturally for a closed quantum system, i.e. one that undergoes Hamiltonian evolution. Open-system effects enter only later, when one separates the full Hilbert space into system and environment and integrates out the environmental degrees of freedom.

The natural starting point is therefore the Hamiltonian evolution of a density matrix. Let the Hilbert space of the full closed system be  $\mathcal{H}$ , and let the Hamiltonian be  $H$ . The density matrix evolves as

$$\rho(t) = U(t, t_0) \rho(t_0) U^\dagger(t, t_0), \quad U(t, t_0) := e^{-iH(t-t_0)}, \quad (4.1)$$

or, equivalently,

$$i \frac{d}{dt} U(t, t_0) = H U(t, t_0), \quad -i \frac{d}{dt} U^\dagger(t, t_0) = U^\dagger(t, t_0) H. \quad (4.2)$$

This already exhibits the basic doubling that lies at the heart of Schwinger–Keldysh (SK): the density matrix has a ket and a bra and therefore evolves both forward in time with  $U$ , for the ket, and backward in time with  $U^\dagger$ , for the bra. This should be contrasted with other objects we are familiar with, such as the wavefunction and time-ordered correlation functions, usually computed in the context of scattering amplitudes, both of which only display forward-in-time evolution. In jargon, we say that the density matrix is an *in-in* object, while the wavefunction or time-ordered correlators are *in-out* objects.

For pedagogical reasons, we will introduce the SK path integral in three steps:

- First, we focus on computing the expectation value of a single simple operator.
- Second, we introduce the formalism of external currents  $J$ .
- Third, we discuss the insertion of operators at different times.

**The SK path integral for a simple operator** The expectation value of an Heisenberg-picture operator<sup>18</sup>  $\hat{x}(t)^n$  in the Schrödinger picture reads

$$\langle \hat{x}(t)^n \rangle = \text{Tr} \left( U(t, t_0) \rho(t_0) U^\dagger(t, t_0) \hat{x}^n \right), \quad (4.3)$$

where

$$U(t, t_0) := T \exp \left[ -i \int_{t_0}^t dt H \right]. \quad (4.4)$$

and  $T$  denotes time ordering. The Schwinger–Keldysh path integral is simply a path-integral representation of this same real-time evolution.

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<sup>18</sup>As we will see shortly, in this section there will be many different objects denoted by the letter “ $x$ ”. For clarity, we sometimes use a hat when  $\hat{x}$  is a quantum operator.

For simplicity, let us consider a single degree of freedom  $x$  with Lagrangian  $L(x, \dot{x})$ . The key idea is to use the resolution of the identity,

$$\mathbb{1} = \int_{\mathbb{R}} dx |x\rangle \langle x|, \quad (4.5)$$

three times inside (4.3). This gives

$$\begin{aligned} \langle \hat{x}(t)^n \rangle &= \int_{\mathbb{R}} dx_{+,0} \int_{\mathbb{R}} dx_{-,0} \int_{\mathbb{R}} dx_f \\ &\times \text{Tr} \left[ U(t, t_0) |x_{+,0}\rangle \langle x_{+,0}| \rho(t_0) |x_{-,0}\rangle \langle x_{-,0}| U^\dagger(t, t_0) |x_f\rangle \langle x_f| \hat{x}(t)^n \right] \end{aligned} \quad (4.6)$$

We note that  $\hat{x} |x_f\rangle = x_f |x_f\rangle$  with  $x_f$  a number. Hence, we can use the cyclicity of the trace to move  $|x_f\rangle$  from the left-hand to the right-hand side<sup>19</sup>. One object that appears is the matrix element of the time-evolution operator  $U$ . This is represented by the usual path integral,

$$\langle x_f | U(t, t_0) | x_{+,0} \rangle = \int_{x_+(t_0)=x_{+,0}}^{x_+(t)=x_f} \mathcal{D}x_+ e^{iS[x_+]}, \quad (4.7)$$

where  $S$  is the standard action, which we choose to write in Lagrangian language as

$$S[x] := \int_{t_0}^t dt L(x, \dot{x}). \quad (4.8)$$

The other, related matrix element is

$$\langle x_{-,0} | U^\dagger(t, t_0) | x_f \rangle = \langle x_f | U(t, t_0) | x_{-,0} \rangle^* = \int_{x_-(t_0)=x_{-,0}}^{x_-(t)=x_f} \mathcal{D}x_- e^{-iS[x_-]}, \quad (4.9)$$

Note that  $x_+(t)$  and  $x_-(t)$  are just integration variables for the path integrals. We could call them whatever we want. Here we have chosen to call them different names so we can easily distinguish which one refers to the first path integral describing forward time evolution of the ket and which one describes backward time evolution of the bra. The fact that two path integral variables  $x_+$  and  $x_-$  are needed to describe a single quantum operator  $\hat{x}$  is sometimes referred to as the “doubling of fields” in the SK formalism. Here we understand that this doubling is just a book-keeping trick to avoid confusion between the two path-integration variables.

Inserting these representations into (4.6), and writing the initial density matrix in the position basis as

$$\rho(x_{+,0}, x_{-,0}; t_0) := \langle x_{+,0} | \rho(t_0) | x_{-,0} \rangle, \quad (4.10)$$

we obtain

$$\begin{aligned} \langle \hat{x}(t)^n \rangle &= \int dx_{+,0} dx_{-,0} dx_f \rho(x_{+,0}, x_{-,0}; t_0) \\ &\times \int_{x_+(t_0)=x_{+,0}}^{x_+(t)=x_f} \mathcal{D}x_+ e^{iS[x_+]} \int_{x_-(t_0)=x_{-,0}}^{x_-(t)=x_f} \mathcal{D}x_- e^{-iS[x_-]} x_\pm(t)^n. \end{aligned} \quad (4.11)$$

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<sup>19</sup>Alternatively, we perform the trace as a fourth integral  $\int dx |x\rangle \langle x|$  and use  $\langle x_f | x \rangle = \delta(x_f - x)$  to perform this integral, giving the same result.

This is the Schwinger–Keldysh path integral in the  $\pm$  basis. Note that since  $x_+(t) = x_-(t) = x_f$ , it doesn't matter if we write  $x_+(t)^n$  or  $x_-(t)^n$  above. The difference between these two choices is crucial for unequal time correlators as we discuss in detail in the next subsection.

We see explicitly that the path integral contains two copies of the dynamical variable,  $x_+$  and  $x_-$ , corresponding to the forward and backward branches. Their final values are identified because of the trace over the final Hilbert space. Averaging over  $x_f$  effectively stitches the forward and backward contours together (see Appendix A of [?] for a nice discussion of this).

Two aspects of the notation are worth stressing. First, there are three very different objects, all involving the letter “ $x$ ”:

- $\hat{x}$  is a quantum operator,
- $x_{\pm}(t)$  are arbitrary functions of time (not quantum operators!) over which we path-integrate,
- $x_{+,0}$ ,  $x_{-,0}$  and  $x_f$  are real numbers over which we integrate.

Second, it's important to appreciate the difference between the standard 1d integral  $\int dx_{\pm,0}$  over the value of the functions  $x_{\pm}(t)$  at time  $t_0$  and the two path integrals  $\mathcal{D}x_{\pm}$  over all possible histories  $x_{\pm}(t)$  of the functions  $x_{\pm}(t)$ .

To keep the notation light, it is often convenient not to display explicitly the integrations over the initial and final boundary data. We will therefore use the shorthand  $\int_{\text{I.C.}}$  to denote the path integral with Initial Conditions (I.C.) weighted by the initial density matrix  $\rho(x_{+,0}, x_{-,0}; t_0)$ , it being understood that the two branches are sewn together continuously at the final time  $t$ , i.e.  $x_+(t) = x_-(t)$  as required by the trace. In other words,  $\int_{\text{I.C.}}$  stands for the explicit integrations over  $x_{+,0}$ ,  $x_{-,0}$ , and over the common final value  $x_f$ , together with the corresponding boundary conditions on the path integral histories. Because of this, the values of  $x_+$  and  $x_-$  at  $t$  are equal but arbitrary.

With this convention, the Schwinger–Keldysh representation of the expectation value may be written more compactly as

$$\langle \hat{x}(t)^n \rangle = \int_{\text{I.C.}} \mathcal{D}x_+ \int_{\text{I.C.}} \mathcal{D}x_- \exp \left[ iS[x_+] - iS[x_-] \right] x_{\pm}(t)^n. \quad (4.12)$$

It is worth emphasizing again the relation to the operator formalism. The path integral has not introduced any new dynamics and nothing intrinsically dissipative has happened yet: this is still just Hamiltonian evolution of a closed system, written in a way adapted to expectation values rather than transition amplitudes. It is merely a different representation of the same operator expression (4.3). The forward branch encodes the action of  $U$ , the backward branch encodes the action of  $U^\dagger$ , and the trace over the final state produces the closed contour. In this sense, the Schwinger–Keldysh path integral is the path-integral representation of the Hamiltonian evolution of the density matrix.

**External sources** Instead of defining a different path integral for each expectation value that we want to compute, we can define a single *generating functional*<sup>20</sup>  $Z$  and compute from it all possible expectation values. As we do in the usual path integral, this is achieved by coupling

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<sup>20</sup>This is sometimes also called a *partition function* because of the close analogy to statistical physics for a thermal state with some conserved charges. This nomenclature is perhaps more common in Euclidean signature, while the term generating functional is more common in Lorentzian signature. However, in practice, they are often used interchangeably.

our variables to external sources. Since the SK path integral has two branches, we need sources both on the forward and backward branch.

For the generating function we define

$$Z[J_+, J_-] := \text{Tr} \left( U_{J_+}(t_f, t_0) \rho(t_0) U_{J_-}^\dagger(t_f, t_0) \right), \quad (4.13)$$

where, in the operator formalism, the time evolution in the presence of sources is

$$U_{J_\pm}(t_f, t_0) := T \exp \left[ -i \int_{t_0}^{t_f} dt (H - J_\pm(t)x) \right]. \quad (4.14)$$

Here  $t_f$  is just some arbitrary time. We'll argue around (4.23) that physical predictions are independent of  $t_f$  as long as it is chosen to be later than any insertion of operators. The source  $J_+(t)$  appears on the forward branch, while  $J_-(t)$  on the backward branch. Again, let's use the example of a single quantum mechanical particle in 1 dimension. We will move to quantum field theory later on. For a single coordinate  $x$ , the time evolution is represented by a path integral, this time with a source

$$\langle x_f | U_J(t_f, t_0) | x_{+,0} \rangle = \int_{x_+(t_0)=x_{+,0}}^{x_+(t_f)=x_f} \mathcal{D}x_+ \exp \left[ iS[x_+] + i \int dt J(t)x_+(t) \right], \quad (4.15)$$

$$\langle x_{-,0} | U_J^\dagger(t_f, t_0) | x_f \rangle = \langle x_f | U_J(t_f, t_0) | x_{-,0} \rangle^* \quad (4.16)$$

$$= \int_{x_-(t_0)=x_{-,0}}^{x_-(t_f)=x_f} \mathcal{D}x_- \exp \left[ -iS[x_-] - i \int dt J(t)x_-(t) \right], \quad (4.17)$$

We can think of  $J_\pm$  as just another coupling constant in the theory<sup>21</sup>, except that we allow it to be time-dependent (and spacetime dependent in field theory). Using the condensed notation introduced previously we can write the generating functional as

$$\begin{aligned} Z[J_+, J_-] = \int_{\text{I.C.}} \mathcal{D}x_+ \int_{\text{I.C.}} \mathcal{D}x_- \exp \left[ iS[x_+] - iS[x_-] \right. \\ \left. + i \int dt J_+(t)x_+(t) - i \int dt J_-(t)x_-(t) \right]. \end{aligned} \quad (4.18)$$

The point is that functional differentiation of  $Z[J_+, J_-]$  with respect to  $J_\pm$  generates various correlation functions. To write this down explicitly and avoid confusion, we need a small bit of useful notation. We will use ordinary angle brackets  $\langle \dots \rangle$  for quantum expectation values of operators in Hilbert space, as we have done throughout so far. Moreover, we introduce double angle brackets  $\langle\langle \dots \rangle\rangle$  for averages computed by the Schwinger–Keldysh path integral. Thus, for example,

$$\langle \mathcal{O} \rangle := \text{Tr}(\rho(t) \mathcal{O}) \quad (4.19)$$

is an operator expectation value, whereas

$$\langle\langle F[x_+, x_-] \rangle\rangle := \int \mathcal{D}x_+ \mathcal{D}x_- F[x_+, x_-] e^{iS[x_+] - iS[x_-]} \quad (4.20)$$

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<sup>21</sup>In fact, one can introduce sources for all possible operators. For example, introducing a source for  $x_\pm^2$  leads, eventually, to the so-called 2PI effective action, while, in field theory, coupling to a spacetime metric  $g_{\mu\nu}$  is tantamount to introducing a source for the energy-momentum tensor.

is a path-integral average over the doubled variables. As we discuss in detail in the next subsection, the two averages are related but they should not be confused:  $x_{\pm}$  are classical commuting variables, while  $\mathcal{O} \sim \hat{x}(t)^n$  denotes quantum operators.

Varying  $Z$  with respect to its sources around  $J_{\pm} = 0$  gives

$$\left[ \prod_i \frac{\delta}{J_+(t_i)} \prod_j \frac{\delta}{J_-(t_j)} Z[J_+, J_-] \right]_{J_{\pm}=0} = \left\langle \left[ \prod_i (ix_+(t_i)) \right] \left[ \prod_j (-ix_-(t_j)) \right] \right\rangle. \quad (4.21)$$

We will see in the next section what this path integral average means in the operator language. Before concluding it is worth noticing that setting  $J_+ = J_-$  gives

$$Z[J, J] = \text{Tr } \rho(t_0) = 1, \quad (4.22)$$

provided the density matrix is normalized. This simple identity will later become the basic normalization condition of the Schwinger-Keldysh action. Note that this doubling of the sources and the subsequent doubling of the fields is simply bookkeeping: it allows us to compute path integral averages involving both  $x_+(t)$  and  $x_-(t)$ .

## 4.2 Path ordering, time ordering and the Keldysh basis

Having introduced the Schwinger-Keldysh generating functional in the  $\pm$  basis, we now discuss three related issues. First, we show that the SK path integral is independent of the choice of turning point of the contour under some assumptions. Second, we translate path integral averages into expectation values of quantum operators. Third, we introduce the Keldysh or  $r/a$  basis, in which response and fluctuation are more transparently organized.

**Turn-around point** Before, we imposed time evolution until the time  $t$  at which we inserted the operator  $\hat{x}(t)^n$ , while later we extended the path integral to an arbitrary time  $t_f$  in defining the generating functional  $Z$ . Here we observe that we are indeed allowed to extend the contour to any time  $t_f$  as long as this is later than the time  $t$  of operator insertion. This simply follows from the fact that unitary forward time evolution without an operator insertion is exactly cancelled by unitary backwards evolution,

$$U(t_f, t)U^\dagger(t_f, t) = \mathbb{1}. \quad (4.23)$$

In the case when many operators are inserted at different times, the turn-around point  $t_f$  must be later than the latest time of insertion. These observations allow us to cleanly separate the time of insertion of operators from the turning point of the contour.

Henceforth, in the condensed notation of (4.12), we will simply assume that the upper boundary condition of the path integral is imposed at a sufficiently late time  $t_f$ , the choice of which does not affect the result of the calculation. The resulting contour is depicted in the Figure 2.

**The closed-time contour** The goal in life of the path integral is to compute path-ordered correlation functions. If the path runs straight in time, as it is the case for the path integral we use to compute in-out correlators used in the calculation of amplitudes, then path and time ordering coincide. If the path turns around and comes back in time, as for the SK path integral, then *path ordering* and *time ordering* are distinct. On the forward branch, path ordering coincides with the usual time ordering; on the backward branch, path ordering coincides with anti-time ordering. When one insertion lies on each branch, there is no ordinary time ordering between them: the contour itself decides the ordering.

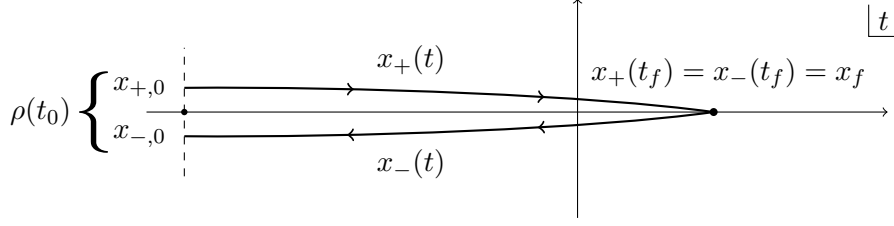


Figure 2: The SK contour on the complex  $t$  plane. The parenthesis after  $\rho(t_0)$  indicates that the boundary conditions  $x_{\pm,0}$  of the path integral should be averaged over the initial density matrix. The forward and backward contours are slightly separated in the imaginary direction for clarity.

To make this distinction clear it is sometimes useful to think of the two branches in the SK path integral as just two parts of a single path known as the *closed-time contour*. This is possible because forward and backward branches are stitched together by the regularity condition  $x_+(t_f) = x_-(t_f)$  at the turning point  $t_f$ . In other words, the contour is closed in time by the trace. Indeed, one could simply rewrite the two-branch path integral as a single path integral

$$\int_{\text{I.C.}} \mathcal{D}x_+ \int_{\text{I.C.}} \mathcal{D}x_- \rightarrow \int_{\text{I.C.}}^{\text{I.C.}} \mathcal{D}X \quad (4.24)$$

where  $X(\lambda)$  is a new integration variable encompassing both  $x_+$  and  $x_-$  and  $\lambda$  is a path variable that is related to time. More concretely, if we conveniently shift time such that the turning point is at  $t_f = 0 \geq t_0$ , and all operator insertions are prior to that, then we have  $t_0 \leq \lambda \leq |t_0|$  and the identification

$$\begin{cases} X(\lambda) = x_+(\lambda) & \text{for } t_0 \leq \lambda \leq 0 \\ X(\lambda) = x_-(-\lambda) & \text{for } 0 \leq \lambda \leq |t_0|. \end{cases} \quad (4.25)$$

Now path ordering on the  $X(\lambda)$  variable is the usual one with insertions at an earlier  $\lambda_1$  coming to the right of an insertion at a later  $\lambda_2 > \lambda_1$ . This simple path ordering in  $\lambda$  leads to a rich set of different time orderings.

Let's see this in an explicit example. For two-point functions one finds

$$\langle\langle x_+(t) x_+(t') \rangle\rangle = \langle T x(t) x(t') \rangle, \quad (4.26)$$

$$\langle\langle x_-(t) x_-(t') \rangle\rangle = \langle \bar{T} x(t) x(t') \rangle, \quad (4.27)$$

$$\langle\langle x_+(t) x_-(t') \rangle\rangle = \langle x(t') x(t) \rangle, \quad (4.28)$$

$$\langle\langle x_-(t) x_+(t') \rangle\rangle = \langle x(t) x(t') \rangle. \quad (4.29)$$

Thus fields on the  $+$  branch are time ordered among themselves, fields on the  $-$  branch are anti-time-ordered among themselves, and fields on different branches give non-time-ordered correlators, often called Wightman correlators. As an example, the graphical representation of (4.26) is shown in Figure 3.

More generally, we can consider the average over any number of operators on the plus and minus branch. This gives the product of a time-ordered block and an anti-time ordered block

$$\langle\langle \prod_{i=1}^n \mathcal{O}_+(t_i) \prod_{j=1}^m \mathcal{O}_-(t_j) \rangle\rangle = \langle \bar{T} \left[ \prod_{j=1}^m \mathcal{O}(t_j) \right] T \left[ \prod_{i=1}^n \mathcal{O}(t_i) \right] \rangle. \quad (4.30)$$

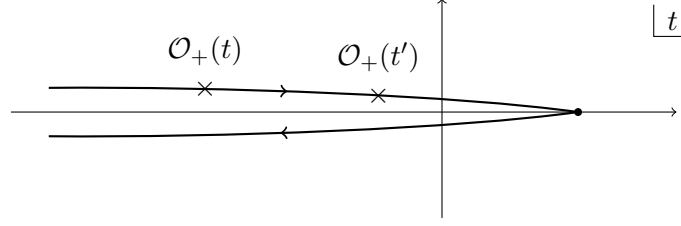


Fig 3: The SK contour featuring the insertion of two operators on the plus branch corresponding to a time-ordered 2-point function. All labels of Figure 2 are left implicit for clarity.

Even more generally, one may be interested in calculating correlators of operators in an arbitrary order [?, ?]. These are sometimes referred to as Out-of-Time-Order Correlators (OTOC) and have attracted recent attention as a diagnostic of quantum chaos. These more general orderings can be computed from a generalization of the SK path integral with more and more branches that alternate backward and forward evolution, but each stitched together in a continuous way to the previous and following branch.

**The retarded and advanced basis** We now introduce the *Keldysh basis*,

$$x_r := \frac{x_+ + x_-}{2}, \quad x_a := x_+ - x_-, \quad (4.31)$$

with inverse relations

$$x_+ = x_r + \frac{x_a}{2}, \quad x_- = x_r - \frac{x_a}{2}. \quad (4.32)$$

The variable  $x_r$  is the mean of the two contour copies and is often called the *retarded* component. The variable  $x_a$  is their difference and is often called the *advanced* component. We will soon discuss how and when they are related to the classical and quantum contributions, but we will avoid using that terminology in these notes. In this basis, the non-equilibrium constraints discussed earlier take a particularly simple form, and the distinction between fluctuation and response becomes much more transparent.

The two-point functions in the  $r/a$  basis are obtained by simple linear combinations of the  $\pm$  correlators. First,

$$\begin{aligned} \langle\langle x_r(t) x_r(t') \rangle\rangle &= \frac{1}{4} \langle\langle (x_+(t) + x_-(t)) (x_+(t') + x_-(t')) \rangle\rangle \\ &= \frac{1}{4} \left[ \langle\langle x_+(t) x_+(t') \rangle\rangle + \langle\langle x_+(t) x_-(t') \rangle\rangle \right. \\ &\quad \left. + \langle\langle x_-(t) x_+(t') \rangle\rangle + \langle\langle x_-(t) x_-(t') \rangle\rangle \right] \\ &= \frac{1}{4} \left[ \langle T \hat{x}(t) \hat{x}(t') \rangle + \langle \hat{x}(t') \hat{x}(t) \rangle \right. \\ &\quad \left. + \langle \hat{x}(t) \hat{x}(t') \rangle + \langle \tilde{T} \hat{x}(t) \hat{x}(t') \rangle \right] \\ &= \frac{1}{4} \left[ \theta(t - t') \langle \hat{x}(t) \hat{x}(t') \rangle + \theta(t' - t) \langle \hat{x}(t') \hat{x}(t) \rangle \right. \\ &\quad \left. + \langle \hat{x}(t') \hat{x}(t) \rangle + \langle \hat{x}(t) \hat{x}(t') \rangle \right. \\ &\quad \left. + \theta(t' - t) \langle \hat{x}(t) \hat{x}(t') \rangle + \theta(t - t') \langle \hat{x}(t') \hat{x}(t) \rangle \right] \\ &= \frac{1}{2} \left[ \langle \hat{x}(t) \hat{x}(t') \rangle + \langle \hat{x}(t') \hat{x}(t) \rangle \right] = \frac{1}{2} \langle \{ \hat{x}(t), \hat{x}(t') \} \rangle. \end{aligned} \quad (4.33)$$

namely the anti-commutator of the fields. This is the symmetrized, or *Keldysh correlator*. Next, we can calculate the other two-point function in a similar fashion. Leaving the intermediate steps as an exercise for the reader, one finds

$$\begin{aligned}\langle\langle x_r(t)x_a(t')\rangle\rangle &= \frac{1}{2}\langle\langle (x_+(t) + x_-(t))(x_+(t') - x_-(t'))\rangle\rangle \\ &= \theta(t - t') \langle[\hat{x}(t), \hat{x}(t')]\rangle.\end{aligned}\quad (4.34)$$

This is the retarded correlator, up to the conventional factor of  $-i$  often included in field theory. Similarly,

$$\begin{aligned}\langle\langle x_a(t)x_r(t')\rangle\rangle &= \frac{1}{2}\langle\langle (x_+(t) - x_-(t))(x_+(t') + x_-(t'))\rangle\rangle \\ &= \theta(t' - t) \langle[\hat{x}(t'), \hat{x}(t)]\rangle,\end{aligned}\quad (4.35)$$

which is the advanced correlator. Notice that the Heaviside theta places the time appearing in the advanced variable always before the time appearing in the retarded variable, hence justifying their names. Finally,

$$\langle\langle x_a(t)x_a(t')\rangle\rangle = \langle\langle [x_+(t) - x_-(t)][x_+(t') - x_-(t')] \rangle\rangle \quad (4.36)$$

$$= \langle T\hat{x}(t)\hat{x}(t') \rangle + \langle \tilde{T}\hat{x}(t)\hat{x}(t') \rangle - \langle \hat{x}(t)\hat{x}(t') \rangle - \langle \hat{x}(t')\hat{x}(t) \rangle = 0. \quad (4.37)$$

Thus the four basic two-point functions in the  $r/a$  basis are

$$G_K(t, t') := \langle\langle x_r(t)x_r(t')\rangle\rangle = \frac{1}{2}\langle\{\hat{x}(t), \hat{x}(t')\}\rangle, \quad (4.38)$$

$$G_R(t, t') := \langle\langle x_r(t)x_a(t')\rangle\rangle = \theta(t - t')\langle[\hat{x}(t), \hat{x}(t')]\rangle, \quad (4.39)$$

$$G_A(t, t') := \langle\langle x_a(t)x_r(t')\rangle\rangle = \theta(t' - t)\langle[\hat{x}(t'), \hat{x}(t)]\rangle, \quad (4.40)$$

$$\langle\langle x_a(t)x_a(t')\rangle\rangle = 0. \quad (4.41)$$

These are the Keldysh, retarded, advanced, and vanishing  $aa$  propagators, respectively. This is one of the main reasons the  $r/a$  basis is so useful: response functions are immediately identified with commutators multiplied by Heaviside functions, while fluctuations are captured by the symmetrized correlator.

To compute path-integral expectation values of the product of a retarded and advanced fields, it is convenient to replace the sources  $J_\pm$  coupled to the two branches by their Keldysh combinations

$$J_r := \frac{J_+ + J_-}{2}, \quad J_a := J_+ - J_-, \quad (4.42)$$

or equivalently

$$J_+ = J_r + \frac{J_a}{2}, \quad J_- = J_r - \frac{J_a}{2}. \quad (4.43)$$

Here  $J_r$  is called the retarded, or average, source, while  $J_a$  is called the advanced, or difference, source. In the Keldysh basis the source term becomes

$$J_+x_+ - J_-x_- = J_ax_r + J_rx_a, \quad (4.44)$$

so that the Schwinger–Keldysh generating functional may be written as

$$Z[J_r, J_a] = \int \mathcal{D}x_r \mathcal{D}x_a \exp\left(iS_{\text{SK}}[x_r, x_a] + i \int dt (J_a x_r + J_r x_a)\right). \quad (4.45)$$

A somewhat confusing consequence of the reasonable convention in (4.42) is that insertions of  $x_r$  are generated by differentiation with respect to  $J_a$ , while insertions of  $x_a$  are generated by differentiation with respect to  $J_r$ .

The vanishing of the  $aa$  two-point function is not an accident. More generally, correlation functions built only from advanced fields vanish:

$$\langle\langle x_a(t_1) \cdots x_a(t_n) \rangle\rangle = 0 \quad \text{for all } n \geq 1. \quad (4.46)$$

This follows directly from the normalization condition of the Schwinger–Keldysh generating functional<sup>22</sup>. In the  $r/a$  basis one has

$$Z[J_r, J_a = 0] = 1$$

for arbitrary  $J_r$ . Since insertions of  $x_a$  are generated by differentiation with respect to  $J_r$ , (4.46) follows immediately.

At higher order, the  $r/a$  basis organizes correlation functions into nested commutators. The most important general family is obtained by taking one  $r$  insertion and any number of  $a$  insertions. These are the higher retarded correlators. For ordered times

$$t_1 > t_2 > \cdots > t_n, \quad (4.48)$$

one finds

$$\langle\langle x_r(t_1) x_a(t_2) \cdots x_a(t_n) \rangle\rangle = \langle [\cdots [[\hat{x}(t_1), \hat{x}(t_2)], \hat{x}(t_3)] \cdots, \hat{x}(t_n)] \rangle, \quad (4.49)$$

while for arbitrary time arguments this is completed by the appropriate product of Heaviside functions enforcing retarded support,

$$\begin{aligned} \langle\langle x_r(t_1) x_a(t_2) \cdots x_a(t_n) \rangle\rangle &= \theta(t_1 - t_2) \theta(t_2 - t_3) \cdots \theta(t_{n-1} - t_n) \\ &\times \langle [\cdots [[\hat{x}(t_1), \hat{x}(t_2)], \hat{x}(t_3)] \cdots, \hat{x}(t_n)] \rangle + \text{permutations}. \end{aligned} \quad (4.50)$$

Thus each additional  $a$  insertion adds one further level of nesting in the commutator. This is the precise sense in which the advanced field is a response field: differentiating with respect to  $J_r$  generates response functions.

To summarize, the  $\pm$  basis makes the contour structure manifest: it directly encodes time ordering on the forward branch, anti-time ordering on the backward branch, and Wightman functions across branches. The  $r/a$  basis, on the other hand, makes response and fluctuation manifest: the  $rr$  correlator gives the symmetrized two-point function, the  $ra$  and  $ar$  correlators give the retarded and advanced propagators, and correlators built only from  $a$  fields vanish identically.

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<sup>22</sup>An alternative derivation uses the relation  $x_a = x_+ - x_-$ . Expanding a product of  $n$  advanced fields gives a signed sum over all  $2^n$  ways of assigning each insertion to the forward or backward branch. The result (4.46) is then equivalent to the identity that this alternating sum of contour-ordered correlators vanishes

$$\langle\langle \prod_{m=1}^n x_a(t_m) \rangle\rangle = \sum_{I \subseteq \{1, \dots, n\}} (-1)^{n-|I|} \left\langle \bar{T} \prod_{j \notin I} \hat{x}(t_j) T \prod_{i \in I} \hat{x}(t_i) \right\rangle = 0. \quad (4.47)$$

### 4.3 Integrating out the environment and the Feynman–Vernon influence functional

We now move from a closed system to an open one, i.e. from Hamiltonian to generically non-Hamiltonian evolution. The conceptual starting point is still a fully Hamiltonian theory, but now the total Hilbert space is split into a subsystem of interest  $S$  and an environment  $E$ ,

$$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E. \quad (4.51)$$

We denote by  $x$  the degrees of freedom of the system and by  $q$  those of the environment. Without loss of generality, the total action is correspondingly decomposed as

$$S_{\text{tot}}[x, q] = S_S[x] + S_E[q] + S_{\text{int}}[x, q]. \quad (4.52)$$

At this stage the full theory is still closed and evolves unitarily according to some Hamiltonian. The Schwinger–Keldysh generating functional  $Z[J_+, J_-]$  for the full system is therefore

$$\begin{aligned} Z = & \int dx_{+,0} dx_{-,0} dq_{+,0} dq_{-,0} dx_f dq_f \rho_{SE}(x_{+,0}, q_{+,0}; x_{-,0}, q_{-,0}; t_0) \\ & \int_{x_+(t_0)=x_{+,0}, q_+(t_0)=q_{+,0}}^{x_+(t_f)=x_f, q_+(t_f)=q_f} \mathcal{D}x_+ \mathcal{D}q_+ \exp \left[ iS_S[x_+] + iS_E[q_+] + iS_{\text{int}}[x_+, q_+] + i \int dt J_+ x_+ \right] \\ & \int_{x_-(t_0)=x_{-,0}, q_-(t_0)=q_{-,0}}^{x_-(t_f)=x_f, q_-(t_f)=q_f} \mathcal{D}x_- \mathcal{D}q_- \exp \left[ -iS_S[x_-] - iS_E[q_-] - iS_{\text{int}}[x_-, q_-] - i \int dt J_- x_- \right]. \end{aligned}$$

As we did previously, to avoid excessively large formulas, we will henceforth omit the upper boundary value of the path integral, leaving it implicit that it should always be continuous between the plus and minus branch. We will also use the shorthand notation  $\int_\rho$  to indicate the fact that one should average over all possible lower boundary conditions of the path integral with weight given by the density matrix  $\rho$ . So we write more compactly

$$\begin{aligned} Z[J_+, J_-] = & \int_{\rho_{SE}(t_0)} \mathcal{D}x_+ \mathcal{D}x_- \mathcal{D}q_+ \mathcal{D}q_- \exp \left( iS_S[x_+] - iS_S[x_-] + iS_E[q_+] - iS_E[q_-] \right. \\ & \left. + iS_{\text{int}}[x_+, q_+] - iS_{\text{int}}[x_-, q_-] + i \int dt J_+ x_+ - i \int dt J_- x_- \right). \end{aligned} \quad (4.53)$$

Here  $\rho_{SE}(t_0)$  is the initial density matrix of the full system. In the simplest and most common setup one assumes that the initial state factorizes,

$$\rho_{SE}(t_0) = \rho_S(t_0) \otimes \rho_E(t_0), \quad (4.54)$$

although the formal manipulations below can be generalized.

The crucial step is now to integrate out the environment variables  $q_\pm$ . This defines the Feynman–Vernon *influence functional*<sup>23</sup> [?]. Explicitly, we define

$$e^{iS_{\text{IF}}[x_+, x_-]} := \int_{\rho_E(t_0)} \mathcal{D}q_+ \mathcal{D}q_- \exp \left( iS_E[q_+] - iS_E[q_-] + iS_{\text{int}}[x_+, q_+] - iS_{\text{int}}[x_-, q_-] \right). \quad (4.55)$$

This object depends only on the system’s histories  $x_+$  and  $x_-$ , but it contains the full effect of the environment on the subsystem. As we will see in the next subsection, the influence functional can be written as an overlap amplitude between two environment states evolved in

<sup>23</sup>Sometimes one uses the word Feynman–Vernon influence *functional* to denote  $e^{iS_{\text{IF}}}$ , where  $S_{\text{IF}}$  is called the influence *action*. I do not make this distinction in these notes.

the presence of two different histories of the system. Once this definition is made, the generating functional becomes

$$Z[J_+, J_-] = \int \mathcal{D}x_+ \mathcal{D}x_- \exp\left(iS_S[x_+] - iS_S[x_-] + iS_{\text{IF}}[x_+, x_-] + i \int dt J_+ x_+ - i \int dt J_- x_- \right) \times \rho_S(x_{+,0}, x_{-,0}; t_0). \quad (4.56)$$

It is therefore natural to define the open-system effective action

$$S_{\text{eff}}[x_+, x_-] := S_S[x_+] - S_S[x_-] + S_{\text{IF}}[x_+, x_-]. \quad (4.57)$$

This is the central result. For a closed system the Schwinger–Keldysh action is simply the difference  $S[x_+] - S[x_-]$ . For an open system, integrating out the environment generates the additional term  $S_{\text{IF}}[x_+, x_-]$ , which in general couples the two branches of the contour. These branch-mixing terms have no counterpart in a purely Hamiltonian closed-system evolution, and they are precisely what encode the effects of the environment on the reduced dynamics.

The influence functional has a clear physical meaning. It summarizes, in a single functional of  $x_+$  and  $x_-$ , all effects mediated by the environment. Depending on the problem, these include shifts of the system parameters, dissipation, stochastic fluctuations, and memory effects. In general,  $S_{\text{IF}}$  is nonlocal in time: the environment remembers part of the past history of the system, and the reduced dynamics is therefore non-Markovian. In this sense, the Feynman–Vernon formalism provides an exact real-time description of open quantum dynamics before any Markovian approximation is made. In Section 4.5 we will explicitly compute the influence functional in a simple setup.

It is often convenient to pass immediately to the Keldysh basis,

$$x_r := \frac{x_+ + x_-}{2}, \quad x_a := x_+ - x_-. \quad (4.58)$$

Then the influence functional becomes  $S_{\text{IF}}[x_r, x_a]$ , and its structure can be interpreted more directly, as we will see shortly.

#### 4.4 Constraints on the Schwinger–Keldysh effective action

Before studying explicit examples, it is important to understand the general constraints that any Schwinger–Keldysh effective action must satisfy in order to describe the evolution of a density matrix. These constraints follow directly from the defining properties of a density matrix,

$$\text{Tr } \rho = 1, \quad \rho^\dagger = \rho, \quad \rho \geq 0. \quad (4.59)$$

They are the path-integral counterparts of trace preservation, Hermiticity preservation, and positivity. We will postpone the stronger requirement of complete positivity to the next subsection. In this subsection we derive the corresponding conditions on the effective action, namely

$$S_{\text{eff}}[x_+, x_+] = 0, \quad S_{\text{eff}}[x_+, x_-] = -S_{\text{eff}}^*[x_-, x_+], \quad \text{Im } S_{\text{eff}}[x_+, x_-] \geq 0. \quad (4.60)$$

These constraints are trivially satisfied for Hamiltonian evolution where  $S_{\text{eff}} = S[x_+] - S[x_-]$  with  $S$  a real functional of its variables by unitarity. The non-trivial fact is that these conditions are satisfied by the influence functional  $S_{\text{IF}}$ , on which we focus henceforth.

The logic of the proof is the following. One starts from a unitary evolution of the full system, made of the subsystem of interest  $x$  and an environment  $q$ , and then rewrites the influence functional as an overlap of two environment states evolved with  $x_+$  and  $x_-$  treated as external sources. In this form, the standard Schwinger–Keldysh constraints follow almost immediately. This derivation is a straightforward adaptation of Appendices A of Ref. [?, ?].

Assuming an initially factorized density matrix as in (4.54), the influence functional is

$$e^{iS_{\text{IF}}[x_+, x_-]} = \int dq dq_i dq'_i \int_{q_i}^q \mathcal{D}q_+ \int_{q'_i}^q \mathcal{D}q_- e^{iS_{\text{tot}}[x_+, q_+] - iS_{\text{tot}}[x_-, q_-]} \rho_q^{(0)}(q_i, q'_i). \quad (4.61)$$

The crucial observation is now to regard  $x_+$  and  $x_-$  as external sources for the environment. Denoting by  $U_q(t, t_0; \{x_+\})$  the environment evolution operator in the presence of the prescribed background history  $x_+$ , and similarly for  $x_-$ , one may rewrite the two environment path integrals as matrix elements of sourced evolution operators,

$$\int_{q_i}^q \mathcal{D}q_+ e^{iS_{\text{tot}}[x_+, q_+]} = \langle q | U_q(t, t_0; \{x_+\}) | q_i \rangle, \quad (4.62)$$

$$\int_{q'_i}^q \mathcal{D}q_- e^{-iS_{\text{tot}}[x_-, q_-]} = \left[ \langle q | U_q(t, t_0; \{x_-\}) | q'_i \rangle \right]^\dagger. \quad (4.63)$$

To keep the discussion as simple as possible, in the following we will assume that the environment is initially in a pure state

$$\rho_q^{(0)}(q_i, q'_i) = |\Omega_q\rangle \langle \Omega_q|.$$

Substituting the above expressions into the influence functional and inserting the identity in the environment Hilbert space then gives

$$e^{iS_{\text{IF}}[x_+, x_-]} = \langle \Omega_q | U_q^\dagger(t, t_0; \{x_-\}) U_q(t, t_0; \{x_+\}) | \Omega_q \rangle. \quad (4.64)$$

Equivalently, defining the sourced-evolved environment states

$$|\Omega_q^{\{x_+\}}(t)\rangle := U_q(t, t_0; \{x_+\}) | \Omega_q \rangle, \quad \langle \Omega_q^{\{x_-\}}(t) | := \langle \Omega_q | U_q^\dagger(t, t_0; \{x_-\}), \quad (4.65)$$

one finds the compact expression

$$e^{iS_{\text{IF}}[x_+, x_-]} = \langle \Omega_q^{\{x_-\}}(t) | \Omega_q^{\{x_+\}}(t) \rangle. \quad (4.66)$$

Thus the influence functional is an overlap amplitude between two environment states evolved with two different histories of the system. This is the key result from which the constraints follow.

The first consequence is positivity of the imaginary part. Since the states in (4.66) are normalized, the Cauchy–Schwarz inequality implies

$$|e^{iS_{\text{IF}}[x_+, x_-]}| = |\langle \Omega_q^{\{x_-\}}(t) | \Omega_q^{\{x_+\}}(t) \rangle| \leq 1. \quad (4.67)$$

Hence

$$e^{-\text{Im } S_{\text{IF}}[x_+, x_-]} \leq 1 \quad \Rightarrow \quad \text{Im } S_{\text{IF}}[x_+, x_-] \geq 0. \quad (4.68)$$

This is the path-integral form of the positivity constraint.

The second consequence is Hermiticity. Taking the complex conjugate of (4.66), one obtains

$$e^{-iS_{\text{IF}}^*[x_+, x_-]} = \langle \Omega_q^{\{x_+\}}(t) | \Omega_q^{\{x_-\}}(t) \rangle = e^{iS_{\text{IF}}[x_-, x_+]}, \quad (4.69)$$

and therefore

$$S_{\text{IF}}[x_+, x_-] = -S_{\text{IF}}^*[x_-, x_+]. \quad (4.70)$$

This is the Schwinger–Keldysh reality condition implied by Hermiticity of the reduced density matrix.

The third consequence is the normalization or trace-preservation condition. If the two histories coincide,  $x_+ = x_-$ , then the two sourced evolutions of the environment are identical, and (4.66) reduces to the norm of a single state,

$$e^{iS_{\text{IF}}[x_+, x_+]} = \langle \Omega_q^{\{x_+\}}(t) | \Omega_q^{\{x_+\}}(t) \rangle = 1. \quad (4.71)$$

Hence

$$S_{\text{IF}}[x_+, x_+] = 0. \quad (4.72)$$

This is the precise derivation of the statement that the Schwinger–Keldysh effective action vanishes when the two branches are identified.

Passing to the Keldysh basis, and combining  $S_{\text{IF}}$  with the Hamiltonian part of the evolution to form the full SK action  $S_{\text{eff}}$ , these constraints become

$$S_{\text{eff}}[x_r, x_a = 0] = 0, \quad (4.73)$$

$$S_{\text{eff}}[x_r, x_a] = -S_{\text{eff}}^*[x_r, -x_a], \quad (4.74)$$

$$\text{Im } S_{\text{eff}}[x_r, x_a] \geq 0. \quad (4.75)$$

They hold non-perturbatively and are inherited directly from the underlying Hamiltonian evolution of the full theory. They are the basic unitarity constraints on any Schwinger–Keldysh effective action, and they are sometimes referred to as non-equilibrium constraints. A longer but perhaps more accurate name would be “constraints from the unitarity of the full system”, highlighting the fact that if these constraints are violated, then the open system dynamics cannot arise from a unitary full system by integrating out an environment, under the assumption (4.54).

#### 4.5 Gaussian environment linearly coupled to the system

We now discuss an explicit model in which the influence functional can be computed exactly. The purpose of this example is to show in a concrete setting how integrating out the environment generates a non-trivial Schwinger–Keldysh effective action for the system, and how the characteristic branch-mixing structures arise. We consider a system degree of freedom  $x$  linearly coupled to an environment described by a Gaussian field  $q$ .

Let the total action be

$$S_{\text{tot}}[x, q] = S_S[x] + S_E[q] + S_{\text{int}}[x, q], \quad (4.76)$$

with interaction

$$S_{\text{int}}[x, q] = \int dt g x(t) q(t). \quad (4.77)$$

For definiteness, we may think of  $q$  either as a single harmonic oscillator or, more generally, as a collection of harmonic oscillators. The crucial assumption is that the environment action  $S_E[q]$  is quadratic in  $q$ , so that the path integral over  $q$  is Gaussian and can be performed exactly.

In the Schwinger–Keldysh formalism, the interaction term becomes

$$S_{\text{int}}[x_+, q_+] - S_{\text{int}}[x_-, q_-] = g \int dt (x_+(t)q_+(t) - x_-(t)q_-(t)). \quad (4.78)$$

The influence functional is therefore

$$e^{iS_{\text{IF}}[x_+, x_-]} = \int_{\rho_E(t_0)} \mathcal{D}q_+ \mathcal{D}q_- \exp \left\{ iS_E[q_+] - iS_E[q_-] + ig \int dt (x_+(t)q_+(t) - x_-(t)q_-(t)) \right\}. \quad (4.79)$$

Since the environment is Gaussian, this is a Gaussian functional integral with linear sources. It can therefore be evaluated exactly by completing the square. The result is quadratic in the system histories  $x_+$  and  $x_-$ :

$$S_{\text{IF}}[x_+, x_-] = i\frac{g^2}{2} \int dt dt' \left[ x_+(t) G_{++}(t, t') x_+(t') - x_+(t) G_{+-}(t, t') x_-(t') - x_-(t) G_{-+}(t, t') x_+(t') + x_-(t) G_{--}(t, t') x_-(t') \right], \quad (4.80)$$

where  $G_{ab}(t, t')$ , with  $a, b \in \{+, -\}$ , are the Schwinger–Keldysh two-point functions of the environment operator  $q$

$$G_{++}(t, t') := \langle\langle q_+(t)q_+(t') \rangle\rangle = \langle T \hat{q}(t)\hat{q}(t') \rangle, \quad (4.81)$$

$$G_{+-}(t, t') := \langle\langle q_+(t)q_-(t') \rangle\rangle = \langle \hat{q}(t')\hat{q}(t) \rangle, \quad (4.82)$$

$$G_{-+}(t, t') := \langle\langle q_-(t)q_+(t') \rangle\rangle = \langle \hat{q}(t)\hat{q}(t') \rangle, \quad (4.83)$$

$$G_{--}(t, t') := \langle\langle q_-(t)q_-(t') \rangle\rangle = \langle \bar{T} \hat{q}(t)\hat{q}(t') \rangle. \quad (4.84)$$

In other words, the exact influence functional is determined entirely by the environment two-point correlators. This is the simplest explicit illustration of the general fact that a Gaussian environment linearly coupled to the system produces a quadratic influence functional.

It is now useful to pass to the Keldysh basis. After some algebra, the influence functional takes the standard form

$$S_{\text{IF}}[x_r, x_a] = -g^2 \int dt dt' x_a(t) D_R(t, t') x_r(t') + \frac{ig^2}{2} \int dt dt' x_a(t) N(t, t') x_a(t'). \quad (4.85)$$

Before expressing  $D_R$  and  $N$  in terms of correlators of  $q$ , we can already note a few properties. First, any antisymmetric part of  $N(t, t')$  does not contribute to (4.85). Second, we may decompose

$$D_R(t, t') = D_R^{(S)}(t, t') + D_R^{(A)}(t, t'), \quad (4.86)$$

where

$$D_R^{(S)}(t, t') = D_R^{(S)}(t', t), \quad D_R^{(A)}(t, t') = -D_R^{(A)}(t', t). \quad (4.87)$$

Using  $x_a = x_+ - x_-$  and  $x_r = (x_+ + x_-)/2$ , the contribution of the symmetric part becomes

$$\int dt dt' x_a(t) D_R^{(S)}(t, t') x_r(t') = \frac{1}{2} \int dt dt' \left[ x_+(t) D_R^{(S)}(t, t') x_+(t') - x_-(t) D_R^{(S)}(t, t') x_-(t') \right].$$

Here the mixed  $+/-$  terms cancel because  $D_R^{(S)}$  is symmetric. This contribution is therefore of the form

$$S_{\text{cons}}[x_+] - S_{\text{cons}}[x_-], \quad (4.88)$$

and can be absorbed into a redefinition of the conservative system action.

By contrast, the diagonal terms involving  $D_R^{(A)}$  vanish, while the mixed terms add:

$$-g^2 \int dt dt' x_a(t) D_R^{(A)}(t, t') x_r(t') = -g^2 \int dt dt' x_+(t) D_R^{(A)}(t, t') x_-(t'). \quad (4.89)$$

This term couples the two branches of the Schwinger–Keldysh contour and cannot be written as the difference of two single-branch actions. It therefore represents the genuinely dissipative part of the influence functional.

Equation (4.80) to (4.85) one finds

$$D_R(t, t') := -i \theta(t - t') \langle [q(t), q(t')] \rangle, \quad (4.90)$$

which is the retarded kernel of the environment. It is real  $D_R(t, t')^* = D_R(t, t')$ . Also

$$N(t, t') := \frac{1}{2} \langle \{q(t), q(t')\} \rangle \quad (4.91)$$

which is its Keldysh correlator. It is real and symmetric,

$$N(t, t') = N(t', t) = N^*(t, t').$$

Up to our conventions and the overall factor of  $g^2$ , these are the two basic kernels that appear in the effective action.

Equation (4.85) is worth pausing over. The exact Gaussian influence functional contains two distinct structures. First, there is a term mixing  $x_a$  and  $x_r$ , weighted by the retarded kernel  $D_R$ . Second, there is a term quadratic in  $x_a$ , weighted by the symmetric kernel  $N$ . Their interpretation in terms of drift and noise will be explained soon.

A key feature of (4.85) is its *nonlocality* in time. Even though the microscopic theory may be local, integrating out the environment generates memory kernels  $D_R(t, t')$  and  $N(t, t')$ , so that the reduced dynamics at time  $t$  depends on the past history of the system. In general, therefore, the exact open-system effective action is non-Markovian. The local-in-time Schwinger–Keldysh actions discussed in the previous subsection arise only after an additional approximation, for example when the environment correlation time is much shorter than the characteristic time scale of the system.

In a suitable local limit, the kernels may be approximated by local distributions. If moreover the environment state is stationary, so that these kernels depend only on the time difference  $t - t'$ , this immediately constrains their local derivative expansion. The Keldysh correlator admits only even powers of derivatives<sup>24</sup>,

$$N(t, t') \sim n_0 \delta(t - t') + n_2 \partial_t^2 \delta(t - t') + \dots, \quad (4.92)$$

The local derivative expansion of the retarded kernel takes the form

$$D_R(t, t') \sim c_0 \delta(t - t') + c_1 \partial_t \delta(t - t') + c_2 \partial_t^2 \delta(t - t') + c_3 \partial_t^3 \delta(t - t') + \dots. \quad (4.93)$$

Under the exchange  $t \leftrightarrow t'$ , the terms containing an even number of derivatives are symmetric, whereas those containing an odd number of derivatives are antisymmetric. As shown above, the symmetric terms describe conservative local corrections, such as shifts of the potential and of the kinetic terms, and may therefore be absorbed into a redefinition of the conservative part of

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<sup>24</sup>In principle, negative powers of the derivative may also appear.

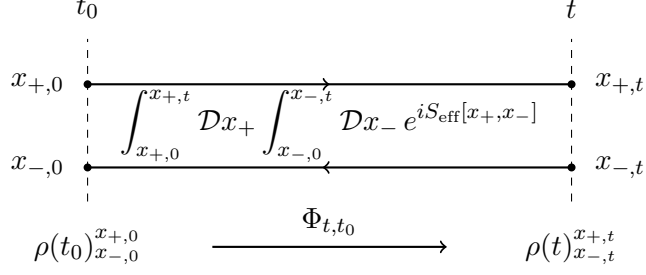


Fig 4: Cutting open the closed-time path turns the SK path integral into a dynamical map that evolves the density matrix,  $\Phi_{t,t_0} : \rho(t_0) \mapsto \rho(t)$ . Note that, for a generic open system, the time evolution on the two branches is not independent but may be mixed by the influence functional. In this precise sense, the two branches of the SK path integral evolve the two indices of the density matrix.

the effective action. After doing so, the genuinely dissipative part of the retarded kernel admits an expansion involving only odd derivatives,

$$D_R^{(A)}(t, t') \sim c_1 \partial_t \delta(t - t') + c_3 \partial_t^3 \delta(t - t') + \dots \quad (4.94)$$

This is why, in a local approximation, the dissipative  $x_a x_r$  sector naturally begins with a single time derivative and leads to a dissipative term such as  $x_a \dot{x}_r$ , while the  $x_a x_a$  sector naturally begins with a zeroth-order term and leads to a local fluctuation term proportional to  $i x_a^2$ .

Finally, let us stress that in general the kernels  $D_R$  and  $N$  are independent. Only if the environment is in thermal equilibrium are they related by a fluctuation–dissipation relation, or equivalently by KMS conditions. Since our eventual interest includes cosmological applications far from equilibrium, we do not impose such relations here.

## 5 The Schwinger Keldysh dynamical map

### 5.1 Cutting open the contour and the SK dynamical map

So far we have mostly used the Schwinger–Keldysh path integral on the closed-time contour, where the two branches are sewn together at the final time and one computes traces of the density matrix times operators, i.e. expectation values. To discuss complete positivity (CP), CP divisibility and the relation to Lindblad evolution, it is useful to take one step back. First, we will introduce the Schwinger–Keldysh path integral with a cut-open contour as a dynamical map that evolves the reduced density matrix. Second, we will prove that if the SK action is local in time, then this map obeys the composition law. Finally, we will extract from the SK action the Liouvillian generator  $\mathcal{L}_t$ , which is generically explicitly time dependent, and show that the SK path integral obeys CP divisibility if and only if this generator is of the GKSL form.

**The SK dynamical map** The initial and final reduced density matrices at times  $t_0$  and  $t_f$  can be related as in (1.199), where we used the Kraus representation. Using the notation of this section this relation becomes

$$\rho_S(x_f^+, x_f^-; t_f) = \int dx_i^+ dx_i^- \mathcal{K}(t_f, t_0; x_f^+, x_f^-; x_i^+, x_i^-) \rho_S(x_i^+, x_i^-; t_0), \quad (5.1)$$

for some kernel  $\mathcal{K}$ . Since the Kraus operators are matrix elements of the time evolution operator, see (1.198), they can be represented by a path integral. Then, the kernel above can be written as a path integral on a cut-open contour

$$\mathcal{K}(t_f, t_0; x_f^+, x_f^-; x_i^+, x_i^-) := \int_{x_i^+}^{x_f^+} \mathcal{D}x_+ \int_{x_i^-}^{x_f^-} \mathcal{D}x_- e^{iS_{\text{eff}}[x_+, x_-]}. \quad (5.2)$$

Here the notation indicates that the  $+$  and  $-$  histories begin at  $x_i^\pm$  and end at  $x_f^\pm$ , with no sewing imposed at the final time. In this sense, one has “cut open” the closed-time contour. We still refer to this as a “Schwinger-Keldysh path integral”, except that it takes place over a different, in this case disjoint, contour.

Equation (5.1) makes it clear that the Schwinger-Keldysh path integral does not merely compute expectation values, it may also be used to define a dynamical map on density matrices. Indeed, writing

$$\rho_S(t_f) =: \Phi_{t_f, t_0}[\rho_S(t_0)], \quad (5.3)$$

the kernel  $\mathcal{K}$  is simply the field-basis representation of the map  $\Phi_{t_f, t_0}$ , which we will call the *Schwinger-Keldysh (SK) dynamical map*. This is depicted in Figure 4.

**The composition law** We recall from previous sections that in the presence of explicit time-dependence, the semi-group property does not apply. Instead now the dynamical map has two time arguments and one should impose the composition law in (2.141), which we prove here follows directly if the effective action is local in time. Suppose that

$$S_{\text{eff}}[x_+, x_-] = \int_{t_0}^{t_f} dt L(x_+(t), x_-(t), \dot{x}_+(t), \dot{x}_-(t), \dots; t), \quad (5.4)$$

for some function  $L$ . Then, for any intermediate time  $t_1$  with  $t_f > t_1 > t_0$ , the action splits additively,

$$S_{\text{eff}}^{[t_0, t_f]}[x_+, x_-] = S_{\text{eff}}^{[t_0, t_1]}[x_+, x_-] + S_{\text{eff}}^{[t_1, t_f]}[x_+, x_-]. \quad (5.5)$$

In the path integral, one may therefore insert a resolution of the identity at time  $t_1$ , integrate over the intermediate boundary data  $\tilde{x}^\pm := x_\pm(t_1)$ , and factorize the kernel into two consecutive pieces. One obtains

$$\begin{aligned} \mathcal{K}(t_f, t_0; x_f^+, x_f^-; x_i^+, x_i^-) &= \int d\tilde{x}^+ d\tilde{x}^- \mathcal{K}(t_f, t_1; x_f^+, x_f^-; \tilde{x}^+, \tilde{x}^-) \\ &\quad \times \mathcal{K}(t_1, t_0; \tilde{x}^+, \tilde{x}^-; x_i^+, x_i^-). \end{aligned} \quad (5.6)$$

Hence, at the operator level, one finds

$$\Phi_{t_f, t_0} = \Phi_{t_f, t_1} \circ \Phi_{t_1, t_0}. \quad (5.7)$$

Thus locality in time of the Schwinger-Keldysh effective action is a sufficient condition for the composition property of the dynamical map.

Notice that this is weaker than the semigroup property. Only in the further special case in which the local effective action is time-translation invariant does one recover the semigroup property discussed in the previous section.

**CP divisibility** Once locality in time has been assumed, it is natural to extract from the short-time kernel a time-local generator, as we did in the proof of the GKSL theorem. Let  $\delta t$  be infinitesimal. Then the kernel over the interval  $[t, t + \delta t]$  takes the form

$$\begin{aligned} \mathcal{K}(t + \delta t, t; x_f^+, x_f^-; x_i^+, x_i^-) &= \delta(x_f^+ - x_i^+) \delta(x_f^- - x_i^-) \\ &+ \delta t \mathcal{L}_t(x_f^+, x_f^-; x_i^+, x_i^-) + \mathcal{O}(\delta t^2), \end{aligned} \quad (5.8)$$

where  $\mathcal{L}_t$  is the kernel representation of a time-local generator, which we may call the time-dependent Liouvillian. Equivalently, assuming differentiability of the map, one defines

$$\partial_{t_f} \Phi_{t_f, t_0} = \mathcal{L}_{t_f} \circ \Phi_{t_f, t_0}. \quad (5.9)$$

As we have seen, the solution is the time-ordered exponential of the superoperator  $\mathcal{L}$ :

$$\Phi_{t_f, t_0} = T \exp \left( \int_{t_0}^{t_f} ds \mathcal{L}_s \right). \quad (5.10)$$

To establish CP divisibility, one must then check whether this generator is of GKSL form for every time  $t$ . Under the standard assumptions of differentiability and invertibility of the map, CP divisibility is equivalent to the statement that

$$\mathcal{L}_t[\rho] = -i[H(t), \rho] + \sum_{\alpha, \beta} c_{\alpha\beta}(t) \left( L_\alpha(t) \rho L_\beta^\dagger(t) - \frac{1}{2} \{L_\beta^\dagger(t) L_\alpha(t), \rho\} \right), \quad (5.11)$$

with the Kossakowski matrix  $c_{\alpha\beta}(t)$  positive semidefinite for every  $t$ , for some jump operators  $L_\alpha$ . We will give an explicit example of this procedure for a noisy and dissipative harmonic oscillator in a later section.

## 5.2 Example: the damped harmonic oscillator

To make the connection between the Schwinger–Keldysh action and the time-local generator more explicit, let us consider a single quantum-mechanical degree of freedom  $x$ . Since the GKSL theorem is naturally formulated in operator phase space, it is convenient to begin directly with a first-order action for the canonical variables  $x$  and  $p$ . We set the mass equal to one throughout, since it plays no essential role in the discussion. We restrict to a local-in-time Gaussian theory with at most one time derivative acting on each variable. A convenient representative is<sup>25</sup>

$$\begin{aligned} S_{\text{SK}}[x_r, x_a, p_r, p_a] &= \int dt \left[ p_a \dot{x}_r + p_r \dot{x}_a - p_r p_a - \Omega^2(t) x_r x_a - \kappa(t) x_a p_r \right. \\ &\quad \left. + \frac{i}{2} N_x(t) x_a^2 + \frac{i}{2} N_p(t) p_a^2 + i N_{xp}(t) x_a p_a \right]. \end{aligned} \quad (5.12)$$

Here  $\Omega(t)$  is the instantaneous oscillator frequency,  $\kappa(t)$  is a dissipative coefficient, and  $N_x(t)$ ,  $N_p(t)$ , and  $N_{xp}(t)$  are the three independent entries of the symmetric phase-space noise matrix. The terms proportional to  $N_x$  and  $N_p$  describe noise in the two canonical directions, while  $N_{xp}$  describes their cross-correlation.

<sup>25</sup>One may integrate out the momentum variables and obtain an equivalent second-order action for  $x_r$  and  $x_a$ . In particular, integration over  $p_r$  imposes  $p_a = \dot{x}_a - \kappa x_a$ . The phase-space noise terms then generate terms proportional to  $\dot{x}_a^2$ ,  $\dot{x}_a^2$ , and  $x_a \dot{x}_a$ . For constant coefficients, the last of these is a total derivative and can be traded for a boundary term. Consequently, in the second-order description the information is distributed between bulk coefficients and boundary or contact terms. The complete-positivity constraints are therefore most transparent in the first-order formulation, which we use throughout the remainder of this subsection.

Over an infinitesimal interval  $[t, t + \delta t]$ , the kernel takes the form

$$\mathcal{K}(t + \delta t, t) = \mathbb{1} + \delta t \mathcal{L}_t + \mathcal{O}(\delta t^2), \quad (5.13)$$

where  $\mathcal{L}_t$  is the time-local generator. We compare this with the expansion of the Schwinger–Keldysh kernel,

$$e^{iS_{\text{eff}}^{[t, t+\delta t]}} = 1 + iS_{\text{eff}}^{[t, t+\delta t]} + \mathcal{O}(\delta t^2). \quad (5.14)$$

The term linear in  $S_{\text{eff}}^{[t, t+\delta t]}$  determines the Liouvillian  $\mathcal{L}_t$ . The resulting expressions may be translated into operator language using the Schwinger–Keldysh dictionary

$$x_a = x_+ - x_- \longleftrightarrow [x, \rho], \quad (5.15)$$

$$p_a = p_+ - p_- \longleftrightarrow [p, \rho], \quad (5.16)$$

$$x_r = \frac{x_+ + x_-}{2} \longleftrightarrow \frac{1}{2}\{x, \rho\}, \quad (5.17)$$

$$p_r = \frac{p_+ + p_-}{2} \longleftrightarrow \frac{1}{2}\{p, \rho\}. \quad (5.18)$$

Thus one factor of an advanced variable translates into a commutator, while one factor of a retarded variable translates into an anticommutator. When dealing with products of advanced and retarded variables, one needs to specify an operator ordering to translate path integral variables into an operator statement. For us, this is necessary only for the term  $x_a p_r$  discussed below, in which case we adopt Weyl ordering and use the above dictionary for the symmetrised product.

In particular, the imaginary terms quadratic in the advanced fields give

$$\frac{i}{2}N_x(t) x_a^2 \longrightarrow -\frac{N_x(t)}{2}[x, [x, \rho]], \quad (5.19)$$

$$\frac{i}{2}N_p(t) p_a^2 \longrightarrow -\frac{N_p(t)}{2}[p, [p, \rho]], \quad (5.20)$$

$$iN_{xp}(t) x_a p_a \longrightarrow -\frac{N_{xp}(t)}{2}[x, [p, \rho]] = -\frac{N_{xp}(t)}{2}[p, [x, \rho]] \quad (5.21)$$

where we included the correct factor of  $i$  in going from the action to the Liouvillian and we massaged the last expression using Jacobi identities and  $[x, p] = i$ . Note that no operator ordering specification is needed in any of these three expressions. The mixed term is represented by the trace-preserving symbol<sup>26</sup>

$$-\kappa(t) x_a p_r \longrightarrow -\frac{i\kappa(t)}{2}[x, \{p, \rho\}]. \quad (5.22)$$

It is convenient to decompose this term as

$$-\frac{i\kappa(t)}{2}[x, \{p, \rho\}] = -i \left[ \frac{\kappa(t)}{4}\{x, p\}, \rho \right] - \frac{i\kappa(t)}{2}(x\rho p - p\rho x) + \frac{\kappa(t)}{2}\rho. \quad (5.23)$$

The master equation can therefore be written as

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<sup>26</sup>This relation may be taken as the definition of the operator ordering for the symbol associated with  $x_a p_r$ . Equivalently, it follows by imposing Weyl ordering on  $x_a$  and  $p_r$  and including a contact term required by trace preservation.

$$\begin{aligned} \frac{d\rho}{dt} = & -i[H(t), \rho] - \frac{i\kappa(t)}{2}(x\rho p - p\rho x) + \frac{\kappa(t)}{2}\rho \\ & - \frac{N_x(t)}{2}[x, [x, \rho]] - \frac{N_p(t)}{2}[p, [p, \rho]] - \frac{N_{xp}(t)}{2}([x, [p, \rho]] + [p, [x, \rho]]), \end{aligned} \quad (5.24)$$

where

$$H(t) = \frac{p^2}{2} + \frac{1}{2}\Omega^2(t)x^2 + \frac{\kappa(t)}{4}\{x, p\}. \quad (5.25)$$

We may now compare (5.24) with the general Gaussian GKSL generator obtained in (3.144), allowing the Hamiltonian and all coefficients to depend on time. In the Hermitian operator basis

$$F_1 = p, \quad F_2 = x, \quad (5.26)$$

the dissipative part of (5.24) is generated by the Kossakowski matrix

$$C(t) = \begin{pmatrix} N_p(t) & N_{xp}(t) + \frac{i}{2}\kappa(t) \\ N_{xp}(t) - \frac{i}{2}\kappa(t) & N_x(t) \end{pmatrix}. \quad (5.27)$$

The real off-diagonal part produces the mixed diffusion term proportional to  $N_{xp}$ , while the imaginary antisymmetric part produces the non-Hamiltonian part of the friction term,

$$-\frac{i\kappa(t)}{2}(x\rho p - p\rho x) + \frac{\kappa(t)}{2}\rho. \quad (5.28)$$

Together with the contribution  $\kappa(t)\{x, p\}/4$  included in  $H(t)$ , it reconstructs  $-i\kappa(t)[x, \{p, \rho\}]/2$ . Complete positivity requires

$$C(t) \geq 0 \quad (5.29)$$

at every time. For a  $2 \times 2$  Hermitian matrix, this is equivalent to

$$N_x(t) \geq 0, \quad N_p(t) \geq 0, \quad N_x(t)N_p(t) - N_{xp}(t)^2 \geq \frac{\kappa(t)^2}{4}. \quad (5.30)$$

These inequalities are the Gaussian open-system manifestation of complete positivity and, when they hold instantaneously for a time-local evolution, of CP divisibility. They show explicitly that a non-zero friction coefficient requires a minimum amount of accompanying noise: friction cannot arise in a completely positive Markovian evolution without sufficiently large fluctuations. It is useful to contrast this result with the weaker constraint obtained directly from Schwinger–Keldysh unitarity. The condition

$$\text{Im } S_{\text{eff}} \geq 0 \quad (5.31)$$

requires the real symmetric noise matrix

$$N(t) = \begin{pmatrix} N_p(t) & N_{xp}(t) \\ N_{xp}(t) & N_x(t) \end{pmatrix} \quad (5.32)$$

to be positive semidefinite, and hence implies only

$$N_x(t) \geq 0, \quad N_p(t) \geq 0, \quad N_x(t)N_p(t) - N_{xp}(t)^2 \geq 0. \quad (5.33)$$

Complete positivity of the time-local generator is stronger because it constrains the noise relative to the dissipative coefficient  $\kappa(t)$ , leading to the lower bound in (5.30). In the second-order formulation the same constraint is present, but it is distributed among bulk terms, endpoint contributions, and contact terms generated by integrating out the momentum. This is why its phase-space origin is most clearly visible in the first-order action.

### 5.3 Jump operators and positivity-improved influence functionals

In realistic applications, the environmental degrees of freedom are usually too complicated to be integrated out exactly as we did in Section 4.5. One must therefore adopt the Effective Field Theory (EFT) philosophy and model the SK functional using the degrees of freedom of the system, the symmetries of the problem, and an appropriate power counting. In this subsection we explain how the jump-operator structure familiar from the Lindblad equation provides a systematic parametrization of local-in-time influence functionals, which can be used as an alternative to a direct expansion of the full SK functional. Besides making complete positivity constraints transparent, this organization can preserve exact positivity under truncation and provides a physically admissible starting point for late-time resummation.

**From the GKSL equation to the influence functional.** Consider a local Markovian master equation of the form

$$\dot{\rho} = -i[H, \rho] + C_{\alpha\beta} \left( L_{\beta}\rho L_{\alpha}^{\dagger} - \frac{1}{2} \{ L_{\alpha}^{\dagger} L_{\beta}, \rho \} \right), \quad (5.34)$$

where repeated indices are summed and the Kossakowski matrix satisfies

$$C_{\alpha\beta} = C_{\beta\alpha}^*, \quad C \geq 0. \quad (5.35)$$

For simplicity, suppose that the jump operators can be rewritten in terms of local-in-time functions  $L_{\alpha}(x)$  of the path integral variables, as it was the case in the toy model in Section 5.2. Denoting

$$L_{\alpha,+} := L_{\alpha}(x_+), \quad L_{\alpha,-} := L_{\alpha}(x_-), \quad (5.36)$$

the dissipative part of the infinitesimal evolution and using the dictionary  $iS_{\text{IF}} \sim \int dt \mathcal{L}$ , discussed around (5.13), gives the Schwinger–Keldysh influence functional

$$S_{\text{IF}} = -i \int dt C_{\alpha\beta} \left[ L_{\alpha,-}^* L_{\beta,+} - \frac{1}{2} L_{\alpha,+}^* L_{\beta,+} - \frac{1}{2} L_{\alpha,-}^* L_{\beta,-} \right]. \quad (5.37)$$

This expression automatically vanishes when the two branches coincide,  $S_{\text{IF}}[x, x] = 0$ , as required by the normalization of the Schwinger–Keldysh generating functional.

To make the separation of  $S_{\text{IF}}$  into real and imaginary parts more explicit, it is useful to define

$$\Delta L_{\alpha} := L_{\alpha,+} - L_{\alpha,-}. \quad (5.38)$$

Consider expanding the following quadratic form:

$$\Delta L_{\alpha}^* C_{\alpha\beta} \Delta L_{\beta} = L_{\alpha,+}^* C_{\alpha\beta} L_{\beta,+} + L_{\alpha,-}^* C_{\alpha\beta} L_{\beta,-} - L_{\alpha,+}^* C_{\alpha\beta} L_{\beta,-} - L_{\alpha,-}^* C_{\alpha\beta} L_{\beta,+}. \quad (5.39)$$

Since the Kossakowski matrix is Hermitian,  $C_{\alpha\beta} = C_{\beta\alpha}^*$ , one has

$$L_{\alpha,+}^* C_{\alpha\beta} L_{\beta,-} = (L_{\alpha,-}^* C_{\alpha\beta} L_{\beta,+})^*. \quad (5.40)$$

It follows that

$$\begin{aligned}
& L_{\alpha,-}^* C_{\alpha\beta} L_{\beta,+} - \frac{1}{2} L_{\alpha,+}^* C_{\alpha\beta} L_{\beta,+} - \frac{1}{2} L_{\alpha,-}^* C_{\alpha\beta} L_{\beta,-} \\
&= -\frac{1}{2} \Delta L_{\alpha}^* C_{\alpha\beta} \Delta L_{\beta} + \frac{1}{2} [L_{\alpha,-}^* C_{\alpha\beta} L_{\beta,+} - (L_{\alpha,-}^* C_{\alpha\beta} L_{\beta,+})^*] \\
&= -\frac{1}{2} \Delta L_{\alpha}^* C_{\alpha\beta} \Delta L_{\beta} + i \operatorname{Im} (L_{\alpha,-}^* C_{\alpha\beta} L_{\beta,+}).
\end{aligned} \tag{5.41}$$

Substituting this identity into the influence functional yields

$$S_{\text{IF}} = \int dt \operatorname{Im} (L_{\alpha,-}^* C_{\alpha\beta} L_{\beta,+}) + \frac{i}{2} \int dt \Delta L_{\alpha}^* C_{\alpha\beta} \Delta L_{\beta}. \tag{5.42}$$

The first term is manifestly real, while the second is purely imaginary by virtue of (5.35). Moreover, since  $C \geq 0$ ,

$$\operatorname{Im} S_{\text{IF}} = \frac{1}{2} \int dt \Delta L_{\alpha}^* C_{\alpha\beta} \Delta L_{\beta} \geq 0. \tag{5.43}$$

We discovered that the last constraint in (4.73), imposed by the unitarity of the full system, follows directly from the positivity of the Kossakowski matrix,  $C \geq 0$ .

Notice that for a single Hermitian jump operator, or more generally for Hermitian jump operators with a real symmetric Kossakowski matrix, the real part of the influence functional vanishes. For general non-Hermitian jump operators, and also for Hermitian jumps before absorbing the imaginary antisymmetric part of the Kossakowski matrix into the Hamiltonian, one may instead have

$$\operatorname{Re} S_{\text{IF}} = \int dt \operatorname{Im} (L_{\alpha,-}^* C_{\alpha\beta} L_{\beta,+}). \tag{5.44}$$

Note that the argument of the imaginary part is real when  $x_+ = x_-$ , as it should be.

**Expansion of jump operators versus expansion of the influence action.** Let  $d$  denote the short-distance scale suppressing nonlinear operators, and suppose that the jump operators admit an expansion in powers of  $x/d$ . As a simple example, consider

$$L_1(x) = x + \frac{x^2}{d}, \quad L_2(x) = \frac{x^2}{d}, \tag{5.45}$$

and assume, for simplicity, a diagonal Kossakowski matrix with positive rates  $\gamma_1$  and  $\gamma_2$ . The corresponding imaginary part of the influence action is

$$\operatorname{Im} S_{\text{IF}} = \frac{1}{2} \int dt [\gamma_1 |\Delta L_1|^2 + \gamma_2 |\Delta L_2|^2]. \tag{5.46}$$

Suppose that we wish to construct the theory only through order  $d^{-1}$ . There are two natural prescriptions. The first is to truncate the expansion of the jump operators, discarding  $L_2$  but retaining the full quadratic contribution of  $L_1$ :

$$\operatorname{Im} S_{\text{IF}}^{\text{imp}} = \frac{\gamma_1}{2} \int dt \left[ \Delta x + \frac{1}{d} \Delta(x^2) \right]^2. \tag{5.47}$$

Expanding this expression gives

$$\operatorname{Im} S_{\text{IF}}^{\text{imp}} = \frac{\gamma_1}{2} \int dt \left[ (\Delta x)^2 + \frac{2}{d} \Delta x \Delta(x^2) + \frac{1}{d^2} (\Delta(x^2))^2 \right]. \tag{5.48}$$

Although the final term is formally of order  $d^{-2}$ , it is retained because it completes the positive square.

The second prescription is to expand the influence action itself and truncate it directly at order  $d^{-1}$ . One then obtains

$$\text{Im } S_{\text{IF}}^{(1/d)} = \frac{\gamma_1}{2} \int dt \left[ (\Delta x)^2 + \frac{2}{d} \Delta x \Delta(x^2) \right]. \quad (5.49)$$

The two prescriptions agree through order  $d^{-1}$  and differ only at order  $d^{-2}$ . Neither expression is the complete answer at order  $d^{-2}$ : additional jump operators, such as  $L_2$ , as well as other symmetry-allowed terms, may also contribute.

The first prescription nevertheless has a structural advantage. By retaining the complete square associated with the truncated jump operator, it guarantees  $\text{Im } S_{\text{IF}}^{\text{imp}} \geq 0$  for arbitrary field configurations. Equivalently, the corresponding truncated Liouvillian remains of GKSL form and generates a completely positive and CP-divisible evolution. By contrast, the influence action truncated directly at order  $d^{-1}$  is not sign definite for arbitrary values of  $x_+$  and  $x_-$ ; its positivity is meaningful only perturbatively, within the regime in which  $x/d$  is small.

One may refer to the first prescription as a *positivity-improved* truncation. It does not determine the full answer at order  $d^{-2}$ , but it provides a particular higher-order completion that preserves complete positivity exactly. This distinction is immaterial in a strictly fixed-order calculation, but can become useful when the truncated generator is evolved non-perturbatively in time, as in Section 3.1, since the resulting resummation then remains completely positive. This situation is structurally similar to what happens when one uses the renormalization group to improve some fixed-order loop calculations.

## 5.4 Semiclassical limit, HS transform, and Langevin dynamics

Let us begin with a general local Schwinger–Keldysh action  $S_{\text{eff}}[x_r, x_a]$  written in the  $r/a$  basis. We note that

$$\langle\langle x_r \rangle\rangle = \frac{1}{2}(\langle\langle x_+ \rangle\rangle + \langle\langle x_- \rangle\rangle) = \langle x \rangle, \quad \langle\langle x_a \rangle\rangle = \langle\langle x_+ \rangle\rangle - \langle\langle x_- \rangle\rangle = 0. \quad (5.50)$$

This suggests that  $x_r$  captures the average, classical part of the operator  $x$  while  $x_a$  captures the fluctuation in the path integral away from the classical saddle, which can be quantum or stochastic in origin.

To make this more concrete, we further note that, because of the normalization condition  $S_{\text{eff}}[x_r, 0] = 0$ , the action contains at least one power of  $x_a$ . In a regime in which  $x_a$  is small, it is natural to expand the action as

$$S_{\text{eff}}[x_r, x_a] = \int dt x_a(t) \mathcal{E}[x_r](t) + \frac{i}{2} \int dt dt' x_a(t) N(t, t') x_a(t') + \mathcal{O}(x_a^3). \quad (5.51)$$

Here  $\mathcal{E}[x_r](t)$  is some functional of the retarded field, and  $N(t, t')$  is a real symmetric kernel. The term linear in  $x_a$  is the part that survives if one varies with respect to the advanced field and then sets  $x_a = 0$ . The quadratic term is the leading imaginary contribution allowed by the Schwinger–Keldysh positivity conditions. Higher powers of  $x_a$  encode genuinely quantum or non-Gaussian corrections to the effective dynamics.

The semiclassical stochastic limit consists in truncating (5.51) at quadratic order in  $x_a$ ,

$$S_{\text{eff}}[x_r, x_a] \approx \int dt x_a(t) \mathcal{E}[x_r](t) + \frac{i}{2} \int dt dt' x_a(t) N(t, t') x_a(t'). \quad (5.52)$$

At this stage, the Hubbard–Stratonovich transformation becomes useful. Since the kernel  $N(t, t')$  is positive semidefinite, one may use the Gaussian identity

$$\exp\left[-\frac{1}{2}\int dt dt' x_a(t) N(t, t') x_a(t')\right] \propto \int \mathcal{D}\xi \exp\left[-\frac{1}{2}\int dt dt' \xi(t) N^{-1}(t, t') \xi(t') + i \int dt \xi(t) x_a(t)\right]. \quad (5.53)$$

Here  $\xi(t)$  is an auxiliary field. The transformation (5.53) is exact, provided the action has been truncated to quadratic order in  $x_a$ . After inserting it into the path integral, the effective action becomes linear in the advanced field,

$$S_{\text{eff}}[x_r, x_a, \xi] = \int dt x_a(t) (\mathcal{E}[x_r](t) - \xi(t)), \quad (5.54)$$

up to the Gaussian weight for  $\xi$ . The path integral over  $x_a$  can then be performed exactly and produces a functional Dirac delta, which imposes

$$\mathcal{E}[x_r](t) = \xi(t). \quad (5.55)$$

This is the Langevin-type stochastic equation associated with the truncated Schwinger–Keldysh action.

The auxiliary field  $\xi$  is Gaussian distributed with zero mean and two-point function

$$\langle \xi(t) \xi(t') \rangle_\xi = N(t, t'), \quad (5.56)$$

where the average is taken with respect to the Gaussian measure appearing in (5.53). Thus the kernel  $N(t, t')$  becomes the noise correlator of the stochastic process. In this way, the term quadratic in the advanced field is reinterpreted as Gaussian noise, while the term linear in  $x_a$  becomes the deterministic drift or equation of motion.

This logic may be illustrated with the Gaussian example discussed above. Suppose the local effective action takes the form

$$S_{\text{eff}}[x_r, x_a] = \int dt \left[ x_a (\ddot{x}_r + \kappa \dot{x}_r + \Omega^2 x_r) + \frac{i}{2} N x_a^2 \right], \quad (5.57)$$

where, for simplicity, we have written a local white-noise kernel  $N(t, t') = N \delta(t - t')$ . The Hubbard–Stratonovich transformation then introduces a Gaussian field  $\xi(t)$  with correlator

$$\langle \xi(t) \xi(t') \rangle_\xi = N \delta(t - t'), \quad (5.58)$$

and the path integral over  $x_a$  yields

$$\ddot{x}_r + \kappa \dot{x}_r + \Omega^2 x_r = \xi(t). \quad (5.59)$$

This is the ordinary *Langevin equation* for a damped harmonic oscillator driven by Gaussian white noise. The Schwinger–Keldysh description therefore reproduces the familiar stochastic description once the effective action is truncated to quadratic order in the advanced field.

It is important to keep track of what is exact and what is approximate in this construction. The Hubbard–Stratonovich transformation is an exact identity for a quadratic form. What is approximate is the truncation of the Schwinger–Keldysh action to at most quadratic order in  $x_a$ . If higher powers of  $x_a$  are present, then the simple Gaussian noise representation is no longer exact. Such terms correspond to non-Gaussian fluctuations or genuinely quantum corrections beyond ordinary Langevin dynamics.

**From the Langevin equation to the Fokker–Planck equation** For each realization of the noise  $\xi(t)$ , the Langevin equation provides a realization of the time evolution of  $x(t)$ . For a distribution of realizations of the noise, this gives a probability distribution  $P(x, t)$  for the value of  $x$  at a given time  $t$ . Here we derive the differential equation satisfied by  $P(x, t)$ . For simplicity, restrict ourselves to a first-order Langevin equation with additive Gaussian white noise, to be specified below. This restriction allows us to avoid the distinction between the Itô and Stratonovich prescriptions, which becomes relevant for multiplicative noise. Langevin equations containing higher time derivatives may be brought into first-order form by enlarging the space of dynamical variables, for example by treating the position and velocity as independent variables. Consider the stochastic equation

$$\dot{x}(t) = A(x(t)) + \xi(t), \quad (5.60)$$

where  $A(x)$  is the deterministic drift and  $\xi(t)$  is Gaussian white noise satisfying

$$\langle \xi(t) \rangle_\xi = 0, \quad \langle \xi(t) \xi(t') \rangle_\xi = N \delta(t - t'). \quad (5.61)$$

The Langevin equation describes individual stochastic trajectories. Our goal is instead to determine the probability density  $P(x, t)$  for the stochastic variable to take the value  $x$  at time  $t$ .

Over a short interval  $\Delta t$ , equation (5.60) gives

$$\Delta x = A(x) \Delta t + \Delta W, \quad (5.62)$$

where

$$\Delta W \equiv \int_t^{t+\Delta t} dt' \xi(t') \quad (5.63)$$

is a Gaussian random variable with

$$\langle \Delta W \rangle_\xi = 0, \quad \langle (\Delta W)^2 \rangle_\xi = N \Delta t. \quad (5.64)$$

It follows that

$$\langle \Delta x \rangle_\xi = A(x) \Delta t, \quad (5.65)$$

$$\langle (\Delta x)^2 \rangle_\xi = N \Delta t + \mathcal{O}((\Delta t)^2). \quad (5.66)$$

Higher moments of  $\Delta x$  are of higher order than  $\Delta t$  and do not contribute in the continuum limit.

Let  $f(x)$  be an arbitrary smooth test function. During the short interval  $\Delta t$ ,

$$f(x + \Delta x) = f(x) + f'(x) \Delta x + \frac{1}{2} f''(x) (\Delta x)^2 + \dots \quad (5.67)$$

Averaging over the noise and keeping terms of order  $\Delta t$ , we obtain

$$\langle f(x + \Delta x) - f(x) \rangle_\xi = \left[ A(x) f'(x) + \frac{N}{2} f''(x) \right] \Delta t + \mathcal{O}((\Delta t)^2). \quad (5.68)$$

Taking the continuum limit gives

$$\frac{d}{dt} \langle f(x) \rangle = \left\langle A(x) f'(x) + \frac{N}{2} f''(x) \right\rangle. \quad (5.69)$$

The expectation value of  $f$  may also be written in terms of the probability density,

$$\langle f(x) \rangle = \int dx P(x, t) f(x). \quad (5.70)$$

Therefore,

$$\frac{d}{dt} \langle f(x) \rangle = \int dx f(x) \frac{\partial P(x, t)}{\partial t}. \quad (5.71)$$

On the other hand, using (5.69),

$$\frac{d}{dt} \langle f(x) \rangle = \int dx P(x, t) \left[ A(x) f'(x) + \frac{N}{2} f''(x) \right]. \quad (5.72)$$

Integrating by parts and assuming that the boundary terms vanish,

$$\frac{d}{dt} \langle f(x) \rangle = \int dx f(x) \left[ -\frac{\partial}{\partial x} (A(x) P(x, t)) + \frac{N}{2} \frac{\partial^2 P(x, t)}{\partial x^2} \right]. \quad (5.73)$$

Since  $f(x)$  is arbitrary, comparison with (5.71) yields the *Fokker-Planck equation*

$$\frac{\partial P(x, t)}{\partial t} = -\frac{\partial}{\partial x} [A(x) P(x, t)] + \frac{N}{2} \frac{\partial^2 P(x, t)}{\partial x^2}. \quad (5.74)$$

The first term transports probability according to the deterministic drift  $A(x)$ , while the second term describes diffusion generated by the stochastic noise. We will use this result to describe light scalar fields in de Sitter in Section ??.

It is often useful to write (5.74) as a continuity equation,

$$\frac{\partial P}{\partial t} = -\frac{\partial J}{\partial x}, \quad (5.75)$$

with probability current

$$J(x, t) = A(x) P(x, t) - \frac{N}{2} \frac{\partial P(x, t)}{\partial x}. \quad (5.76)$$

For completeness, consider a system of first-order Langevin equations

$$\dot{X}^I = A^I(X) + \xi^I(t), \quad (5.77)$$

with additive Gaussian noise

$$\langle \xi^I(t) \xi^J(t') \rangle_\xi = N^{IJ} \delta(t - t'). \quad (5.78)$$

The associated probability density  $P(X, t)$  obeys

$$\frac{\partial P}{\partial t} = -\frac{\partial}{\partial X^I} [A^I(X) P] + \frac{1}{2} N^{IJ} \frac{\partial^2 P}{\partial X^I \partial X^J}. \quad (5.79)$$

A higher-order Langevin equation may be treated by introducing additional variables until the dynamics is written as such a first-order system.

**The Martin–Siggia–Rose construction** It is useful to briefly review the Martin–Siggia–Rose construction [?] for a generic stochastic ordinary differential equation, in order to better understand its relation to the semiclassical limit of the SK path integral. Consider a set of real variables  $\phi^I(t)$ , with  $I = 1, \dots, N_r$ , obeying

$$E^i[\phi](t) = \xi^i(t), \quad (5.80)$$

where  $E^i[\phi]$ , for  $i = 1, \dots, N_a$ , is some, possibly non-linear, differential operator and  $\xi^i(t)$  is a stochastic noise. For definiteness, let the noise be Gaussian with vanishing mean and covariance

$$\langle \xi^i(t) \xi^j(t') \rangle = N^{ij}(t, t'). \quad (5.81)$$

One is then interested in predicting statistical properties of the fields  $\phi^I$ , which must be averaged over the realizations of the noise. There is no quantum mechanics in this setting. This is a classical stochastic problem.

The central idea of Martin, Siggia, and Rose is to rewrite this stochastic average as a functional integral. One first inserts into the noise average a functional delta imposing the stochastic equation<sup>27</sup>,

$$1 = \int \mathcal{D}\phi \, \delta(E[\phi] - \xi) \, \det\left(\frac{\delta E}{\delta \phi}\right), \quad (5.82)$$

where the determinant is the Jacobian associated with the change of variables from  $\phi$  to  $E[\phi]$ . The delta functional may then be exponentiated by introducing an auxiliary field  $\hat{\phi}_i(t)$ , often called the response field,

$$\delta(E[\phi] - \xi) \propto \int \mathcal{D}\hat{\phi} \, \exp\left[i \int dt \, \hat{\phi}_i(t) (E^i[\phi](t) - \xi^i(t))\right]. \quad (5.83)$$

At this stage the path integral is still exact.

If the noise is Gaussian, one may perform the average over  $\xi$  explicitly. This produces a term quadratic in the response field and yields the Martin–Siggia–Rose action

$$\begin{aligned} S_{\text{MSR}}[\phi, \hat{\phi}] &= \int dt \, \hat{\phi}_i(t) E^i[\phi](t) + \frac{i}{2} \int dt \, dt' \, \hat{\phi}_i(t) N^{ij}(t, t') \hat{\phi}_j(t') \\ &\quad + S_{\text{Jac}}[\phi, \hat{\phi}]. \end{aligned} \quad (5.84)$$

Here  $S_{\text{Jac}}$  denotes the contribution of the Jacobian, which in many simple cases may be absorbed into a normalization or represented by ghosts if needed. The resulting functional integral generates the stochastic correlation functions of the original Langevin system. The term linear in  $\hat{\phi}_i$  imposes the deterministic part of the equations of motion, while the quadratic term encodes the statistics of the noise. In this sense, the response fields are Lagrange multipliers dressed by the noise average.

The construction is more general than the simplest Langevin equation. It applies to non-linear differential equations, to multiplicative noise with appropriate care about discretization, and to non-local kernels  $N^{ij}(t, t')$ . The key point is always the same: the stochastic dynamics is represented by a path integral over the original variables together with one auxiliary response field for each equation, and the noise average generates terms quadratic and, for non-Gaussian noise, higher order in the response fields.

It is worth emphasizing the relation to the semiclassical limit of the Schwinger–Keldysh formalism. In that context, one expands the effective action in powers of the advanced fields.

<sup>27</sup>For simplicity, in this derivation we now assume  $N_r = N_a$ , so that the Jacobian is just a determinant. However, the MSR formalism can be generalized to  $N_a \neq N_r$ .

Truncating the action to terms linear and quadratic in those fields yields precisely the same structural form as the Martin–Siggia–Rose action: the linear term imposes an effective equation of motion, while the quadratic term can be interpreted as Gaussian noise after a Hubbard–Stratonovich transformation. In this sense, the Martin–Siggia–Rose formalism may be viewed as the stochastic semiclassical counterpart of the Schwinger–Keldysh path integral.

## 5.5 A dissipative Gaussian scalar field and its propagators

From this point onward, it is convenient to pass from a single variable  $x(t)$  to a field  $\phi(t, \mathbf{x})$ . If the background is homogeneous and isotropic, it is natural to Fourier transform in space,

$$\phi(t, \mathbf{x}) = \int \frac{d^d k}{(2\pi)^d} e^{i\mathbf{k}\cdot\mathbf{x}} \phi_{\mathbf{k}}(t). \quad (5.85)$$

Then different momenta decouple in a quadratic theory, and each Fourier mode  $\phi_{\mathbf{k}}(t)$  behaves as the analogue of the single variable discussed in the previous subsections. In this sense, a free or Gaussian field theory may be viewed as a collection of decoupled oscillators labelled by  $\mathbf{k}$ . Of course, quantum field theory comes with additional complications on top of quantum mechanics, and we may mention some of them, such as loop divergences and renormalization, when we get to it.

Accordingly, the Schwinger–Keldysh doubled variables become  $\phi_{\pm}(t, \mathbf{x})$ , or equivalently

$$\phi_r(t, \mathbf{x}) := \frac{\phi_+(t, \mathbf{x}) + \phi_-(t, \mathbf{x})}{2}, \quad \phi_a(t, \mathbf{x}) := \phi_+(t, \mathbf{x}) - \phi_-(t, \mathbf{x}). \quad (5.86)$$

We now consider the simplest dissipative Gaussian model in Minkowski spacetime. In the  $r/a$  basis, let the quadratic Schwinger–Keldysh action be

$$S_{\text{SK}} = \int dt \frac{d^d k}{(2\pi)^d} \left[ \phi_a(-\mathbf{k}, t) \left( \partial_t^2 + \gamma \partial_t + \omega_{\mathbf{k}}^2 \right) \phi_r(\mathbf{k}, t) + \frac{i}{2} \mathcal{N}_{\mathbf{k}} \phi_a(-\mathbf{k}, t) \phi_a(\mathbf{k}, t) \right], \quad (5.87)$$

where

$$\omega_{\mathbf{k}}^2 := \mathbf{k}^2 + m^2. \quad (5.88)$$

Here  $\gamma$  is a damping coefficient, while  $\mathcal{N}_{\mathbf{k}} \geq 0$  is the Keldysh or noise kernel. Because of the constraints inherited from a unitary full system, in particular

$$-S_{\text{SK}}[\phi_r, -\phi_a]^* = S_{\text{SK}}[\phi_r, \phi_a],$$

both  $\gamma$  and  $\mathcal{N}$  must be real. This action is local in time, quadratic, and translation invariant in space and time. It therefore provides a particularly simple laboratory in which all propagators can be computed explicitly.

In frequency space,

$$\phi(\omega, \mathbf{k}) = \int dt e^{i\omega t} \phi(t, \mathbf{k}), \quad (5.89)$$

the action becomes

$$S_{\text{SK}} = \int \frac{d\omega}{2\pi} \frac{d^d k}{(2\pi)^d} \left[ \phi_a(-\omega, -\mathbf{k}) \mathcal{D}_R(\omega, \mathbf{k}) \phi_r(\omega, \mathbf{k}) + \frac{i}{2} \mathcal{N}_{\mathbf{k}} \phi_a(-\omega, -\mathbf{k}) \phi_a(\omega, \mathbf{k}) \right], \quad (5.90)$$

with retarded inverse kernel

$$\mathcal{D}_R(\omega, \mathbf{k}) = -\omega^2 - i\gamma\omega + \omega_{\mathbf{k}}^2. \quad (5.91)$$

The advanced kernel is its conjugate,

$$\mathcal{D}_A(\omega, \mathbf{k}) = -\omega^2 + i\gamma\omega + \omega_{\mathbf{k}}^2 = \mathcal{D}_R(\omega, \mathbf{k})^*. \quad (5.92)$$

Since the theory is Gaussian, the propagators are obtained by inverting the quadratic kernel. This is worked out in detail in Appendix ???. In the  $r/a$  basis one finds<sup>28</sup>

$$G_R(\omega, \mathbf{k}) = \langle\langle \phi_r(\omega, \mathbf{k}) \phi_a(-\omega, -\mathbf{k}) \rangle\rangle = \frac{1}{\mathcal{D}_R(\omega, \mathbf{k})} = \frac{1}{-\omega^2 - i\gamma\omega + \omega_{\mathbf{k}}^2}, \quad (5.93)$$

$$G_A(\omega, \mathbf{k}) = \langle\langle \phi_a(\omega, \mathbf{k}) \phi_r(-\omega, -\mathbf{k}) \rangle\rangle = \frac{1}{\mathcal{D}_A(\omega, \mathbf{k})} = \frac{1}{-\omega^2 + i\gamma\omega + \omega_{\mathbf{k}}^2}, \quad (5.94)$$

$$\begin{aligned} G_K(\omega, \mathbf{k}) &= \langle\langle \phi_r(\omega, \mathbf{k}) \phi_r(-\omega, -\mathbf{k}) \rangle\rangle = -i\mathcal{N}_{\mathbf{k}} G_R(\omega, \mathbf{k}) G_A(\omega, \mathbf{k}) \\ &= -i\mathcal{N}_{\mathbf{k}} \frac{1}{(-\omega^2 - i\gamma\omega + \omega_{\mathbf{k}}^2)(-\omega^2 + i\gamma\omega + \omega_{\mathbf{k}}^2)}, \end{aligned} \quad (5.95)$$

$$G_{aa}(\omega, \mathbf{k}) = \langle\langle \phi_a(\omega, \mathbf{k}) \phi_a(-\omega, -\mathbf{k}) \rangle\rangle = 0. \quad (5.96)$$

Thus the retarded and advanced propagators are determined entirely by the deterministic part of the action, which is linear in  $\phi_a$ , while the Keldysh propagator depends also on the noise kernel  $\mathcal{N}_{\mathbf{k}}$ .

It is useful to rewrite these propagators in the  $\pm$  basis:

$$G_{++} := \langle\langle \phi_+ \phi_+ \rangle\rangle = G_K + \frac{1}{2}G_R + \frac{1}{2}G_A, \quad (5.97)$$

$$G_{--} := \langle\langle \phi_- \phi_- \rangle\rangle = G_K - \frac{1}{2}G_R - \frac{1}{2}G_A, \quad (5.98)$$

$$G_{+-} := \langle\langle \phi_+ \phi_- \rangle\rangle = G_K - \frac{1}{2}G_R + \frac{1}{2}G_A, \quad (5.99)$$

$$G_{-+} := \langle\langle \phi_- \phi_+ \rangle\rangle = G_K + \frac{1}{2}G_R - \frac{1}{2}G_A. \quad (5.100)$$

In other words, once  $G_R$ ,  $G_A$ , and  $G_K$  are known, all contour-ordered propagators in the  $\pm$  basis are determined.

Let us now discuss the time-domain retarded propagator. By definition, it is the retarded Green's function of

$$\left(\partial_t^2 + \gamma\partial_t + \omega_{\mathbf{k}}^2\right)G_R(t, \mathbf{k}) = \delta(t), \quad G_R(t, \mathbf{k}) = 0 \quad \text{for } t < 0. \quad (5.101)$$

By Fourier transforming back to the time domain

<sup>29</sup>

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<sup>28</sup>Notice a difference with the free-field propagators in closed quantum field theory. There, the inverse quadratic operator typically develops zeros on the mass shell, so its inversion requires an additional prescription, such as the familiar  $i\epsilon$ . Here, by contrast, the retarded inverse kernel already contains a finite imaginary part proportional to  $\gamma$ , so the denominator does not vanish for real kinematics, and the poles are automatically displaced away from the real axis into the lower half-plane. The inversion is therefore unambiguous.

<sup>29</sup>For a finite physical damping coefficient  $\gamma > 0$ , the poles of  $G_R$  are displaced into the lower half of the complex  $\omega$ -plane and, except for the massless zero mode, they do not lie on the real-frequency integration contour. No additional  $i\epsilon$  prescription is therefore required for the Fourier transform. If  $\gamma = 0$ , the poles return to the real axis and the Fourier transform must instead be defined with the usual retarded prescription,

$$\mathcal{D}_R(\omega, \mathbf{k}) = -\omega^2 + \mathbf{k}^2 + m^2 - i\epsilon\omega, \quad \epsilon \rightarrow 0^+. \quad (5.102)$$

Thus the retarded  $i\epsilon$  prescription may be viewed as introducing an infinitesimal positive damping coefficient.

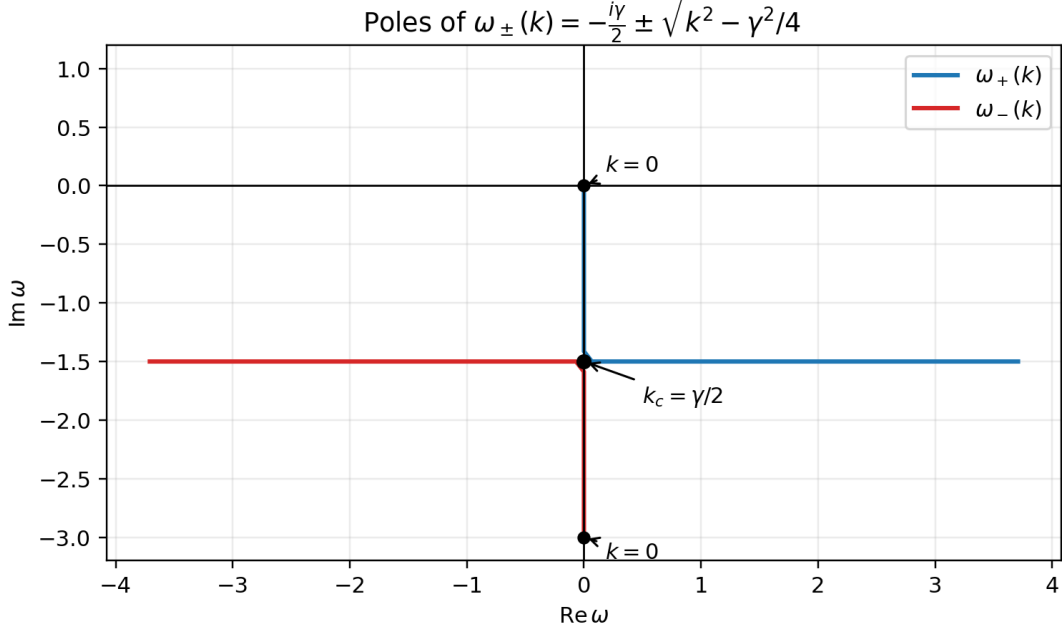


FIG 5: Poles of the retarded propagator in the complex frequency plane for the massless dissipative mode. For  $k < \gamma/2$ , both poles lie on the negative imaginary axis. At  $k = \gamma/2$  they merge, while for  $k > \gamma/2$  they move horizontally with constant imaginary part  $-\gamma/2$ .

(see Appendix ?? for details), one finds, in the underdamped regime  $\gamma \leq 2\omega_k$ ,

$$G_R(t, \mathbf{k}) = \theta(t) e^{-\gamma t/2} \frac{\sin(\Omega_{\mathbf{k}} t)}{\Omega_{\mathbf{k}}}, \quad \Omega_{\mathbf{k}} := \sqrt{\mathbf{k}^2 + m^2 - \frac{\gamma^2}{4}}. \quad (5.103)$$

and in the overdamped regime  $\gamma \geq 2\omega_k$ ,

$$G_R(t, \mathbf{k}) = \theta(t) \frac{e^{-(\gamma/2 - \Lambda_{\mathbf{k}})t} - e^{-(\gamma/2 + \Lambda_{\mathbf{k}})t}}{2\Lambda_{\mathbf{k}}}, \quad \Lambda_{\mathbf{k}} := \sqrt{\frac{\gamma^2}{4} - \mathbf{k}^2 - m^2}. \quad (5.104)$$

Thus every mode with  $\mathbf{k}^2 + m^2 > 0$  retains only exponentially suppressed memory of the past. In the underdamped regime the envelope decays on the timescale  $2/\gamma$ , whereas in the overdamped regime the late-time decay is controlled by the mode-dependent rate

$$\Gamma_{\text{slow}} = \frac{\gamma}{2} - \Lambda_{\mathbf{k}} \simeq \frac{\mathbf{k}^2 + m^2}{\gamma}, \quad (5.105)$$

where the last expression applies for  $\mathbf{k}^2 + m^2 \ll \gamma^2$ . The corresponding memory time therefore becomes arbitrarily long as  $\mathbf{k}^2 + m^2 \rightarrow 0$ . For the exactly massless homogeneous mode,  $m = k = 0$ , one finds

$$G_R(t, \mathbf{0}) = \theta(t) \frac{1 - e^{-\gamma t}}{\gamma}, \quad (5.106)$$

which approaches a constant rather than decaying at late times.

**The dispersion relation** The poles of the retarded propagator, which define the relevant dispersion relation, are given by the zeros of  $\mathcal{D}_R(\omega, \mathbf{k})$ ,

$$-\omega^2 - i\gamma\omega + \omega_{\mathbf{k}}^2 = 0. \quad (5.107)$$

They are therefore

$$\omega_{\pm}(\mathbf{k}) = -\frac{i\gamma}{2} \pm \sqrt{\omega_{\mathbf{k}}^2 - \frac{\gamma^2}{4}}. \quad (5.108)$$

These poles always lie in the lower half of the complex  $\omega$ -plane, as required for a retarded propagator. This is shown in Figure 5 in the massless case. Their precise form distinguishes different regimes.

In the underdamped regime,

$$\gamma < 2\omega_{\mathbf{k}}, \quad (5.109)$$

the square root in (5.108) is real, and the poles have non-zero real parts. The corresponding modes oscillate with frequency  $\Omega_{\mathbf{k}}$  while decaying at rate  $\gamma/2$ .

At critical damping,

$$\gamma = 2\omega_{\mathbf{k}}, \quad (5.110)$$

the two poles coincide,

$$\omega_+(\mathbf{k}) = \omega_-(\mathbf{k}) = -i\gamma/2. \quad (5.111)$$

In the overdamped regime,

$$\gamma > 2\omega_{\mathbf{k}}, \quad (5.112)$$

the square root becomes purely imaginary, and both poles are purely imaginary:

$$\omega_{\pm}(\mathbf{k}) = -\frac{i\gamma}{2} \pm i\sqrt{\frac{\gamma^2}{4} - \omega_{\mathbf{k}}^2}. \quad (5.113)$$

In this case the two modes are no longer oscillatory. They are purely relaxational. In the low-momentum limit and for  $m = 0$  we find a low-frequency “diffusive” mode,

$$\omega(\mathbf{k}) \sim -iD\mathbf{k}^2 + \dots. \quad (5.114)$$

It is also useful to comment on the real and imaginary parts of the retarded propagator. Writing

$$G_R(\omega, \mathbf{k}) = \frac{\omega_{\mathbf{k}}^2 - \omega^2 + i\gamma\omega}{(\omega_{\mathbf{k}}^2 - \omega^2)^2 + \gamma^2\omega^2}, \quad (5.115)$$

one finds

$$\text{Re } G_R(\omega, \mathbf{k}) = \frac{\omega_{\mathbf{k}}^2 - \omega^2}{(\omega_{\mathbf{k}}^2 - \omega^2)^2 + \gamma^2\omega^2}, \quad (5.116)$$

$$\text{Im } G_R(\omega, \mathbf{k}) = \frac{\gamma\omega}{(\omega_{\mathbf{k}}^2 - \omega^2)^2 + \gamma^2\omega^2}. \quad (5.117)$$

The real part is dispersive and changes sign near the resonance, while the imaginary part is odd in  $\omega$  and measures dissipative broadening. In particular, the finite width of the pole is controlled by  $\gamma$ .

Finally, the Keldysh propagator takes the explicit form

$$G_K(\omega, \mathbf{k}) = -i\mathcal{N}_{\mathbf{k}} \frac{1}{(\omega_{\mathbf{k}}^2 - \omega^2)^2 + \gamma^2\omega^2}. \quad (5.118)$$

It is even in frequency and its amplitude is directly proportional to the noise kernel.

At this stage  $\mathcal{N}_{\mathbf{k}}$  is arbitrary. Only if one further assumes thermal equilibrium does it become related to the retarded and advanced propagators by a fluctuation–dissipation or KMS relation, a point that we will discuss later.

To summarize, the dissipative Gaussian scalar field provides a simple but instructive example in which all Schwinger–Keldysh propagators can be computed explicitly. The retarded and advanced propagators are controlled by the inverse kernels  $\mathcal{D}_R$  and  $\mathcal{D}_A$ , the Keldysh propagator encodes the noise spectrum, and the contour-ordered propagators in the  $\pm$  basis are reconstructed from these. The location of the retarded poles reveals whether the excitations are underdamped, critically damped, or overdamped, while a genuinely diffusive mode appears only in an additional low-frequency and low-momentum limit.

## 6 Cosmology and inflation

We now turn to the cosmological setting that will be the main arena for the rest of these notes. Our ultimate goal is to understand how the tools of open quantum systems, master equations, and Schwinger–Keldysh effective actions can be applied to the primordial universe, and in particular to inflation. Before doing so, however, it is useful to review the corresponding closed-system description. Inflation provides a remarkable playground for quantum field theory: the background spacetime is time dependent, the relevant observables are late-time correlators rather than scattering amplitudes, and the fluctuations generated during the early universe are believed to be the seeds of the structures observed in the sky today.

The aim of this section is therefore to introduce the basic cosmological ingredients and the formalism needed for later developments. We begin with a brief review of homogeneous cosmological backgrounds and the motivations for inflation, then discuss quasi-de Sitter expansion and simple single-field realizations. We next explain how cosmological observables are computed as equal-time correlators, introduce the corresponding perturbative Feynman rules, and briefly review primordial non-Gaussianity through the bispectrum. This will set the stage for the effective field theory of inflation in the next section, where we will first study the unitary case and then generalize it to an open-system framework.

### 6.1 Background cosmology

We begin with a brief review of the homogeneous cosmological backgrounds relevant for inflation. In general relativity, the spacetime metric is dynamical and obeys the Einstein equations

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{1}{M_{\text{Pl}}^2}T_{\mu\nu}, \quad (6.1)$$

which follow from the action

$$S = \int d^4x \sqrt{-g} \left[ \frac{M_{\text{Pl}}^2}{2}R + \mathcal{L}_M \right]. \quad (6.2)$$

The energy-momentum tensor  $T_{\mu\nu}$  is defined as the response of the matter action to a variation of the spacetime metric,

$$T^{\mu\nu} := \frac{2}{\sqrt{-g}} \frac{\delta S_M}{\delta g_{\mu\nu}}. \quad (6.3)$$

Exact solutions are in general difficult to obtain, but the high degree of symmetry observed on sufficiently large scales suggests focusing on homogeneous and isotropic spacetimes.

The simplest such metric is the Friedmann–Lemaître–Robertson–Walker geometry,

$$ds^2 = -dt^2 + a(t)^2 \frac{\delta_{ij} dx^i dx^j}{(1 + Kx^2/4)^2}, \quad (6.4)$$

where  $a(t)$  is the scale factor and  $K$  parameterizes the spatial curvature. We use the convention that the scale factor is set to unity today

$$a(t_0) = a_0 = 1. \quad (6.5)$$

It is often convenient to introduce conformal time  $\tau$ , defined by  $dt = a d\tau$ , so that

$$ds^2 = a(\tau)^2 \left[ -d\tau^2 + \frac{\delta_{ij} dx^i dx^j}{(1 + Kx^2/4)^2} \right]. \quad (6.6)$$

The rate of expansion is measured by the Hubble parameter

$$H := \frac{\dot{a}}{a}. \quad (6.7)$$

Homogeneity and isotropy require the background energy-momentum tensor to take the perfect-fluid form

$$T^\mu{}_\nu = \text{diag}(-\rho, p, p, p), \quad (6.8)$$

with energy density  $\rho(t)$  and pressure  $p(t)$ . Covariant conservation of  $T_{\mu\nu}$  then implies the continuity equation

$$\dot{\rho} + 3H(\rho + p) = 0. \quad (6.9)$$

For a component with equation of state

$$p = w\rho, \quad (6.10)$$

the continuity equation

$$\dot{\rho} + 3H(\rho + p) = 0 \quad (6.11)$$

integrates to

$$\rho(a) = \rho_0 a^{-3(1+w)}, \quad (6.12)$$

where  $\rho_0$  is the value of the energy density today. This immediately gives the familiar scalings

$$\rho_{\text{rad}}(a) = \rho_{\text{rad},0} a^{-4}, \quad \rho_{\text{mat}}(a) = \rho_{\text{mat},0} a^{-3}, \quad \rho_\Lambda(a) = \rho_{\Lambda,0}. \quad (6.13)$$

The 00 component of Einstein's equations gives the *Friedmann equation*

$$3M_{\text{Pl}}^2 \left( H^2 + \frac{K}{a^2} \right) = \sum_\alpha \rho_\alpha, \quad (6.14)$$

while a second useful combination yields the acceleration equation

$$M_{\text{Pl}}^2 \frac{\ddot{a}}{a} = -\frac{1}{6}(\rho + 3p), \quad (6.15)$$

and, equivalently the *second Friedmann equation*

$$-M_{\text{Pl}}^2 \left( \dot{H} - \frac{K}{a^2} \right) = \frac{1}{2}(\rho + p). \quad (6.16)$$

These equations summarize the homogeneous dynamics of the universe. In the standard cosmological history, different components dominate at different times. Radiation domination extends from the end of reheating until matter-radiation equality, which occurs at redshift

$$z_{\text{eq}} \simeq 3.4 \times 10^3, \quad (6.17)$$

when the universe was about  $t_{\text{eq}} \sim 5 \times 10^4$  yr old. During this era one has  $w = 1/3$ , and therefore

$$a(t) = \left( \frac{t}{t_0} \right)^{1/2}. \quad (6.18)$$

Matter domination then lasts approximately from  $z_{\text{eq}}$  down to the epoch at which matter and dark energy become comparable,

$$z_{\Lambda\text{eq}} \sim 0.3, \quad (6.19)$$

corresponding to a cosmic time of order  $t_{\Lambda\text{eq}} \sim 1 \times 10^{10}$  yr. During matter domination one has  $w \simeq 0$ , and so

$$a(t) = \left( \frac{t}{t_0} \right)^{2/3}. \quad (6.20)$$

Finally, in the late-time dark-energy dominated era, which takes over after  $z \sim 0.3$ , the equation of state is approximately  $w = -1$ , so the scale factor grows approximately exponentially,

$$a(t) = \exp[H(t - t_0)]. \quad (6.21)$$

The transition to accelerated expansion actually begins somewhat earlier, at redshift  $z \sim 0.7$ . Here  $t_0 \simeq 1.4 \times 10^{10}$  yr denotes the present age of the universe.

**Motivations for inflation** Despite the success of the hot Big Bang description after nucleosynthesis, it leaves several striking questions unanswered. The first is the *horizon problem*: in a purely decelerating universe, regions of the Cosmic Microwave Background that appear statistically similar were never in causal contact. The second is the *curvature problem*, namely the fact that current observations are consistent with a universe that is extremely close to spatially flat

$$\Omega_{K,0} = 0.000 \pm 0.005. \quad (6.22)$$

On the other hand, in a decelerating expanding universe the curvature contribution grows relative to radiation and matter

$$\frac{d}{dt}|\Omega_K| = -2|\Omega_K|\frac{\ddot{a}}{\dot{a}} = |\Omega_K|H(1+3w), \quad (6.23)$$

Therefore, for ordinary radiation- or matter-dominated evolution,  $\Omega_K$  increases with time, so the fact that it is so small today implies that it must have been even smaller in the early universe. The curvature problem is precisely the question of why the initial curvature had to be so extraordinarily close to zero.

The third is the *phase coherence problem*: cosmological perturbations are observed to oscillate coherently even on scales that would be super-horizon in the standard hot Big Bang evolution. These three puzzles all point toward the existence of a primordial phase of accelerated expansion, before the hot big bang, which we call inflation, during which  $\dot{a} > 0$  and  $\ddot{a} > 0$ .

The last motivation for inflation is the scale invariance problem. Observations indicate that the primordial perturbations have approximately the same amplitude on all cosmological scales that we can probe. In the language of equal-time correlators, scale invariance means that for every  $\lambda > 0$ ,

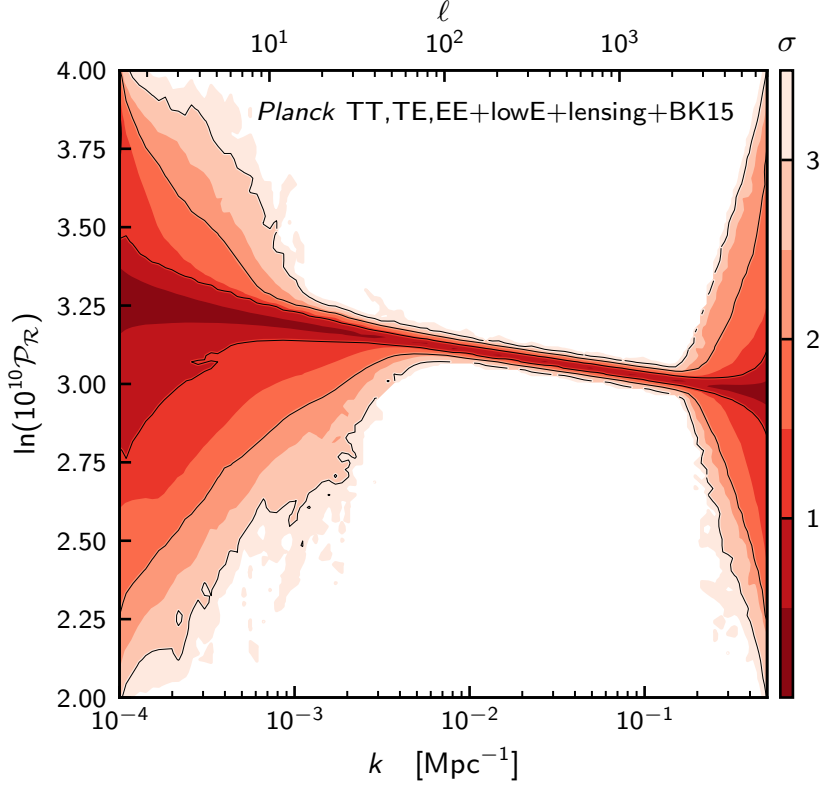


Figure 6: Observations from Planck 2018 [?]. The plot shows the constraints on the primordial power spectrum, which is scale invariant up to a 4% “red tilt”.

$$\langle \phi(x_1)\phi(x_2)\cdots\phi(x_n) \rangle = \langle \phi(\lambda x_1)\phi(\lambda x_2)\cdots\phi(\lambda x_n) \rangle. \quad (6.24)$$

In particular, the observed two-point function is very close to scale invariant. This is shown, for example, in Figure 6. Note that this is extremely surprising. Our general intuition, built upon the calculation of, for example, correlators in Minkowski, would tell us that the two-point function should decay as two points are separated. Scale invariance says that the correlation is independent of distance, an extremely remarkable fact that deserves a clear explanation.

A very attractive possibility is that this property follows from a symmetry of the primordial background rather than from an unexplained coincidence. A simple and elegant possibility is that the early universe was approximately described by de Sitter spacetime,

$$ds^2 = -\frac{d\tau^2 - dx^i dx^i}{H^2 \tau^2} = -dt^2 + e^{2Ht} dx^i dx^i, \quad (6.25)$$

which is maximally symmetric. In particular, de Sitter in flat slicing admits the dilation isometry

$$\tau \rightarrow \lambda \tau, \quad x^i \rightarrow \lambda x^i. \quad (6.26)$$

If the other background quantities respect this symmetry at least approximately, then it provides a natural explanation for the approximate scale invariance of primordial correlators. This is In the following, we will assume an extended period during which the spacetime is well approximated by de Sitter.

## 6.2 A prolonged phase of quasi-de Sitter expansion and an example

The problems reviewed above suggest that the early universe underwent a prolonged phase of accelerated expansion, together with an approximately scale-invariant background. The spacetime with these properties is de Sitter,

$$ds^2 = -\frac{d\tau^2 - dx^i dx^i}{H^2 \tau^2} = -dt^2 + e^{2Ht} dx^i dx^i, \quad (6.27)$$

whose scale factor is

$$a(\tau) = -\frac{1}{H\tau} = e^{Ht}. \quad (6.28)$$

However, exact de Sitter is eternal, whereas inflation must end. We are therefore led to consider an approximately de Sitter phase, in which the Hubble parameter varies slowly with time.

It is useful to rewrite the condition of accelerated expansion as

$$\frac{\ddot{a}}{a} = \dot{H} + H^2 = H^2(1 - \epsilon) > 0, \quad (6.29)$$

where we introduced the *first Hubble slow-roll parameter*

$$\epsilon := -\frac{\dot{H}}{H^2}. \quad (6.30)$$

Thus accelerated expansion requires  $\epsilon < 1$ , while a quasi-de Sitter stage corresponds to

$$0 < \epsilon \ll 1. \quad (6.31)$$

This condition captures the idea that  $H$  varies only slowly while the universe expands rapidly. A second important requirement is that inflation last long enough. To quantify this, one introduces the *number of e-foldings*  $N$ ,

$$dN := d \log a = H dt, \quad (6.32)$$

so that

$$N - N_0 = \log \left( \frac{a}{a_0} \right). \quad (6.33)$$

Solving the horizon and curvature problems requires a sufficiently long interval of accelerated expansion,

$$\Delta N_{\text{infl}} := N_e - N_i > 50 + \log \left( \frac{T_{\text{reh}}}{10^{10} \text{ GeV}} \right), \quad (6.34)$$

where  $T_{\text{reh}}$  is temperature of the universe at the end of reheating, a.k.a. the beginning of the hot big bang. For phenomenologically relevant reheating temperatures this is often summarized as

$$\Delta N_{\text{infl}} \sim 25\text{--}60. \quad (6.35)$$

In rough estimates one often simply uses  $\Delta N_{\text{infl}} \sim 50$ .

Since observations indicate approximate scale invariance over several e-foldings, it is natural to require that the departure from de Sitter remain small during the whole inflationary era. This is conveniently expressed in terms of the higher Hubble slow-roll parameters,

$$\epsilon = -\partial_N \log H, \quad \eta = \frac{\dot{\epsilon}}{H\epsilon} = \partial_N \log \epsilon, \quad \xi_{n \geq 3} := \partial_N \log \xi_{n-1}, \quad (6.36)$$

with  $\xi_2 \equiv \eta$ . Expanding  $\epsilon(N)$  around some reference time  $N_*$ , one finds schematically

$$\begin{aligned}\epsilon(N) - \epsilon(N_*) &= \left. \frac{\partial \epsilon}{\partial N} \right|_{N_*} (N - N_*) + \left. \frac{\partial^2 \epsilon}{\partial N^2} \right|_{N_*} \frac{(N - N_*)^2}{2} + \dots \\ &= \epsilon \left[ \eta(N - N_*) + \eta(\eta + \xi_3) \frac{(N - N_*)^2}{2} + \dots \right].\end{aligned}\quad (6.37)$$

This motivates the standard slow-roll conditions

$$\epsilon, \eta, \xi_n \ll 1. \quad (6.38)$$

These conditions ensure not only accelerated expansion, but also that the background remains close to de Sitter over many e-foldings. It should be noted that this description of inflation is somewhat different from what is encountered in introductory courses. In particular, note that we have not yet invoked a scalar field or a potential. Moreover, the slow-roll parameters have nothing to do with those ingredients and have been introduced here exclusively as a property of the spacetime. In this sense, this description is so far agnostic about what drove inflation.

**A class of inflationary models** A broad class of explicit realizations is provided by single-field  $P(X, \phi)$  theories,

$$S = \int d^4x \sqrt{-g} \left[ \frac{1}{2} M_{\text{Pl}}^2 R + P(X, \phi) \right], \quad (6.39)$$

where  $X$  denotes the familiar canonical kinetic term

$$X := -\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi = \frac{1}{2} \left( \dot{\phi}^2 - \frac{\partial_i \phi \partial^i \phi}{a^2} \right). \quad (6.40)$$

A canonical scalar corresponds to the special case

$$P(X, \phi) = X - V(\phi). \quad (6.41)$$

We will not discuss here more general non-minimal couplings such as  $R\phi^2$  or curvature couplings with additional derivatives.

The energy-momentum tensor derived from this action is

$$T_{\mu\nu} = P_{,X} \partial_\mu \phi \partial_\nu \phi + g_{\mu\nu} P(X, \phi). \quad (6.42)$$

This takes the perfect-fluid form

$$T_{\mu\nu} = (\rho + p) u_\mu u_\nu + p g_{\mu\nu} \quad (6.43)$$

under the identifications

$$\rho = 2X P_{,X} - P, \quad p = P, \quad u_\mu = -\frac{\partial_\mu \phi}{\sqrt{2X}}. \quad (6.44)$$

Thus  $P(X, \phi)$  theories provide a natural field-theoretic realization of the homogeneous cosmological fluid.

Focusing on a homogeneous background  $\bar{\phi}(t)$ , the scalar equation of motion becomes

$$\ddot{\bar{\phi}} (P_{,X} + 2X P_{,XX}) + 3H \dot{\bar{\phi}} P_{,X} + (2X P_{,X\phi} - P_{,\phi}) = 0, \quad (6.45)$$

while the Friedmann and acceleration equations read

$$3M_{\text{Pl}}^2 H^2 = 2XP_{,X} - P, \quad -M_{\text{Pl}}^2 \dot{H} = XP_{,X}. \quad (6.46)$$

The detailed background evolution depends on the specific choice of the function  $P$ , but for our purposes we will not need to solve these equations explicitly.

It is enough to note that the first slow-roll parameter can be written as

$$\epsilon = \frac{3XP_{,X}}{2XP_{,X} - P}. \quad (6.47)$$

This shows that many choices of  $P(X, \phi)$  can support a prolonged phase of slow-roll inflation, namely a regime with  $\epsilon, \eta \ll 1$ . In this sense,  $P(X, \phi)$  theories provide a simple and flexible framework in which the quasi-de Sitter inflationary backgrounds discussed above can be realized dynamically.

### 6.3 Cosmology and in-in correlators

Here we explain that the main objects of interest in cosmology are so-called cosmological correlators, namely expectation values of the product of local operators at the same time, usually taken to be the end of inflation. Luckily for us, this is exactly the object that is computed by the Schwinger-Keldysh formalism. In this section, we describe this formalism for a closed system in a pure state. This is the standard assumption in the vast majority of the literature. In the next section, we will extend this to open systems and density matrices, where we will see that the diagrammatic Feynman rules of perturbation theories are essentially the same.

In cosmology, we are interested in expectation values of the form

$$\langle \Omega | O(\tau) | \Omega \rangle, \quad (6.48)$$

where  $O(\tau)$  is a product of local Heisenberg-picture operators evaluated at the same conformal time  $\tau$ , and  $|\Omega\rangle$  denotes the interacting vacuum. In inflationary applications one is often interested in the late-time limit

$$\tau \rightarrow 0^-, \quad (6.49)$$

which in de Sitter corresponds to future infinity, or equivalently to the future conformal boundary. This is the natural cosmological analogue of sending asymptotic times to  $\pm\infty$  in scattering theory.

To compute such correlators, it is convenient to work in the interaction picture. For a closed unitary system, let the Hamiltonian be split as

$$\hat{H}(\tau) = \hat{H}_0(\tau) + \hat{H}_{\text{int}}(\tau). \quad (6.50)$$

The free part  $\hat{H}_0$  defines the interaction-picture fields, while  $\hat{H}_{\text{int}}$  is treated perturbatively. If  $|0\rangle$  denotes the free vacuum, then the interacting vacuum  $|\Omega\rangle$  may be obtained by adiabatically switching on the interactions in the infinite past. The resulting expectation value may be written as

$$\begin{aligned} \langle O(\tau) \rangle &:= \langle \Omega | O(\tau) | \Omega \rangle \\ &= \frac{\langle 0 | \bar{T} \exp\left(i \int_{-\infty}^{\tau} d\tau' \hat{H}_{\text{int}}(\tau')\right) O(\tau) T \exp\left(-i \int_{-\infty}^{\tau} d\tau' \hat{H}_{\text{int}}(\tau')\right) | 0 \rangle}{\langle 0 | \bar{T} \exp\left(i \int_{-\infty}^{\tau} d\tau' \hat{H}_{\text{int}}(\tau')\right) T \exp\left(-i \int_{-\infty}^{\tau} d\tau' \hat{H}_{\text{int}}(\tau')\right) | 0 \rangle}. \end{aligned} \quad (6.51)$$

Here  $T$  and  $\bar{T}$  denote time ordering and anti-time ordering, respectively.

The denominator is in fact equal to one once the correct projection onto the interacting vacuum is implemented. To make this projection precise, one slightly deforms the integration contour into the complex plane. The final result is

$$\langle O(\tau) \rangle = \langle 0 | \left[ \bar{T} e^{i \int_{-\infty(1+i\epsilon)}^{\tau} d\tau' \hat{H}_{\text{int}}(\tau')} \right] O(\tau) \left[ T e^{(-i \int_{-\infty(1-i\epsilon)}^{\tau} d\tau' \hat{H}_{\text{int}}(\tau'))} \right] | 0 \rangle. \quad (6.52)$$

The small imaginary tilt of the contour ensures that the free vacuum in the far past is projected onto the interacting vacuum. This expression can be rewritten in terms of a density matrix simply as

$$\langle O(\tau) \rangle = \text{Tr}(\rho(\tau) O(\tau)), \quad (6.53)$$

where

$$\rho(\tau) = U(\tau, -\infty) | 0 \rangle \langle 0 | U^\dagger(\tau, -\infty), \quad (6.54)$$

$$U(\tau, -\infty) = T \exp \left( -i \int_{-\infty(1-i\epsilon)}^{\tau} d\tau' \hat{H}_{\text{int}}(\tau') \right). \quad (6.55)$$

In the previous section, we have translated this operator expression into the SK path integral formalism. Here below we show an equivalent way to study it while remaining in the operator formalism, which is what is often done in the cosmology literature and referred to as *in-in formalism*. We do this to further clarify the connection between these two formalisms. The end point is the derivation of diagrammatic Feynman rules for perturbative calculations. Those, of course, are the same irrespective of whether they are derived in the operator or path integral formalism.

Expanding the exponentials perturbatively gives

$$\langle O \rangle = \left\langle \bar{T} \left[ 1 + i \int H_{\text{int}} - \frac{1}{2} \left( \int H_{\text{int}} \right)^2 + \dots \right] O T \left[ 1 - i \int H_{\text{int}} - \frac{1}{2} \left( \int H_{\text{int}} \right)^2 + \dots \right] \right\rangle,$$

where the shorthand  $\int H_{\text{int}}$  denotes the appropriate time integral of  $\hat{H}_{\text{int}}$ . Writing out the first few terms explicitly, one finds schematically

$$\begin{aligned} \langle O(\tau) \rangle &= \langle O(\tau) \rangle_0 + i \left\langle \int H_{\text{int}} O(\tau) - O(\tau) \int H_{\text{int}} \right\rangle_0 \\ &\quad - \frac{1}{2} \left\langle \left( \int H_{\text{int}} \right)^2 O(\tau) + O(\tau) \left( \int H_{\text{int}} \right)^2 \right\rangle_0 + \left\langle \left( \int H_{\text{int}} \right) O(\tau) \left( \int H_{\text{int}} \right) \right\rangle_0 + \dots, \end{aligned} \quad (6.56)$$

where  $\langle \dots \rangle_0$  denotes expectation values in the free vacuum. This is sometimes called the *factorized form* of the perturbative expansion.

**The commutator form** There is an equivalent representation that is often more convenient. It may be written as

$$\begin{aligned} \langle O(\tau) \rangle &= \sum_{N=0}^{\infty} i^N \int_{-\infty}^{\tau} d\tau_N \int_{-\infty}^{\tau_N} d\tau_{N-1} \dots \int_{-\infty}^{\tau_2} d\tau_1 \\ &\quad \langle 0 | [\hat{H}_{\text{int}}(\tau_1), [\hat{H}_{\text{int}}(\tau_2), \dots [\hat{H}_{\text{int}}(\tau_N), O(\tau)] \dots]] | 0 \rangle. \end{aligned} \quad (6.57)$$

This is the *commutator form* of the in-in perturbative expansion. The first two orders make the equivalence particularly transparent. At zeroth order,

$$\langle O(\tau) \rangle^{(0)} = \langle 0 | O(\tau) | 0 \rangle. \quad (6.58)$$

At first order,

$$\begin{aligned} \langle O(\tau) \rangle^{(1)} &= \left\langle 0 \left| \left[ i \int d\tau' \hat{H}_{\text{int}}(\tau') \right] O(\tau) + O(\tau) \left[ -i \int d\tau' \hat{H}_{\text{int}}(\tau') \right] \right| 0 \right\rangle \\ &= i \int_{-\infty}^{\tau} d\tau' \langle 0 | [\hat{H}_{\text{int}}(\tau'), O(\tau)] | 0 \rangle. \end{aligned} \quad (6.59)$$

Thus the two branches of the factorized formula combine into a single commutator. At higher orders the same mechanism produces the nested commutators appearing in (6.57). The equivalence to (6.56) can be proven by induction [?].

Using the formalism of open systems, there is a nice way to think about the above expression. First, recall that in the interaction picture the density matrix obeys

$$\rho' = -i[H_{\text{int}}, \rho] =: \mathcal{L}\rho, \quad (6.60)$$

where we introduce the Liouvillian superoperator  $\mathcal{L}$ . The formal solution is

$$\rho(\tau) = T \exp \left( \int_{\tau_0}^{\tau} \mathcal{L} \right) \rho(\tau_0).$$

Inserting into the trace we can write

$$\langle O(\tau) \rangle = \text{Tr}(\rho(\tau) O(\tau)) = \text{Tr} \left( T \exp \left( \int_{\tau_0}^{\tau} \mathcal{L} \right) \rho(\tau_0) O(\tau) \right) \quad (6.61)$$

$$= \text{Tr} \left( \rho(\tau_0) \bar{T} \exp \left( \int_{\tau_0}^{\tau} \mathcal{L}^\dagger \right) O(\tau) \right), \quad (6.62)$$

where the adjoint superoperator  $\mathcal{L}^\dagger$  now act on  $O(\tau)$  as  $i[H_{\text{int}}, \dots]$ . Using the pure state initial condition  $\rho(\tau_0) = |0\rangle \langle 0|$  as  $\tau_0 \rightarrow -\infty$  and using the cyclicity of the trace this reproduces (6.57). The factorized form (6.52) and the commutator form (6.57) are completely equivalent, but each is useful in different circumstances. The factorized form is often closer to diagrammatic perturbation theory and to the Schwinger–Keldysh contour description, while the commutator form makes causality and cancellations between the two branches more manifest. In practice, one chooses whichever form is most convenient for the correlator under consideration.

## 6.4 Feynman rules for in-in correlators: the $\pm$ basis

As in the case of scattering amplitudes, perturbative calculations of cosmological correlators are most efficiently organized in terms of diagrams. The resulting diagrammatics are sometimes referred to as Feynman–Witten rules, reflecting their resemblance to Witten diagrams in AdS. The rest of this section follows closely the discussion in [?, ?].

Because of translation invariance, any momentum-space correlator contains one or more delta functions imposing momentum conservation. Correlators proportional to more than one delta function are called disconnected. By subtracting all possible products of lower-point functions, one isolates the connected part, which carries a single overall delta function. For connected correlators we write

$$\left\langle \prod_{a=1}^n \phi(\mathbf{k}_a) \right\rangle_c = (2\pi)^3 \delta^{(3)} \left( \sum_{a=1}^n \mathbf{k}_a \right) B_n(\mathbf{k}_1, \dots, \mathbf{k}_n). \quad (6.63)$$

The main object of interest is then the reduced correlator  $B_n$ .

The perturbative rules for  $B_n$  are derived simply by performing all possible Wick contractions of the interaction picture operators appearing in the expansion we discussed in the previous section. Since in this closed system setting the time evolution is Hamiltonian, the final Feynman rule will have a natural interpretation in terms of the plus and minus basis of the path integral, as we see shortly. In the next section, we will develop the equivalent Feynman rules in the language of retarded and advanced fields, starting from the path integral and generalizing to open systems.

1. For an  $n$ -point correlator, draw all diagrams with  $n$  external lines ending on the late-time boundary  $\tau \rightarrow 0$ , together with some number  $V$  of bulk interaction vertices and  $I$  internal lines connecting pairs of vertices. Time runs from the infinite past  $\tau \rightarrow -\infty$  to the future conformal boundary  $\tau \rightarrow 0$ .
2. Each interaction vertex can lie on either branch of the in-in contour. We therefore label each vertex either as a  $+$  vertex, associated with the time-ordered evolution operator, or as a  $-$  vertex, associated with the anti-time-ordered evolution operator. One must sum over all  $2^V$  assignments of  $+/-$  labels to the  $V$  vertices.
3. Assign to each external line a momentum  $\mathbf{k}_a$ , with  $a = 1, \dots, n$ , all taken to flow from the bulk vertex toward the future boundary. Assign to each internal line an integration momentum  $\mathbf{p}_m$ , with  $m = 1, \dots, I$ , and integrate over all such momenta.
4. For each bulk vertex, include the corresponding interaction Hamiltonian density, evaluated at the time  $\tau_i$  of that vertex, together with an integral over  $\tau_i$ . A  $+$  vertex contributes a factor

$$-i \int_{-\infty(1-i\epsilon)}^0 d\tau_i H_{\text{int}}(\tau_i), \quad (6.64)$$

whereas a  $-$  vertex contributes

$$+i \int_{-\infty(1+i\epsilon)}^0 d\tau_i H_{\text{int}}(\tau_i). \quad (6.65)$$

The opposite signs and opposite  $i\epsilon$  prescriptions reflect the origin of these terms from  $U$  and  $U^\dagger$  in the time evolution of the bra and the ket. In the language of the Schwinger-Keldysh path integral, these are the two branches of the closed time contour.

5. For every external line connecting a bulk vertex at time  $\tau_i$  to the boundary operator at time  $\tau = 0$ , insert the appropriate bulk-to-boundary propagator  $G_\pm$ . Concretely, if the field operator is expanded in free modes as

$$\phi_{\mathbf{k}}(\tau) = u_{\mathbf{k}}(\tau) a_{\mathbf{k}} + u_{\mathbf{k}}^*(\tau) a_{-\mathbf{k}}^\dagger, \quad (6.66)$$

then an external line contributes either  $u_{\mathbf{k}}(\tau_i)$  or  $u_{\mathbf{k}}^*(\tau_i)$ , depending on whether it is attached to a  $+$  or  $-$  vertex

$$G_+(k; \tau_i, 0) = u_{\mathbf{k}}(\tau_i) u_{\mathbf{k}}^*(0), \quad G_-(k; \tau_i, 0) = u_{\mathbf{k}}^*(\tau_i) u_{\mathbf{k}}(0). \quad (6.67)$$

6. For every internal line joining vertices at times  $\tau_i$  and  $\tau_j$ , insert the free two-point contraction between the corresponding interaction-picture fields. The precise propagator depends on whether the two endpoints lie on the same branch or on different branches of the contour:

$$G_{++}(\tau_i, \tau_j; \mathbf{p}) = \langle T \phi_{\mathbf{p}}(\tau_i) \phi_{-\mathbf{p}}(\tau_j) \rangle, \quad (6.68)$$

$$G_{--}(\tau_i, \tau_j; \mathbf{p}) = \langle \bar{T} \phi_{\mathbf{p}}(\tau_i) \phi_{-\mathbf{p}}(\tau_j) \rangle, \quad (6.69)$$

$$G_{+-}(\tau_i, \tau_j; \mathbf{p}) = \langle \phi_{-\mathbf{p}}(\tau_j) \phi_{\mathbf{p}}(\tau_i) \rangle, \quad (6.70)$$

$$G_{-+}(\tau_i, \tau_j; \mathbf{p}) = \langle \phi_{\mathbf{p}}(\tau_i) \phi_{-\mathbf{p}}(\tau_j) \rangle. \quad (6.71)$$

We will later see how one may work in the  $r/a$  basis, where the same information is encoded in the retarded, advanced and Keldysh propagators.

7. At each vertex impose momentum conservation. After all vertex delta functions have been used, a single overall factor

$$(2\pi)^3 \delta \left( \sum_{a=1}^n \mathbf{k}_a \right) \quad (6.72)$$

remains, as required for a connected correlator.

8. Divide by the appropriate symmetry factor of the diagram. For details see [?, ?].
9. Finally, sum over all distinct diagrams and over all assignments of  $+/-$  labels.

These rules are the cosmological analogue of ordinary Feynman rules. Their main novelty is the doubled contour structure, which forces one to distinguish  $+$  and  $-$  vertices and to keep track of the corresponding propagators and signs. In practice, the factorized form of the in-in expansion leads naturally to the  $+/-$  basis, while the  $r/a$  basis is often more efficient for organizing causal properties and studying the semiclassical limit.

A number of examples of the application of these rules can be found in [?]. Here we will not repeat these and move on to the introduction of the main class of theories we will study.

## 6.5 Primordial non-Gaussianity and the bispectrum

A Gaussian random field is entirely characterized by its two-point function. Equivalently, all connected correlators beyond the two-point function vanish, while the full higher even-point functions are determined by Wick's theorem in terms of the two-point function. Non-Gaussianity is therefore encoded in connected correlators beyond the two-point function, the leading example being the three-point function, or bispectrum. Here we briefly discuss and summarize the main aspects of the study of non-Gaussianity produced during inflation, often known as primordial non-Gaussianity.

For any connected  $n$ -point function of a statistically homogeneous field  $\zeta$ , translation invariance implies the momentum-space form

$$\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \cdots \zeta_{\mathbf{k}_n} \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{k}_1 + \mathbf{k}_2 + \cdots + \mathbf{k}_n) B_n(\mathbf{k}_1, \mathbf{k}_2, \dots, \mathbf{k}_n). \quad (6.73)$$

The overall delta function enforces momentum conservation, while the reduced correlator  $B_n$  contains the physical information. Since the  $n$  momenta carry  $3n$  real components, translation invariance removes three of them. Rotational invariance removes three more. Therefore, before

imposing any additional symmetry, the reduced  $n$ -point function depends on  $3n - 6$  independent variables, for  $n \geq 3$ .

If the field is approximately scale invariant, then the correlators satisfy

$$\langle \zeta(\lambda \mathbf{x}_1) \zeta(\lambda \mathbf{x}_2) \cdots \zeta(\lambda \mathbf{x}_n) \rangle = \langle \zeta(\mathbf{x}_1) \zeta(\mathbf{x}_2) \cdots \zeta(\mathbf{x}_n) \rangle, \quad (6.74)$$

or equivalently in momentum space

$$B_n(\lambda \mathbf{k}_1, \lambda \mathbf{k}_2, \dots, \lambda \mathbf{k}_n) = \lambda^{-3(n-1)} B_n(\mathbf{k}_1, \mathbf{k}_2, \dots, \mathbf{k}_n). \quad (6.75)$$

Thus scale invariance removes one further degree of freedom, leaving  $3n - 7$  independent variables.

**The bispectrum** In particular, for the bispectrum  $B_3$ , the most general rotationally invariant configuration is specified by three numbers, which may be taken to be the lengths of the three sides of a triangle,

$$k_1 := |\mathbf{k}_1|, \quad k_2 := |\mathbf{k}_2|, \quad k_3 := |\mathbf{k}_3|, \quad \mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3 = 0. \quad (6.76)$$

Under the additional assumption of scale invariance, only two independent shape variables remain, which may be chosen for example as the ratios  $k_2/k_1$  and  $k_3/k_1$  [?].

Because the field  $\zeta$  is identical in each insertion, the bispectrum is also symmetric under permutations of  $k_1$ ,  $k_2$ , and  $k_3$ . It is therefore natural to think of the bispectrum as a function on the space of momentum triangles modulo permutations. One often separates the overall amplitude from the dependence on the shape of the triangle by writing

$$B_3(k_1, k_2, k_3) = A \frac{S(k_1, k_2, k_3)}{(k_1 k_2 k_3)^2}, \quad (6.77)$$

where  $A$  is an overall amplitude and  $S$  is a shape function. A normalized shape function may then be defined by fixing its value in a preferred configuration, for example by requiring that

$$S(k, k, k) = 1 \quad (6.78)$$

in the equilateral configuration. In this way, the amplitude and the dependence on triangle shape are clearly disentangled. A variety of normalization conditions have been chosen in the literature, so one should check the convention being used before interpreting any discussion about the value of  $A$ .

Different physical mechanisms generate bispectra that peak in different regions of triangle space. A useful way to organize the discussion is therefore to focus on a few special limits.

The first is the *squeezed limit*,

$$k_1 \ll k_2 \simeq k_3. \quad (6.79)$$

This probes the coupling between a long-wavelength mode and two short-wavelength modes. A bispectrum that is large in this limit indicates strong long-short mode coupling. In simple single-clock attractor models this limit is highly constrained [?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?], whereas a sizeable signal in the squeezed configuration can point to the presence of additional light degrees of freedom during inflation, or more generally to a departure from the simplest single-field adiabatic picture.

The second is the *equilateral limit*,

$$k_1 = k_2 = k_3. \quad (6.80)$$

This configuration is naturally enhanced by derivative self-interactions of the inflaton sector. As we will see later, the effective field theory of inflation in the decoupling limit gives rise precisely to this type of non-Gaussianity.

The third is the flattened, or *folded limit*,

$$k_1 + k_2 = k_3, \quad (6.81)$$

up to permutations. In this case the area of the momentum triangle tends to zero. Signals peaking in such configurations can arise from non-standard physical effects, for example from non-trivial initial states [?, ?] or, as we will later discuss, from open-system effects in an effective theory of inflation [?, ?, ?, ?, ?].

We will see that the standard effective field theory of inflation predicts a bispectrum of predominantly equilateral type, while later we will study how this picture is modified in open effective theories of inflation, where contributions to the folded limit generically arise.

## 7 The Open Effective Field Theory of Inflation

We are now ready to combine the two main threads developed so far: inflation and the general framework of open quantum systems. In the unitary Effective Field Theory (EFT) approach to inflation, one focuses on the Goldstone boson of broken time translations and studies its dynamics as a closed system, possibly coupled to gravity. In practice, however, the observable sector may interact with additional degrees of freedom that are heavy, short-lived, or otherwise inaccessible. It is therefore natural to ask how the inflationary dynamics is modified when the Goldstone mode is treated as an open system interacting with an unknown environment. The purpose of this section is to construct the corresponding open effective field theory and to study its leading observational consequences.

Our goal in the first two subsections is to derive the EFT of inflation for the field  $\pi$  alone, (7.20), where gravity is a fixed, non-dynamical background. To reach this goal, one can work first in flat spacetime and construct the theory of the Goldstone boson  $\pi$  of spontaneously broken time translations. Then one can minimally couple this to a fixed FLRW background spacetime. This approach, which we outline in Section 7.1, is very intuitive but misses some constraints on the Wilsonian coefficients from the background Einstein equations. A longer but more precise derivation starts from a general theory of the metric alone, where time diffs are spontaneously broken. Here,  $\pi$  emerges as a Stückelberg field that makes the theory 4d-diff invariant. The final action (7.20) is then obtained by invoking the decoupling of  $\pi$  from metric fluctuations.

Next, we generalize this construction to an open-system setting by doubling the fields, imposing the non-equilibrium consistency conditions, discussing the fate of global symmetries on the Schwinger-Keldysh contour and organizing the resulting theory in the retarded/advanced basis. Finally, we study the quadratic theory, its propagators and power spectrum, and then turn to the cubic interactions and the bispectrum. This will allow us to identify characteristic signatures of open inflationary dynamics, in particular the enhancement of the signal near folded configurations.

### 7.1 The Goldstone boson of broken time translations in flat space

Before introducing the effective field theory of inflation in the presence of gravity, it is useful to consider a simpler non-gravitational system in flat spacetime<sup>30</sup>. The purpose of this warm-up is to isolate the symmetry logic behind the appearance of the Goldstone mode associated with the spontaneous breaking of time translations.

Consider a Poincaré invariant theory of a single scalar field  $U(x)$ , with the most general local Lagrangian

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<sup>30</sup>This subsection introduces the construction of the EFT of inflation [?, ?] paralleling the discussion in [?].

$$\mathcal{L} = \mathcal{L}(U, \partial_\mu U, \partial_\mu \partial_\nu U, \dots). \quad (7.1)$$

Suppose that this theory admits a homogeneous but time-dependent background solution  $\bar{U}(t)$ . Perturbations around this background are then conveniently parameterized as

$$U(x) = \bar{U}(t + \pi(x)). \quad (7.2)$$

The field  $\pi(x)$  is the Goldstone boson associated with the spontaneous breaking of time translations. It *transforms non-linearly* under the broken spacetime symmetries.

Under a Poincaré transformation  $x^\mu \rightarrow \Lambda^\mu{}_\nu x^\nu + \alpha^\mu$ , the scalar field  $U(x)$  transforms linearly, but the Goldstone field  $\pi$  transforms non-linearly. Writing the result explicitly,

$$\pi(x) \rightarrow \pi(\Lambda x + \alpha) + \alpha^0 + \Lambda^0{}_\mu x^\mu - t. \quad (7.3)$$

This expression makes manifest that the background has broken time translations and boosts, while spatial translations and rotations remain linearly realized. This non-linear transformation should be contrasted with the more familiar transformation of a covariant scalar field

$$\phi(x) \rightarrow \phi(\Lambda x + \alpha),$$

where the transformed expression is *linear* in the field  $\phi$ . This structure is the hallmark of spontaneous symmetry breaking.

The effective action for  $\pi$  is obtained by writing the most general local action compatible with these non-linearly realized symmetries, organized in a derivative expansion. Assuming that  $\bar{U}(t)$  is monotonic, one may always perform a field redefinition so that

$$\bar{U}(t) = t. \quad (7.4)$$

Then, at lowest order in derivatives, the first covariant building block is

$$\partial_\mu U \partial^\mu U = -1 - 2\dot{\pi} + \partial_\mu \pi \partial^\mu \pi. \quad (7.5)$$

It is often convenient to remove the background by adding 1 to the above building block so that this quantity starts linear in perturbations. The second covariant building block is  $U = t + \pi$ , so the Wilson coefficients may be arbitrary functions of  $(t + \pi(t, \mathbf{x}))$ . For any Poincaré transformation acting on  $\pi$  as in (7.3) one finds

$$t + \pi(t, \mathbf{x}) \rightarrow t' + \pi(t', \mathbf{x}'). \quad (7.6)$$

In building the effective action all Wilsonian coefficients can be arbitrary functions of  $(t + \pi(t))$ . Therefore, up to terms with at most one derivative per field, the effective action takes the form

$$S = \int d^4x \sum_{n=0}^{\infty} \frac{d_n(t + \pi)}{n!} \left( -2\dot{\pi} + \partial_\mu \pi \partial^\mu \pi \right)^n. \quad (7.7)$$

More generally, one may also include higher-derivative operators. Following the notation of the paper, one useful example is the quantity

$$K = \frac{\square U}{\sqrt{-\partial_\mu U \partial^\mu U}} + \frac{\partial^\mu U \partial_\nu U \partial_\mu \partial_\nu U}{(-\partial_\mu U \partial^\mu U)^{3/2}}, \quad (7.8)$$

which, through quadratic order in  $\pi$ , becomes

$$K = -\partial_i \partial_i \pi + \dot{\pi} \partial_i \partial_i \pi + 2 \partial_i \pi \partial_i \dot{\pi} + \mathcal{O}(\pi^3). \quad (7.9)$$

Including such terms, the effective action may be written as

$$S_{\text{h.d.}} = \int d^4x \sum_{n=0}^{\infty} \frac{\hat{d}_n(t+\pi)}{n!} \left( -2\dot{\pi} + \partial_\mu \pi \partial^\mu \pi \right)^n K + \dots \quad (7.10)$$

Henceforth we will drop these higher-derivative terms.

It is useful to rewrite the action so that two facts become manifest: first, tadpole cancellation implies that the action begins at quadratic order in perturbations; second, when the coefficients are time independent, the Goldstone mode decouples at zero momentum, equivalently  $\pi = \text{const}$  is always a solution. In this form one finds

$$S = \int d^4x \left[ d_1(t+\pi) \partial_\mu \pi \partial^\mu \pi + \sum_{n=2}^{\infty} \frac{d_n(t+\pi)}{n!} \left( -2\dot{\pi} + \partial_\mu \pi \partial^\mu \pi \right)^n \right] \quad (7.11)$$

This action describes the universal low-energy sector of any flat-space system that spontaneously breaks time translations and boosts, independently of the microscopic details.

The lesson for cosmology is immediate. In any cosmology, the homogeneous background also breaks time translations, and the corresponding Goldstone boson will again be introduced by shifting the clock variable. The main difference is that in the gravitational system time translations are part of diffeomorphism invariance, so the Goldstone mode is most naturally introduced after fixing unitary gauge and then restoring time diffeomorphisms through a Stückelberg field. The flat-space construction above provides the simplest setting in which this logic can be seen without the additional complications of gravity.

## 7.2 The EFT of inflation in the decoupling limit

We now introduce the class of theories for which we will perform explicit calculations: the effective field theory of inflation. In this subsection we restrict throughout to the unitary, closed-system theory, and we keep only the operators with the lowest number of derivatives per field. In particular, we neglect operators involving the extrinsic curvature and higher-derivative geometric quantities.

The conceptual starting point is that an inflationary background selects a preferred slicing of spacetime, namely the slices of constant background clock field. As a result, time diffeomorphisms are spontaneously broken by the background, while spatial diffeomorphisms remain unbroken. The effective field theory is therefore the most general action built from geometric quantities that respect spatial diffeomorphisms only.

One may choose to use time diffeomorphisms to set the fluctuation of the clock field to zero, so that all scalar perturbations are encoded in the metric. This is called *unitary gauge*. In this gauge, the theory is simply described by a theory of the spacetime metric that is invariant under time-dependent spatial diffeomorphisms, but not necessarily under time diffeomorphisms. Restricting to the lowest-derivative sector, the action takes the form

$$S = \int d^4x \sqrt{-g} \left[ \frac{M_{\text{Pl}}^2}{2} R + d_0(t) + d_1(t) \delta g^{00} + \sum_{n=2}^{\infty} \frac{d_n(t)}{n!} (\delta g^{00})^n \right], \quad (7.12)$$

where we work in cosmological time  $t$ , so that  $\bar{g}_{00} = -1$ , and we removed the background contribution by defining

$$\delta g^{00} := g^{00} - \bar{g}^{00} = g^{00} + 1. \quad (7.13)$$

The first two terms after the Einstein-Hilbert piece are tadpoles. Their role is simply to ensure that the chosen FLRW background solves the equations of motion. They are therefore fixed by the background evolution and are not independent interaction coefficients.

For a flat FLRW background,

$$ds^2 = -dt^2 + a(t)^2 d\mathbf{x}^2, \quad (7.14)$$

the tadpole coefficients are fixed by the Friedmann equations. It is convenient to rewrite them as

$$d_0(t) = -M_{\text{Pl}}^2(3H^2 + 2\dot{H}), \quad d_1(t) = M_{\text{Pl}}^2\dot{H}, \quad (7.15)$$

so that the action becomes

$$S = \int d^4x \sqrt{-g} \left[ \frac{M_{\text{Pl}}^2}{2} R - M_{\text{Pl}}^2(3H^2 + \dot{H}) + M_{\text{Pl}}^2\dot{H} g^{00} + \sum_{n=2}^{\infty} \frac{d_n(t)}{n!} (\delta g^{00})^n \right]. \quad (7.16)$$

At lowest order in derivatives, all departures from the minimal slow-roll theory are encoded in the coefficients  $d_n(t)$ .

While one could work directly with this action, it is very convenient to rewrite it in a way that is invariant under the full four-dimensional diffs, Which makes contact with the way we usually study general relativity. One can achieve this through the Stückelberg trick. As we will see shortly, this achieves much more, though. The Stückelberg trick makes explicit the presence of an additional scalar degree of freedom and the high-energy regime in which it decouples from the gravitational fluctuations.

In practice, one performs a broken time diffeomorphism with a gauge parameter  $\pi(x)$  that is a new field of the theory,

$$t \rightarrow t + \pi(x). \quad (7.17)$$

We interpret  $\pi(x)$  as the Goldstone boson of the broken time translations. Under this transformation, the metric component  $g^{00}$  transforms non-linearly

$$\delta g^{00} \rightarrow \delta g^{00} + 2g^{0\mu} \partial_\mu \pi + g^{\mu\nu} \partial_\mu \pi \partial_\nu \pi. \quad (7.18)$$

**The decoupling limit** A particularly useful simplification is the *decoupling limit*. In the full theory, the Goldstone mode  $\pi$  mixes with metric fluctuations because time diffeomorphisms are broken by the inflationary background. However, at sufficiently high energies this mixing becomes negligible, and one may study the dynamics of  $\pi$  on the unperturbed FLRW background alone. In practice, this means that one evaluates the metric on its homogeneous background and restores the Goldstone through the replacement  $t \rightarrow t + \pi(x)$ , without keeping the fluctuations of the lapse, shift, or spatial metric. The resulting action captures the leading scalar dynamics while greatly simplifying the calculation.

The decoupling limit is expected to be reliable at energies of order  $H$  or larger. Physically, this is because the mixing of  $\pi$  with gravity is controlled by the same scales that govern the slow time dependence of the background, whereas the self-interactions of  $\pi$  that we retain remain unsuppressed in this regime. Deviations from the decoupling limit are therefore expected to be slow-roll suppressed, and hence to produce corrections that are parametrically smaller than the leading effects we keep here, under our assumptions. In the decoupling limit, one may simply evaluate the transformation (7.18) on the unperturbed FLRW metric. One then finds

$$\delta g^{00} \rightarrow -2\dot{\pi} - \dot{\pi}^2 + \frac{(\partial_i \pi)^2}{a^2}. \quad (7.19)$$

This single replacement is enough to determine the Goldstone action to the order relevant for us. Substituting (7.19) into (7.16) and expanding to cubic order in  $\pi$ , one obtains

$$S_\pi = \int d^4x a^3 \left[ -M_{\text{Pl}}^2 \dot{H} \left( \dot{\pi}^2 - \frac{(\partial_i \pi)^2}{a^2} \right) + 2d_2 \dot{\pi}^2 + \left( 2d_2 - \frac{4}{3}d_3 \right) \dot{\pi}^3 - 2d_2 \dot{\pi} \frac{(\partial_i \pi)^2}{a^2} + \dots \right]. \quad (7.20)$$

The ellipsis denotes higher-order terms and terms suppressed in the decoupling-limit expansion. It is customary to parametrize the quadratic action in terms of the speed of sound  $c_s$ . Combining the first two quadratic terms, one may write

$$S_\pi^{(2)} = \int d^4x a^3 \left[ -M_{\text{Pl}}^2 \dot{H} c_s^{-2} \dot{\pi}^2 + M_{\text{Pl}}^2 \dot{H} \frac{(\partial_i \pi)^2}{a^2} \right], \quad (7.21)$$

with

$$\frac{1}{c_s^2} = 1 - \frac{2d_2}{M_{\text{Pl}}^2 \dot{H}}. \quad (7.22)$$

Since during inflation one has  $\dot{H} < 0$ , the coefficient of the kinetic term is positive for stable theories.

Using (7.22), the quadratic action becomes

$$S_\pi^{(2)} = \int d^4x a^3 \left[ -M_{\text{Pl}}^2 \dot{H} \left( \frac{1}{c_s^2} \dot{\pi}^2 - \frac{(\partial_i \pi)^2}{a^2} \right) \right]. \quad (7.23)$$

This shows that a non-trivial coefficient  $d_2$  modifies the propagation speed of the Goldstone mode away from the relativistic value  $c_s = 1$ .

At cubic order, the two leading interactions are

$$S_\pi^{(3)} = \int d^4x a^3 \left[ \left( 2d_2 - \frac{4}{3}d_3 \right) \dot{\pi}^3 - 2d_2 \dot{\pi} \frac{(\partial_i \pi)^2}{a^2} \right]. \quad (7.24)$$

The operator  $\dot{\pi}(\partial_i \pi)^2$  is not an independent addition to the theory once we restrict to the lowest-derivative EFT built from powers of  $\delta g^{00}$ . Rather, it is fixed by the same non-linearly realized symmetry that dictated the replacement (7.19). In particular, it arises from the same operator  $(\delta g^{00})^2$  that modifies the quadratic kinetic term and hence the speed of sound. This is why its coefficient is determined by  $c_s$ , whereas the coefficient of  $\dot{\pi}^3$  also depends on the next EFT parameter  $d_3$ .

Using (7.22), the mixed cubic interaction may be written as

$$-2d_2 \dot{\pi} \frac{(\partial_i \pi)^2}{a^2} = -M_{\text{Pl}}^2 \dot{H} \left( 1 - \frac{1}{c_s^2} \right) \dot{\pi} \frac{(\partial_i \pi)^2}{a^2}. \quad (7.25)$$

Thus lowering the speed of sound automatically enhances this interaction.

Finally, the Goldstone mode  $\pi$  is directly related to the curvature perturbation  $\zeta$  at linear order,

$$\zeta = -H\pi + \dots, \quad (7.26)$$

so the action above provides the starting point for computing inflationary correlators of the observable scalar perturbations. In the next subsection, we will use the in-in formalism developed above to evaluate the correlators generated by the cubic interactions in (7.24).

### 7.3 Primordial non-Gaussianity from the leading EFT operators

We now compute the tree-level bispectrum generated by the two leading cubic interactions in the decoupling-limit EFT of inflation,

$$S_\pi^{(3)} = \int d^4x a^3 \left[ C_{\dot{\pi}^3} \dot{\pi}^3 + C_{\dot{\pi}(\partial\pi)^2} \dot{\pi} \frac{(\partial_i \pi)^2}{a^2} \right], \quad (7.27)$$

where

$$C_{\dot{\pi}(\partial\pi)^2} = -M_{\text{Pl}}^2 \dot{H} \left( 1 - \frac{1}{c_s^2} \right), \quad (7.28)$$

while  $C_{\dot{\pi}^3}$  depends on the next EFT coefficient as discussed in the previous subsection.

For these purely unitary tree-level computations, the  $+/-$  basis is the most economical one, because the answer is simply the sum of a  $+$  contact diagram and a  $-$  contact diagram. The corresponding calculation in the  $r/a$  basis is possible but less compact, since the interaction is split into several terms. By contrast, the  $r/a$  basis will become much more useful later on in the open-system setting.

At this order only bulk-to-boundary propagators are needed. Using the free massless de Sitter mode function for the Goldstone field,

$$u_k(\tau) = \frac{i}{2M_{\text{Pl}}\sqrt{\epsilon c_s} k^3} (1 + i k c_s \tau) e^{-i k c_s \tau}, \quad (7.29)$$

the bulk-to-boundary propagator is

$$G_+(k; \tau) := u_k(\tau) u_k^*(0) = G_-(k; \tau)^*. \quad (7.30)$$

We also need its time and space derivatives,

$$\partial_\tau u_k(\tau) = -\frac{i}{2M_{\text{Pl}}\sqrt{\epsilon c_s} k^3} c_s^2 k^2 \tau e^{-i k c_s \tau}, \quad \partial_i \rightarrow i k_i \quad \text{in Fourier space.} \quad (7.31)$$

Finally, at linear order

$$\zeta = -H\pi, \quad (7.32)$$

so once the three-point function of  $\pi$  has been computed, the corresponding bispectrum of  $\zeta$  follows immediately.

Since the  $+$  and  $-$  contact diagrams are complex conjugates of each other, the total result is twice the real part of the  $+$  contribution. Equivalently, one may write it as

$$\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle' = 2 \text{Re} \left[ -i \int_{-\infty(1-i\epsilon)}^0 d\tau \langle \zeta_{\mathbf{k}_1}(0) \zeta_{\mathbf{k}_2}(0) \zeta_{\mathbf{k}_3}(0) H_{\text{int}}(\tau) \rangle \right], \quad (7.33)$$

where the prime indicates that the overall momentum-conserving delta function has been stripped off. The small  $i\epsilon$  tilt guarantees convergence at  $\tau \rightarrow -\infty$ .

**The  $\dot{\pi}^3$  interaction.** For

$$H_{\text{int}}^{\dot{\pi}^3}(\tau) = - \int d^3x a^3 C_{\dot{\pi}^3} \dot{\pi}^3, \quad (7.34)$$

the factors of the scale factor cancel against the three time derivatives, so the  $+$  diagram is proportional to

$$-i C_{\dot{\pi}^3} \int_{-\infty(1-i\epsilon)}^0 d\tau \partial_\tau G_+(\tau, k_1) \partial_\tau G_+(\tau, k_2) \partial_\tau G_+(\tau, k_3). \quad (7.35)$$

Using the explicit mode functions, the time integral is elementary and one finds

$$B_{\zeta}^{\dot{\pi}^3}(k_1, k_2, k_3) = \mathcal{A}_{\dot{\pi}^3} \frac{1}{(k_1 k_2 k_3)^3} S_{\dot{\pi}^3}(k_1, k_2, k_3), \quad (7.36)$$

where  $\mathcal{A}_{\dot{\pi}^3}$  is an overall amplitude depending on  $C_{\dot{\pi}^3}$ ,  $H$ ,  $\epsilon$ ,  $c_s$ , and  $M_{\text{Pl}}$ , and

$$S_{\dot{\pi}^3}(k_1, k_2, k_3) = \frac{k_1^2 k_2^2 k_3^2}{k_T^3}, \quad k_T := k_1 + k_2 + k_3. \quad (7.37)$$

The symmetric combination  $k_T$  of the lengths of the three momenta is known as the *total energy*. Thus the shape generated by  $\dot{\pi}^3$  is supported when the three momenta are comparable.

**The  $\dot{\pi}(\partial_i \pi)^2$  interaction.** For

$$H_{\text{int}}^{\dot{\pi}(\partial\pi)^2}(\tau) = - \int d^3x a^3 C_{\dot{\pi}(\partial\pi)^2} \dot{\pi} \frac{(\partial_i \pi)^2}{a^2}, \quad (7.38)$$

the scale factors again cancel, and the + contact diagram is proportional to

$$-i C_{\dot{\pi}(\partial\pi)^2} \int_{-\infty(1-i\epsilon)}^0 d\tau \left[ \partial_\tau G_+(\tau, k_1) (-\mathbf{k}_2 \cdot \mathbf{k}_3) G_+(\tau, k_2) G_+(\tau, k_3) + 2 \text{ perms.} \right]. \quad (7.39)$$

Performing the time integral and using

$$\mathbf{k}_2 \cdot \mathbf{k}_3 = \frac{1}{2} (k_1^2 - k_2^2 - k_3^2) \quad (7.40)$$

together with permutations, one obtains

$$B_{\zeta}^{\dot{\pi}(\partial\pi)^2}(k_1, k_2, k_3) = \mathcal{A}_{\dot{\pi}(\partial\pi)^2} \frac{1}{(k_1 k_2 k_3)^3} S_{\dot{\pi}(\partial\pi)^2}(k_1, k_2, k_3), \quad (7.41)$$

with

$$S_{\dot{\pi}(\partial\pi)^2}(k_1, k_2, k_3) = -\frac{1}{k_T} \sum_{i < j} k_i^2 k_j^2 + \frac{1}{2k_T^2} \sum_{i \neq j} k_i^2 k_j^3 + \frac{1}{8} \sum_i k_i^3. \quad (7.42)$$

Again, the overall amplitude  $\mathcal{A}_{\dot{\pi}(\partial\pi)^2}$  depends on  $C_{\dot{\pi}(\partial\pi)^2}$ ,  $H$ ,  $\epsilon$ ,  $c_s$ , and  $M_{\text{Pl}}$ , but the shape dependence is fixed. Both shapes of non-Gaussianity, as well as all other tree-level shapes to any order in derivatives, can be also obtained using the boostless cosmological bootstrap [?, ?].

**Squeezed limit.** To see that both operators are predominantly equilateral, let us study the squeezed limit

$$k_1 = q, \quad k_2 = k_3 = k, \quad x := \frac{q}{k} \ll 1. \quad (7.43)$$

Dropping the overall scale  $k^3$ , the first shape becomes

$$S_{\dot{\pi}^3}(x, 1, 1) = \frac{x^2}{(2+x)^3} = \frac{x^2}{8} - \frac{3x^3}{16} + \mathcal{O}(x^4). \quad (7.44)$$

Thus both the constant and linear terms vanish exactly.

For the second operator one finds

$$S_{\dot{\pi}(\partial\pi)^2}(x, 1, 1) = -\frac{1+2x^2}{2+x} + \frac{1+x^2+x^3}{(2+x)^2} + \frac{2+x^3}{8} = -\frac{11}{16}x^2 + \frac{9}{16}x^3 + \mathcal{O}(x^4). \quad (7.45)$$

Again, the constant and linear terms vanish exactly. In this sense, neither operator produces a local-type signal in the squeezed configuration. Both shapes are instead mostly supported

on configurations in which the three momenta are comparable, namely on equilateral triangles. We conclude that, at lowest order in derivatives and in the decoupling limit, the unitary EFT of inflation generates bispectra of predominantly equilateral type.

## 7.4 Constructing the open effective field theory of inflation

We now move from the unitary effective field theory of inflation to its open-system generalization. The physical motivation is that the Goldstone boson of broken time translations, which we denote by  $\pi$ , may interact with additional degrees of freedom that are not directly observed. This idea has been discussed many times in the literature. Earlier on open system dynamics emerged from warm inflation [?, ?, ?, ?, ?, ?]. An EFT description of open dynamics in a different formalism was developed in [?]. Here we follow the discussion of [?], which builds upon [?, ?, ?, ?].

After tracing over an unknown environment, the appropriate object is no longer an ordinary action for a single field, but a Schwinger–Keldysh effective functional depending on two copies of the Goldstone field,

$$S_{\text{eff}}[\pi_+, \pi_-] = S_\pi[\pi_+] - S_\pi[\pi_-] + S_{\text{IF}}[\pi_+, \pi_-], \quad (7.46)$$

where  $S_\pi$  denotes the unitary part of the dynamics and  $S_{\text{IF}}$  is the influence functional induced by the environment. Here we are using the different font  $\pi$  for the Greek letter pi because we will reserve the other font  $\pi$  for the normalized version of the field, which we will introduce shortly.

It is convenient to pass to the Keldysh basis,

$$\pi_r := \frac{\pi_+ + \pi_-}{2}, \quad \pi_a := \pi_+ - \pi_-, \quad \pi_\pm = \pi_r \pm \frac{1}{2}\pi_a. \quad (7.47)$$

As usual, the open effective functional must satisfy three general non-equilibrium constraints in (4.73).

These conditions encode, respectively, normalization, Hermiticity, and positivity of the reduced density matrix. In practice, they imply that the effective action starts at linear order in  $\pi_a$ , that odd powers of  $\pi_a$  come with real coefficients, and that even powers of  $\pi_a$  come with purely imaginary coefficients.

The symmetry structure is also modified relative to the unitary theory. Before tracing out the environment, the doubled closed system admits independent time translations on the  $+$  and  $-$  branches. Once the environment is integrated out, only the diagonal subgroup survives. As a result, in the Keldysh basis the diagonal time translation acts as

$$\pi_r(t, \mathbf{x}) \rightarrow \pi_r(t + \epsilon_r, \mathbf{x}) + \epsilon_r, \quad \pi_a(t, \mathbf{x}) \rightarrow \pi_a(t + \epsilon_r, \mathbf{x}). \quad (7.48)$$

Similarly, under the diagonal Lorentz boosts one has

$$\pi_r(x) \rightarrow \pi_r(\Lambda_r x) + \Lambda_r^0{}_\mu x^\mu - t, \quad \pi_a(x) \rightarrow \pi_a(\Lambda_r x). \quad (7.49)$$

The key point is that  $\pi_r$  transforms non-linearly, as the Goldstone boson of broken time translations, whereas  $\pi_a$  transforms linearly, like an ordinary matter field. This is the basic symmetry principle underlying the open EFT.

To construct the effective action, we assume locality in space and time and work directly in the decoupling limit. We further restrict to operators with at most one derivative per field. The invariant building blocks are then  $\pi_a$ , its derivative  $\partial_\mu \pi_a$ , and the combination<sup>31</sup>

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<sup>31</sup>The notation here is chosen to match that of [?].

$$P_\mu := \partial_\mu(t + \pi_r) = \delta_\mu^0 + \partial_\mu \pi_r. \quad (7.50)$$

From  $P_\mu$  one can form the usual invariant

$$P_\mu P^\mu + 1 = -2\dot{\pi}_r + (\partial_\mu \pi_r)^2, \quad (7.51)$$

which is the direct open-system analogue of the unitary EFT building block. The most general local effective Lagrangian may then be organized as an expansion in powers of  $\pi_a$ ,

$$\mathcal{L}_{\text{eff}} = \sum_{n=1}^{\infty} \mathcal{L}_n, \quad \mathcal{L}_n = \mathcal{O}(\pi_a^n). \quad (7.52)$$

At first order in  $\pi_a$ , the most general local structure with at most one derivative per field is

$$\mathcal{L}_1 = \sum_{n=0}^{\infty} (P_\mu P^\mu + 1)^n [\gamma_n \pi_a - \alpha_n P^\mu \partial_\mu \pi_a]. \quad (7.53)$$

The coefficients  $\alpha_n$  and  $\gamma_n$  are real. The terms proportional to  $\alpha_n$  contain the unitary kinetic structure and its higher-order symmetry completions, while the terms proportional to  $\gamma_n$  encode dissipative effects. In particular,  $\alpha_0$  is the analogue of the usual kinetic coefficient,  $\alpha_1$  controls the sound speed, and  $\gamma_1$  gives the leading dissipative term. Expanding to quadratic and cubic order, one finds the contributions

$$\mathcal{L}_1^{(2)} = (\alpha_0 - 2\alpha_1) \dot{\pi}_r \dot{\pi}_a - \alpha_0 \partial_i \pi_r \partial^i \pi_a - 2\gamma_1 \dot{\pi}_r \pi_a \quad (7.54)$$

$$\begin{aligned} \mathcal{L}_1^{(3)} = & (4\alpha_2 - 3\alpha_1) \dot{\pi}_r^2 \dot{\pi}_a + \alpha_1 (\partial_i \pi_r)^2 \dot{\pi}_a + 2\alpha_1 \dot{\pi}_r \partial_i \pi_r \partial^i \pi_a \\ & + (4\gamma_2 - \gamma_1) \dot{\pi}_r^2 \pi_a + \gamma_1 (\partial_i \pi_r)^2 \pi_a \end{aligned} \quad (7.55)$$

At second order in  $\pi_a$ , the most general local structure takes the schematic form

$$\mathcal{L}_2 = i \left[ \beta_1 \pi_a^2 + \beta_2 (\partial_\mu \pi_a)^2 + \beta_3 (-\dot{\pi}_a + \partial^\mu \pi_r \partial_\mu \pi_a) \pi_a + \beta_4 (-\dot{\pi}_a + \partial^\mu \pi_r \partial_\mu \pi_a)^2 + \dots \right] \quad (7.56)$$

where the coefficients  $\beta_i$  are real and the overall factor of  $i$  follows from the non-equilibrium constraints. These are the leading noise terms of the EFT. The coefficient  $\beta_1$  corresponds to the usual local noise, while  $\beta_2$  and  $\beta_4$  generate derivative corrections to the noise kernel. Expanding again to quadratic and cubic order, one finds the contributions

$$\mathcal{L}_2^{(2)} = i \left[ \beta_1 \pi_a^2 - (\beta_2 - \beta_4) \dot{\pi}_a^2 + \beta_2 (\partial_i \pi_a)^2 \right], \quad (7.57)$$

$$\begin{aligned} \mathcal{L}_2^{(3)} = & i \left[ -(\beta_3 - 2\beta_7) \dot{\pi}_r \dot{\pi}_a \pi_a + \beta_3 \partial_i \pi_r \partial^i \pi_a \pi_a + 2(\beta_4 + \beta_6 - \beta_8) \dot{\pi}_r \dot{\pi}_a^2 \right. \\ & \left. - 2\beta_4 \partial_i \pi_r \partial^i \pi_a \dot{\pi}_a - 2\beta_5 \dot{\pi}_r \pi_a^2 - 2\beta_6 \dot{\pi}_r (\partial_i \pi_a)^2 \right]. \end{aligned} \quad (7.58)$$

Finally, we can write the term that is cubic in the advanced field,

$$\begin{aligned} \mathcal{L}_3 = & \delta_1 \pi_a^3 + \delta_2 (\partial_\mu \pi_a)^2 \pi_a + \delta_3 (-\dot{\pi}_a + \partial^\mu \pi_r \partial_\mu \pi_a) \pi_a^2 + \delta_4 (-\dot{\pi}_a + \partial^\mu \pi_r \partial_\mu \pi_a) (\partial_\nu \pi_a)^2 \\ & + \delta_5 (-\dot{\pi}_a + \partial^\mu \pi_r \partial_\mu \pi_a)^2 \pi_a + \delta_6 (-\dot{\pi}_a + \partial^\mu \pi_r \partial_\mu \pi_a)^3 + \dots, \end{aligned} \quad (7.59)$$

This contributes only to cubic order, giving

$$\mathcal{L}_3^{(3)} = \delta_1 \pi_a^3 + (\delta_5 - \delta_2) \dot{\pi}_a^2 \pi_a + \delta_2 (\partial_i \pi_a)^2 \pi_a - \delta_4 (\partial_i \pi_a)^2 \dot{\pi}_a + (\delta_4 - \delta_6) \dot{\pi}_a^3, \quad (7.60)$$

## 7.5 The quadratic action, propagators and the power spectrum

Combining all the quadratic terms we have found so far, we obtain the action

$$S_{\text{eff}}^{(2)} = \int d^4x \left\{ (\alpha_0 - 2\alpha_1) a^2 \pi'_r \pi'_a - \alpha_0 a^2 \partial_i \pi_r \partial^i \pi_a \right. \\ \left. - 2a^3 \gamma_1 \pi'_r \pi_a + i \left[ \beta_1 a^4 \pi_a^2 - (\beta_2 - \beta_4) a^2 \pi_a'^2 + \beta_2 a^2 (\partial_i \pi_a)^2 \right] \right\}, \quad (7.61)$$

Because this is the most general quadratic open theory that is local in time, it must also contain the unitary EFT of inflation, which we discussed in previous sections. Indeed, that unitary theory is obtained by setting to zero the second line and re-defining

$$c_s^2 = \frac{\alpha_0}{\alpha_0 - 2\alpha_1}. \quad (7.62)$$

It is convenient to work with canonically normalized fields. To that end, we define the energy scale

$$f_\pi^4 \equiv \alpha_0 - 2\alpha_1. \quad (7.63)$$

We use this scale to define a canonically normalized Goldstone boson

$$\pi \equiv f_\pi^2 \pi. \quad (7.64)$$

Then, we also rescale the coefficients by

$$c_s^2 \equiv \frac{\alpha_0}{f_\pi^4}, \quad \gamma \equiv \frac{2\gamma_1}{f_\pi^4}, \quad \beta_i \equiv \frac{\beta_i}{f_\pi^4} \quad \text{for } i = 1, \dots, 8. \quad (7.65)$$

to obtain the quadratic action in its final form

$$S_{\text{eff}}^{(2)} = \int d^4x \left\{ a^2 \pi'_r \pi'_a - c_s^2 a^2 \partial_i \pi_r \partial^i \pi_a \right. \\ \left. - a^3 \gamma \pi'_r \pi_a + i \left[ \beta_1 a^4 \pi_a^2 - (\beta_2 - \beta_4) a^2 \pi_a'^2 + \beta_2 a^2 (\partial_i \pi_a)^2 \right] \right\}. \quad (7.66)$$

**Propagators** After integrating the quadratic action by parts, one may write it in bilinear form as

$$S_{\text{eff}}^{(2)} = -\frac{1}{2} \int d^4x \begin{pmatrix} \pi_r & \pi_a \end{pmatrix} \begin{pmatrix} 0 & \hat{D}_A \\ \hat{D}_R & -2i \hat{D}_K \end{pmatrix} \begin{pmatrix} \pi_r \\ \pi_a \end{pmatrix}. \quad (7.67)$$

The corresponding differential operators are

$$\hat{D}_A = a^2 \partial_\eta^2 + a^3 \left( \frac{2H}{a} - \gamma \right) \partial_\eta - a^2 c_s^2 \partial_i^2 - 3a^3 H \gamma, \quad (7.68)$$

$$\hat{D}_R = a^2 \partial_\eta^2 + a^3 \left( \frac{2H}{a} + \gamma \right) \partial_\eta - a^2 c_s^2 \partial_i^2, \quad (7.69)$$

$$\hat{D}_K = a^4 \beta_1 + a^2 (\beta_2 - \beta_4) (\partial_\eta^2 + 2H \partial_\eta) - a^2 \beta_2 \partial_i^2. \quad (7.70)$$

In the following, and as in the paper, it is convenient to set  $c_s = 1$  for simplicity.

The retarded and advanced propagators are defined by

$$\widehat{D}_R(x) G_R(x, y) = -i\delta(x - y), \quad \widehat{D}_A(x) G_A(x, y) = -i\delta(x - y), \quad (7.71)$$

with

$$G_R(x, y) = G_A(y, x). \quad (7.72)$$

In Fourier space, the retarded Green function obeys

$$\left( \partial_{\eta_1}^2 - \frac{2 + \gamma/H}{\eta_1} \partial_{\eta_1} + k^2 \right) G_R(k; \eta_1, \eta_2) = H^2 \eta_1^2 \delta(\eta_1 - \eta_2). \quad (7.73)$$

Defining

$$\nu_\gamma := \frac{3}{2} + \frac{\gamma}{2H}, \quad z_i := -k\eta_i, \quad (7.74)$$

the solution may be written as

$$G_R(k; \eta_1, \eta_2) = \frac{\pi^2 H^2}{k^3} \left( \frac{z_1}{z_2} \right)^{\nu_\gamma} z_2^3 \left[ Y_{\nu_\gamma}(z_1) J_{\nu_\gamma}(z_2) - J_{\nu_\gamma}(z_1) Y_{\nu_\gamma}(z_2) \right] \theta(\eta_1 - \eta_2), \quad (7.75)$$

Where  $Y$  and  $J$  are Bessel functions. The Keldysh propagator is not a Green function of a homogeneous equation, but is instead determined by the noise kernel,

$$G_K(x, y) = 2i \int d^4 z \sqrt{-g(z)} G_R(x, z) \widehat{D}_K(z) G_R(y, z). \quad (7.76)$$

Restricting to the first noise term proportional to  $\beta_1$ , one finds in Fourier space

$$G_{K,1}(k; \eta_1, \eta_2) = \frac{i\pi^2 \beta_1}{4k^3} (z_1 z_2)^{\nu_\gamma} \left[ Y_{\nu_\gamma}(z_1) Y_{\nu_\gamma}(z_2) A_{\nu_\gamma}^{(1)}(z_2) + J_{\nu_\gamma}(z_1) J_{\nu_\gamma}(z_2) C_{\nu_\gamma}^{(1)}(z_2) \right] \quad (7.77)$$

$$- (J_{\nu_\gamma}(z_1) Y_{\nu_\gamma}(z_2) + J_{\nu_\gamma}(z_2) Y_{\nu_\gamma}(z_1)) B_{\nu_\gamma}^{(1)}(z_2) \Big] + (1 \leftrightarrow 2), \quad (7.78)$$

where

$$A_{\nu_\gamma}^{(1)}(z) := \int_z^\infty dz' z'^{2-2\nu_\gamma} J_{\nu_\gamma}(z')^2, \quad (7.79)$$

$$B_{\nu_\gamma}^{(1)}(z) := \int_z^\infty dz' z'^{2-2\nu_\gamma} J_{\nu_\gamma}(z') Y_{\nu_\gamma}(z'), \quad (7.80)$$

$$C_{\nu_\gamma}^{(1)}(z) := \int_z^\infty dz' z'^{2-2\nu_\gamma} Y_{\nu_\gamma}(z')^2. \quad (7.81)$$

**The power spectrum** The power spectrum is obtained from the coincident-time limit of the Keldysh propagator. Using

$$\zeta = -\frac{H}{f_\pi^2} \pi, \quad (7.82)$$

one defines

$$\Delta_\zeta^2(k) := \frac{k^3}{2\pi^2} P_\zeta(k), \quad \langle \zeta_{\mathbf{k}} \zeta_{\mathbf{k}'} \rangle = (2\pi)^3 \delta(\mathbf{k} + \mathbf{k}') P_\zeta(k). \quad (7.83)$$

For the  $\beta_1$  contribution, the exact result is

$$P_\zeta(k) = \frac{\pi^2 \beta_1}{8k^3} \frac{H^2}{f_\pi^4} z^{2\nu_\gamma} \left[ Y_{\nu_\gamma}(z)^2 A_{\nu_\gamma}^{(1)}(z) + J_{\nu_\gamma}(z)^2 C_{\nu_\gamma}^{(1)}(z) - 2J_{\nu_\gamma}(z) Y_{\nu_\gamma}(z) B_{\nu_\gamma}^{(1)}(z) \right]. \quad (7.84)$$

Explicit results for the contributions from other noise terms and the case of  $c_s \neq 1$  can be found in [?]. In the super-Hubble regime  $z \ll 1$ , the power spectrum freezes to

$$\Delta_\zeta^2(k) = \frac{1}{4} \frac{\beta_1}{H^2} \frac{H^4}{f_\pi^4} 2^{2\nu_\gamma} \frac{\Gamma(\nu_\gamma - 1) \Gamma(\nu_\gamma)^2}{\Gamma(\nu_\gamma - \frac{1}{2}) \Gamma(2\nu_\gamma - \frac{1}{2})}. \quad (7.85)$$

Expanding this result in the two opposite dissipation regimes gives

$$\Delta_\zeta^2(k) \propto \begin{cases} \frac{\beta_1}{H^2} \frac{H^4}{f_\pi^4} + \mathcal{O}\left(\frac{\gamma}{H}\right), & \gamma \ll H, \\ \frac{\beta_1}{H^2} \frac{H^4}{f_\pi^4} \sqrt{\frac{H}{\gamma}} \left[ 1 + \mathcal{O}\left(\frac{H}{\gamma}\right) \right], & \gamma \gg H. \end{cases} \quad (7.86)$$

If one further assumes thermal equilibrium of the environment, the fluctuation-dissipation relation imposes  $\beta_1 = 2\pi\gamma T$ , so that in the strongly dissipative regime

$$\Delta_\zeta^2(k) \propto \frac{T}{H} \frac{H^4}{f_\pi^4} \sqrt{\frac{\gamma}{H}}. \quad (7.87)$$

These expressions make the physical interpretation transparent:  $G_R$  captures the dissipative response of the Goldstone mode, while  $G_K$  captures the fluctuations injected by the environment. The late-time power spectrum is determined by the coincident-time Keldysh propagator and remains nearly scale invariant, but with an amplitude controlled by the interplay of noise and dissipation. Note that since the system has finite memory of the past, whatever initial conditions were imposed in the infinite past are quickly forgotten, and the power spectrum is driven to the above value by the noise fluctuations of the environment. This is a conceptually different paradigm for the generation of primordial perturbations and it stands in stark contrast to the more often discussed quantum fluctuations of the Bunch-Davies state.

## 7.6 Feynman rules for in-in correlators: the $r/a$ basis

The same perturbative expansion discussed in the previous subsection can be reorganized in the Keldysh, or retarded/advanced, basis. Being just a change of basis, this is always completely equivalent to the  $+/-$  description, but it reshuffles the perturbative expansion into terms that make causality and the distinction between response and fluctuations more manifest. In particular,  $r$ -fields are associated with anti-commutators and statistical fluctuations, while  $a$ -fields are associated with commutators and retarded response. In a unitary closed system this is often a less economical basis for calculation unless one is particularly interested in the semiclassical limit or certain aspects of causality. In that case, the plus and minus basis is much more widely used for calculations. Conversely, the  $r/a$  basis is quite convenient for calculations in open theories, and therefore we briefly review it below. The rest of this section follows closely our previous discussion in Section 6.4.

The perturbative rules for  $B_n$  now follow by rewriting the interaction vertices and propagators in the  $r/a$  basis.

1. For an  $n$ -point correlator, draw all diagrams with  $n$  external lines ending on the late-time boundary  $\tau \rightarrow 0$ , together with some number  $V$  of bulk interaction vertices and  $I$  internal lines connecting pairs of vertices. Time runs from the infinite past  $\tau \rightarrow -\infty$  to the future conformal boundary  $\tau \rightarrow 0$ .

2. Rewrite every interaction vertex in the  $r/a$  basis. For a unitary closed system, each vertex contains an odd number of  $a$ -fields. For example, a cubic interaction gives vertices of the schematic form  $\phi_r^2\phi_a$  and  $\phi_a^3$ , while a quartic interaction gives  $\phi_r^3\phi_a$  and  $\phi_r\phi_a^3$ . Each such term is treated as a separate vertex, with the coefficient inherited from the expansion of the action. Notice that because of the unitarity constraints, every vertex has at least one power of the advanced fields.
3. Assign to each external line a momentum  $\mathbf{k}_a$ , with  $a = 1, \dots, n$ , all taken to flow from the bulk vertex toward the future boundary. Assign to each internal line an integration momentum  $\mathbf{p}_m$ , with  $m = 1, \dots, I$ , and integrate over all such momenta.
4. For each bulk vertex, include the corresponding interaction coefficient together with the integral over its time  $\tau_i$ ,

$$\int_{-\infty}^0 d\tau_i. \quad (7.88)$$

The  $i\epsilon$  prescription is inherited from the original  $+/-$  contour representation, but in the  $r/a$  basis it is usually left implicit once the retarded and Keldysh propagators are used.

5. For every external line connecting a bulk vertex at time  $\tau_i$  to a boundary operator at  $\tau = 0$ , insert the appropriate bulk-to-boundary propagator in the  $r/a$  basis. Starting from

$$\phi_{\mathbf{k}}(\tau) = u_{\mathbf{k}}(\tau)a_{\mathbf{k}} + u_{\mathbf{k}}^*(\tau)a_{-\mathbf{k}}^\dagger, \quad (7.89)$$

one defines

$$G_r(k; \tau_i, 0) = \frac{1}{2} \left( u_{\mathbf{k}}(\tau_i)u_{\mathbf{k}}^*(0) + u_{\mathbf{k}}^*(\tau_i)u_{\mathbf{k}}(0) \right), \quad (7.90)$$

$$G_a(k; \tau_i, 0) = u_{\mathbf{k}}(\tau_i)u_{\mathbf{k}}^*(0) - u_{\mathbf{k}}^*(\tau_i)u_{\mathbf{k}}(0). \quad (7.91)$$

6. For every internal line joining vertices at times  $\tau_i$  and  $\tau_j$ , insert the appropriate free two-point function in the  $r/a$  basis. These are

$$G_K(\tau_i, \tau_j; \mathbf{p}) = \langle\langle \phi_r(\tau_i, \mathbf{p})\phi_r(\tau_j, -\mathbf{p}) \rangle\rangle = \frac{1}{2} \langle\{ \phi_{\mathbf{p}}(\tau_i), \phi_{-\mathbf{p}}(\tau_j) \}\rangle, \quad (7.92)$$

$$G_R(\tau_i, \tau_j; \mathbf{p}) = \langle\langle \phi_r(\tau_i, \mathbf{p})\phi_a(\tau_j, -\mathbf{p}) \rangle\rangle = -i\theta(\tau_i - \tau_j)\langle[\phi_{\mathbf{p}}(\tau_i), \phi_{-\mathbf{p}}(\tau_j)]\rangle. \quad (7.93)$$

Thus  $G_R$  encodes causal response, while  $G_K$  encodes statistical fluctuations.

7. Because  $G_{aa} = 0$ , any diagram containing an internal  $a$ - $a$  line vanishes. More generally, the  $r/a$  basis makes causality manifest: every non-vanishing connected diagram must contain enough retarded or advanced propagators to connect each  $a$ -field to the rest of the graph. In practice, this strongly reduces the number of non-zero diagrams compared with the  $+/-$  basis.
8. At each vertex impose momentum conservation. After using all vertex delta functions, a single overall factor

$$(2\pi)^3 \delta\left(\sum_{a=1}^n \mathbf{k}_a\right) \quad (7.94)$$

remains, as required for a connected correlator.

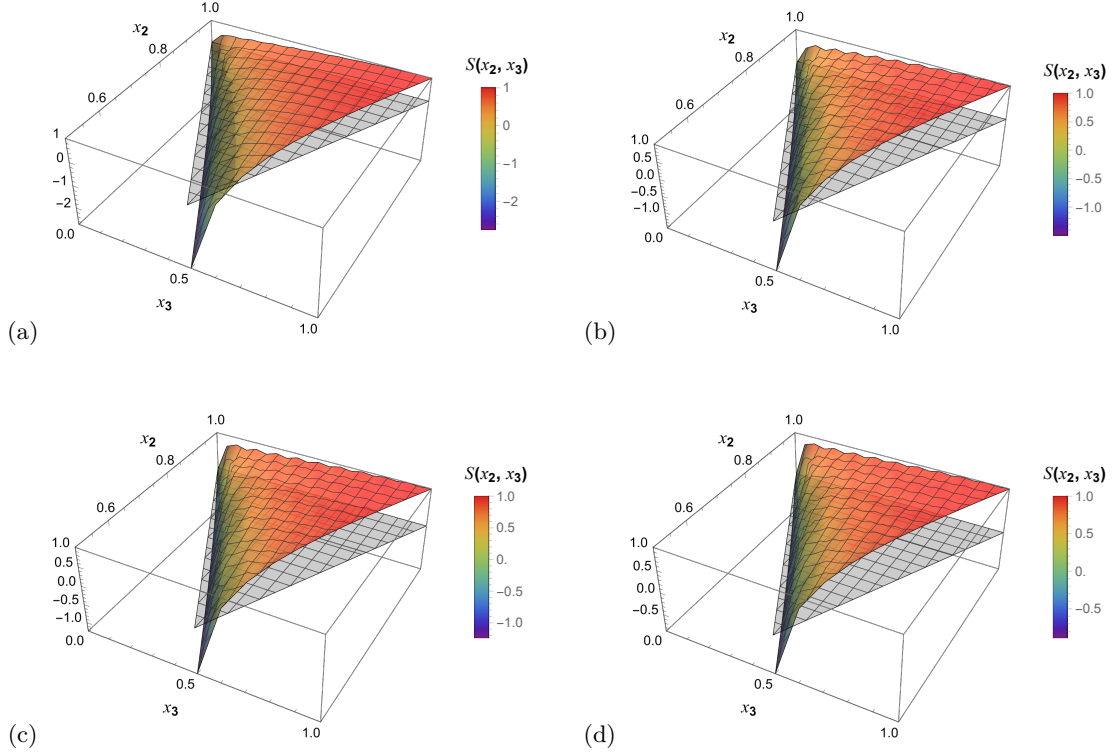


图 7: The figure shows numerical evaluations of the shape of the bispectrum for small dissipation,  $\gamma = 0.001H$  (small oscillations are numerical artifacts). The axes are the ratio of external momenta,  $x_{2,3} := k_{2,3}/k_1$ . **a.**  $(\partial_i \pi_r)^2 \pi_a$  operator; **b.**  $\dot{\pi}_r^2 \pi_a$  operator; **c.**  $\dot{\pi}_r \pi_a^2$  operator; **d.**  $\pi_a^3$  operator. Figure taken from [?].

9. Divide by the appropriate symmetry factor of the diagram. As in the  $+/-$  basis, this removes the overcounting associated with identical vertices and identical internal contractions. For details see [?, ?].
10. Finally, sum over all distinct non-vanishing  $r/a$  labelings of propagators obtained from the expansion of the interaction vertices.

We will now use these Feynman rules to compute primordial non-Gaussianities in the open EFT of inflation.

## 7.7 Non-Gaussianity

Combining all cubic terms we found in the construction of the open EFT of inflation and rescaling them to the canonical basis, we find the following cubic action

$$\begin{aligned}
 S_{\text{eff}}^{(3)} = \frac{1}{f_\pi^2} \int d^4x \Big\{ & \left[ 4\alpha_2 - \frac{3}{2}(c_s^2 - 1) \right] a \pi_r'^2 \pi_a' + \frac{1}{2}(c_s^2 - 1) a \left[ (\partial_i \pi_r)^2 \pi_a' + 2\pi_r' \partial_i \pi_r \partial^i \pi_a \right] \\
 & + \left( 4\gamma_2 - \frac{\gamma}{2} \right) a^2 \pi_r'^2 \pi_a + \frac{\gamma}{2} a^2 (\partial_i \pi_r)^2 \pi_a \\
 & + i \left[ (2\beta_7 - \beta_3) a^2 \pi_r' \pi_a' \pi_a + \beta_3 a^2 \partial_i \pi_r \partial^i \pi_a \pi_a + 2(\beta_4 + \beta_6 - \beta_8) a \pi_r' \pi_a'^2 \right. \\
 & \quad \left. - 2\beta_4 a \partial_i \pi_r \partial^i \pi_a \pi_a' - 2\beta_5 a^3 \pi_r' \pi_a'^2 - 2\beta_6 a \pi_r' (\partial_i \pi_a)^2 \right] \\
 & + \delta_1 a^4 \pi_a^3 + (\delta_5 - \delta_2) a^2 \pi_a'^2 \pi_a + \delta_2 a^2 (\partial_i \pi_a)^2 \pi_a - \delta_4 a (\partial_i \pi_a)^2 \pi_a' + (\delta_4 - \delta_6) a \pi_a'^3 \Big\},
 \end{aligned} \tag{7.95}$$

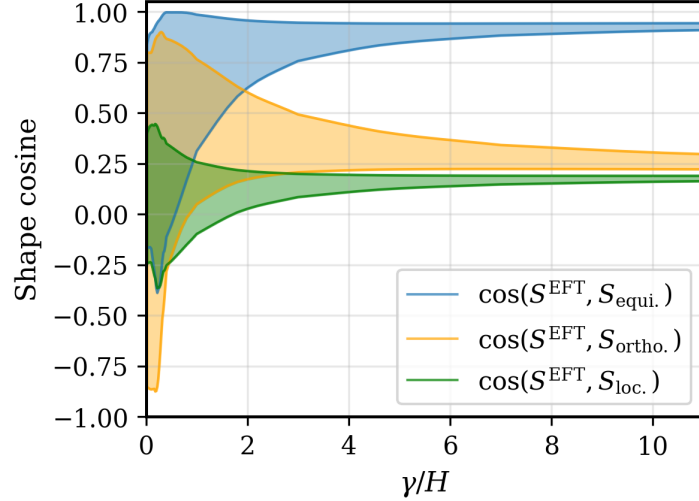


Figure 8: Shape correlation of the Open EFToI bispectrum with the standard *equilateral*, *orthogonal* and *local* templates, as used in *Planck*’s non-Gaussianity analysis. The shaded regions show the envelope obtained by varying  $c_s$  between 0.01 and 1. At strong dissipation, the signal correlates with the equilateral template at the 90% level. At weak dissipation, the signal peaks in a strongly folded configuration whose correlations with standard templates depend on  $c_s$  and are not generically orthogonal. Figure taken from [?].

This clearly contains a large number of new effective parameters to be constrained by data. However, a particularly remarkable fact is that the size of some of these interaction terms is fixed by coefficients appearing in the quadratic action, namely the speed of sound  $c_s$  and the dissipation  $\gamma$  [?]. Focusing on these we have

$$S_{\text{eff}}^{(3)} = \int d^4x \sqrt{-g} \left\{ \frac{c_s^2 - 1}{2f_\pi^2} \left[ (\partial_i \pi_r)^2 \dot{\pi}_a + 2\dot{\pi}_r \partial_i \pi_r \partial^i \pi_a - 3\dot{\pi}_r^2 \dot{\pi}_a \right] + \frac{\gamma}{2f_\pi^2} \left[ (\partial_i \pi_r)^2 - \dot{\pi}_r^2 \right] \pi_a \right\},$$

The tree-level bispectrum can be computed using the Feynman rules in the previous section from this set of operators, or the enlarged set written above. Unfortunately, only the operators that are cubic in the advanced fields can be computed analytically, while for all others one needs to resort to numerics. Representative numerical results are shown below. In Figure ??, we show some numerical results for a variety of cubic couplings. These shapes are particularly interesting because, in the weak-dissipation regime, they have a large signal for folded configurations. This makes them particularly different from the non-Gaussianity encountered from self and gravitational interactions of the inflaton in the Bunch-Davies state with unitary evolution. This difference can be captured quantitatively using a cosine [?]. A high value, positive or negative, indicates very similar shapes, while a low value means shapes that would look very different in the data and can be discriminated effectively. The cosines with the most commonly studied shapes, which are those arising in unitary single-field slow-roll inflation with a Bunch-Davies state, are plotted in Figure ?? [?].

## 8 Stochastic inflation and late-time behaviour in de Sitter

This section introduces stochastic inflation as an effective description of the long-wavelength dynamics of light scalar fields in de Sitter space. The central idea is to separate the field into long- and short-wavelength modes using a time-dependent coarse-graining scale of order the Hubble radius. Because modes continuously cross this scale as the universe expands, the

long-wavelength sector behaves as an open system: its evolution contains both a deterministic drift and a stochastic force generated by the short modes entering the coarse-grained sector. We first illustrate this mechanism for free massless and light massive fields, for which the exact mode functions make the secular growth and its eventual saturation transparent. We then derive the corresponding Langevin and Fokker–Planck equations and use them to study interacting theories, with particular emphasis on a quartic potential. This provides a simple setting in which perturbative secular growth, stochastic resummation, and the non-perturbative late-time equilibrium distribution can be compared explicitly. This review is elementary and captures exclusively the leading stochastic behavior. For a more in-depth discussion covering the literature, see for example [?, ?, ?]. The open-system perspective adopted here and my own introduction to the subject owes a particular debt to C. P. Burgess, R. Holman and collaborators, whose work connected inflationary decoherence and stochastic inflation to master equations, open effective field theories and late-time resummation [?, ?, ?].

## 8.1 A time-dependent split between system and environment

We consider a light spectator scalar field propagating on a fixed, exact de Sitter background. The scalar is assumed not to contribute appreciably to the stress-energy tensor and therefore does not affect the geometry. In spatially flat coordinates,

$$ds^2 = -dt^2 + a^2(t) d\mathbf{x}^2, \quad a(t) = e^{Ht}, \quad (8.1)$$

where the Hubble parameter  $H$  is constant. The action for the scalar is

$$S = - \int d^4x \sqrt{-g} \left[ \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + V(\phi) \right]. \quad (8.2)$$

We shall eventually be interested in the quartic potential

$$V(\phi) = \frac{\lambda}{4!} \phi^4, \quad (8.3)$$

but it is useful first to consider the free massless theory and then a free field with a small mass. The latter provides an exactly solvable example in which the origin and interpretation of secular terms can be seen particularly clearly.

To turn this Hamiltonian quantum field theory into an open system problem, we have to separate system and environment. Throughout these notes, we have considered only open system effects that emerge because of interactions between a system and its environment. However, there is a conceptually different way in which open system dynamics may emerge. The key idea is to consider a bipartition between system and environment that is time dependent. This is, in fact, the only element that is relevant to understanding the *leading order* description of stochastic inflation as originally understood by Starobinsky and Yokoyama [?, ?]. Additional open system effects do arise from genuine interactions, and have been studied in an extensive body of literature. That is the challenging and hotly debated part of the problem and will not be discussed in these notes.

We separate the field into long- and short-wavelength components according to

$$\phi(t, \mathbf{x}) = \phi_L(t, \mathbf{x}) + \phi_S(t, \mathbf{x}), \quad (8.4)$$

$$\phi_L(t, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} W\left(\frac{k}{\epsilon a H}\right) \phi_{\mathbf{k}}(t) e^{i\mathbf{k}\cdot\mathbf{x}}, \quad (8.5)$$

where  $W$  is a window function and  $\epsilon \lesssim 1$ . The precise definition of the window will not be important for the leading infrared behavior. Now we define our system to consist of the long modes  $\phi_L$ , and our environment to consist of the short modes  $\phi_S$ . This separation is

time dependent because of the argument of the window function  $W$ , which roughly separates long modes  $k_L < \epsilon aH$  from short modes  $k_S > \epsilon aH$ . Even if at any given time long and short modes did not interact, open effects still emerge from the fact that short modes drift into our system as the universe expands and the scale factor  $a$  grows. When neglecting system-environment interactions, this phenomenon is sufficiently simple that we will be able to describe it non-perturbatively and compare the result with the misleading results of perturbation theory. This whole section studies the effect of a time-dependent bipartition in a series of examples of increasing complexity.

**The massless free scalar** As the first example, we consider a massless free scalar in the Bunch–Davies state. The exact mode function is

$$\phi_{\mathbf{k}}(\eta) = \frac{H}{\sqrt{2k^3}} (1 + ik\eta) e^{-ik\eta}, \quad \eta = -\frac{1}{aH}. \quad (8.6)$$

Its absolute square is therefore

$$|\phi_{\mathbf{k}}(\eta)|^2 = \frac{H^2}{2k^3} (1 + k^2\eta^2). \quad (8.7)$$

At late times, when  $k \ll aH$ , this approaches

$$|\phi_{\mathbf{k}}(t)|^2 \simeq \frac{H^2}{2k^3}. \quad (8.8)$$

Our system comprises of the long-wavelength modes. To focus on the phenomenon that is relevant for us, namely the time dependence of the system-environment bipartition, we introduce an artificial deep infrared (IR) cutoff  $k_{\text{IR}}$  and remove all modes longer than this from our system. The equal-point variance is then given by the integral

$$\langle \phi_L^2(t, \mathbf{x}) \rangle \simeq \int_{k_{\text{IR}}}^{\epsilon a(t)H} \frac{d^3k}{(2\pi)^3} \frac{H^2}{2k^3}, \quad (8.9)$$

and hence

$$\langle \phi_L^2(t, \mathbf{x}) \rangle \simeq \frac{H^2}{4\pi^2} \ln \left( \frac{\epsilon a(t)H}{k_{\text{IR}}} \right). \quad (8.10)$$

Equivalently, measuring the growth relative to some initial time  $t_0$ ,

$$\langle \phi_L^2(t, \mathbf{x}) \rangle - \langle \phi_L^2(t_0, \mathbf{x}) \rangle \simeq \frac{H^3}{4\pi^2} (t - t_0) = \frac{H^2}{4\pi^2} N, \quad (8.11)$$

where

$$N = \ln \left( \frac{a(t)}{a(t_0)} \right) = H(t - t_0) \quad (8.12)$$

is the number of elapsed e-folds. This result tells us that the two-point function grows without bound as time progresses, an instance of a secular effect.

The growth in (??) has a simple physical interpretation. During each Hubble time, a new shell of Fourier modes crosses the bipartition scale  $\epsilon aH$ , leaves the environment and becomes part of our system. Each such shell contributes an approximately independent fluctuation of characteristic magnitude extracted from (??),

$$\delta\phi_L \sim \frac{H}{2\pi}. \quad (8.13)$$

After  $N$  e-folds, the long field  $\phi_L$  in our system has therefore received approximately  $N$  independent kicks. As in an ordinary random walk, the mean displacement remains zero,  $\langle \phi_L \rangle = 0$ , while the variance grows in proportion to the number of steps,

$$\langle \phi_L^2 \rangle \sim N \left( \frac{H}{2\pi} \right)^2. \quad (8.14)$$

This reproduces (??). The long-wavelength scalar thus undergoes diffusion in field space. In this free massless theory, the linear growth of the variance is an exact, non-perturbative result. It reflects the continual accumulation of independent modes in the long-wavelength sector. We stress again that, since this is a free theory on a homogeneous background, this effect emerges even in the absence of any instantaneous interaction between long (system) and short (environment) wavelength modes.

## 8.2 A light massive scalar

We now consider a free scalar field with a small positive mass,

$$V(\phi) = \frac{1}{2} m^2 \phi^2, \quad 0 < m^2 \ll H^2. \quad (8.15)$$

This theory is again free and hence provides a particularly simple example in which the late-time behavior can be determined exactly and can be contrasted with a perturbative treatment. It will later serve as a useful toy model for understanding how secular terms may arise when an otherwise regular result is expanded perturbatively.

We'll work in conformal time,  $\eta = -1/(aH)$  for  $-\infty < \eta < 0$ . Expanding the field as

$$\phi(\eta, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \left[ a_{\mathbf{k}} \phi_k(\eta) e^{i\mathbf{k}\cdot\mathbf{x}} + a_{\mathbf{k}}^\dagger \phi_k^*(\eta) e^{-i\mathbf{k}\cdot\mathbf{x}} \right], \quad (8.16)$$

the mode functions obey

$$\phi_k'' - \frac{2}{\eta} \phi_k' + \left( k^2 + \frac{m^2}{H^2 \eta^2} \right) \phi_k = 0. \quad (8.17)$$

The Bunch–Davies solution is

$$\phi_k(\eta) = \frac{\sqrt{\pi}}{2} e^{i\frac{\pi}{2}(\nu+\frac{1}{2})} H(-\eta)^{3/2} H_\nu^{(1)}(-k\eta), \quad \nu = \sqrt{\frac{9}{4} - \frac{m^2}{H^2}}. \quad (8.18)$$

The phase has been chosen so that at early times, when  $-k\eta \gg 1$ , the canonically normalized field approaches the positive-frequency Minkowski mode.

For a light scalar it is useful to introduce

$$\Delta \equiv \frac{3}{2} - \nu. \quad (8.19)$$

When  $m^2 \ll H^2$ ,

$$\Delta = \frac{m^2}{3H^2} + \mathcal{O}\left(\frac{m^4}{H^4}\right). \quad (8.20)$$

Using the small-argument behavior of the Hankel function, the super-Hubble mode function becomes

$$\phi_k(\eta) \simeq -i \frac{2^{\nu-1} \Gamma(\nu)}{\sqrt{\pi}} \frac{H}{k^{3/2}} \left( \frac{k}{aH} \right)^\Delta, \quad k \ll aH. \quad (8.21)$$

Its absolute value is therefore

$$|\phi_k(\eta)|^2 \simeq \frac{H^2}{2k^3} \mathcal{C}(\nu) \left( \frac{k}{aH} \right)^{2\Delta}, \quad (8.22)$$

where

$$\mathcal{C}(\nu) = \frac{2^{2\nu-1} \Gamma^2(\nu)}{\pi} \xrightarrow{m \rightarrow 0} 1. \quad (8.23)$$

In the following we shall retain the leading infrared behavior and set  $\mathcal{C}(\nu) \simeq 1$ . The important difference from the massless case is the factor

$$\left( \frac{k}{aH} \right)^{2\Delta}. \quad (8.24)$$

At late times, the amplitude of a mode  $k$  is no longer constant but decays slowly as  $a^{-\Delta}$ . The mass therefore causes long modes, which are inside our system, to contribute progressively less as time progresses. As we will see now, this provides a restoring force that leads to a finite late-time behavior.

Let's consider again our bipartition between system and environment. Again field modes with  $k_{\text{IR}} < k < \epsilon a(t)H$ , define our system, with  $k_{\text{IR}}$  a fixed infrared cutoff and  $\epsilon \lesssim 1$ . Using (??), the variance for the modes in our system is

$$\langle \phi_L^2(t, \mathbf{x}) \rangle \simeq \int_{k_{\text{IR}}}^{\epsilon aH} \frac{d^3k}{(2\pi)^3} \frac{H^2}{2k^3} \left( \frac{k}{aH} \right)^{2\Delta} = \frac{H^2}{4\pi^2} \int_{k_{\text{IR}}}^{\epsilon aH} \frac{dk}{k} \left( \frac{k}{aH} \right)^{2\Delta} \quad (8.25)$$

$$= \frac{H^2}{8\pi^2 \Delta} \left[ \epsilon^{2\Delta} - \left( \frac{k_{\text{IR}}}{a(t)H} \right)^{2\Delta} \right] = \frac{H^2 \epsilon^{2\Delta}}{8\pi^2 \Delta} [1 - e^{-2N\Delta}] \quad (8.26)$$

where in the last step we chose  $k_{\text{IR}} = \epsilon a(t_0)H$  for some sufficiently early time  $t_0$ . At leading order in the light-mass limit, we may approximate  $\epsilon^{2\Delta} \simeq 1$ , yielding

$$\langle \phi_L^2(t, \mathbf{x}) \rangle \simeq \frac{H^2}{8\pi^2 \Delta} [1 - e^{-2N\Delta}]. \quad (8.27)$$

Using (??), this can equivalently be written as

$$\langle \phi_L^2(t, \mathbf{x}) \rangle \simeq \frac{3H^4}{8\pi^2 m^2} \left[ 1 - \exp \left( -\frac{2m^2}{3H} (t - t_0) \right) \right]. \quad (8.28)$$

This non-perturbative results, valid in the superHubble regime for a sharp window function, tell us that the late-time divergence we encountered in the massless free scalar, (??), is absent in the presence of a small but finite mass. Intuitively, the restoring force of the quadratic potential balances the spreading due to new modes entering into the long-wavelength system from the short-wavelength environment at every time step.

**Perturbative expansion and secular growth** We now compare the non-perturbative result above with what would have been obtained by treating the mass term perturbatively. The simplest way to do so is to expand our non-perturbative result in powers of  $\Delta$ , or equivalently in powers of  $m^2/H^2$ . To set the expectation, we notice that this is an expansion around  $m = 0$ , which, as we just saw, has a genuine divergence at late times. This warns us that perturbation theory will likely feature these secular divergences as well.

Expanding (??) one finds

$$\langle \phi_L^2(t, \mathbf{x}) \rangle = \frac{H^2}{4\pi^2} \left[ N - N^2 \Delta + \frac{2}{3} N^3 \Delta^2 + \dots \right] \quad (8.29)$$

$$\simeq \frac{H^2}{4\pi^2} \left[ N - \frac{m^2}{3H^2} N^2 + \frac{2m^4}{27H^4} N^3 + \dots \right], \quad (8.30)$$

where we used  $\Delta \simeq m^2/(3H^2)$ . At early times, when  $N\Delta \ll 1$ , this expansion agrees with the non-perturbative result. At later times, however, the secular powers of  $N$  compensate the smallness of the mass, and perturbation theory breaks down when

$$N\Delta \sim 1, \quad \text{or equivalently} \quad N \sim \frac{H^2}{m^2}. \quad (8.31)$$

We conclude that a calculation performed perturbatively from the outset would therefore appear to predict an increasingly large correction to the massless result.

The non-perturbative solution shows that this apparent breakdown is not a physical instability but an artefact of perturbation theory. The full variance (??) remains finite and approaches

$$\lim_{t \rightarrow \infty} \langle \phi_L^2(t, \mathbf{x}) \rangle = \frac{H^2}{8\pi^2 \Delta} \simeq \frac{3H^4}{8\pi^2 m^2}. \quad (8.32)$$

The secular terms arise from expanding the factor  $e^{-2\Delta N}$  in powers of  $\Delta$ . Keeping this exponential unexpanded resums the entire series of terms enhanced by powers of  $N\Delta$  and restores the correct regular late-time behavior. This provides a canonical example in which secular growth is an artefact of strict perturbation theory rather than a property of the theory.

So far, our discussion of light fields in de Sitter was connected to open systems because of the bipartition of the Hilbert space, but did not employ any of the techniques we have developed in these notes. This is because we have so far only studied three theories for which a non-perturbative solution is readily available.

We will now move on to interacting theories, or the full solution is unknown. To make progress there, we will develop a technique that sums the effect of the time-dependent partition between system and environment. As we mentioned above, we will effectively neglect all instantaneous interactions between system and environment, Since these do not contribute to leading order.

### 8.3 A Langevin equation for the long-wavelength field

We now turn to the massless interacting theory,

$$V(\phi) = \frac{\lambda}{4!} \phi^4, \quad \lambda > 0. \quad (8.33)$$

As in the massive example, strict perturbation theory produces terms that grow secularly with the number of elapsed e-folds. For instance, the leading infrared terms in the equal-point two-point function are [?]

$$\langle \phi_L^2 \rangle = \frac{H^2}{4\pi^2} \left[ N - \frac{\lambda}{36\pi^2} N^3 + \frac{\lambda^2}{720\pi^4} N^5 - \frac{53\lambda^3}{544320\pi^6} N^7 + \dots \right]. \quad (8.34)$$

The first term is tree-level, the second is at one loop, and so on. More generally,

$$\langle \phi_L^2 \rangle_{n\text{-loop}} \sim H^2 \lambda^n N^{2n+1}. \quad (8.35)$$

The effective expansion parameter is therefore  $\lambda N^2$ , and perturbation theory ceases to be reliable when  $N \sim \lambda^{-1/2}$ . We shall derive an evolution equation that captures these leading infrared effects without expanding the evolution perturbatively in time.

**Coarse graining in phase space** A systematic derivation keeps both the coarse-grained field and its velocity as phase-space variables [?, ?]; for a recent open-system formulation, see also [?]. At leading order in the stochastic approximation, the short modes that cross the coarse-graining scale may be evaluated using the free Bunch–Davies mode functions. We define

$$\phi_L(t, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} W_t(k) \left[ a_{\mathbf{k}} u_k(t) e^{i\mathbf{k}\cdot\mathbf{x}} + a_{\mathbf{k}}^\dagger u_k^*(t) e^{-i\mathbf{k}\cdot\mathbf{x}} \right], \quad (8.36)$$

$$v_L(t, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} W_t(k) \left[ a_{\mathbf{k}} \dot{u}_k(t) e^{i\mathbf{k}\cdot\mathbf{x}} + a_{\mathbf{k}}^\dagger \dot{u}_k^*(t) e^{-i\mathbf{k}\cdot\mathbf{x}} \right], \quad (8.37)$$

where

$$W_t(k) = W\left(\frac{k}{k_c(t)}\right), \quad k_c(t) = \epsilon a(t) H. \quad (8.38)$$

Because the window is time dependent,  $v_L$  is not equal to  $\dot{\phi}_L$ . The microscopic Heisenberg field obeys

$$\ddot{\phi}(t, \mathbf{x}) + 3H\dot{\phi}(t, \mathbf{x}) - \frac{1}{a^2(t)} \nabla^2 \phi(t, \mathbf{x}) + V'(\phi(t, \mathbf{x})) = 0. \quad (8.39)$$

Differentiating the two definitions and using the microscopic field equation gives, at leading zeroth order in spatial gradients and in instantaneous long–short interactions,

$$\dot{\phi}_L = v_L + \hat{\xi}_\phi, \quad (8.40)$$

$$\dot{v}_L = -3Hv_L - V'(\phi_L) + \hat{\xi}_v, \quad (8.41)$$

where the two noise operators are

$$\hat{\xi}_\phi(t, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \dot{W}_t(k) \left[ a_{\mathbf{k}} u_k(t) e^{i\mathbf{k}\cdot\mathbf{x}} + a_{\mathbf{k}}^\dagger u_k^*(t) e^{-i\mathbf{k}\cdot\mathbf{x}} \right], \quad (8.42)$$

$$\hat{\xi}_v(t, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \dot{W}_t(k) \left[ a_{\mathbf{k}} \dot{u}_k(t) e^{i\mathbf{k}\cdot\mathbf{x}} + a_{\mathbf{k}}^\dagger \dot{u}_k^*(t) e^{-i\mathbf{k}\cdot\mathbf{x}} \right]. \quad (8.43)$$

The noise is thus present throughout the phase-space evolution: it enters because modes cross the moving boundary between the short- and long-wavelength sectors.

**Noise correlations** It is convenient to write  $U_{\phi,k} = u_k$  and  $U_{v,k} = \dot{u}_k$ . In the Bunch–Davies state the full phase-space noise kernel is

$$\begin{aligned} \frac{1}{2} \left\langle \left\{ \hat{\xi}_A(t, \mathbf{x}), \hat{\xi}_B(t', \mathbf{x}') \right\} \right\rangle &= \text{Re} \int \frac{d^3k}{(2\pi)^3} \dot{W}_t(k) \dot{W}_{t'}(k) \\ &\times U_{A,k}(t) U_{B,k}^*(t') e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')}, \quad A, B \in \{\phi, v\}. \end{aligned} \quad (8.44)$$

and  $\langle \hat{\xi}_A \rangle = 0$ . For definiteness, consider the sharp window

$$W_t(k) = \Theta(k_c(t) - k), \quad \dot{W}_t(k) = H k_c(t) \delta(k - k_c(t)). \quad (8.45)$$

The two delta functions in (??) imply

$$\delta(k_c(t) - k_c(t')) = \frac{1}{H k_c(t)} \delta(t - t'), \quad (8.46)$$

so the noise is local in time. For the massless Bunch–Davies mode function, one obtains

$$\frac{1}{2} \left\langle \left\{ \hat{\xi}_\phi(t, \mathbf{x}), \hat{\xi}_\phi(t', \mathbf{x}') \right\} \right\rangle = \frac{H^3}{4\pi^2} (1 + \epsilon^2) \frac{\sin(k_c(t)r)}{k_c(t)r} \delta(t - t'), \quad r = |\mathbf{x} - \mathbf{x}'|. \quad (8.47)$$

Moreover, at the crossing scale  $k = \epsilon a H$ ,

$$\frac{\dot{u}_k}{H u_k} = -\frac{\epsilon^2}{1 - i\epsilon}. \quad (8.48)$$

Consequently, the correlators involving  $\hat{\xi}_v$  are suppressed by powers of  $\epsilon^2$ : if  $D_{AB}$  denotes the local phase-space noise matrix, then

$$\frac{D_{\phi v}}{H D_{\phi\phi}} = \mathcal{O}(\epsilon^2), \quad \frac{D_{vv}}{H^2 D_{\phi\phi}} = \mathcal{O}(\epsilon^4). \quad (8.49)$$

At coincident points and to leading order for  $\epsilon \ll 1$ , the field noise therefore satisfies

$$\langle \xi_\phi(t, \mathbf{x}) \xi_\phi(t', \mathbf{x}) \rangle_\xi = \frac{H^3}{4\pi^2} \delta(t - t'), \quad (8.50)$$

where we have used the super-Hubble classicalization of the modes to replace the operator noise by a classical Gaussian stochastic variable.

**Overdamped reduction** Equations (??) and (??) are the appropriate starting point for the long-wavelength dynamics. Combining them into a single second-order equation would instead give

$$\ddot{\phi}_L + 3H\dot{\phi}_L + V'(\phi_L) = \dot{\xi}_\phi + 3H\xi_\phi + \xi_v. \quad (8.51)$$

For a sharp window,  $\xi_\phi$  is white noise, so it is preferable to perform the overdamped reduction directly in phase space rather than manipulate  $\dot{\xi}_\phi$ .

The velocity relaxes on the Hubble timescale, while a light field satisfying  $|V''| \ll H^2$  evolves more slowly. Adiabatically eliminating  $v_L$  from (??) gives

$$v_L \simeq -\frac{V'(\phi_L)}{3H} + \frac{\xi_v}{3H}. \quad (8.52)$$

Substituting this result into (??), and using the suppression (??), yields the leading stochastic equation

$$\dot{\phi}_L(t, \mathbf{x}) = -\frac{V'(\phi_L)}{3H} + \xi(t, \mathbf{x}), \quad \xi \equiv \xi_\phi, \quad (8.53)$$

with

$$\langle \xi(t, \mathbf{x}) \rangle_\xi = 0, \quad \langle \xi(t, \mathbf{x}) \xi(t', \mathbf{x}) \rangle_\xi = \frac{H^3}{4\pi^2} \delta(t - t'). \quad (8.54)$$

The drift describes the slow classical motion in the potential, while the noise describes the continual addition of modes across the time-dependent system–environment boundary. This is the leading Starobinsky Langevin equation [?, ?].

## 8.4 Fokker–Planck evolution in stochastic inflation

We have shown that, at leading order in the stochastic approximation, the long-wavelength field obeys the Langevin equation in (??) with Gaussian white noise. Henceforth we suppress the spatial dependence.

Using the general relation between a first-order Langevin equation with additive noise and the corresponding Fokker–Planck equation derived in (5.74), the probability density  $P(\phi, t)$  obeys

$$\frac{\partial P(\phi, t)}{\partial t} = \frac{1}{3H} \frac{\partial}{\partial \phi} [V'(\phi)P(\phi, t)] + \frac{H^3}{8\pi^2} \frac{\partial^2 P(\phi, t)}{\partial \phi^2}. \quad (8.55)$$

The first term describes the deterministic drift generated by the potential, while the second describes diffusion in field space due to modes continually entering the long-wavelength sector. To gain intuition, we may write (??) as a continuity equation, where the probability current is

$$J(\phi, t) = -\frac{V'(\phi)}{3H} P(\phi, t) - \frac{H^3}{8\pi^2} \frac{\partial P(\phi, t)}{\partial \phi}. \quad (8.56)$$

The first contribution transports probability along the classical slow-roll flow, while the second transports probability from regions of larger probability density to regions of smaller probability density.

Equation (??) is a first-order evolution equation in time,

$$\frac{\partial P}{\partial t} = \mathcal{L}_{\text{FP}} P, \quad (8.57)$$

with Fokker–Planck generator

$$\mathcal{L}_{\text{FP}} = \frac{1}{3H} \frac{\partial}{\partial \phi} V'(\phi) + \frac{H^3}{8\pi^2} \frac{\partial^2}{\partial \phi^2}, \quad (8.58)$$

where the first term is understood to act on the product  $V'(\phi)P$ . Once this generator has been obtained, its evolution can be solved without expanding the solution perturbatively in time. This was the key insight of Starobinsky and allows us to understand the true nature of the secular effects encountered in perturbation theory. Note the similarity of (??) to our discussion in Section 3.1, Which can be thought of as the stochastic equivalent of the quantum master equation.

**Free fields** As a first check, consider the free massless theory,  $V(\phi) = 0$ . The Fokker–Planck equation reduces to the ordinary diffusion equation,

$$\frac{\partial P}{\partial t} = \frac{H^3}{8\pi^2} \frac{\partial^2 P}{\partial \phi^2}. \quad (8.59)$$

For the sharply localized initial condition

$$P(\phi, t_0) = \delta(\phi - \phi_0), \quad (8.60)$$

the solution is<sup>32</sup>

$$P(\phi, t) = \frac{1}{\sqrt{4\pi D(t - t_0)}} \exp \left[ -\frac{(\phi - \phi_0)^2}{4D(t - t_0)} \right], \quad D = \frac{H^3}{8\pi^2}. \quad (8.61)$$

The mean remains fixed,

$$\langle \phi \rangle = \phi_0, \quad (8.62)$$

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<sup>32</sup>The observant reader will recognise in (??) the Wick rotation of the Schrödinger equation and in this solution the ground state of the quantum harmonic oscillator.

while the variance grows as

$$\langle (\phi - \phi_0)^2 \rangle = 2D(t - t_0) = \frac{H^3}{4\pi^2}(t - t_0). \quad (8.63)$$

For  $\phi_0 = 0$ , this reproduces the secular growth obtained directly from the non-perturbative mode functions in (??). The Fokker–Planck equation therefore reproduces the random-walk interpretation of the free massless field.

We next consider the free massive field with quadratic potential

$$V(\phi) = \frac{1}{2}m^2\phi^2. \quad (8.64)$$

Equation (??) becomes

$$\frac{\partial P}{\partial t} = \frac{m^2}{3H} \frac{\partial}{\partial \phi} [\phi P] + \frac{H^3}{8\pi^2} \frac{\partial^2 P}{\partial \phi^2}. \quad (8.65)$$

This is the Fokker–Planck equation for the Ornstein–Uhlenbeck process.

Rather than solving for the full probability distribution, we may directly derive the evolution of the second moment. Multiplying (??) by  $\phi^2$ , integrating over  $\phi$ , and assuming that the boundary terms vanish, one finds

$$\frac{d}{dt} \langle \phi^2 \rangle = -\frac{2m^2}{3H} \langle \phi^2 \rangle + \frac{H^3}{4\pi^2}. \quad (8.66)$$

For the initial condition

$$\langle \phi^2(t_0) \rangle = 0, \quad (8.67)$$

the solution is

$$\langle \phi^2(t) \rangle = \frac{3H^4}{8\pi^2 m^2} \left[ 1 - \exp\left(-\frac{2m^2}{3H}(t - t_0)\right) \right]. \quad (8.68)$$

This agrees with the result obtained directly from the exact massive mode functions in (??). The stochastic description therefore reproduces both the unbounded diffusion of the massless field and the finite late-time variance of the light massive field. The beauty of the Fokker–Planck equation is that it extends this ability of resumming the effects of a time-dependent system–environment split to an interacting theory.

**Stationary distribution** We now return to a generic potential  $V(\phi)$ . A stationary solution satisfies

$$\frac{\partial P_{\text{eq}}}{\partial t} = -\frac{\partial J_{\text{eq}}}{\partial \phi} = 0, \quad (8.69)$$

and therefore has a probability current independent of  $\phi$ . For a normalizable equilibrium distribution on the full real line, the current must vanish,  $J_{\text{eq}}(\phi) = 0$ . Using (??), this condition becomes

$$\frac{H^3}{8\pi^2} \frac{dP_{\text{eq}}}{d\phi} = -\frac{V'(\phi)}{3H} P_{\text{eq}}(\phi). \quad (8.70)$$

Solving this gives

$$P_{\text{eq}}(\phi) = \frac{1}{Z} \exp\left[-\frac{8\pi^2}{3H^4} V(\phi)\right], \quad (8.71)$$

where

$$Z = \int_{-\infty}^{+\infty} d\phi \exp \left[ -\frac{8\pi^2}{3H^4} V(\phi) \right] \quad (8.72)$$

is fixed by normalization.

A stationary probability distribution exists only if this integral converges. The free massless theory therefore has no normalizable equilibrium distribution, in agreement with its unbounded diffusion. By contrast, any potential that grows sufficiently rapidly as  $|\phi| \rightarrow \infty$  admits a normalizable stationary distribution. In such cases, the deterministic drift towards smaller values of the potential eventually balances the continual diffusion induced by modes entering the long-wavelength sector.

## 8.5 The quartic potential

Let's now focus on the quartic potential

$$V(\phi) = \frac{\lambda}{4!} \phi^4. \quad (8.73)$$

In this case, we can solve the Fokker-Planck equation in two complementary regimes: at early times in an expansion in  $\lambda N^2$ , and at very late times, around the equilibrium distribution we found previously.

**Perturbative solutions** The Fokker-Planck equation (??) implies a coupled hierarchy of equations for the moments of the probability distribution. Multiplying (??) by  $\phi^n$ , integrating over  $\phi$ , and assuming that boundary terms vanish, one finds

$$\frac{d}{dN} \langle \phi^n \rangle = -\frac{n\lambda}{18H^2} \langle \phi^{n+2} \rangle + \frac{n(n-1)H^2}{8\pi^2} \langle \phi^{n-2} \rangle, \quad (8.74)$$

where we switched to the number of e-folds  $N = H(t - t_0)$  as the time variable. This equation is not closed: the two-point function depends on the four-point function, whose evolution in turn depends on the six-point function, and so on. Nevertheless, the hierarchy may be solved recursively as an expansion in  $\lambda$ .

For a distribution initially localized at  $\phi = 0$ , the free massless theory gives

$$\langle \phi^{2n} \rangle_0 = (2n-1)!! \left( \frac{H^2 N}{4\pi^2} \right)^n, \quad \langle \phi^{2n+1} \rangle_0 = 0. \quad (8.75)$$

Substituting these moments into (??) for  $n = 2$  gives the first perturbative correction, and the procedure may then be iterated order by order. For the two-point function, the resulting expansion takes the form

$$\langle \phi^2(N) \rangle = H^2 \sum_{p=0}^{\infty} c_p \lambda^p N^{2p+1}, \quad (8.76)$$

where the coefficients  $c_p$  are determined recursively by the moment hierarchy. The first terms are

$$\langle \phi^2(N) \rangle = \frac{H^2}{4\pi^2} \left[ N - \frac{\lambda}{36\pi^2} N^3 + \mathcal{O}(\lambda^2 N^5) \right]. \quad (8.77)$$

These terms reproduce the leading infrared secular contributions obtained in perturbative quantum field theory. More generally, the stochastic hierarchy generates at order  $\lambda^p$  the leading contribution proportional to

$$\lambda^p N^{2p+1}. \quad (8.78)$$

As anticipated, the perturbative expansion is therefore controlled by the combination  $\lambda N^2$ , rather than by  $\lambda$  alone. Even for arbitrarily small coupling, strict perturbation theory eventually breaks down when  $N \sim \lambda^{-1/2}$ .

**Non-perturbative equilibrium solution** The moment expansion describes the early-time solution of the Fokker–Planck equation at fixed order in  $\lambda$ , but breaks down at late times. From the non-perturbative derivation of the stationary distribution, we already suspect that these secular effects may be an artefact of perturbation theory. To determine the late-time behavior therefore we instead look at the stationary distribution (??). For the quartic potential, it is

$$P_{\text{eq}}(\phi) = \frac{1}{Z} \exp \left[ -\frac{\pi^2 \lambda}{9H^4} \phi^4 \right], \quad (8.79)$$

which is normalizable for  $\lambda > 0$ , showing that the quartic interaction arrests the unbounded diffusion of the free massless field.

All equal-time moments of the long-wavelength field can now be evaluated without expanding in  $\lambda$ . Since the equilibrium distribution is even, all odd moments vanish,

$$\langle \phi^{2n+1} \rangle_{\text{eq}} = 0. \quad (8.80)$$

The even moments are

$$\langle \phi^{2n} \rangle_{\text{eq}} = \left( \frac{\pi^2 \lambda}{9H^4} \right)^{-n/2} \frac{\Gamma\left(\frac{2n+1}{4}\right)}{\Gamma\left(\frac{1}{4}\right)}. \quad (8.81)$$

In particular,

$$\langle \phi^2 \rangle_{\text{eq}} = \frac{3H^2}{\pi\sqrt{\lambda}} \frac{\Gamma\left(\frac{3}{4}\right)}{\Gamma\left(\frac{1}{4}\right)}, \quad \langle \phi^4 \rangle_{\text{eq}} = \frac{9H^4}{4\pi^2\lambda}. \quad (8.82)$$

The equilibrium two-point function is finite, but it is nonanalytic in the coupling. It therefore cannot be reproduced at any finite order in perturbation theory. Instead, it represents the result of resumming the entire series of secular terms generated by the moment hierarchy.

The relation between the perturbative and equilibrium descriptions can be made manifest by dimensional analysis. The full time-dependent solution must take the scaling form

$$\langle \phi^2(N) \rangle = \frac{H^2}{\sqrt{\lambda}} F\left(\sqrt{\lambda} N\right), \quad (8.83)$$

for some dimensionless function  $F$ . At early times,  $\sqrt{\lambda} N \ll 1$ , the perturbative hierarchy determines its Taylor expansion,

$$F(z) = \frac{z}{4\pi^2} - \frac{z^3}{144\pi^4} + \dots. \quad (8.84)$$

At late times, the equilibrium distribution determines its asymptotic value,

$$F(z) \xrightarrow{z \rightarrow \infty} \frac{3}{\pi} \frac{\Gamma\left(\frac{3}{4}\right)}{\Gamma\left(\frac{1}{4}\right)}. \quad (8.85)$$

The full Fokker–Planck evolution provides the interpolation between these two regimes. The time-dependent evolution close to the equilibrium distribution can be studied via the eigenvectors of the Fokker–Planck generator, but we omit this discussion here and refer the interested reader to the literature.

## 9 Conclusions and outlook

These lectures developed two complementary descriptions of open quantum systems. In the operator formalism, density matrices, partial traces and completely positive dynamical maps led, under Markovian and time-homogeneous assumptions, to the Gorini–Kossakowski–Sudarshan–Lindblad master equation and its Hamiltonian and dissipative contributions.

We then introduced the Schwinger–Keldysh path integral. Its doubled fields encode time- and anti-time-ordered products, while the  $r/a$  basis separates response from fluctuations. Integrating out an environment yields the Feynman–Vernon influence functional, which describes dissipation, noise and decoherence. Although its quadratic semiclassical limit often admits a Langevin description, the full effective action may contain genuinely quantum interactions beyond Gaussian stochastic dynamics.

We applied these ideas to inflation by constructing an open effective field theory for the Goldstone boson of broken time translations. This framework systematically incorporates environmental dissipation and fluctuations and can be used to calculate the power spectrum and primordial non-Gaussianity, including interactions with no purely classical stochastic counterpart.

Finally, a time-dependent split between long and short modes in de Sitter led to stochastic evolution for light fields. For interacting fields, the resulting Fokker–Planck equation resums perturbative contributions that grow secularly with time and may approach a regular late-time distribution. Such secular growth need not signal a physical instability; it may instead indicate that an evolution equation has been expanded beyond the regime of validity of perturbation theory.

**Operator and path-integral perspectives.** One motivation for presenting the operator and path-integral formalisms together is that each makes manifest structures that are comparatively obscure in the other. The density-matrix language is particularly well adapted to questions concerning states, entropy, quantum channels and complete positivity. The Schwinger–Keldysh language, by contrast, is naturally suited to locality, symmetry, effective field theory and the computation of general multi-time correlation functions. In principle, the two descriptions encode the same physics, but they organize the relevant information in very different ways. Several examples in these lectures illustrate what may be learned by bringing the two perspectives together. The elementary Schwinger–Keldysh conditions, such as the normalization of the generating functional and the positivity of the imaginary part of the effective action, already place important restrictions on an influence functional. Nevertheless, when a local-in-time influence functional admits a description in terms of a GKSL generator, complete positivity imposes stronger constraints. In the Gaussian examples studied above, positivity of the Kossakowski matrix relates the dissipative and noise coefficients and shows that a non-zero amount of friction requires a minimum amount of accompanying fluctuation. More generally, expanding the jump operators rather than directly expanding the influence functional offers a way of constructing positivity-improved effective theories in which complete positivity can remain exact even after truncation.

Conversely, the Schwinger–Keldysh formalism naturally generates a much broader class of observables than those most directly accessed by evolving a density matrix and evaluating equal-time expectation values. Functional differentiation with respect to the doubled sources produces correlators with different time orderings, while the  $r/a$  basis efficiently organizes response and fluctuation functions. This is particularly valuable in cosmology, where the main observables are correlation functions of quantum fields rather than the density matrix itself. The operator formalism supplies powerful structural constraints on the admissible evolution, while the path integral provides the language in which these constraints can be combined with spacetime

symmetries and effective field theory.

The applications considered here have been restricted almost entirely to the early universe and, more specifically, to inflation. They should therefore be viewed only as an initial illustration of the much wider role that open-system methods may play in cosmology. In many cosmological problems one observes only a restricted range of scales, species or collective variables, while the remaining degrees of freedom act as an environment. Coarse-graining then generically produces dissipative terms, fluctuations, memory effects and an evolution that need not be unitary when restricted to the retained variables.

An well studied example is the effective field theory of large-scale structure. Short-wavelength nonlinear modes are integrated out in order to obtain an effective description of the long-wavelength matter distribution. The result contains effective stresses and stochastic contributions and is therefore, in this sense, an open statistical theory. It is traditionally formulated directly in terms of equations of motion and their correlation functions, but an equivalent path-integral formulation should make it possible to organize its symmetries, response fields, stochastic terms and renormalization properties in a unified way.

There are many further possible applications throughout cosmology and astroparticle physics. The thermal history of the universe involves repeated examples of subsystems interacting with an evolving medium, including reheating, thermalization, particle production and freeze-out. Neutrino oscillations and transport, dark-matter kinetics, baryogenesis, gravitational-wave propagation through matter and the evolution of cosmological perturbations in the presence of additional sectors may all, in suitable regimes, benefit from an open-system description. Not every such problem requires a fully quantum treatment, but the operator and Schwinger–Keldysh methods provide a common framework in which quantum, statistical and semiclassical limits can be distinguished.

**Open quantum field theories.** At a more formal level, a central open problem is to develop a systematic understanding of open quantum field theories. Effective field theory teaches us to write the most general local action compatible with the degrees of freedom, symmetries and power counting of a problem. For an open theory this prescription must be supplemented by the Schwinger–Keldysh constraints associated with normalization, Hermiticity and positivity, and, whenever a Markovian description is appropriate, by the requirement of complete positivity. Additional constraints should follow from fundamental properties of the full system evolution, such as unitarity, locality and causality.

In particular, it would be useful to characterize which local Schwinger–Keldysh actions define admissible quantum evolutions. The condition  $\text{Im } S_{\text{eff}} \geq 0$  is necessary but is not, in general, sufficient to guarantee complete positivity of a corresponding dynamical map. The jump-operator construction discussed in these notes gives one possible starting point, but several questions remain. Complete positivity can become obscured by derivative expansions, field redefinitions, the integration of auxiliary variables or the truncation of an operator basis. It would be valuable to formulate criteria that are both sufficiently general for field theory and sufficiently practical to be imposed on an effective action.

Generic open systems are also non-Markovian. Integrating out an environment ordinarily produces kernels that are nonlocal in time and retain information about the past history of the system. A local master equation or local influence functional is justified only when the environmental correlation time is sufficiently short compared with the scales being probed. Understanding the systematic expansion around this limit, and developing an effective-field-theory power counting for memory effects, would extend the framework substantially. In cosmology this issue is particularly important because the background is time dependent and a clean separation of timescales may hold only during part of the evolution.

A related question concerns the dependence of the effective theory on the choice of system–environment split. Which variables are retained, and which are integrated out, is not unique.

In stochastic inflation the separation between long and short modes evolves continuously as modes cross the coarse-graining scale. More generally, changing this split should induce a transformation of the effective dissipative, noise and interaction coefficients. Clarifying this dependence may lead to a notion of renormalization-group evolution acting not only on ordinary couplings but also on the choice of reduced description itself.

**Gravity and gauge theories.** A particularly important challenge is the systematic formulation of dynamical gravity as an open quantum theory. The Schwinger–Keldysh path integral appears to be a natural starting point because diffeomorphism invariance and spacetime locality can be implemented directly at the level of a doubled action. Nevertheless, the operator formalism suggests that additional constraints related to positivity and the structure of the reduced state must also be present.

Several conceptual difficulties arise already in defining the problem. In gravity, the division of the Hilbert space into a subsystem and an environment may be complicated by gauge constraints, and local regions do not generally admit a simple tensor-product decomposition. The Hamiltonian and momentum constraints must remain compatible with the reduced dynamics, while the distinction between system and environment may itself depend on the geometry. Horizons provide a natural source of inaccessible degrees of freedom, but they also introduce questions concerning observer dependence, entropy and information loss. A satisfactory open effective theory of gravity should reconcile these features with locality, causality, diffeomorphism invariance and, whenever appropriate, complete positivity.

Such a framework could have applications to semiclassical gravity, black-hole evaporation, cosmological horizons and the stochastic description of metric fluctuations. It could also help clarify when the semiclassical Einstein equations should be supplemented by noise and dissipation, and how those terms are constrained by the quantum state of matter. More ambitiously, it may provide an effective language for discussing situations in which only a restricted set of gravitational observables is operationally accessible.

**Thermal systems, hydrodynamics and quantum information.** Another major direction that we have not developed is the relation between open systems and thermal physics. When the environment is in equilibrium, the Kubo–Martin–Schwinger condition imposes nontrivial relations between response and fluctuation functions. At the level of the Schwinger–Keldysh effective action and for local equilibrium, these relations are encoded by a dynamical KMS symmetry. The resulting fluctuation–dissipation relations connect coefficients that would otherwise appear independent and provide an important additional organizing principle for open effective theories.

These ideas play a central role in dissipative hydrodynamics. Hydrodynamics is a universal effective description of systems close to local thermal equilibrium, including systems for which no weakly coupled microscopic description is available. Its Schwinger–Keldysh formulation naturally incorporates dissipation, fluctuations, conservation laws and the dynamical KMS symmetry. It therefore provides one of the most developed examples of an interacting open effective field theory. Since hydrodynamics also describes strongly coupled quantum field theories and admits a gravitational realization through holography, open-system methods have become increasingly relevant to the study of gauge–gravity duality.

There are equally important connections with quantum information. Decoherence, thermalization, scrambling and the growth of quantum chaos concern the way information is redistributed among degrees of freedom that may no longer be individually accessible. Quantum channels and reduced density matrices provide a natural language for these processes, while generalized Schwinger–Keldysh contours can be used to compute out-of-time-ordered correlators and other diagnostics of operator growth. It would be interesting to understand more systematically how the effective description of open quantum fields constrains entanglement generation, information

transport and scrambling.

The quantum-to-classical transition deserves particular care in cosmological applications and has been repeatedly studied over the years. The emergence of a Langevin or Fokker–Planck equation does not by itself establish that the underlying fluctuations have become classical. Squeezing, decoherence, the suppression of interference and the emergence of a positive probabilistic description are related but distinct phenomena. A framework that keeps both the density matrix and the Schwinger–Keldysh effective action in view may help disentangle these notions and identify which genuinely quantum correlations are discarded in a semiclassical stochastic approximation.

**Renormalization and universality.** Finally, much remains to be understood about renormalization in open quantum field theory. Radiative corrections will generically generate every operator compatible with the symmetries and Schwinger–Keldysh constraints. It is therefore important to determine whether physically interesting subclasses of open theories are closed under renormalization. For example, one may ask whether Markovianity, complete positivity or a particular jump-operator structure can be preserved along a renormalization-group flow, or whether integrating out additional scales inevitably produces non-locality and more general multi-advanced-field interactions.

It would also be interesting to identify the endpoints of open-system renormalization-group flows. In closed quantum field theory, fixed points and universality classes provide a powerful classification of long-distance behaviour. Open systems may possess a richer set of possibilities, including thermal, driven, dissipative and genuinely non-equilibrium fixed points. One would like to know which quantities characterize these endpoints, whether analogues of monotonicity theorems exist, and how positivity, entropy production and information loss constrain the flow. Understanding these questions could eventually place open quantum field theory on a footing comparable to that of ordinary effective field theory.

The general lesson of these lectures is that open-system effects should not be regarded merely as small corrections caused by an imperfect isolation of an otherwise closed system. In many settings, including cosmology, the observables of interest intrinsically refer only to a subset of the available degrees of freedom. Their effective dynamics is then naturally described by a reduced state, an influence functional or both. The operator and path-integral formalisms emphasize different aspects of this description, and their combination offers a powerful route towards effective theories that incorporate dissipation, fluctuations and quantum correlations while remaining consistent with the fundamental principles of quantum mechanics. The examples studied here, which represent only a glimpse of what has been studied in literature, suggest that open quantum systems may provide an increasingly important language for quantum field theory, gravity and cosmology.

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## A Inverting the quadratic Schwinger–Keldysh kernel

In this appendix we show in some detail how the propagators of a quadratic Schwinger–Keldysh theory are obtained by inverting the corresponding kernel. We work in momentum space, where translation invariance makes different Fourier modes independent. For simplicity we suppress the momentum label  $(\omega, \mathbf{k})$  whenever no confusion can arise.

Consider a generic Gaussian Schwinger–Keldysh action for a real field  $\phi$ , written in the  $r/a$  basis as

$$S_{\text{SK}} = \int \frac{d\omega}{2\pi} \frac{d^d k}{(2\pi)^d} \left[ \phi_a(-\omega, -\mathbf{k}) \mathcal{D}_R(\omega, \mathbf{k}) \phi_r(\omega, \mathbf{k}) + \phi_r(-\omega, -\mathbf{k}) \mathcal{D}_A(\omega, \mathbf{k}) \phi_a(\omega, \mathbf{k}) + \frac{i}{2} \phi_a(-\omega, -\mathbf{k}) \mathcal{N}(\omega, \mathbf{k}) \phi_a(\omega, \mathbf{k}) \right]. \quad (\text{A.1})$$

Here  $\mathcal{D}_R$  and  $\mathcal{D}_A$  are the inverse retarded and advanced kernels, while  $\mathcal{N}$  is the Keldysh or noise kernel. For a real field, one has

$$\mathcal{D}_A(\omega, \mathbf{k}) = \mathcal{D}_R(\omega, \mathbf{k})^*, \quad \mathcal{N}(\omega, \mathbf{k})^* = \mathcal{N}(\omega, \mathbf{k}). \quad (\text{A.2})$$

It is convenient to combine the two fields into the column vector

$$\Phi(\omega, \mathbf{k}) := \begin{pmatrix} \phi_r(\omega, \mathbf{k}) \\ \phi_a(\omega, \mathbf{k}) \end{pmatrix}. \quad (\text{A.3})$$

Then the action may be written as

$$S_{\text{SK}} = \frac{1}{2} \int \frac{d\omega}{2\pi} \frac{d^d k}{(2\pi)^d} \Phi^\dagger(-\omega, -\mathbf{k}) \mathbb{K}(\omega, \mathbf{k}) \Phi(\omega, \mathbf{k}), \quad (\text{A.4})$$

with kernel matrix

$$\mathbb{K}(\omega, \mathbf{k}) = \begin{pmatrix} 0 & \mathcal{D}_A(\omega, \mathbf{k}) \\ \mathcal{D}_R(\omega, \mathbf{k}) & i \mathcal{N}(\omega, \mathbf{k}) \end{pmatrix}. \quad (\text{A.5})$$

The vanishing of the  $rr$  entry is the usual Schwinger–Keldysh normalization constraint that  $S_{\text{SK}}[\phi_r, 0] = 0$ .

**Propagator matrix** The propagators are obtained by inverting the quadratic kernel. Let

$$\mathbb{G}(\omega, \mathbf{k}) := \begin{pmatrix} G_K(\omega, \mathbf{k}) & G_R(\omega, \mathbf{k}) \\ G_A(\omega, \mathbf{k}) & G_{aa}(\omega, \mathbf{k}) \end{pmatrix}, \quad (\text{A.6})$$

where

$$G_K(\omega, \mathbf{k}) := \langle\langle \phi_r(\omega, \mathbf{k}) \phi_r(-\omega, -\mathbf{k}) \rangle\rangle, \quad (\text{A.7})$$

$$G_R(\omega, \mathbf{k}) := \langle\langle \phi_r(\omega, \mathbf{k}) \phi_a(-\omega, -\mathbf{k}) \rangle\rangle, \quad (\text{A.8})$$

$$G_A(\omega, \mathbf{k}) := \langle\langle \phi_a(\omega, \mathbf{k}) \phi_r(-\omega, -\mathbf{k}) \rangle\rangle, \quad (\text{A.9})$$

$$G_{aa}(\omega, \mathbf{k}) := \langle\langle \phi_a(\omega, \mathbf{k}) \phi_a(-\omega, -\mathbf{k}) \rangle\rangle. \quad (\text{A.10})$$

At the formal Gaussian level,  $\mathbb{G}$  is the inverse of  $\mathbb{K}$ , up to the conventional factors of  $i$  already absorbed in our definitions. Thus

$$\mathbb{K}(\omega, \mathbf{k}) \mathbb{G}(\omega, \mathbf{k}) = 1. \quad (\text{A.11})$$

Writing this out explicitly gives

$$\begin{pmatrix} 0 & \mathcal{D}_A \\ \mathcal{D}_R & i\mathcal{N} \end{pmatrix} \begin{pmatrix} G_K & G_R \\ G_A & G_{aa} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (\text{A.12})$$

Multiplying the matrices gives four equations:

$$\mathcal{D}_A G_A = 1, \quad (\text{A.13})$$

$$\mathcal{D}_A G_{aa} = 0, \quad (\text{A.14})$$

$$\mathcal{D}_R G_K + i\mathcal{N} G_A = 0, \quad (\text{A.15})$$

$$\mathcal{D}_R G_R + i\mathcal{N} G_{aa} = 1. \quad (\text{A.16})$$

Assuming  $\mathcal{D}_R$  and  $\mathcal{D}_A$  are invertible, this immediately yields

$$G_R(\omega, \mathbf{k}) = \frac{1}{\mathcal{D}_R(\omega, \mathbf{k})}, \quad (\text{A.17})$$

$$G_A(\omega, \mathbf{k}) = \frac{1}{\mathcal{D}_A(\omega, \mathbf{k})}, \quad (\text{A.18})$$

$$G_{aa}(\omega, \mathbf{k}) = 0, \quad (\text{A.19})$$

$$G_K(\omega, \mathbf{k}) = -i\mathcal{N}(\omega, \mathbf{k}) \frac{1}{\mathcal{D}_R(\omega, \mathbf{k})\mathcal{D}_A(\omega, \mathbf{k})}. \quad (\text{A.20})$$

This is the general Gaussian result quoted in the main text.

### A.1 Time-domain retarded propagator

Let us now derive the retarded propagator in the time domain. Starting from

$$G_R(\omega, \mathbf{k}) = \frac{1}{-\omega^2 - i\gamma\omega + \omega_{\mathbf{k}}^2}, \quad \omega_{\mathbf{k}}^2 := \mathbf{k}^2 + m^2, \quad (\text{A.21})$$

we define

$$G_R(t, \mathbf{k}) = \int \frac{d\omega}{2\pi} e^{-i\omega t} G_R(\omega, \mathbf{k}). \quad (\text{A.22})$$

The poles are located at

$$\omega_{\pm}(\mathbf{k}) = -\frac{i\gamma}{2} \pm \sqrt{\omega_{\mathbf{k}}^2 - \frac{\gamma^2}{4}}, \quad (\text{A.23})$$

so that

$$G_R(\omega, \mathbf{k}) = -\frac{1}{(\omega - \omega_+)(\omega - \omega_-)}. \quad (\text{A.24})$$

For  $\gamma > 0$ , both poles lie in the closed lower half of the complex  $\omega$ -plane. Therefore, when  $t < 0$ , the contour may be closed in the upper half-plane and no poles are enclosed, giving

$$G_R(t, \mathbf{k}) = 0, \quad t < 0. \quad (\text{A.25})$$

For  $t > 0$ , the contour is instead closed in the lower half-plane and the retarded propagator is obtained by summing the residues of the enclosed poles. Its explicit form depends on whether the mode is underdamped or overdamped. To make this explicit, we separate the different regimes.

**Underdamped regime.** Suppose first that

$$\omega_{\mathbf{k}}^2 > \frac{\gamma^2}{4}. \quad (\text{A.26})$$

It is convenient to define

$$\Omega_{\mathbf{k}} := \sqrt{\omega_{\mathbf{k}}^2 - \frac{\gamma^2}{4}}, \quad (\text{A.27})$$

so that

$$\omega_{\pm} = -\frac{i\gamma}{2} \pm \Omega_{\mathbf{k}}. \quad (\text{A.28})$$

The propagator may then be decomposed as

$$G_R(\omega, \mathbf{k}) = -\frac{1}{2\Omega_{\mathbf{k}}} \left( \frac{1}{\omega - \omega_+} - \frac{1}{\omega - \omega_-} \right). \quad (\text{A.29})$$

For  $t > 0$ , the residue theorem gives

$$G_R(t, \mathbf{k}) = -i\theta(t) \left[ -\frac{1}{2\Omega_{\mathbf{k}}} e^{-i\omega_+ t} + \frac{1}{2\Omega_{\mathbf{k}}} e^{-i\omega_- t} \right] = \theta(t) e^{-\gamma t/2} \frac{\sin(\Omega_{\mathbf{k}} t)}{\Omega_{\mathbf{k}}}. \quad (\text{A.30})$$

Thus, in the underdamped regime, the response oscillates with frequency  $\Omega_{\mathbf{k}}$  while its amplitude decays as  $e^{-\gamma t/2}$ .

**Overdamped regime.** Suppose instead that

$$\omega_{\mathbf{k}}^2 < \frac{\gamma^2}{4}. \quad (\text{A.31})$$

For clarity, we define

$$\Lambda_{\mathbf{k}} := \sqrt{\frac{\gamma^2}{4} - \omega_{\mathbf{k}}^2} \in \mathbb{R}, \quad (\text{A.32})$$

so that

$$\omega_{\pm} = -\frac{i\gamma}{2} \pm i\Lambda_{\mathbf{k}} = -i \left( \frac{\gamma}{2} \mp \Lambda_{\mathbf{k}} \right). \quad (\text{A.33})$$

The two poles are now purely imaginary and lie in the lower half-plane. Repeating the same contour argument gives

$$\begin{aligned}
G_R(t, \mathbf{k}) &= \theta(t) e^{-\gamma t/2} \frac{\sinh(\Lambda_{\mathbf{k}} t)}{\Lambda_{\mathbf{k}}} \\
&= \theta(t) \frac{e^{-(\gamma/2 - \Lambda_{\mathbf{k}})t} - e^{-(\gamma/2 + \Lambda_{\mathbf{k}})t}}{2\Lambda_{\mathbf{k}}}.
\end{aligned} \tag{A.34}$$

The response is therefore a sum of two decaying exponentials, with decay rates

$$\Gamma_{\pm} = \frac{\gamma}{2} \pm \Lambda_{\mathbf{k}}. \tag{A.35}$$

The late-time behaviour is controlled by the slower rate

$$\Gamma_{\text{slow}} = \frac{\gamma}{2} - \Lambda_{\mathbf{k}}. \tag{A.36}$$

In the strongly overdamped regime,  $\omega_{\mathbf{k}} \ll \gamma$ , this becomes

$$\Gamma_{\text{slow}} = \frac{\gamma}{2} - \sqrt{\frac{\gamma^2}{4} - \omega_{\mathbf{k}}^2} \simeq \frac{\omega_{\mathbf{k}}^2}{\gamma}. \tag{A.37}$$

For a massless field, this gives the diffusive scaling

$$\Gamma_{\text{slow}} \simeq \frac{\mathbf{k}^2}{\gamma}. \tag{A.38}$$

The critically damped case,  $\omega_{\mathbf{k}}^2 = \gamma^2/4$ , is obtained continuously by taking  $\Omega_{\mathbf{k}} \rightarrow 0$  in the underdamped expression, or equivalently  $\Lambda_{\mathbf{k}} \rightarrow 0$  in the overdamped expression. Using  $\sin z/z \rightarrow 1$  or  $\sinh z/z \rightarrow 1$ , one finds

$$G_R(t, \mathbf{k}) = \theta(t) t e^{-\gamma t/2}. \tag{A.39}$$

These expressions make causality manifest through the factor  $\theta(t)$ . For every mode with  $\omega_{\mathbf{k}}^2 > 0$ , the retarded response decays at late times, although the relaxation timescale is mode dependent and can become parametrically long in the overdamped regime. The exceptional massless homogeneous mode has

$$m = 0, \quad \mathbf{k} = 0, \quad \omega_{\mathbf{k}} = 0, \tag{A.40}$$

for which

$$G_R(t, \mathbf{0}) = \theta(t) \frac{1 - e^{-\gamma t}}{\gamma}. \tag{A.41}$$

This approaches a constant rather than vanishing at late times.